

THE SPECIAL THEORY OF RELATIVITY



This 12-foot tall statue of Albert Einstein is located at the headquarters of the National Academy of Sciences in Washington DC. The page in his hand shows three equations that he discovered: the fundamental equation of general relativity, which revolutionized our understanding of gravity; the equation for the photoelectric effect, which opened the path to the development of quantum mechanics; and the equation for mass-energy equivalence, which is the cornerstone of his special theory of relativity. ROGER L. WOLLENBERG / UPI / Newscom

Einstein's special theory of relativity and Planck's quantum theory burst forth on the physics scene almost simultaneously during the first decade of the 20th century. Both theories caused profound changes in the way we view our universe at its most fundamental level.

In this chapter, we study the special theory of relativity.* This theory has a completely undeserved reputation as being so exotic that few people can understand it. On the contrary, special relativity is basically a system of kinematics and dynamics, based on a set of postulates that are different from those of classical physics. The resulting formalism is not much more complicated than Newton's laws, but it does lead to several predictions that seem to go against our common sense. Even so, the special theory of relativity has been carefully and thoroughly tested by experiment and found to be correct in all its predictions.

We first review the classical relativity of Galileo and Newton, and then we show why Einstein proposed to replace it. We then discuss the mathematical aspects of special relativity, the predictions of the theory, and finally some experimental tests.

2.1 CLASSICAL RELATIVITY

A “theory of relativity” is in effect a way for observers in different frames of reference to compare the results of their observations. For example, consider an observer in a car parked by a highway near a large rock. To this observer, the rock is at rest. Another observer, who is moving along the highway in a car, sees the rock rush past as the car drives by. To this observer, the rock appears to be moving. A theory of relativity provides the conceptual framework and mathematical tools that enable the two observers to transform a statement such as “rock is at rest” in one frame of reference to the statement “rock is in motion” in another frame of reference. More generally, relativity gives a means for expressing the laws of physics in different frames of reference.

The mathematical basis for comparing the two descriptions is called a *transformation*. Figure 2.1 shows an abstract representation of the situation.

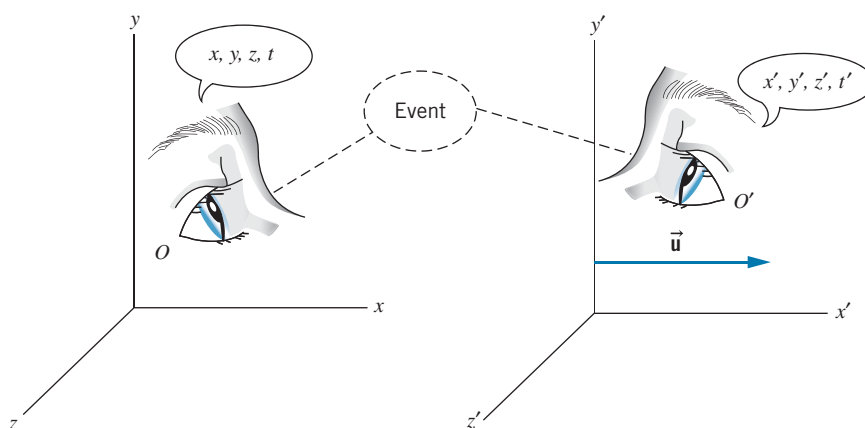


FIGURE 2.1 Two observers O and O' observe the same event. O' moves relative to O with a constant velocity $\vec{u}$.

*The *general* theory of relativity, which is covered briefly in Chapter 15, deals with “curved” coordinate systems, in which gravity is responsible for the curvature. Here we discuss the *special* case of the more familiar “flat” coordinate systems.

Two observers O and O' are each at rest in their own frames of reference but move relative to one another with constant velocity $\vec{u}$. (O and O' refer both to the observers and their reference frames or coordinate systems.) They observe the same *event*, which happens at a particular point in space and a particular time, such as a collision between two particles. According to O , the space and time coordinates of the event are x, y, z, t , while according to O' the coordinates of the *same event* are x', y', z', t' . The two observers use calibrated meter sticks and synchronized clocks, so any differences between the coordinates of the two events are due to their different frames of reference and not to the measuring process. We simplify the discussion by assuming that the relative velocity $\vec{u}$ always lies along the common xx' direction, as shown in Figure 2.1, and we let $\vec{u}$ represent the velocity of O' as measured by O (and thus O' would measure velocity $-\vec{u}$ for O).

In this discussion, we make a particular choice of the kind of reference frames inhabited by O and O' . We assume that each observer has the capacity to test Newton's laws and finds them to hold in that frame of reference. For example, each observer finds that an object at rest or moving with a constant velocity remains in that state unless acted upon by an external force (Newton's first law, the law of inertia). Such frames of reference are called *inertial frames*. An observer in interstellar space floating in a nonrotating rocket with the engines off would be in an inertial frame of reference. An observer at rest on the surface of the Earth is *not* in an inertial frame, because the Earth is rotating about its axis and orbiting about the Sun; however, the accelerations associated with those motions are so small that we can usually regard our reference frame as approximately inertial. (The noninertial reference frame at the Earth's surface does produce important and often spectacular effects, such as the circulation of air around centers of high or low pressure.) An observer in an accelerating car, a rotating merry-go-round, or a descending roller coaster is *not* in an inertial frame of reference!

We now derive the classical or *Galilean* transformation that relates the coordinates x, y, z, t to x', y', z', t' . We assume as a postulate of classical physics that $t = t'$, that is, time is the same for all observers. We also assume for simplicity that the coordinate systems are chosen so that their origins coincide at $t = 0$. Consider an object in O' at the coordinates x', y', z' (Figure 2.2). According to O , the y and z coordinates are the same as those in O' . Along the x direction, O would observe the object at $x = x' + ut$. We therefore have the *Galilean coordinate transformation*

$$x' = x - ut \quad y' = y \quad z' = z \quad (2.1)$$

To find the velocities of the object as observed by O and O' , we take the derivatives of these expressions with respect to t' on the left and with respect to t on the right (which we can do because we have assumed $t' = t$). This gives the *Galilean velocity transformation*

$$v'_x = v_x - u \quad v'_y = v_y \quad v'_z = v_z \quad (2.2)$$

In a similar fashion, we can take the derivatives of Eq. 2.2 with respect to time and obtain relationships between the accelerations

$$a'_x = a_x \quad a'_y = a_y \quad a'_z = a_z \quad (2.3)$$

Equation 2.3 shows again that Newton's laws hold for both observers. As long as u is constant ($du/dt = 0$), the observers measure identical accelerations and agree on the results of applying $\vec{F} = m\vec{a}$.

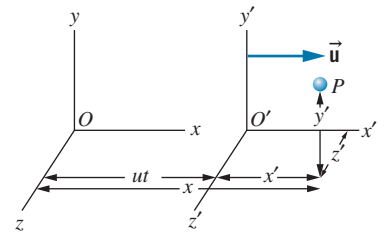


FIGURE 2.2 An object or event at point P is at coordinates x', y', z' with respect to O' . The x coordinate measured by O is $x = x' + ut$. The y and z coordinates in O are the same as those in O' .

Example 2.1

Two cars are traveling at constant speed along a road in the same direction. Car A moves at 60 km/h and car B moves at 40 km/h, each measured relative to an observer on the ground (Figure 2.3a). What is the speed of car A relative to car B?

Solution

Let O be the observer on the ground, who observes car A to move at $v_x = 60$ km/h. Assume O' to be moving with car B at $u = 40$ km/h. Then

$$\begin{aligned} v'_x &= v_x - u = 60 \text{ km/h} - 40 \text{ km/h} \\ &= 20 \text{ km/h} \end{aligned}$$

Figure 2.3b shows the situation as observed by O' .

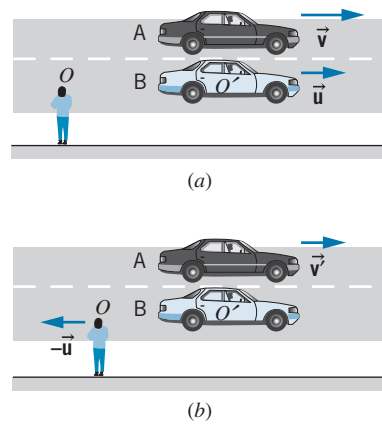


FIGURE 2.3 Example 2.1. (a) As observed by O at rest on the ground. (b) As observed by O' in car B.

Example 2.2

An airplane is flying due east relative to still air at a speed of 320 km/h. There is a 65 km/h wind blowing toward the north, as measured by an observer on the ground. What is the velocity of the plane measured by the ground observer?

Solution

Let O be the observer on the ground, and let O' be an observer who is moving with the wind, for example, a balloonist (Figure 2.4). Then $u = 65$ km/h, and (because our equations are set up with $\vec{u}$ in the xx' direction) we must choose the xx' direction to be to the north. In this case, we know the velocity with respect to O' ; taking the y direction to the east, we have $v'_x = 0$ and $v'_y = 320$ km/h. Using Eq. 2.2 we obtain

$$\begin{aligned} v_x &= v'_x + u = 0 + 65 \text{ km/h} = 65 \text{ km/h} \\ v_y &= v'_y = 320 \text{ km/h} \end{aligned}$$

Relative to the ground, the plane flies in a direction determined by $\phi = \tan^{-1}(65 \text{ km/h})/(320 \text{ km/h}) = 11.5^\circ$, or 11.5° north of east.

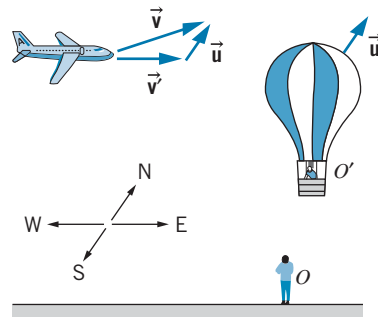


FIGURE 2.4 Example 2.2. As observed by O at rest on the ground, the balloon drifts north with the wind, while the plane flies north of east.

Example 2.3

A swimmer capable of swimming at a speed c in still water is swimming in a stream in which the current is u (which we assume to be less than c). Suppose the swimmer swims upstream a distance L and then returns downstream to the starting point. Find the time necessary to make the round trip, and compare it with the time to swim across the stream a distance L and return.

Solution

Let the frame of reference of O be the ground and the frame of reference of O' be the water, moving at speed u (Figure 2.5a). The swimmer always moves at speed c relative to the water, and thus $v'_x = -c$ for the upstream swim. (Remember that u always defines the *positive* x direction.) According to Eq. 2.2, $v'_x = v_x - u$, so

$v_x = v'_x + u = u - c$. (As expected, the velocity relative to the ground has magnitude smaller than c ; it is also *negative*, since the swimmer is swimming in the negative x direction, so $|v_x| = c - u$.) Therefore, $t_{\text{up}} = L/(c - u)$. For the downstream swim, $v'_x = c$, so $v_x = u + c$, $t_{\text{down}} = L/(c + u)$, and the total time is

$$\begin{aligned} t &= \frac{L}{c + u} + \frac{L}{c - u} = \frac{L(c - u) + L(c + u)}{c^2 - u^2} \\ &= \frac{2Lc}{c^2 - u^2} = \frac{2L}{c} \frac{1}{1 - u^2/c^2} \end{aligned} \quad (2.4)$$

To swim directly across the stream, the swimmer's efforts must be directed somewhat upstream to counter the effect of the current (Figure 2.5b). That is, in the frame of reference of O we would like to have $v_x = 0$, which requires $v'_x = -u$ according to Eq. 2.2. Since the speed relative to the water is always c , $\sqrt{v'^2_x + v'^2_y} = c$; thus $v'_y = \sqrt{c^2 - v'^2_x} = \sqrt{c^2 - u^2}$, and the round-trip time is

$$t = 2t_{\text{across}} = \frac{2L}{\sqrt{c^2 - u^2}} = \frac{2L}{c} \frac{1}{\sqrt{1 - u^2/c^2}} \quad (2.5)$$

Notice the difference *in form* between this result and the result for the upstream–downstream swim, Eq. 2.4.

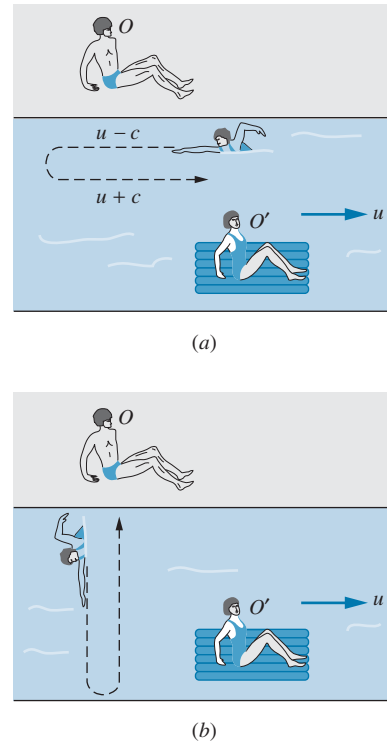
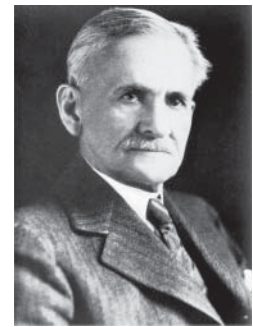


FIGURE 2.5 Example 2.3. The motion of a swimmer as seen by observer O at rest on the bank of the stream. Observer O' moves with the stream at speed u .

2.2 THE MICHELSON–MORLEY EXPERIMENT

We have seen how Newton's laws remain valid with respect to a Galilean transformation that relates the description of the motion of an object in one reference frame to that in another reference frame. It is then interesting to ask whether the same transformation rules apply to the motion of a light beam. According to the Galilean transformation, a light beam moving relative to observer O' in the x' direction at speed $c = 299,792,458$ m/s would have a speed of $c + u$ relative to O . Direct high-precision measurements of the speed of light beams have become possible in recent years (as we discuss later in this chapter), but in the 19th century it was necessary to devise a more indirect measurement of the speed of light according to different observers in relative motion.

Suppose the swimmer in Example 2.3 is replaced by a light beam. Observer O' is in a frame of reference in which the speed of light is c , and the frame of reference of observer O' is in motion relative to observer O . What is the speed of light as measured by observer O ? If the Galilean transformation is correct, we should expect to see a difference between the speed of the light beam according to O and O' and therefore a time difference between the upstream–downstream and cross-stream times, as in Example 2.3.



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Albert A. Michelson (1852–1931, United States). He spent 50 years doing increasingly precise experiments with light, for which he became the first U.S. citizen to win the Nobel Prize in physics (1907).

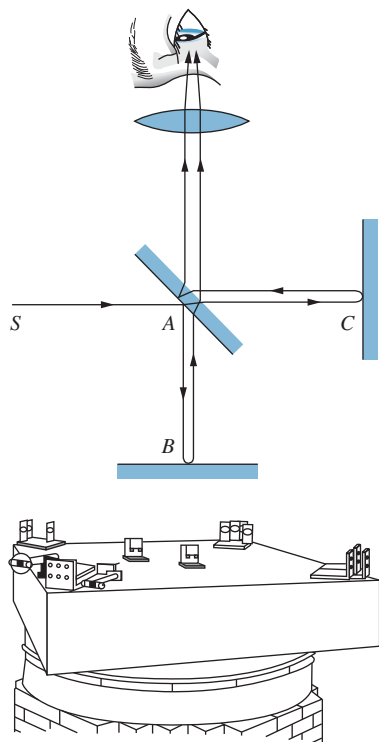


FIGURE 2.6 (Top) Beam diagram of Michelson interferometer. Light from source S is split at A by the half-silvered mirror; one part is reflected by the mirror at B and the other is reflected at C . The beams are then recombined for observation of the interference. (Bottom) Michelson's apparatus. To improve sensitivity, the beams were reflected to travel each leg of the apparatus eight times, rather than just twice. To reduce vibrations from the surroundings, the interferometer was mounted on a 1.5-m square stone slab floating in a pool of mercury.

Physicists in the 19th century postulated just such a situation—a preferred frame of reference in which the speed of light has the precise value of c and other frames in relative motion in which the speed of light would differ, according to the Galilean transformation. The preferred frame, like that of observer O' in Example 2.3, is one that is at rest with respect to the medium in which light propagates at c (like the water of that example). What is the medium of propagation for light waves? It was inconceivable to physicists of the 19th century that a wave disturbance could propagate without a medium (consider mechanical waves such as sound or seismic waves, for example, which propagate due to mechanical forces in the medium). They postulated the existence of an invisible, massless medium, called the *ether*, which filled all space, was undetectable by any mechanical means, and existed solely for the propagation of light waves. It seemed reasonable then to obtain evidence for the ether by measuring the velocity of the Earth moving through the ether. This could be done in the geometry of Figure 2.5 by measuring the difference between the upstream–downstream and cross-stream times for a light wave. The calculation based on Galilean relativity would then give the relative velocity $\vec{u}$ between O (in the Earth's frame of reference) and the ether.

The first detailed and precise search for the preferred frame was performed in 1887 by the American physicist Albert A. Michelson and his associate E. W. Morley. Their apparatus consisted of a specially designed Michelson interferometer, illustrated in Figure 2.6. A monochromatic beam of light is split in two; the two beams travel different paths and are then recombined. Any phase difference between the combining beams causes bright and dark bands or “fringes” to appear, corresponding, respectively, to constructive and destructive interference, as shown in Figure 2.7.

There are two contributions to the phase difference between the beams. The first contribution comes from the path difference $AB - AC$; one of the beams may travel a longer distance. The second contribution, which would still be present even if the path lengths were equal, comes from the time difference between the upstream–downstream and cross-stream paths (as in Example 2.3) and indicates the motion of the Earth through the ether. Michelson and Morley used a clever method to isolate this second contribution—they rotated the entire apparatus by 90° ! The rotation doesn't change the first contribution to the phase difference (because the lengths AB and AC don't change), but the second contribution changes sign, because what was an upstream–downstream path before the rotation becomes a cross-stream path after the rotation. As the apparatus is rotated through 90° , the fringes should change from bright to dark and back again as the phase difference changes. Each change from bright to dark represents a phase change of 180° (a half cycle), which corresponds to a time difference of a half period (about 10^{-15} s for visible light). Counting the number of fringe changes thus gives a measure of the time difference between the paths, which in turn gives the relative velocity u . (See Problem 3.)

When Michelson and Morley performed their experiment, there was no observable change in the fringe pattern—they deduced a shift of less than 0.01 fringe, corresponding to a speed of the Earth through the ether of at most 5 km/s. As a last resort, they reasoned that perhaps the orbital motion of the Earth just happened to cancel out the overall motion through the ether.

If this were true, six months later (when the Earth would be moving in its orbit in the opposite direction) the cancellation should not occur. When they repeated the experiment six months later, they again obtained a null result. In no experiment were Michelson and Morley able to detect the motion of the Earth through the ether.

In summary, we have seen that there is a direct chain of reasoning that leads from Galileo's principle of inertia, through Newton's laws with their implicit assumptions about space and time, ending with the failure of the Michelson–Morley experiment to observe the motion of the Earth relative to the ether. Although several explanations were offered for the unobservability of the ether and the corresponding failure of the upstream–downstream and cross-stream velocities to add in the expected way, the most novel, revolutionary, and ultimately successful explanation is given by Einstein's special theory of relativity, which requires a serious readjustment of our traditional concepts of space and time, and therefore alters some of the very foundations of physics.

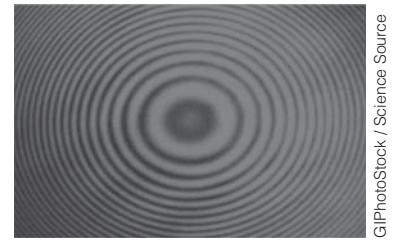


FIGURE 2.7 Interference fringes as observed with the Michelson interferometer of Figure 2.6. When the path length ACA changes by one-half wavelength relative to ABA , all light areas turn dark and all dark areas turn light.

2.3 EINSTEIN'S POSTULATES

The *special theory of relativity* is based on two postulates proposed by Albert Einstein in 1905:

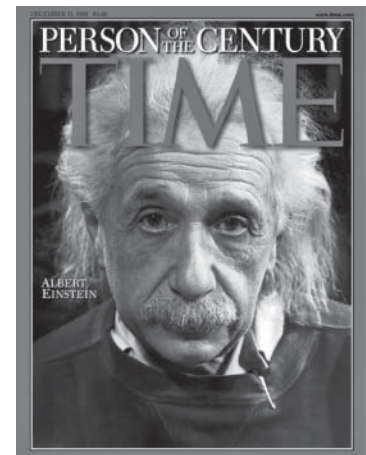
The principle of relativity: *The laws of physics are the same in all inertial reference frames.*

The principle of the constancy of the speed of light: *The speed of light in free space has the same value c in all inertial reference frames.*

The first postulate declares that the laws of physics are absolute, universal, and the same for all inertial observers. Laws that hold for one inertial observer cannot be violated for *any* inertial observer.

The second postulate is more difficult to accept because it seems to go against our “common sense,” which is based on the Galilean kinematics we observe in everyday experiences. Consider three observers A , B , and C . Observer B is at rest, while A and C move away from B in opposite directions each at a speed of $c/4$. B fires a light beam in the direction of A . According to the Galilean transformation, if B measures a speed of c for the light beam, then A measures a speed of $c - c/4 = 3c/4$, while C measures a speed of $c + c/4 = 5c/4$. Einstein's second postulate, on the other hand, requires all three observers to measure the same speed of c for the light beam! This postulate immediately explains the failure of the Michelson–Morley experiment—the upstream–downstream and cross-stream speeds are identical (both are equal to c), so there is no phase difference between the two beams.

The two postulates also allow us to dispose of the ether hypothesis. The first postulate does not permit a preferred frame of reference (all inertial frames are equivalent), and the second postulate does not permit only a single frame of reference in which light moves at speed c , because light moves at speed c in *all* frames. The ether, as a preferred reference frame in which light has a unique speed, is therefore unnecessary.



Albert Einstein (1879–1955, Germany–United States). A gentle philosopher and pacifist, he was the intellectual leader of two generations of theoretical physicists and left his imprint on nearly every field of modern physics.

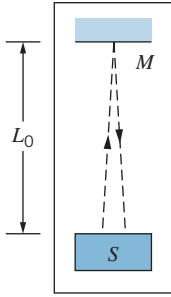


FIGURE 2.8 The clock ticks at intervals Δt_0 determined by the time for a light flash to travel the distance $2L_0$ from the light source S to the mirror M and back to the source where it is detected. (We assume the emission and detection occur at the same location, so the beam travels perpendicular to the mirror.)

2.4 CONSEQUENCES OF EINSTEIN'S POSTULATES

Among their many consequences, Einstein's postulates require a new consideration of the fundamental nature of time and space. In this section, we discuss how the postulates affect measurements of time and length intervals by observers in different frames of reference.

The Relativity of Time

To demonstrate the relativity of time, we use the timing device illustrated in Figure 2.8. It consists of a flashing light source S that is a distance L_0 from a mirror M . A flash of light from the source is reflected by the mirror, and when the light returns to S the clock ticks and triggers another flash. The time interval between ticks is the distance $2L_0$ (assuming the light travels perpendicular to the mirror) divided by the speed c :

$$\Delta t_0 = 2L_0/c \quad (2.6)$$

This is the time interval that is measured when the clock is at rest with respect to the observer.

We consider two observers: O is at rest on the ground, and O' moves with speed u . Each observer carries a timing device. Figure 2.9 shows a sequence of events that O observes for the clock carried by O' . According to O , the flash is emitted when the clock of O' is at A , reflected when it is at B , and detected at C . In this interval Δt , O observes the clock to move forward a distance of $u\Delta t$ from the point at which the flash was emitted, and O concludes that the light beam travels a distance $2L$, where $L = \sqrt{L_0^2 + (u\Delta t/2)^2}$, as shown in Figure 2.9. Because O observes the light beam to travel at speed c (as required by Einstein's second postulate) the time interval measured by O is

$$\Delta t = \frac{2L}{c} = \frac{2\sqrt{L_0^2 + (u\Delta t/2)^2}}{c} \quad (2.7)$$

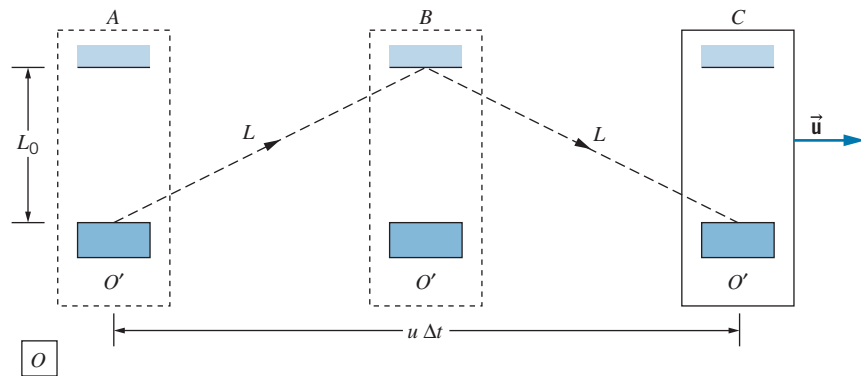


FIGURE 2.9 In the frame of reference of O , the clock carried by O' moves with speed u . The dashed line, of length $2L$, shows the path of the light beam according to O .

Substituting for L_0 from Eq. 2.6 and solving Eq. 2.7 for Δt , we obtain

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - u^2/c^2}} \quad (2.8)$$

According to Eq. 2.8, observer O measures a longer time interval than O' measures. This is a general result of special relativity, which is known as *time dilation*. An observer O' is at rest relative to a device that produces a time interval Δt_0 . For this observer, the beginning and end of the time interval occur at the same location, and so the interval Δt_0 is known as the *proper time*. An observer O , relative to whom O' is in motion, measures a longer time interval Δt for the same device. The dilated time interval Δt is always longer than the proper time interval Δt_0 , no matter what the magnitude or direction of $\vec{u}$.

This is a real effect that applies not only to clocks based on light beams but also to time itself; all clocks run more slowly according to an observer in relative motion, biological clocks included. Even the growth, aging, and decay of living systems are slowed by the time dilation effect. However, note that under normal circumstances ($u \ll c$), there is no measurable difference between Δt and Δt_0 , so we don't notice the effect in our everyday activities. Time dilation has been verified experimentally with decaying elementary particles as well as with precise atomic clocks carried aboard aircraft. Some experimental tests are discussed in the last section of this chapter.

Example 2.4

Muons are elementary particles with a (proper) lifetime of $2.2 \mu\text{s}$. They are produced with very high speeds in the upper atmosphere when cosmic rays (high-energy particles from space) collide with air molecules. Take the height L_0 of the atmosphere to be 100 km in the reference frame of the Earth, and find the minimum speed that enables the muons to survive the journey to the surface of the Earth.

Solution

The birth and decay of the muon can be considered as the “ticks” of a clock. In the frame of reference of the Earth (observer O), this clock is moving, and therefore its ticks are slowed by the time dilation effect. If the muon is moving at a speed that is close to c , the time necessary for it to travel from the top of the atmosphere to the surface of the Earth is

$$\Delta t = \frac{L_0}{c} = \frac{100 \text{ km}}{3.00 \times 10^8 \text{ m/s}} = 333 \mu\text{s}$$

If the muon is to be observed at the surface of the Earth, it must live for at least $333 \mu\text{s}$ in the Earth's frame of reference. In the muon's frame of reference, the interval between its birth and decay is a proper time interval of $2.2 \mu\text{s}$. The time intervals are related by Eq. 2.8:

$$333 \mu\text{s} = \frac{2.2 \mu\text{s}}{\sqrt{1 - u^2/c^2}}$$

Solving, we find

$$u = 0.999978c$$

If it were not for the time dilation effect, muons would not survive to reach the Earth's surface. The observation of these muons is a direct verification of the time dilation effect of special relativity.

The Relativity of Length

For this discussion, the moving timing device of O' is turned sideways, so that the light travels parallel to the direction of motion of O' . Figure 2.10 shows the sequence of events that O observes for the moving clock. According to O , the

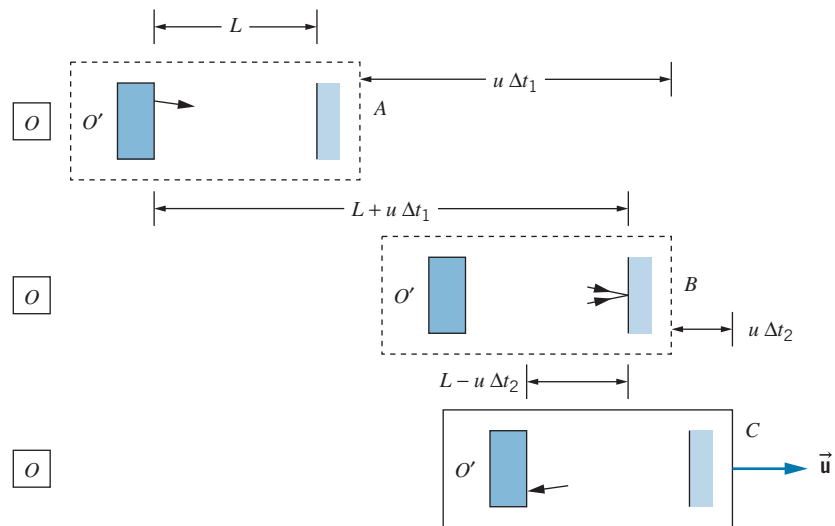


FIGURE 2.10 Here the clock carried by O' emits its light flash in the direction of motion.

length of the clock (distance between the light source and the mirror) is L ; as we shall see, this length is different from the length L_0 measured by O' , relative to whom the clock is at rest.

The flash of light is emitted when the clock of O' is at A and reaches the mirror (position B) at time Δt_1 later. In this time interval, the light travels a distance $c \Delta t_1$, equal to the length L of the clock plus the additional distance $u \Delta t_1$ that the mirror moves forward in this interval. That is,

$$c \Delta t_1 = L + u \Delta t_1 \quad (2.9)$$

The flash of light travels from the mirror to the detector in a time Δt_2 and covers a distance of $c \Delta t_2$, equal to the length L of the clock less the distance $u \Delta t_2$ that the clock moves forward in this interval:

$$c \Delta t_2 = L - u \Delta t_2 \quad (2.10)$$

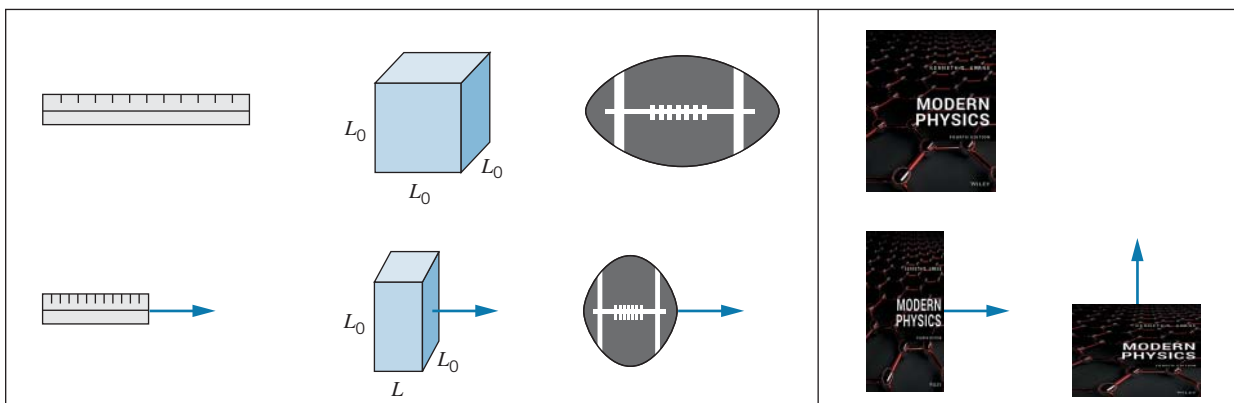


FIGURE 2.11 Some length-contracted objects. Notice that the shortening occurs only in the direction of motion.

Solving Eqs. 2.9 and 2.10 for Δt_1 and Δt_2 , and adding to find the total time interval, we obtain

$$\Delta t = \Delta t_1 + \Delta t_2 = \frac{L}{c-u} + \frac{L}{c+u} = \frac{2L}{c} \frac{1}{1-u^2/c^2} \quad (2.11)$$

From Eq. 2.8,

$$\Delta t = \frac{\Delta t_0}{\sqrt{1-u^2/c^2}} = \frac{2L_0}{c} \frac{1}{\sqrt{1-u^2/c^2}} \quad (2.12)$$

Setting Eqs. 2.11 and 2.12 equal to one another and solving, we obtain

$$L = L_0 \sqrt{1-u^2/c^2} \quad (2.13)$$

Equation 2.13 summarizes the effect known as *length contraction*. Observer O' , who is at rest with respect to the object, measures the *rest length* L_0 (also known as the *proper length*, in analogy with the proper time). All observers relative to whom O' is in motion measure a shorter length, but only along the direction of motion; length measurements transverse to the direction of motion are unaffected (Figure 2.11).

For ordinary speeds ($u \ll c$), the effects of length contraction are too small to be observed. For example, a rocket of length 100 m traveling at the escape speed from the Earth (11.2 km/s) would appear to an observer on the Earth to contract only by about two atomic diameters!

Length contraction suggests that objects in motion are measured to have a shorter length than they do at rest. The objects do not actually shrink; there is merely a difference in the length measured by different observers. For example, to observers on the Earth a high-speed rocket ship would appear to be contracted along its direction of motion (Figure 2.12a), but to an observer on the ship it is the passing Earth that appears to be contracted (Figure 2.12b).

These representations of length-contracted objects are somewhat idealized. The actual appearance of a rapidly moving object is determined by the time at which light leaves the various parts of the object and enters the eye or the camera. The result is that the object appears distorted in shape and slightly rotated.

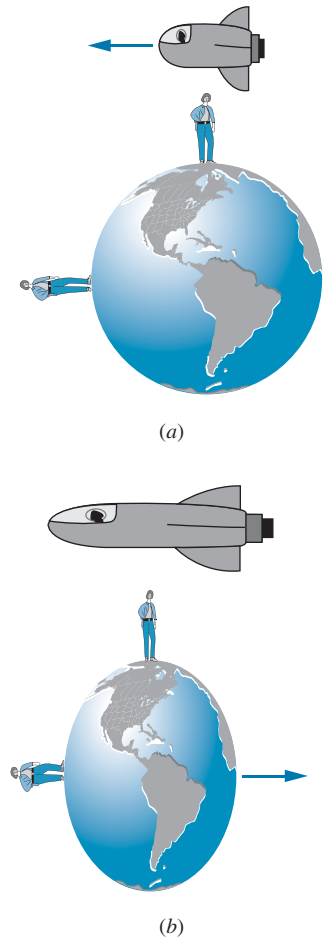


FIGURE 2.12 (a) The Earth views the passing contracted rocket. (b) From the rocket's frame of reference, the Earth appears contracted.

Example 2.5

Consider the point of view of an observer who is moving toward the Earth at the same velocity as the muon in Example 2.4. In this reference frame, what is the apparent thickness of the Earth's atmosphere?

Solution

In this observer's reference frame, the muon is at rest and the Earth is rushing toward it at a speed of $u = 0.999978c$, as we found in Example 2.4. To an observer on the Earth, the height of the atmosphere is its rest

length L_0 of 100 km. To the observer in the muon's rest frame, the moving Earth has an atmosphere of height given by Eq. 2.13:

$$\begin{aligned} L &= L_0 \sqrt{1-u^2/c^2} \\ &= (100 \text{ km}) \sqrt{1-(0.999978)^2} = 0.66 \text{ km} = 660 \text{ m} \end{aligned}$$

This distance is small enough for the muons to reach the Earth's surface within their lifetime.

Note that what appears as a *time dilation* in one frame of reference (the observer on Earth) can be regarded as a *length contraction* in another frame of reference (the observer traveling with the muon). For another example of this effect, let's review again the example of the pion decay discussed in

Section 1.2. A pion at rest has a lifetime of 26.0 ns. According to observer O_1 at rest in the laboratory frame of reference, a pion moving through the laboratory at a speed of $0.913c$ has a longer lifetime, which can be calculated to be 63.7 ns (using Eq. 2.8 for the time dilation). According to observer O_2 , who is traveling through the laboratory at the same velocity as the pion, the pion appears to be at rest and has its proper lifetime of 26.0 ns. Thus O_1 sees a time dilation effect.

O_1 erects two markers in the laboratory, at the locations where the pion is created and decays. To O_1 , the distance between those markers is the pion's speed times its lifetime, which works out to be 17.4 m. Suppose O_1 places a stick of length 17.4 m in the laboratory connecting the two markers. That stick is at rest in the laboratory reference frame and so has its proper length in that frame. In the reference frame of O_2 , the stick is moving at a speed of $0.913c$ and has a shorter length of 7.1 m, which we can find using the length contraction formula (Eq. 2.13). So O_2 measures a distance of 7.1 m between the locations in the laboratory where the pion was created and where it decayed.

Note that O_1 measures the proper length and the dilated time, while O_2 measures the proper time and the contracted length. The proper time and proper length must always be referred to specific observers, who might not be in the same reference frame. The proper time is always measured by an observer according to whom the beginning of the time interval and the end of the time interval occur at the same location. If the time interval is the lifetime of the pion, then O_2 (relative to whom the pion does not move) sees its creation and decay at the same location and thus measures the proper time interval. The proper length, on the other hand, is always measured by an observer according to whom the measuring stick is at rest (O_1 in this case).

Example 2.6

An observer O is standing on a platform of length $D_0 = 65$ m on a space station. A rocket passes at a relative speed of $0.80c$ moving parallel to the edge of the platform. The observer O notes that the front and back of the rocket simultaneously line up with the ends of the platform at a particular instant (Figure 2.13a). (a) According to O , what is the time necessary for the rocket to pass a particular point on the platform? (b) What is the rest length L_0 of the rocket? (c) According to an observer O' on the rocket, what is the length D of the platform? (d) According to O' , how long does it take for observer O to pass the entire length of the rocket? (e) According to O , the ends of the rocket simultaneously line up with the ends of the platform. Are these events simultaneous to O' ?

Solution

(a) According to O , the length L of the rocket matches the length D_0 of the platform. The time for the rocket to pass a particular point is measured by O to be

$$\Delta t_0 = \frac{L}{0.80c} = \frac{65 \text{ m}}{2.40 \times 10^8 \text{ m/s}} = 0.27 \mu\text{s}$$

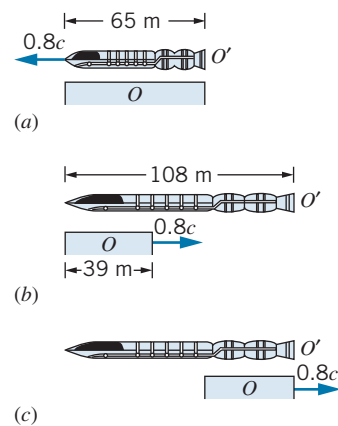


FIGURE 2.13 Example 2.6. (a) From the reference frame of O at rest on the platform, the passing rocket lines up simultaneously with the front and back of the platform. (b, c) From the reference frame O' in the rocket, the passing platform lines up first with the front of the rocket and later with the rear. Note the differing effects of length contraction in the two reference frames.

This is a proper time interval because O measures the interval between two events that occur at the same point in the frame of reference of O (the front of the rocket passes a point, and then the back of the rocket passes the same point).

(b) O measures the contracted length L of the rocket. We can find its proper length L_0 using Eq. 2.13:

$$L_0 = \frac{L}{\sqrt{1 - u^2/c^2}} = \frac{65 \text{ m}}{\sqrt{1 - (0.80)^2}} = 108 \text{ m}$$

(c) According to O the platform is at rest, so 65 m is its proper length D_0 . According to O' , the contracted length of the platform is therefore

$$D = D_0 \sqrt{1 - u^2/c^2} = (65 \text{ m}) \sqrt{1 - (0.80)^2} = 39 \text{ m}$$

(d) For O to pass the entire length of the rocket, O' concludes that O must move a distance equal to its rest length, or 108 m. The time needed to do this is

$$\Delta t' = \frac{108 \text{ m}}{0.80c} = 0.45 \mu\text{s}$$

Note that this is *not* a proper time interval for O' , who determines this time interval using one clock at the front of the rocket to measure the time at which O passes the front of the rocket, and another clock on the rear of the

rocket to measure the time at which O passes the rear of the rocket. The two events therefore occur at different points in O' and so cannot be separated by a proper time in O' . The corresponding time interval measured by O for the same two events, which we calculated in part (a), is a proper time interval for O , because the two events *do* occur at the same point in O . The time intervals measured by O and O' should be related by the time dilation formula, as you should verify.

(e) According to O' , the rocket has a rest length of $L_0 = 108 \text{ m}$ and the platform has a contracted length of $D = 39 \text{ m}$. There is thus no way that O' could observe the two ends of both to align simultaneously. The sequence of events according to O' is illustrated in Figures 2.13b and c. The time interval $\Delta t'$ in O' between the two events that are simultaneous in O can be calculated by noting that, according to O' , the time interval between the situations shown in Figures 2.13b and c must be that necessary for the platform to move a distance of $108 \text{ m} - 39 \text{ m} = 69 \text{ m}$, which takes a time

$$\Delta t' = \frac{69 \text{ m}}{0.80c} = 0.29 \mu\text{s}$$

This result illustrates the relativity of simultaneity: two events at different locations that are simultaneous to O (the lining up of the two ends of the rocket with the two ends of the platform) *cannot* be simultaneous to O' .

Example 2.7

A meter stick in the $x'y'$ plane makes angles of 45° with the x' and y' axes. O' and the meter stick move away from O in the x direction. According to O , the angle that the meter stick makes with the x axis is (a) less than 45° ; (b) equal to 45° ; (c) greater than 45° .

Solution

According to O' , the x' and y' components of the length of the stick are equal. According to O , there is length

contraction only in the x direction, not in the y direction. Since the x component of the length of the stick becomes smaller than the y component, the tangent of the angle the stick makes with the x axis (the ratio of the y component to the x component) is greater than 1, so the angle is greater than 45° .

Relativistic Velocity Addition

The timing device is now modified as shown in Figure 2.14. A source P emits particles that travel at speed v' according to an observer O' at rest with respect to the device. The flashing bulb F is triggered to flash when a particle reaches it. The flash of light makes the return trip to the detector D , and the clock ticks. The time interval Δt_0 between ticks measured by O' is composed of

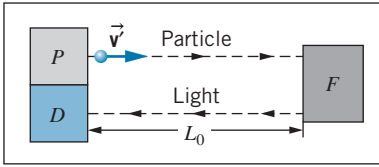


FIGURE 2.14 In this timing device, a particle is emitted by P at a speed v' . When the particle reaches F , it triggers the emission of a flash of light that travels to the detector D .

two parts: one for the particle to travel the distance L_0 at speed v' and another for the light to travel the same distance at speed c :

$$\Delta t_0 = L_0/v' + L_0/c \quad (2.14)$$

According to observer O , relative to whom O' moves at speed u , the sequence of events is similar to that shown in Figure 2.10. The emitted particle, which travels at speed v according to O , reaches F in a time interval Δt_1 after traveling the distance $v \Delta t_1$ equal to the (contracted) length L plus the additional distance $u \Delta t_1$ moved by the clock in that interval:

$$v \Delta t_1 = L + u \Delta t_1 \quad (2.15)$$

In the interval Δt_2 , the light beam travels a distance $c \Delta t_2$ equal to the length L less the distance $u \Delta t_2$ moved by the clock in that interval:

$$c \Delta t_2 = L - u \Delta t_2 \quad (2.16)$$

We now solve Eqs. 2.15 and 2.16 for Δt_1 and Δt_2 , add to find the total interval Δt between ticks according to O , use the time dilation formula, Eq. 2.8, to relate this result to Δt_0 from Eq. 2.14, and finally use the length contraction formula, Eq. 2.13, to relate L to L_0 . After doing the algebra, we find the result

$$v = \frac{v' + u}{1 + v'u/c^2} \quad (2.17)$$

Equation 2.17 is the *relativistic velocity addition law* for velocity components that are in the direction of u . Later in this chapter we use a different method to derive the corresponding results for motion in other directions.

We can also regard Eq. 2.17 as a velocity transformation, enabling us to convert a velocity v' measured by O' to a velocity v measured by O . The corresponding classical law was given by Eq. 2.2: $v = v' + u$. The difference between the classical and relativistic results is the denominator of Eq. 2.17, which reduces to 1 in cases when the speeds are small compared with c . Example 2.8 shows how this factor prevents the measured speeds from exceeding c .

Equation 2.17 gives an important result when O' observes a light beam. For $v' = c$,

$$v = \frac{c + u}{1 + cu/c^2} = c \quad (2.18)$$

That is, when $v' = c$, then $v = c$, *independent of the value of u* . All observers measure the same speed c for light, exactly as required by Einstein's second postulate.

Example 2.8

A spaceship moving away from the Earth at a speed of $0.80c$ fires a missile parallel to its direction of motion (Figure 2.15). The missile moves at a speed of $0.60c$ relative to the ship. What is the speed of the missile as measured by an observer on Earth?

Solution

Here O' is on the ship and O is on the Earth; O' moves with a speed of $u = 0.80c$ relative to O . The missile moves at speed $v' = 0.60c$ relative to O' , and we seek its speed v

relative to O . Using Eq. 2.17, we obtain

$$\begin{aligned} v &= \frac{v' + u}{1 + v'u/c^2} = \frac{0.60c + 0.80c}{1 + (0.60c)(0.80c)/c^2} \\ &= \frac{1.40c}{1.48} = 0.95c \end{aligned}$$

According to classical kinematics (the numerator of Eq. 2.17), an observer on the Earth would see the missile moving at $0.60c + 0.80c = 1.40c$, thereby exceeding the maximum relative speed of c permitted by relativity. You can see how Eq. 2.17 brings about this speed limit.

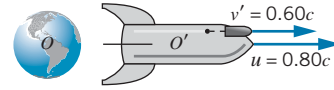


FIGURE 2.15 Example 2.8. A spaceship moves away from the Earth at a speed of $0.80c$. An observer O' on the spaceship fires a missile and measures its speed to be $0.60c$ relative to the ship.

Even if v' were $0.9999 \dots c$ and u were $0.9999 \dots c$, the relative speed v measured by O would remain less than c .

The Relativistic Doppler Effect

In the classical Doppler effect for sound waves, an observer moving relative to a source of waves (sound, for example) detects a frequency different from that emitted by the source. The frequency f' heard by the observer O is related to the frequency f emitted by the source S according to

$$f' = f \frac{v \pm v_O}{v \mp v_S} \quad (2.19)$$

where v is the speed of the waves in the medium (such as still air, in the case of sound waves), v_S is the speed of the source *relative to the medium*, and v_O is the speed of the observer *relative to the medium*. The upper signs in the numerator and denominator are chosen whenever S moves toward O or O moves toward S , while the lower signs apply whenever O and S move away from one another.

The classical Doppler shift for motion of the source differs from that for motion of the observer. For example, suppose the source emits sound waves at $f = 1000$ Hz. If the source moves at 30 m/s toward the observer who is at rest in the medium (which we take to be air, in which sound moves at $v = 340$ m/s), then $f' = 1097$ Hz, while if the source is at rest in the medium and the observer moves toward the source at 30 m/s, the frequency is 1088 Hz. Other possibilities in which the relative speed between S and O is 30 m/s, such as each moving toward the other at 15 m/s, give still different frequencies.

Here we have a situation in which it is not the relative speed of the source and observer that determines the Doppler shift—it is the speed of each with respect to the medium. This cannot occur for light waves, since there is no medium (no “ether”) and no preferred reference frame by Einstein’s first postulate. We therefore require a different approach to the Doppler effect for light waves, an approach that does not distinguish between source motion and observer motion, but involves only the relative motion between the source and the observer.

Consider a source of waves that is at rest in the reference frame of observer O . Observer O' moves relative to the source at speed u . We consider the situation from the frame of reference of O' , as shown in Figure 2.16. Suppose O observes the source to emit N waves at frequency f . According to O , it takes an interval $\Delta t_0 = N/f$ for these N waves to be emitted; this is a proper time interval in the frame of reference of O . The corresponding time

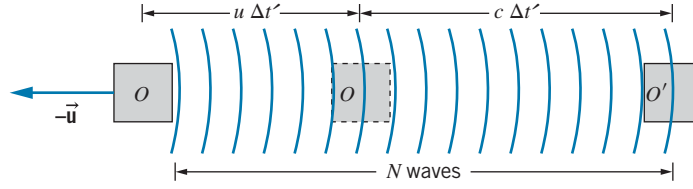


FIGURE 2.16 A source of waves, in the reference frame of O , moves at speed u away from observer O' . In the time $\Delta t'$ (according to O'), O moves a distance $u \Delta t'$ and emits N waves.

interval to O' is $\Delta t'$, during which O moves a distance $u \Delta t'$. The wavelength according to O' is the total length interval occupied by these waves divided by the number of waves:

$$\lambda' = \frac{c \Delta t' + u \Delta t'}{N} = \frac{c \Delta t' + u \Delta t'}{f \Delta t_0} \quad (2.20)$$

The frequency according to O' is $f' = c/\lambda'$, so

$$f' = f \frac{\Delta t_0}{\Delta t'} \frac{1}{1 + u/c} \quad (2.21)$$

and using the time dilation formula, Eq. 2.8, to relate $\Delta t'$ and Δt_0 , we obtain

$$f' = f \frac{\sqrt{1 - u^2/c^2}}{1 + u/c} = f \sqrt{\frac{1 - u/c}{1 + u/c}} \quad (2.22)$$

This is the formula for the *relativistic Doppler shift*, for the case in which the waves are observed in a direction parallel to $\vec{u}$. Note that, unlike the classical formula, it does *not* distinguish between source motion and observer motion; the relativistic Doppler effect depends only on the relative speed u between the source and observer.

Equation 2.22 assumes that the source and observer are separating. If the source and observer are approaching each other, replace u by $-u$ in the formula.

Example 2.9

A distant galaxy is moving away from Earth at such high speed that the blue hydrogen line at a wavelength of 434 nm is recorded at 600 nm in the red range of the spectrum. What is the speed of the galaxy relative to Earth?

Solution

Using Eq. 2.22 with $f = c/\lambda$ and $f' = c/\lambda'$, we obtain

$$\frac{c}{\lambda'} = \frac{c}{\lambda} \sqrt{\frac{1 - u/c}{1 + u/c}}$$

$$\frac{c}{600 \text{ nm}} = \frac{c}{434 \text{ nm}} \sqrt{\frac{1 - u/c}{1 + u/c}}$$

Solving, we find

$$u/c = 0.31$$

Thus the galaxy is moving away from the Earth at a speed of $0.31c = 9.4 \times 10^7 \text{ m/s}$. Evidence obtained in this way indicates that nearly all the galaxies we observe are moving away from us. This suggests that the universe is expanding, and is usually taken to provide evidence in favor of the Big Bang theory of cosmology (see Chapter 15).

2.5 THE LORENTZ TRANSFORMATION

We have seen that the Galilean transformation of coordinates, time, and velocity is not consistent with Einstein's postulates. Although the Galilean transformation agrees with our "common-sense" experience at low speeds, it does not agree with experiment at high speeds. We therefore need a new set of transformation equations that replaces the Galilean set and that is capable of predicting relativistic effects such as time dilation, length contraction, velocity addition, and the Doppler shift.

As before, we seek a transformation that enables observers O and O' in relative motion to compare their measurements of the space and time coordinates of the same event. The transformation equations relate the measurements of O (namely, x, y, z, t) to those of O' (namely, x', y', z', t'). This new transformation must have several properties: It must be linear (depending only on the first power of the space and time coordinates), which follows from the homogeneity of space and time; it must be consistent with Einstein's postulates; and it must reduce to the Galilean transformation when the relative speed between O and O' is small. We again assume that the velocity of O' relative to O is in the positive xx' direction.

This new transformation consistent with special relativity is called the *Lorentz transformation**. Its equations are

$$x' = \frac{x - ut}{\sqrt{1 - u^2/c^2}} \quad (2.23a)$$

$$y' = y \quad (2.23b)$$

$$z' = z \quad (2.23c)$$

$$t' = \frac{t - (u/c^2)x}{\sqrt{1 - u^2/c^2}} \quad (2.23d)$$

It is often useful to write these equations in terms of *intervals* of space and time by replacing each coordinate by the corresponding interval (replace x by Δx , x' by $\Delta x'$, t by Δt , t' by $\Delta t'$).

These equations are written assuming that O' moves *away from* O in the xx' direction. If O' moves *toward* O , replace u with $-u$ in the equations.

The first three equations reduce directly to the Galilean transformation for space coordinates, Eqs. 2.1, when $u \ll c$. The fourth equation, which links the time coordinates, reduces to $t' = t$, which is a fundamental postulate of the Galilean–Newtonian world.

We now use the Lorentz transformation equations to derive some of the predictions of special relativity. The problems at the end of the chapter guide you in some other derivations. The results derived here are identical with those we obtained previously using Einstein's postulates, which shows that the

*H. A. Lorentz (1853–1928) was a Dutch physicist who shared the 1902 Nobel Prize for his work on the influence of magnetic fields on light. In an unsuccessful attempt to explain the failure of the Michelson–Morley experiment, Lorentz developed the transformation equations that are named for him in 1904, a year *before* Einstein published his special theory of relativity. For a derivation of the Lorentz transformation, see R. Resnick and D. Halliday, *Basic Concepts in Relativity* (New York, Macmillan, 1992).

equations of the Lorentz transformation are consistent with the postulates of special relativity.

Length Contraction

A rod of length L_0 is at rest in the reference frame of observer O' . The rod extends along the x' axis from x'_1 to x'_2 ; that is, O' measures the proper length $L_0 = x'_2 - x'_1$. Observer O , relative to whom the rod is in motion, measures the ends of the rod to be at coordinates x_1 and x_2 . For O to determine the length of the moving rod, O must make a *simultaneous* determination of x_1 and x_2 , and then the length is $L = x_2 - x_1$. Suppose the first event is O' setting off a flash bulb at one end of the rod at x'_1 and t'_1 , which O observes at x_1 and t_1 , and the second event is O' setting off a flash bulb at the other end at x'_2 and t'_2 , which O observes at x_2 and t_2 . The equations of the Lorentz transformation relate these coordinates, specifically,

$$x'_1 = \frac{x_1 - ut_1}{\sqrt{1 - u^2/c^2}} \quad x'_2 = \frac{x_2 - ut_2}{\sqrt{1 - u^2/c^2}} \quad (2.24)$$

Subtracting these equations, we obtain

$$x'_2 - x'_1 = \frac{x_2 - x_1}{\sqrt{1 - u^2/c^2}} - \frac{u(t_2 - t_1)}{\sqrt{1 - u^2/c^2}} \quad (2.25)$$

O' must arrange to set off the flash bulbs so that the flashes appear to be simultaneous to O . (They will *not* be simultaneous to O' , as we discuss later in this section.) This enables O to make a simultaneous determination of the coordinates of the endpoints of the rod. If O observes the flashes to be simultaneous, then $t_2 = t_1$, and Eq. 2.25 reduces to

$$x'_2 - x'_1 = \frac{x_2 - x_1}{\sqrt{1 - u^2/c^2}} \quad (2.26)$$

With $x'_2 - x'_1 = L_0$ and $x_2 - x_1 = L$, this becomes

$$L = L_0 \sqrt{1 - u^2/c^2} \quad (2.27)$$

which is identical with Eq. 2.13, which we derived earlier using Einstein's postulates.

Velocity Transformation

If O observes a particle to travel with velocity v (components v_x , v_y , v_z), what velocity v' does O' observe for the particle? The relationship between the velocities measured by O and O' is given by the *Lorentz velocity transformation*:

$$v'_x = \frac{v_x - u}{1 - v_x u/c^2} \quad (2.28a)$$

$$v'_y = \frac{v_y \sqrt{1 - u^2/c^2}}{1 - v_x u/c^2} \quad (2.28b)$$

$$v'_z = \frac{v_z \sqrt{1 - u^2/c^2}}{1 - v_x u/c^2} \quad (2.28c)$$

By solving Eq. 2.28a for v'_x , you can show that it is identical to Eq. 2.17, a result we derived previously based on Einstein's postulates. Note that, in the limit of low speeds ($u \ll c$), the Lorentz velocity transformation reduces to the Galilean velocity transformation, Eq. 2.2. Note also that $v'_y \neq v_y$, even though $y' = y$. This occurs because of the way the Lorentz transformation handles the time coordinate.

We can derive these transformation equations for velocity from the Lorentz coordinate transformation. By way of example, we derive the velocity transformation for $v'_y = dy'/dt'$. Differentiating the coordinate transformation $y' = y$, we obtain $dy' = dy$. Similarly, differentiating the time coordinate transformation (Eq. 2.23d), we obtain

$$dt' = \frac{dt - (u/c^2)dx}{\sqrt{1 - u^2/c^2}}$$

So,

$$\begin{aligned} v'_y &= \frac{dy'}{dt'} = \frac{dy}{[dt - (u/c^2)dx]/\sqrt{1 - u^2/c^2}} = \sqrt{1 - u^2/c^2} \frac{dy}{dt - (u/c^2)dx} \\ &= \sqrt{1 - u^2/c^2} \frac{dy/dt}{1 - (u/c^2)dx/dt} = \frac{v_y \sqrt{1 - u^2/c^2}}{1 - uv_x/c^2} \end{aligned}$$

Similar methods can be used to obtain the transformation equations for v'_x and v'_z . These derivations are left as exercises (Problem 16).

Simultaneity and Clock Synchronization

Under ordinary circumstances, synchronizing one clock with another is a simple matter. But for scientific work, where timekeeping at a precision below the nanosecond range is routine, clock synchronization can present some significant challenges. At very least, we need to correct for the time that it takes for the signal showing the reading on one clock to be transmitted to the other clock. However, for observers who are in motion with respect to each other, special relativity gives yet another way that clocks may appear to be out of synchronization.

Consider the device shown in Figure 2.17. Two clocks are located at $x = 0$ and $x = L$. A flash lamp is located at $x = L/2$, and the clocks are set running when they receive the flash of light from the lamp. The light takes the same interval of time to reach the two clocks, so the clocks start together precisely at a time $L/2c$ after the flash is emitted, and the clocks are exactly synchronized.

Now let us examine the same situation from the point of view of the moving observer O' . In the frame of reference of O , two events occur: the receipt of a light signal by clock 1 at $x_1 = 0$, $t_1 = L/2c$ and the receipt of a light signal by clock 2 at $x_2 = L$, $t_2 = L/2c$. Using Eq. 2.23d, we find that O' observes clock 1 to receive its signal at

$$t'_1 = \frac{t_1 - (u/c^2)x_1}{\sqrt{1 - u^2/c^2}} = \frac{L/2c}{\sqrt{1 - u^2/c^2}} \quad (2.29)$$

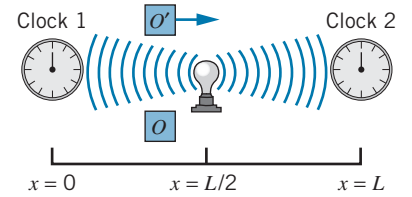


FIGURE 2.17 A flash of light, emitted from a point midway between the two clocks, starts the two clocks simultaneously according to O . Observer O' sees clock 2 start ahead of clock 1.

while clock 2 receives its signal at

$$t'_2 = \frac{t_2 - (u/c^2)x_2}{\sqrt{1 - u^2/c^2}} = \frac{L/2c - (u/c^2)L}{\sqrt{1 - u^2/c^2}} \quad (2.30)$$

Thus t'_2 is smaller than t'_1 and clock 2 appears to receive its signal earlier than clock 1, so that the clocks start at times that differ by

$$\Delta t' = t'_1 - t'_2 = \frac{uL/c^2}{\sqrt{1 - u^2/c^2}} \quad (2.31)$$

according to O' . Keep in mind that this is *not* a time dilation effect—time dilation comes from the *first* term of the Lorentz transformation (Eq. 2.23d) for t' , while the lack of synchronization arises from the *second* term. O' observes *both* clocks to run slow, due to time dilation; O' *also* observes clock 2 to be ahead of clock 1.

We therefore reach the following conclusion: two events that are simultaneous in one reference frame are not simultaneous in another reference frame moving with respect to the first, unless the two events occur at the same point in space. (If $L = 0$, Eq. 2.31 shows that the clocks are synchronized in all reference frames.) Clocks that appear to be synchronized in one frame of reference will not necessarily be synchronized in another frame of reference in relative motion.

It is important to note that this clock synchronization effect does not depend on the *location* of observer O' but only on the *velocity* of O' . In Figure 2.17, the location of O' could have been drawn far to the left side of clock 1 or far to the right side of clock 2, and the result would be the same. In those different locations, the propagation time of the light signal showing clock 1 starting will differ from the propagation time of the light signal showing clock 2 starting. However, O' is assumed to be an “intelligent” observer who is aware of the locations where the light signals showing the two clocks starting are received relative to the locations of the clocks. O' corrects for this time difference, which is due only to the propagation time of the light signals, and *even after making that correction* the clocks still do not appear to be synchronized!

Although the location of O' does not appear in Eq. 2.31, the *direction* of the velocity of O' is important—if O' is moving in the opposite direction, the observed starting order of the two clocks is reversed.

Example 2.10

Two rockets are leaving their space station along perpendicular paths, as measured by an observer on the space station. Rocket 1 moves at $0.60c$ and rocket 2 moves at $0.80c$, both measured relative to the space station. What is the velocity of rocket 2 as observed by rocket 1?

Solution

Observer O is the space station, observer O' is rocket 1 (moving at $u = 0.60c$), and each observes rocket 2, moving (according to O) in a direction perpendicular to rocket 1. We take this to be the y direction of the reference frame of O . Thus O observes rocket 2 to have

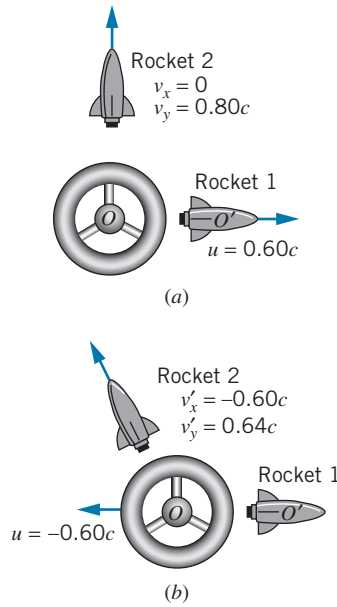


FIGURE 2.18 Example 2.10. (a) As viewed from the reference frame of O . (b) As viewed from the reference frame of O' .

velocity components $v_x = 0$, $v_y = 0.80c$, as shown in Figure 2.18a.

We can find v'_x and v'_y using the Lorentz velocity transformation:

$$v'_x = \frac{v_x - u}{1 - v_x u / c^2} = \frac{0 - 0.60c}{1 - 0(0.60c)/c^2} = -0.60c$$

$$\begin{aligned} v'_y &= \frac{v_y \sqrt{1 - u^2/c^2}}{1 - v_x u / c^2} \\ &= \frac{0.80c \sqrt{1 - (0.60c)^2/c^2}}{1 - 0(0.60c)/c^2} = 0.64c \end{aligned}$$

Thus, according to O' , the situation looks like Figure 2.18b.

The speed of rocket 2 according to O' is $\sqrt{(0.60c)^2 + (0.64c)^2} = 0.88c$, less than c . According to the Galilean transformation, v'_y would be identical with v_y , and thus the speed would be $\sqrt{(0.60c)^2 + (0.80c)^2} = c$. Once again, the Lorentz transformation prevents relative speeds from reaching or exceeding the speed of light.

Example 2.11

In Example 2.6, two events that were simultaneous to O (the lining up of the front and back of the rocket ship with the ends of the platform) were not simultaneous to O' . Find the time interval between these events according to O' .

Solution

According to O , the two simultaneous events are separated by a distance of $L = 65$ m. For $u = 0.80c$, Eq. 2.31 gives

$$\begin{aligned} \Delta t' &= \frac{uL/c^2}{\sqrt{1 - u^2/c^2}} \\ &= \frac{(0.80)(65 \text{ m})/(3.00 \times 10^8 \text{ m/s})}{\sqrt{1 - (0.80)^2}} = 0.29 \mu\text{s} \end{aligned}$$

which agrees with the result calculated in part (e) of Example 2.6.

Example 2.12

Space cadet O is standing in the exact middle of a space platform. Suddenly O observes two asteroids striking the front and back ends of the platform at precisely the same moment. This event instantly triggers an automatic report to the insurance company, which pays extra for damage from such double hits. As luck would have it, the insurance adjuster O' is flying along the platform from back to front and is directly above O when the

report is filed. Does O' confirm that the hits were simultaneous?

Solution

Even though O' is directly above O when O receives the simultaneous light signals from the two hits, the events are *not* simultaneous to O' . From the reference frame of O' , the platform with O is moving such that first the back

end of the platform passes beneath O' , then O at the center of the platform, and finally the front of the platform. Because of the motion of the platform, O' concludes that, since the light signal showing the front impact had

to travel a longer distance at the same speed to reach the midpoint, it must have been emitted earlier, so the front impact occurred before the rear impact.

2.6 THE TWIN PARADOX

We now turn briefly to what has become known as the twin paradox. Suppose there is a pair of twins on the Earth. One, whom we shall call Casper, remains on the Earth, while his twin sister Amelia sets off in a rocket ship on a trip to a distant planet. Casper, based on his understanding of special relativity, knows that his sister's clocks will run slow relative to his own and that therefore she should be younger than he when she returns, as our discussion of time dilation would suggest. However, recalling that discussion, we know that for two observers in relative motion, *each* thinks the *other's* clocks are running slow. We could therefore study this problem from the point of view of Amelia, according to whom Casper and the Earth (accompanied by the solar system and galaxy) make a round-trip journey away from her and back again. Under such circumstances, she will think it is her brother's clocks (which are now in motion relative to her own) that are running slow and will therefore expect her brother to be younger than she when they meet again. While it is possible to disagree over whose clocks are running slow relative to his or her own, which is merely a problem of frames of reference, when Amelia returns to the Earth (or when Earth returns to Amelia), all observers must agree as to which twin has aged less rapidly. This is the paradox—each twin expects the other to be younger.

The resolution of this paradox lies in considering the asymmetric role of the two twins. The laws of special relativity apply only to inertial frames, those moving relative to one another at constant velocity. We may supply Amelia's rockets with sufficient thrust so that they accelerate for a very short length of time, bringing the ship to a speed at which it can coast to the planet, and thus during her outward journey Amelia spends all but a negligible amount of time in a frame of reference moving at constant speed relative to Casper. However, in order to return to the Earth, she must decelerate and reverse her motion. Although this also may be done in a very short time interval, Amelia's return journey occurs in a completely different inertial frame than her outward journey. It is Amelia's jump from one inertial frame to another that causes the asymmetry in the ages of the twins. Only Amelia has the necessity of jumping to a new inertial frame to return, and therefore *all observers will agree* that it is Amelia who is “really” in motion, and that it is her clocks that are “really” running slow; therefore, she is indeed the younger twin on her return.

Let us make this discussion more quantitative with a numerical example. We assume, as discussed above, that the acceleration and deceleration take negligible time intervals, so that all of Amelia's aging is done during the coasting. For simplicity, we assume the distant planet is at rest relative to Earth; this does not change the problem, but it avoids the need to introduce yet another frame of reference. Suppose the planet to be 6 light-years distant from Earth,

and suppose Amelia travels at a speed of $0.6c$. Then according to Casper it takes his sister 10 years ($10 \text{ years} \times 0.6c = 6 \text{ light-years}$) to reach the planet and 10 years to return, and therefore she is gone for a total of 20 years. (However, Casper doesn't know his sister has reached the planet until the light signal carrying news of her arrival reaches Earth. Since light takes 6 years to make the journey, it is 16 years after her departure when Casper sees his sister's arrival at the planet. Four years later she returns to Earth.) From the frame of reference of Amelia aboard the rocket, the distance to the planet is contracted by a factor of $\sqrt{1 - (0.6)^2} = 0.8$, and is therefore $0.8 \times 6 \text{ light-years} = 4.8 \text{ light-years}$. At a speed of $0.6c$, Amelia will measure 8 years for the trip to the planet, for a total round trip time of 16 years. Thus Casper ages 20 years while Amelia ages only 16 years and is indeed the younger on her return.

We can confirm this analysis by having Casper send a light signal to his sister each year on his birthday. We know that the frequency of the signal as received by Amelia will be Doppler shifted. During the outward journey, she will receive signals at the rate of

$$(1/\text{year})\sqrt{\frac{1 - u/c}{1 + u/c}} = 0.5/\text{year}$$

During the return journey, the Doppler-shifted rate will be

$$(1/\text{year})\sqrt{\frac{1 + u/c}{1 - u/c}} = 2/\text{year}$$

Thus for the first 8 years, during Amelia's trip to the planet, she receives 4 signals, and during the return trip of 8 years, she receives 16 signals, for a total of 20. She receives 20 signals, indicating her brother has celebrated 20 birthdays during her 16-year journey.

Spacetime Diagrams

A particularly helpful way of visualizing the journeys of Casper and Amelia uses a *spacetime* diagram. Figure 2.19 shows an example of a spacetime diagram for motion that involves only one spatial direction.

In your introductory physics course, you probably became familiar with plotting motion on a graph in which distance appeared on the vertical axis and time on the horizontal axis. On such a graph, a straight line represents motion at constant velocity; the slope of the line is equal to the velocity. Note that the axes of the spacetime diagram are switched from the traditional graph of particle motion, with time on the vertical axis and space on the horizontal axis.

On a spacetime diagram, the graph that represents the motion of a particle is called its *worldline*. The *inverse* of the slope of the particle's worldline gives its velocity. Equivalently, the velocity is given by the tangent of the angle that the worldline makes with the *vertical* axis (rather than with the horizontal axis, as would be the case with a conventional plot of distance vs. time). Usually, the units of x and t are chosen so that motion at the speed of light is represented by a line with a 45° slope. A vertical line represents a particle that is at the same spatial locations at all times—that is, a particle at rest. Permitted motions with constant velocity are then represented by straight lines between the vertical and the 45° line representing the maximum velocity.

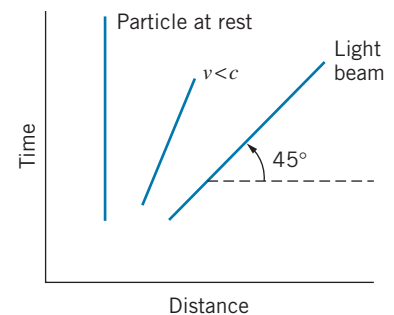


FIGURE 2.19 A spacetime diagram.

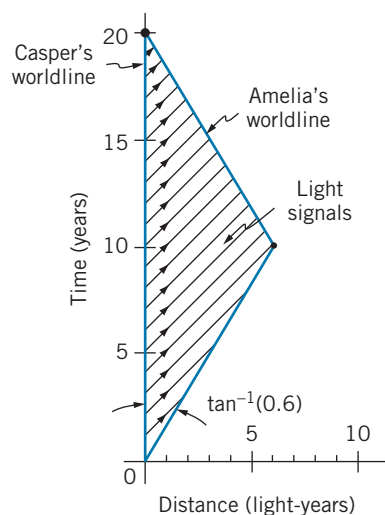


FIGURE 2.20 Casper's spacetime diagram, showing his worldline and Amelia's.

Let's draw the worldlines of Casper and Amelia according to Casper's frame of reference. Casper's worldline is a vertical line, because he is at rest in this frame (Figure 2.20). In Casper's frame of reference, 20 years pass between Amelia's departure and her return, so we can follow Casper's vertical worldline for 20 years.

Amelia is traveling at a speed of $0.6c$, so her worldline makes an angle with the vertical whose tangent is 0.6 (31°). In Casper's frame of reference, the planet visited by Amelia is 6 light-years from Earth. Amelia travels a distance of 6 light-years in a time of 10 years (according to Casper) so that $v = 6 \text{ light-years}/10 \text{ years} = 0.6c$.

The birthday signals that Casper sends to Amelia at the speed of light are represented by the series of 45° lines in Figure 2.20. Amelia receives 4 birthday signals during her outbound journey (the 4th arrives just as she reaches the planet) and 16 birthday signals during her return journey (the 16th is sent and received just as she returns to Earth).

It is left as an exercise (Problems 24 and 26) to consider the situation if it is Amelia who is sending the signals.

2.7 RELATIVISTIC DYNAMICS

We have seen how Einstein's postulates have led to a new "relative" interpretation of previously absolute concepts such as length and time, and that the classical concept of absolute velocity is not valid. It is reasonable then to ask how far this revolution is to go in changing our interpretation of physical concepts. Dynamical quantities, such as momentum and kinetic energy, depend on length, time, and velocity. Do classical laws of momentum and energy conservation remain valid in Einstein's relativity?

Let's test the conservation laws by examining the collision shown in Figure 2.21a. Two particles collide elastically as observed in the reference frame of O' . Particle 1 of mass $m_1 = 2m$ is initially at rest, and particle 2 of mass $m_2 = m$ is moving in the negative x direction with an initial velocity of $v'_{2i} = -0.750c$. Using the classical law of momentum conservation to analyze this collision, O' would calculate the particles to be moving with final

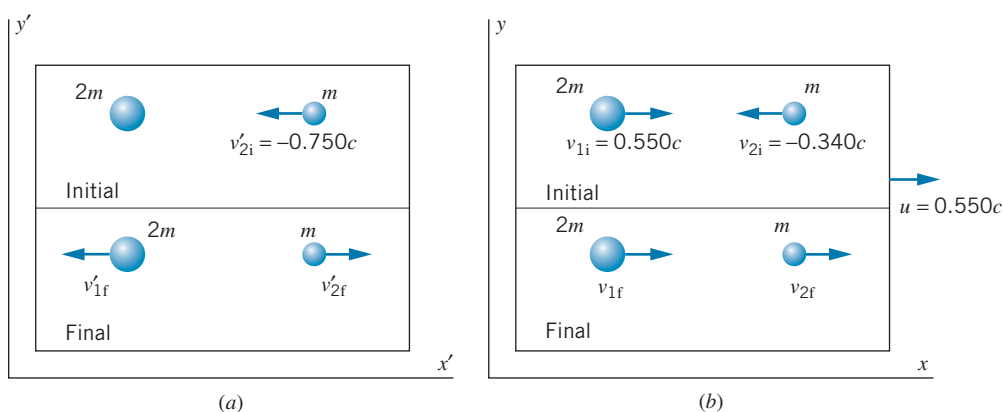


FIGURE 2.21 (a) A collision between two particles as observed from the reference frame of O' . (b) The same collision observed from the reference frame of O .

velocities $v'_{1f} = -0.500c$ and $v'_{2f} = +0.250c$. According to O' , the total initial and final momenta of the particles would be

$$\begin{aligned} p'_i &= m_1 v'_{1i} + m_2 v'_{2i} = (2m)(0) + (m)(-0.750c) = -0.750mc \\ p'_f &= m_1 v'_{1f} + m_2 v'_{2f} = (2m)(-0.500c) + (m)(0.250c) = -0.750mc \end{aligned}$$

The initial and final momenta are equal according to O' , demonstrating that momentum is conserved.

Suppose that the reference frame of O' moves at a velocity of $u = +0.550c$ in the x direction relative to observer O , as in Figure 2.21b. How would observer O analyze this collision? We can find the initial and final velocities of the two particles according to O using the velocity transformation of Eq. 2.17, which gives the initial velocities shown in the figure and the final velocities $v_{1f} = +0.069c$ and $v_{2f} = +0.703c$. Observer O can now calculate the initial and final values of the total momentum of the two particles:

$$\begin{aligned} p_i &= m_1 v_{1i} + m_2 v_{2i} = (2m)(+0.550c) + (m)(-0.340c) = +0.760mc \\ p_f &= m_1 v_{1f} + m_2 v_{2f} = (2m)(+0.069c) + (m)(+0.703c) = +0.841mc \end{aligned}$$

Momentum is therefore *not conserved* according to observer O .

This collision experiment has shown that the law of conservation of linear momentum, with momentum defined as $\vec{p} = m\vec{v}$, does not satisfy Einstein's first postulate (the law must be the same in all inertial frames). We cannot have a law that is valid for some observers but not for others. Therefore, *if we are to retain the conservation of momentum as a general law consistent with Einstein's first postulate, we must find a new definition of momentum.* This new definition of momentum must have two properties: (1) It must yield a law of conservation of momentum that satisfies the principle of relativity; that is, if momentum is conserved according to an observer in one inertial frame, then it is conserved according to observers in all inertial frames. (2) At low speeds, the new definition must reduce to $\vec{p} = m\vec{v}$, which we know works perfectly well in the nonrelativistic case.

These requirements are satisfied by defining the relativistic momentum for a particle of mass m moving with velocity $\vec{v}$ as

$$\vec{p} = \frac{m\vec{v}}{\sqrt{1 - v^2/c^2}} \quad (2.32)$$

In terms of components, we can write Eq. 2.32 as

$$p_x = \frac{mv_x}{\sqrt{1 - v^2/c^2}} \quad \text{and} \quad p_y = \frac{mv_y}{\sqrt{1 - v^2/c^2}} \quad (2.33)$$

The velocity v that appears in the denominator of these expressions is always the velocity of the particle as measured in a particular inertial frame. It is not the velocity of an inertial frame. The velocity in the numerator can be any of the components of the velocity vector.

We can now reanalyze the collision shown in Figure 2.21 using the relativistic definition of momentum. The initial relativistic momentum according to O' is

$$p'_i = \frac{m_1 v'_{1i}}{\sqrt{1 - v'^2_{1i}/c^2}} + \frac{m_2 v'_{2i}}{\sqrt{1 - v'^2_{2i}/c^2}} = \frac{(2m)(0)}{\sqrt{1 - 0^2}} + \frac{(m)(-0.750c)}{\sqrt{1 - (0.750)^2}} = -1.134mc$$

The final velocities according to O' are $v'_{1f} = -0.585c$ and $v'_{2f} = +0.294c$, and the total final momentum is

$$\begin{aligned} p'_f &= \frac{m_1 v'_{1f}}{\sqrt{1 - v'^2_{1f}/c^2}} + \frac{m_2 v'_{2f}}{\sqrt{1 - v'^2_{2f}/c^2}} \\ &= \frac{(2m)(-0.585c)}{\sqrt{1 - (0.585)^2}} + \frac{(m)(0.294c)}{\sqrt{1 - (0.294)^2}} = -1.134mc \end{aligned}$$

Thus $p'_i = p'_f$, and observer O' concludes that momentum is conserved. According to O , the initial relativistic momentum is

$$p_i = \frac{m_1 v_{1i}}{\sqrt{1 - v^2_{1i}/c^2}} + \frac{m_2 v_{2i}}{\sqrt{1 - v^2_{2i}/c^2}} = \frac{(2m)(+0.550c)}{\sqrt{1 - (0.550)^2}} + \frac{(m)(-0.340c)}{\sqrt{1 - (0.340)^2}} = 0.956mc$$

Using the velocity transformation, the final velocities measured by O are $v_{1f} = -0.051c$ and $v_{2f} = +0.727c$, and so O calculates the final momentum to be

$$p_f = \frac{m_1 v_{1f}}{\sqrt{1 - v^2_{1f}/c^2}} + \frac{m_2 v_{2f}}{\sqrt{1 - v^2_{2f}/c^2}} = \frac{(2m)(-0.051c)}{\sqrt{1 - (0.051)^2}} + \frac{(m)(+0.727c)}{\sqrt{1 - (0.727)^2}} = 0.956mc$$

Observer O also concludes that $p_i = p_f$ and that the law of conservation of momentum is valid. Defining momentum according to Eq. 2.32 gives conservation of momentum in *all* reference frames, as required by the principle of relativity.

Example 2.13

What is the momentum of a proton moving at a speed of $v = 0.86c$?

Solution

Using Eq. 2.32, we obtain

$$\begin{aligned} p &= \frac{mv}{\sqrt{1 - v^2/c^2}} \\ &= \frac{(1.67 \times 10^{-27} \text{ kg})(0.86)(3.00 \times 10^8 \text{ m/s})}{\sqrt{1 - (0.86)^2}} \\ &= 8.44 \times 10^{-19} \text{ kg} \cdot \text{m/s} \end{aligned}$$

The units of $\text{kg} \cdot \text{m/s}$ are generally not convenient in solving problems of this type. Instead, we manipulate Eq. 2.32 to obtain

$$\begin{aligned} pc &= \frac{mvc}{\sqrt{1 - v^2/c^2}} = \frac{mc^2(v/c)}{\sqrt{1 - v^2/c^2}} = \frac{(938 \text{ MeV})(0.86)}{\sqrt{1 - (0.86)^2}} \\ &= 1580 \text{ MeV} \end{aligned}$$

Here we have used the proton's *rest energy* mc^2 , which is defined later in this section. The momentum is obtained from this result by dividing by the symbol c (not its numerical value), which gives

$$p = 1580 \text{ MeV}/c$$

The units of MeV/c for momentum are often used in relativistic calculations because, as we show later, the quantity pc often appears in these calculations. You should be able to convert MeV/c to $\text{kg} \cdot \text{m/s}$ and show that the two results obtained for p are equivalent.

Relativistic Kinetic Energy

Like the classical definition of momentum, the classical definition of kinetic energy also causes difficulties when we try to compare the interpretations of

different observers. According to O' , the initial and final kinetic energies in the collision shown in Figure 2.21a are

$$K_i' = \frac{1}{2}m_1 v_{1i}'^2 + \frac{1}{2}m_2 v_{2i}'^2 = (0.5)(2m)(0)^2 + (0.5)(m)(-0.750c)^2 = 0.281mc^2$$

$$K_f' = \frac{1}{2}m_1 v_{1f}'^2 + \frac{1}{2}m_2 v_{2f}'^2 = (0.5)(2m)(-0.500c)^2 + (0.5)(m)(0.250c)^2 = 0.281mc^2$$

and so energy is conserved according to O' . The initial and final kinetic energies observed from the reference frame of O (as in Figure 2.21b) are

$$K_i = \frac{1}{2}m_1 v_{1i}^2 + \frac{1}{2}m_2 v_{2i}^2 = (0.5)(2m)(0.550c)^2 + (0.5)(m)(-0.340c)^2 = 0.360mc^2$$

$$K_f = \frac{1}{2}m_1 v_{1f}^2 + \frac{1}{2}m_2 v_{2f}^2 = (0.5)(2m)(0.069c)^2 + (0.5)(m)(0.703c)^2 = 0.252mc^2$$

Thus energy is *not conserved* in the reference frame of O if we use the classical formula for kinetic energy. This leads to a serious inconsistency—an elastic collision for one observer would not be elastic for another observer. As in the case of momentum, if we want to preserve the law of conservation of energy for all observers, we must replace the classical formula for kinetic energy with an expression that is valid in the relativistic case (but that reduces to the classical formula for low speeds).

In classical physics, we obtained the expression for kinetic energy by using the work–energy theorem, which states that the change in kinetic energy of a particle is equal to the work W done on the particle by a force F . In one dimension, we can write $W = \int F dx = \int (dp/dt) dx = \int (dx/dt) dp = \int v dp = pv - \int p dv$, where the last step comes from integration by parts. If we assume the work–energy theorem can be carried over into the relativistic region, then using Eq. 2.32 for the momentum we obtain

$$K = \frac{mv}{\sqrt{1-v^2/c^2}}v - \int_0^v \frac{mv}{\sqrt{1-v^2/c^2}} dv = \frac{mv^2}{\sqrt{1-v^2/c^2}} + mc^2 \sqrt{1-v^2/c^2} - mc^2$$

Combining terms we get

$$K = \frac{mc^2}{\sqrt{1-v^2/c^2}} - mc^2 \quad (2.34)$$

This is the relativistic form for the kinetic energy. Using Eq. 2.34, you can show that both O and O' will conclude that kinetic energy is conserved. In fact, all observers will agree that the energy conservation law is valid if the relativistic definition of kinetic energy is used.

Equation 2.34 looks very different from the classical form for kinetic energy, and it may not be obvious that the relativistic equation reduces to the classical when $v \ll c$. To show that this is the case we use the binomial expansion $(1+x)^n \approx 1+nx$ to approximate the denominator in Eq. 2.34:

$$K \approx mc^2 \left(1 + \frac{1}{2} \frac{v^2}{c^2} \right) - mc^2 = \frac{1}{2}mv^2$$

The classical expression for kinetic energy also violates the second relativity postulate by allowing speeds in excess of the speed of light. There is no limit (in either classical or relativistic dynamics) to the energy we can give to

a particle. Yet, if we allow the kinetic energy to increase without limit, the classical expression $K = \frac{1}{2}mv^2$ implies that the velocity must correspondingly increase without limit, thereby violating the second postulate. You can also see from the first term of Eq. 2.34 that $K \rightarrow \infty$ as $v \rightarrow c$. Thus we can increase the relativistic kinetic energy of a particle without limit, and its speed will not exceed c .

Relativistic Total Energy and Rest Energy

We can also express Eq. 2.34 as

$$K = E - E_0 \quad (2.35)$$

where the *relativistic total energy* E is defined as

$$E = \frac{mc^2}{\sqrt{1 - v^2/c^2}} \quad (2.36)$$

and the *rest energy* E_0 is defined as

$$E_0 = mc^2 \quad (2.37)$$

The rest energy is in effect the relativistic total energy of a particle measured in a frame of reference in which the particle is at rest.

Sometimes m in Eq. 2.37 is called the *rest mass* m_0 and is distinguished from the “relativistic mass,” which is defined as $m_0/\sqrt{1 - v^2/c^2}$. We choose not to use relativistic mass, because it can be a misleading concept. Whenever we refer to mass, we always mean rest mass.

Equation 2.37 suggests that mass can be expressed in units of energy divided by c^2 , such as MeV/c^2 . For example, a proton has a rest energy of 938 MeV and thus a mass of $938 \text{ MeV}/c^2$. Just like expressing momentum in units of MeV/c , expressing mass in units of MeV/c^2 turns out to be very useful in calculations.

The relativistic total energy is given by Eq. 2.35 as

$$E = K + E_0 \quad (2.38)$$

Collisions of particles at high energies often result in the production of new particles, and thus the final rest energy may not be equal to the initial rest energy (see Example 2.21). Such collisions must be analyzed using conservation of total relativistic energy E ; kinetic energy will *not* be conserved when the rest energy changes in a collision. In the special example of the elastic collision considered in this section, the identities of the particles did not change, and so kinetic energy was conserved. In general, collisions do not conserve kinetic energy—it is the relativistic total energy that is conserved in collisions.

Manipulation of Eqs. 2.32 and 2.36 gives a useful relationship among the total energy, momentum, and rest energy:

$$E = \sqrt{(pc)^2 + (mc^2)^2} \quad (2.39)$$

Figure 2.22 shows a useful way of remembering this relationship, which has the form of the Pythagorean theorem for the sides of a right triangle.

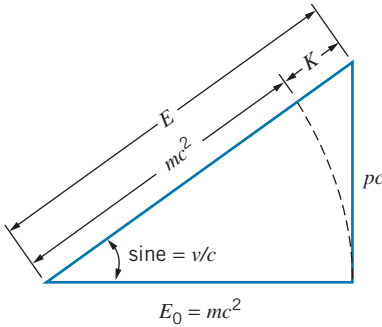


FIGURE 2.22 A useful way of recalling the relationships among E_0 , p , K , and E . Note that to put all variables in energy units, the quantity pc must be used.

When a particle travels at a speed close to the speed of light (say, $v > 0.99c$), which often occurs in high-energy particle accelerators, the particle's kinetic energy is much greater than its rest energy; that is, $K \gg E_0$. In this case, Eq. 2.39 can be written, to a very good approximation,

$$E \cong pc \quad (2.40)$$

This is called the *extreme relativistic approximation* and is often useful for simplifying calculations. As v approaches c , the angle in Figure 2.22 between the bottom leg of the triangle (representing mc^2) and the hypotenuse (representing E) approaches 90° . Imagine in this case a very tall triangle, in which the vertical leg (pc) and the hypotenuse (E) are nearly the same length.

For massless particles (such as photons), Eq. 2.39 becomes exactly

$$E = pc \quad (2.41)$$

All massless particles travel at the speed of light; otherwise, by Eqs. 2.34 and 2.36, their kinetic and total energies would be zero.

Example 2.14

What are the kinetic and relativistic total energies of a proton ($E_0 = 938 \text{ MeV}$) moving at a speed of $v = 0.86c$?

Solution

In Example 2.13, we found the momentum of this particle to be $p = 1580 \text{ MeV}/c$. The total energy can be found from Eq. 2.39:

$$\begin{aligned} E &= \sqrt{(pc)^2 + (mc^2)^2} = \sqrt{(1580 \text{ MeV})^2 + (938 \text{ MeV})^2} \\ &= 1837 \text{ MeV} \end{aligned}$$

The kinetic energy follows from Eq. 2.35:

$$\begin{aligned} K &= E - E_0 \\ &= 1837 \text{ MeV} - 938 \text{ MeV} \\ &= 899 \text{ MeV} \end{aligned}$$

We also could have solved this problem by finding the kinetic energy directly from Eq. 2.34.

Example 2.15

Find the velocity and momentum of an electron ($E_0 = 0.511 \text{ MeV}$) with a kinetic energy of 10.0 MeV .

Solution

The total energy is $E = K + E_0 = 10.0 \text{ MeV} + 0.511 \text{ MeV} = 10.51 \text{ MeV}$. We then can find the momentum from Eq. 2.39:

$$\begin{aligned} p &= \frac{1}{c} \sqrt{E^2 - (mc^2)^2} = \frac{1}{c} \sqrt{(10.51 \text{ MeV})^2 - (0.511 \text{ MeV})^2} \\ &= 10.5 \text{ MeV}/c \end{aligned}$$

Note that in this problem we could have used the extreme relativistic approximation, $p \cong E/c$, from Eq. 2.40. The error we would make in this case would be only 0.1%.

The velocity can be found by solving Eq. 2.36 for v .

$$\begin{aligned} \frac{v}{c} &= \sqrt{1 - \left(\frac{mc^2}{E}\right)^2} = \sqrt{1 - \left(\frac{0.511 \text{ MeV}}{10.51 \text{ MeV}}\right)^2} \\ &= 0.9988 \end{aligned} \quad (2.42)$$

Example 2.16

In the Stanford Linear Collider electrons are accelerated to a kinetic energy of 50 GeV. Find the speed of such an electron as (a) a fraction of c and (b) a difference from c . The rest energy of the electron is $0.511 \text{ MeV} = 0.511 \times 10^{-3} \text{ GeV}$.

Solution

(a) First we solve Eq. 2.34 for v , obtaining

$$v = c \sqrt{1 - \frac{1}{(1 + K/mc^2)^2}} \quad (2.43)$$

and thus

$$\begin{aligned} v &= c \sqrt{1 - \frac{1}{[1 + (50 \text{ GeV})/(0.511 \times 10^{-3} \text{ GeV})]^2}} \\ &= 0.999\,999\,999\,948c \end{aligned}$$

Calculators cannot be trusted to 12 significant digits. Here is a way to avoid this difficulty. We can write

Eq. 2.43 as $v = c(1 + x)^{1/2}$, where $x = -1/(1 + K/mc^2)^2$. Because $K \gg mc^2$, we have $x \ll 1$, and we can use the binomial expansion to write $v \cong c(1 + \frac{1}{2}x)$, or

$$v \cong c \left[1 - \frac{1}{2(1 + K/mc^2)^2} \right]$$

which gives

$$v \cong c(1 - 5.2 \times 10^{-11})$$

This leads to the same value of v given above.

(b) From the above result, we have

$$\begin{aligned} c - v &= 5.2 \times 10^{-11}c \\ &= 0.016 \text{ m/s} \\ &= 1.6 \text{ cm/s} \end{aligned}$$

Example 2.17

At a distance equal to the radius of the Earth's orbit ($1.5 \times 10^{11} \text{ m}$), the Sun's radiation has an intensity of about $1.4 \times 10^3 \text{ W/m}^2$. Find the rate at which the mass of the Sun is decreasing.

Solution

If we assume that the Sun's radiation is distributed uniformly over the surface area $4\pi r^2$ of a sphere of radius $1.5 \times 10^{11} \text{ m}$, then the total radiative power emitted by the Sun is

$$\begin{aligned} 4\pi(1.5 \times 10^{11} \text{ m})^2(1.4 \times 10^3 \text{ W/m}^2) \\ = 4.0 \times 10^{26} \text{ W} = 4.0 \times 10^{26} \text{ J/s} \end{aligned}$$

By conservation of energy, we know that the energy lost by the Sun through radiation must be accounted for by a corresponding loss in its rest energy. The change in mass Δm corresponding to a change in rest energy ΔE_0 of $4.0 \times 10^{26} \text{ J}$ each second is

$$\Delta m = \frac{\Delta E_0}{c^2} = \frac{4.0 \times 10^{26} \text{ J}}{9.0 \times 10^{16} \text{ m}^2/\text{s}^2} = 4.4 \times 10^9 \text{ kg}$$

The Sun loses mass at a rate of about 4 billion kilograms per second! If this rate were to remain constant, the Sun (with a present mass of $2 \times 10^{30} \text{ kg}$) would shine “only” for another 10^{13} years.

2.8 CONSERVATION LAWS IN RELATIVISTIC DECAYS AND COLLISIONS

In all decays and collisions, we must apply the law of conservation of momentum. The only difference between applying this law for collisions at low speed (as we did in Example 1.1) and at high speed is the use of the relativistic expression for momentum (Eq. 2.32) instead of Eq. 1.2. The law of conservation of

momentum for relativistic motion can be stated in exactly the same way as for classical motion:

In an isolated system of particles, the total linear momentum remains constant.

In the classical case, kinetic energy is the only form of energy that is present in elastic collisions, so conservation of energy is equivalent to conservation of kinetic energy. In inelastic collisions or decay processes, the kinetic energy does not remain constant. Total energy is conserved in classical inelastic collisions, but we did not account for the other forms of energy that might be important. This missing energy is usually stored in the particles, perhaps as atomic or nuclear energy.

In the relativistic case, the internal stored energy contributes to the rest energy of the particles. Usually rest energy and kinetic energy are the only two forms of energy that we consider in atomic or nuclear processes (later we'll add the energy of radiation to this balance). A loss of kinetic energy in a collision is thus accompanied by a gain in rest energy, but the total relativistic energy (kinetic energy + rest energy) of all the particles involved in the process doesn't change. For example, in a reaction in which new particles are produced, the loss in kinetic energy of the original reacting particles gives the increase in rest energy of the product particles. On the other hand, in a nuclear decay process such as alpha decay, the initial nucleus gives up some rest energy to account for the kinetic energy carried by the decay products.

The law of energy conservation in the relativistic case is

In an isolated system of particles, the relativistic total energy (kinetic energy plus rest energy) remains constant.

In applying this law to relativistic collisions, we don't have to worry whether the collision is elastic or inelastic, because the inclusion of the rest energy accounts for any loss in kinetic energy.

The following examples illustrate applications of the conservation laws for relativistic momentum and energy.

Example 2.18

Two particles each of mass m are traveling toward each other with equal speeds v in reference frame O . They collide and form a new composite particle of mass M . (a) Is the mass M greater than $2m$, equal to $2m$, or less than $2m$? (b) Observer O' is in motion relative to O with speed v in the same direction as one of the particles, so that particle appears to be at rest in O' . Will O' agree with the answer to question (a)?

Solution

(a) Since the total momentum is zero in O , particle M is formed at rest. Its total energy, which is equal to the

sum of the total energies (rest energy + kinetic energy) of the two colliding particles, is equal to its rest energy. Therefore, the rest energy Mc^2 is greater than $2mc^2$, the difference being the kinetic energies of the two particles. (b) To observer O' , the momenta and kinetic energies of the two original particles differ from those in O , and the composite particle will likewise have different values of momentum and energy. But mass is an invariant in special relativity—all observers will agree on the value of the mass. In fact, no matter what the relative speed of O and O' , they both agree that $M > 2m$.

Example 2.19

A neutral K meson (mass $497.7 \text{ MeV}/c^2$) is moving with a kinetic energy of 77.0 MeV . It decays into a pi meson (mass $139.6 \text{ MeV}/c^2$) and another particle of unknown mass. The pi meson is moving in the direction of the original K meson with a momentum of $381.6 \text{ MeV}/c$. (a) Find the momentum and total relativistic energy of the unknown particle. (b) Find the mass of the unknown particle.

Solution

(a) The total energy and momentum of the K meson are

$$E_K = K_K + m_K c^2 = 77.0 \text{ MeV} + 497.7 \text{ MeV} = 574.7 \text{ MeV}$$

$$\begin{aligned} p_K &= \frac{1}{c} \sqrt{E_K^2 - (m_K c^2)^2} \\ &= \frac{1}{c} \sqrt{(574.7 \text{ MeV})^2 - (497.7 \text{ MeV})^2} \\ &= 287.4 \text{ MeV}/c \end{aligned}$$

and for the pi meson

$$\begin{aligned} E_\pi &= \sqrt{(cp_\pi)^2 + (m_\pi c^2)^2} \\ &= \sqrt{(381.6 \text{ MeV})^2 + (139.6 \text{ MeV})^2} \\ &= 406.3 \text{ MeV} \end{aligned}$$

Conservation of relativistic momentum ($p_{\text{initial}} = p_{\text{final}}$) gives $p_K = p_\pi + p_x$ (where x represents the unknown particle), so

$$\begin{aligned} p_x &= p_K - p_\pi = 287.4 \text{ MeV}/c - 381.6 \text{ MeV}/c \\ &= -94.2 \text{ MeV}/c \end{aligned}$$

and conservation of total relativistic energy ($E_{\text{initial}} = E_{\text{final}}$) gives $E_K = E_\pi + E_x$, so

$$\begin{aligned} E_x &= E_K - E_\pi = 574.7 \text{ MeV} - 406.3 \text{ MeV} \\ &= 168.4 \text{ MeV} \end{aligned}$$

(b) We can find the mass by solving Eq. 2.39 for mc^2 :

$$\begin{aligned} m_x c^2 &= \sqrt{E_x^2 - (cp_x)^2} \\ &= \sqrt{(168.4 \text{ MeV})^2 - (94.2 \text{ MeV})^2} \\ &= 139.6 \text{ MeV} \end{aligned}$$

Thus the unknown particle has a mass of $139.6 \text{ MeV}/c^2$, and its mass shows that it is another pi meson.

Example 2.20

In the reaction $K^- + p \rightarrow \Lambda^0 + \pi^0$, a charged K meson (mass $493.7 \text{ MeV}/c^2$) collides with a proton ($938.3 \text{ MeV}/c^2$) at rest, producing a lambda particle ($1115.7 \text{ MeV}/c^2$) and a neutral pi meson ($135.0 \text{ MeV}/c^2$), as represented in Figure 2.23. The initial kinetic energy of the K meson is 152.4 MeV . After the interaction, the pi meson has a kinetic energy of 254.8 MeV . (a) Find the kinetic energy of the lambda. (b) Find the directions of motion of the lambda and the pi meson.

Solution

(a) The initial and final total energies are

$$\begin{aligned} E_{\text{initial}} &= E_K + E_p = K_K + m_K c^2 + m_p c^2 \\ E_{\text{final}} &= E_\Lambda + E_\pi = K_\Lambda + m_\Lambda c^2 + K_\pi + m_\pi c^2 \end{aligned}$$

In these two equations, the value of every quantity is known except the kinetic energy of the lambda.

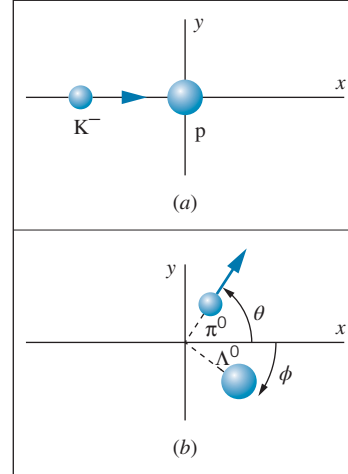


FIGURE 2.23 Example 2.20. (a) A K^- meson collides with a proton at rest. (b) After the collision, a π^0 meson and a Λ^0 are produced.

Using conservation of total relativistic energy, we set $E_{\text{initial}} = E_{\text{final}}$ and solve for K_{Λ} :

$$\begin{aligned} K_{\Lambda} &= K_K + m_K c^2 + m_p c^2 - m_{\Lambda} c^2 - K_{\pi} - m_{\pi} c^2 \\ &= 152.4 \text{ MeV} + 493.7 \text{ MeV} + 938.3 \text{ MeV} \\ &\quad - 1115.7 \text{ MeV} - 254.8 \text{ MeV} - 135.0 \text{ MeV} \\ &= 78.9 \text{ MeV} \end{aligned}$$

(b) To find the directional information we must apply conservation of momentum. The initial momentum is just that of the K meson. From its total energy, $E_K = K_K + m_K c^2 = 152.4 \text{ MeV} + 493.7 \text{ MeV} = 646.1 \text{ MeV}$, we can find the momentum:

$$\begin{aligned} p_{\text{initial}} &= p_K = \frac{1}{c} \sqrt{(E_K)^2 - (m_K c^2)^2} \\ &= \frac{1}{c} \sqrt{(646.1 \text{ MeV})^2 - (493.7 \text{ MeV})^2} \\ &= 416.8 \text{ MeV}/c \end{aligned}$$

A similar procedure applied to the two final particles gives $p_{\Lambda} = 426.9 \text{ MeV}/c$ and $p_{\pi} = 365.7 \text{ MeV}/c$. The total momentum of the two final particles is $p_{x,\text{final}} = p_{\Lambda} \cos \theta + p_{\pi} \cos \phi$ and $p_{y,\text{final}} = p_{\Lambda} \sin \theta - p_{\pi} \sin \phi$. Conservation of momentum in the x and y directions gives

$$p_{\Lambda} \cos \theta + p_{\pi} \cos \phi = p_{\text{initial}} \quad \text{and} \quad p_{\Lambda} \sin \theta - p_{\pi} \sin \phi = 0$$

Here we have two equations with two unknowns (θ and ϕ). We can eliminate θ by writing the first equation as $p_{\Lambda} \cos \theta = p_{\text{initial}} - p_{\pi} \cos \phi$, then squaring both equations and adding them. The resulting equation can be solved for ϕ :

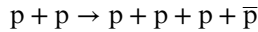
$$\begin{aligned} \phi &= \cos^{-1} \left(\frac{p_{\text{initial}}^2 + p_{\pi}^2 - p_{\Lambda}^2}{2p_{\pi} p_{\text{initial}}} \right) \\ &= \cos^{-1} \left(\frac{(416.8 \text{ MeV}/c)^2 + (365.7 \text{ MeV}/c)^2 - (426.9 \text{ MeV}/c)^2}{2(365.7 \text{ MeV}/c)(416.8 \text{ MeV}/c)} \right) \\ &= 65.7^\circ \end{aligned}$$

From the conservation of momentum equation for the y components, we have

$$\begin{aligned} \theta &= \sin^{-1} \left(\frac{p_{\pi} \sin \phi}{p_{\Lambda}} \right) \\ &= \sin^{-1} \left(\frac{(365.7 \text{ MeV}/c)(\sin 65.7^\circ)}{426.9 \text{ MeV}/c} \right) = 51.3^\circ \end{aligned}$$

Example 2.21

The discovery of the antiproton $\bar{p}$ (a particle with the same rest energy as a proton, 938 MeV, but with the opposite electric charge) took place in 1956 through the following reaction:



in which accelerated protons were incident on a target of protons at rest in the laboratory. The minimum incident kinetic energy needed to produce the reaction is called the *threshold* kinetic energy, for which the final particles move together as if they were a single unit (Figure 2.24). Find the threshold kinetic energy to produce antiprotons in this reaction.

Solution

This problem can be solved by a straightforward application of energy and momentum conservation. Let E_p and p_p represent the total energy and momentum of the

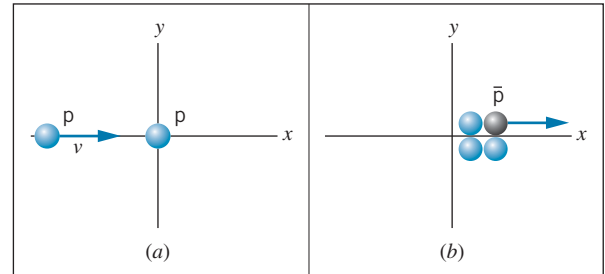


FIGURE 2.24 Example 2.21. (a) A proton moving with velocity v collides with another proton at rest. (b) The reaction produces three protons and an antiproton, which move together as a unit.

incident proton. Thus the initial total energy of the two protons is $E_p + m_p c^2$. Let E'_p and p'_p represent the total energy and momentum of *each* of the four final particles (which move together and thus have the same energy and

momentum). We can then apply conservation of total energy:

$$E_p + m_p c^2 = 4E'_p$$

and conservation of momentum:

$$p_p = 4p'_p$$

We can write the momentum equation as $\sqrt{E_p^2 - (m_p c^2)^2} = 4\sqrt{E_p'^2 - (m_p c^2)^2}$, so now we have two equations in two unknowns (E_p and E_p'). We eliminate E_p' , for example, by solving the energy conservation equation for E_p' and substituting into the momentum equation. The result is

$$E_p = 7m_p c^2$$

from which we can calculate the kinetic energy of the incident proton:

$$K_p = E_p - m_p c^2 = 6m_p c^2 = 6(938 \text{ MeV}) = 5628 \text{ MeV} \\ = 5.628 \text{ GeV}$$

The Bevatron accelerator at the Lawrence Berkeley Laboratory was designed with this experiment in mind, so that it could produce a beam of protons whose energy exceeded 5.6 GeV. The discovery of the antiproton in this reaction was honored with the award of the 1959 Nobel Prize to the experimenters, Emilio Segrè and Owen Chamberlain.

2.9 EXPERIMENTAL TESTS OF SPECIAL RELATIVITY

Because special relativity provided such a radical departure from the notions of space and time in classical physics, it is important to perform detailed experimental tests that can clearly distinguish between the predictions of special relativity and those of classical physics. Many tests of increasing precision have been done since the theory was originally presented, and in every case the predictions of special relativity are upheld. Here we discuss a few of these tests.

Universality of the Speed of Light

The second relativity postulate asserts that the speed of light has the same value c for all observers. This leads to several types of experimental tests, of which we discuss two: (1) Does the speed of light change with the direction of travel? (2) Does the speed of light change with relative motion between source and observer?

The Michelson–Morley experiment provides a test of the first type. This experiment compared the upstream–downstream and cross-stream speeds of light and concluded that they were equal within the experimental error. Equivalently, we may say that the experiment showed that there is no preferred reference frame (no ether) relative to which the speed of light must be measured. If there is an ether, the speed of the Earth through the ether is less than 5 km/s, which is much smaller than the Earth’s orbital speed about the Sun, 30 km/s. We can express their result as a difference Δc between the upstream–downstream and cross-stream speeds; the experiment showed that $\Delta c/c < 3 \times 10^{-10}$.

To reconcile the result of the Michelson–Morley experiment with classical physics, Lorentz proposed the “ether drag” hypothesis, according to which the motion of the Earth through the ether caused an electromagnetic drag that contracted the arm of the interferometer in the direction of motion.

This contraction was just enough to compensate for the difference in the upstream–downstream and cross-stream times predicted by the Galilean transformation. This hypothesis succeeds only when the two arms of the interferometer are of the same length. To test this hypothesis, a similar experiment was done in 1932 by Kennedy and Thorndike; in their experiment, the lengths of the interferometer arms differed by about 16 cm, the maximum distance over which light sources available at that time could remain coherent. The Kennedy–Thorndike experiment in effect tests the second question, whether the speed of light changes due to relative motion. Their result was $\Delta c/c < 3 \times 10^{-8}$, which excludes the Lorentz contraction hypothesis as an explanation for the Michelson–Morley experiment.

In recent years, these fundamental experiments have been repeated with considerably improved precision using lasers as light sources. Experimenters working at the Joint Institute for Laboratory Astrophysics in Boulder, Colorado, built an apparatus that consisted of two He-Ne lasers on a rotating granite platform. By electronically stabilizing the lasers, they improved the sensitivity of their apparatus by several orders of magnitude. Again expressing the result as a difference between the speeds along the two arms of the apparatus, this experiment corresponds to $\Delta c/c < 8 \times 10^{-15}$, an improvement of about 5 orders of magnitude over the original Michelson–Morley experiment. In a similar repetition of the Kennedy–Thorndike experiment using He-Ne lasers, they obtained $\Delta c/c < 1 \times 10^{-10}$, an improvement over the original experiment by a factor of 300. [See A. Brillet and J. L. Hall, *Physical Review Letters* **42**, 549 (1979); D. Hils and J. L. Hall, *Physical Review Letters* **64**, 1697 (1990).] A considerable improvement in the Kennedy–Thorndike type of experiment has been made possible by comparing the oscillation frequency of a crystal with the frequency of a hydrogen maser (a maser is similar to a laser, but it uses microwaves rather than visible light). The experimenters measured for nearly one year, looking for a change in the relative frequencies as the Earth's velocity changed. No effect was observed, leading to a limit of $\Delta c/c < 2 \times 10^{-12}$. [See P. Wolf et al., *Physical Review Letters* **90**, 060402 (2003).]

Another way of testing the second question is to measure the speed of a light beam emitted by a source in motion. Suppose we observe this beam along the direction of motion of the moving source, which might be moving toward us or away from us. In the rest frame of the source, the emitted light travels at speed c . We can express the speed of light in our reference frame as $c' = c + \Delta c$, where Δc is zero according to special relativity ($c' = c$) or is $\pm u$ according to classical physics ($c' = c \pm u$ in the Galilean transformation, depending on whether the motion is toward or away from the observer).

In one experiment of this type, the decay of pi mesons (pions) into gamma rays (a form of electromagnetic waves traveling at c) was observed. When pions (produced in laboratories with large accelerators) emit these gamma rays, they are traveling at speeds close to the speed of light, relative to the laboratory. Thus if Galilean relativity were valid, we should expect to find gamma rays emitted in the direction of motion of the decaying pions traveling at a speed c' in the laboratory of nearly $2c$, rather than always with c as predicted by special relativity. The observed laboratory speed of these gamma rays in one experiment was $(2.9977 \pm 0.0004) \times 10^8$ m/s when the decaying pions were moving at $u/c = 0.99975$. These results give $\Delta c/c < 2 \times 10^{-4}$, and

thus $c' = c$ as expected from special relativity. This experiment shows directly that an object moving at a speed of nearly c relative to the laboratory emits “light” that travels at a speed of c relative to both the object *and* the laboratory, giving direct evidence for Einstein’s second postulate. [See T. Alvagar and et al., *Physical Letters* **12**, 260 (1964).]

Another experiment of this type is to study the X rays emitted by a binary pulsar, a rapidly pulsating source of X rays in orbit about another star, which would eclipse the pulsar as it rotated in its orbit. If the speed of light (in this case, X rays) were to change as the pulsar moved first toward and later away from the Earth in its orbit, the beginning and end of the eclipse would not be equally spaced in time from the midpoint of the eclipse. No such effect is observed, and from these observations, it is concluded that $\Delta c/c < 2 \times 10^{-12}$, in agreement with predictions of special relativity. These experiments were done at $u/c = 10^{-3}$. [See K. Brecher, *Physical Review Letters* **39**, 1051 (1977).]

A different type of test of the limit by which the speed of light changes with direction of travel can be done using the clocks carried aboard the network of Earth satellites that make up the global positioning system (GPS). By comparing the readings of clocks on the GPS satellites with clocks on the ground at different times of day (as the satellites move relative to the ground stations), it is possible to test whether the change in the direction of travel affects the apparent synchronization of the clocks. No effect was observed, and the experimenters were able to set a limit of $\Delta c/c < 5 \times 10^{-9}$ for the difference between the one-way and round-trip speeds of light. [See P. Wolf and G. Petit, *Physical Review A* **56**, 4405 (1997).]

Time Dilation

We have already discussed the time dilation effect on the decay of muons produced by cosmic rays. Muon decay can also be studied in the laboratory. Muons can be produced following collisions in high-energy accelerators, and the decay of the muons can be followed by observing their decay products (ordinary electrons). These muons can either be trapped and decay at rest, or they can be placed in a beam and decay in flight. When muons are observed at rest, their decay lifetime is $2.198 \mu\text{s}$. (As we discuss in Chapter 12, decays generally follow an exponential law. The lifetime is the time after which a fraction $1/e = 0.368$ of the original muons remain.) This is the *proper lifetime*, measured in a frame of reference in which the muon is at rest. In one particular experiment, muons were trapped in a ring and circulated at a momentum of $p = 3094 \text{ MeV}/c$. The decays in flight occurred with a lifetime of $64.37 \mu\text{s}$ (measured in the laboratory frame of reference). For muons of this momentum, Eq. 2.8 gives a dilated lifetime of (see Problem 49) $64.38 \mu\text{s}$, which is in excellent agreement with the measured value and confirms the time dilation effect. [See J. Bailey et al., *Nature* **268**, 301 (1977).]

Another similar experiment was done with pions. The proper lifetime, measured for pions at rest, is known to be 26.0 ns . In one experiment, pions were observed in flight at $u/c = 0.913$, and their lifetime was measured to be 63.7 ns . (Pions decay to muons, so we can follow the exponential radioactive decay of the pions by observing the muons emitted as a result of the decay.) For pions moving at this speed, the expected dilated lifetime is in exact agreement with the measured value, once again confirming the time dilation effect. [See D. S. Ayres et al., *Physical Review D* **3**, 1051 (1971).]

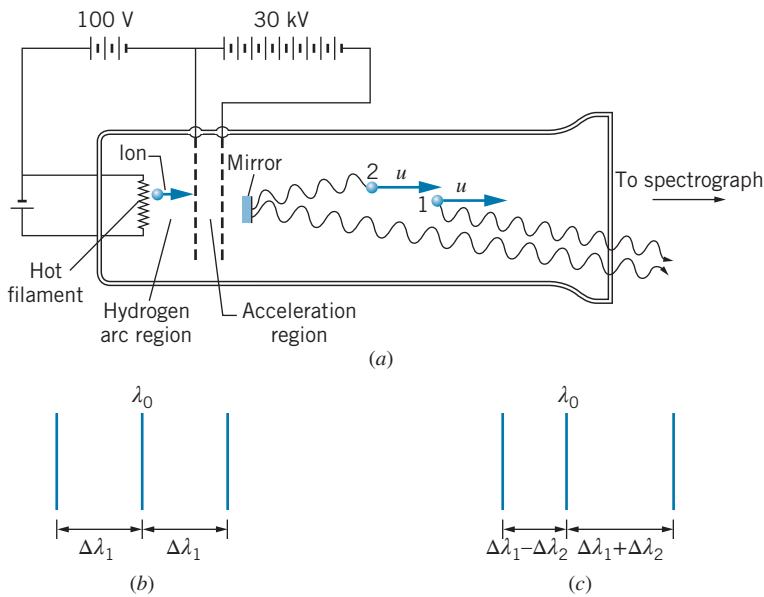


FIGURE 2.25 (a) Apparatus used in the Ives–Stilwell experiment. (b) Line spectrum expected from classical Doppler effect. (c) Line spectrum expected from relativistic Doppler effect.

The Doppler Effect

Confirmation of the relativistic Doppler effect first came from experiments done in 1938 by Ives and Stilwell. They sent a beam of hydrogen atoms, generated in a gas discharge, down a tube at a speed u , as shown in Figure 2.25. They could simultaneously observe light emitted by the atoms in a direction parallel to u (atom 1) and opposite to u (atom 2, reflected from the mirror). Using a spectrograph, the experimenters were able to photograph the characteristic spectral lines from these atoms and also, on the same photographic plate, from atoms at rest. If the classical Doppler formula were valid, the wavelengths of the lines from atoms 1 and 2 would be placed at symmetric intervals $\Delta\lambda_1 = \pm\lambda_0(u/c)$ on either side of the line from the atoms at rest (wavelength λ_0), as in Figure 2.25b. The relativistic Doppler formula, on the other hand, gives a small additional asymmetric shift $\Delta\lambda_2 = +\frac{1}{2}\lambda_0(u/c)^2$, as in Figure 2.25c (computed for $u \ll c$, so that higher-order terms in u/c can be neglected). Figure 2.26 shows the results of Ives and Stilwell for one of the hydrogen lines (the blue line of the Balmer series at $\lambda_0 = 486 \text{ nm}$). The agreement between the observed values and those predicted by the relativistic formula is impressive.

Recent experiments with lasers have verified the relativistic formula at greater accuracy. These experiments are based on the absorption of laser light by an atom; when the radiation is absorbed, the atom changes from its lowest-energy state (the ground state) to one of its excited states. The experiment consists essentially of comparing the laser wavelength needed to excite atoms at rest with that needed for atoms in motion. One experiment used a beam of hydrogen atoms with kinetic energy 800 MeV (corresponding to $u/c = 0.84$) produced in a high-energy proton accelerator. An ultraviolet

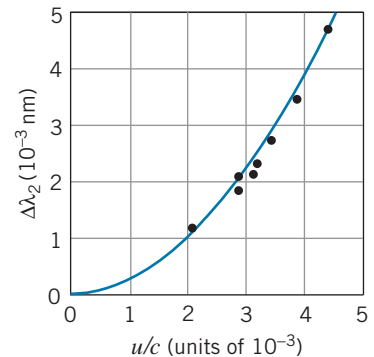


FIGURE 2.26 Results of the Ives–Stilwell experiment. According to classical theory, $\Delta\lambda_2 = 0$, while according to special relativity, $\Delta\lambda_2$ depends on $(u/c)^2$. The solid line, which represents the relativistic formula, gives excellent agreement with the data points.

laser was used to excite the atoms. This experiment verified the relativistic Doppler effect to an accuracy of about 3×10^{-4} . [See D. W. MacArthur et al., *Physical Review Letters* **56**, 282 (1986).] In another experiment, a beam of neon atoms moving with a speed of $u = 0.0036c$ was irradiated with light from a tunable dye laser. This experiment verified the relativistic Doppler shift to a precision of 2×10^{-6} . [See R. W. McGowan et al., *Physical Review Letters* **70**, 251 (1993).] A more recent study used two tunable dye lasers parallel and antiparallel to a beam of lithium atoms moving at $0.064c$. The results of this experiment agreed with the relativistic Doppler formula to within a precision of 2×10^{-7} , improving on the best previous results by an order of magnitude. [See G. Saathoff et al., *Physical Review Letters* **91**, 190403 (2003).]

Experiments with individual trapped atoms have enabled tests of special relativity at unprecedented levels of precision at even ordinary speeds. In one experiment, the frequency of light emitted by an aluminum ion oscillating in an applied electric field was measured by comparison with the light from a stationary trapped ion. According to the Doppler formula, Eq. 2.22, the change in frequency is $\Delta f = f' - f \cong -(u^2)_{\text{av}}/2c^2$, because u_{av} is zero for an oscillating atom. At speeds of the order of 10 m/s (typical of a short-distance sprinter), the relative change in frequency $\Delta f/f$ is of order 10^{-15} , but the precision of an atomic clock with trapped atoms is of order 10^{-17} , which allows a very precise test. The agreement with the special relativity formula at these low speeds is excellent. [See C. W. Chou et al., *Science* **329**, 1630 (2010).]

Relativistic Momentum and Energy

The earliest direct confirmation of the relativistic relationship for energy and momentum came just a few years after Einstein's 1905 paper. Simultaneous measurements were made of the momentum and velocity of high-energy electrons emitted in certain radioactive decay processes (nuclear beta decay, which is discussed in Chapter 12). Figure 2.27 shows the results of several different investigations plotted as p/mv , which should have the value 1 according to classical physics. The results agree with the relativistic formula and disagree

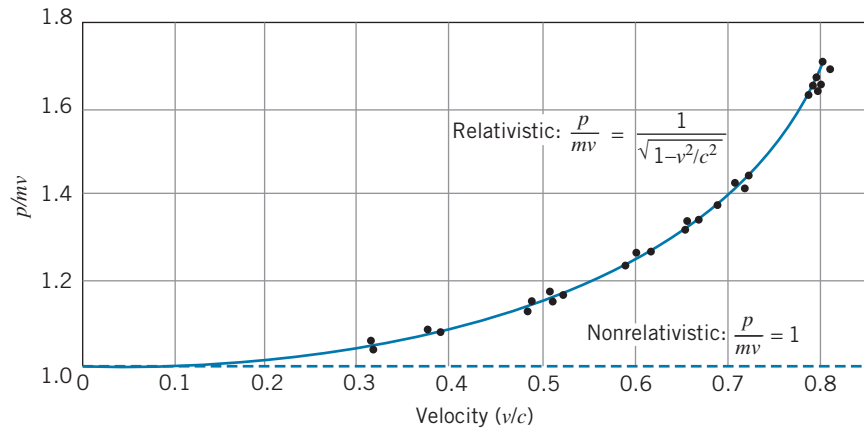


FIGURE 2.27 The ratio p/mv is plotted for electrons of various speeds. The data agree with the relativistic result and not at all with the nonrelativistic result ($p/mv = 1$).

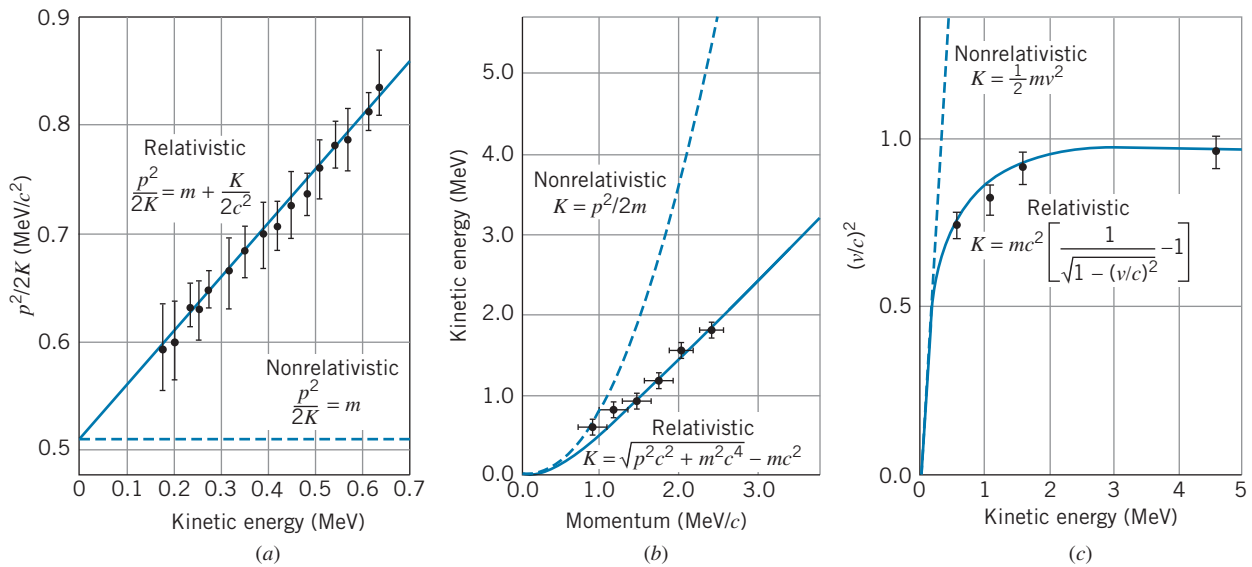


FIGURE 2.28 Confirmation of relativistic kinetic energy relationships. In (a) and (b) the momentum and energy of radioactive decay electrons were measured simultaneously. In these two independent experiments, the data were plotted in different ways, but the results are clearly in good agreement with the relativistic relationships and in poor agreement with the classical, nonrelativistic relationships. In (c) electrons were accelerated to a fixed energy through a large electric field (up to 4.5 million volts, as shown) and the velocities of the electrons were determined by measuring the flight time over 8.4 m. Notice that at small kinetic energies ($K \ll mc^2$), the relativistic and nonrelativistic relationships become identical. [Sources: (a) K. N. Geller and R. Kollarits, *Am. J. Phys.* **40**, 1125 (1972); (b) S. Parker, *Am. J. Phys.* **40**, 241 (1972); (c) W. Bertozzi, *Am. J. Phys.* **32**, 551 (1964).]

with the classical one. Note that the relativistic and classical formulas give the same results at low speeds, and in fact the two cannot be distinguished for speeds below $0.1c$, which accounts for our failure to observe these effects in experiments with ordinary laboratory objects.

Other more recent experiments, in which the kinetic energies of fast electrons were measured, are shown in Figure 2.28. Once again, the data at high speeds agree with special relativity and disagree with the classical equations. In a more extreme example, experimenters at the Stanford Linear Accelerator Center measured the speed of 20 GeV electrons, whose speed is within 5×10^{-10} of the speed of light (or about 0.15 m/s less than c). The measurement was not capable of this level of precision, but it did determine that the speed of the electrons was within 2×10^{-7} of the speed of light (60 m/s). [See Z. G. T. Guiragossian et al., *Physical Review Letters* **34**, 355 (1975).]

Nearly every time the nuclear or particle physicist enters the laboratory, a direct or indirect test of the momentum and energy relationships of special relativity is made. Principles of special relativity must be incorporated in the design of the high-energy accelerators used by nuclear and particle physicists, so even the construction of these projects gives testimony to the validity of the formulas of special relativity.

For example, consider the capture of a neutron by an atom of hydrogen to form an atom of deuterium or “heavy hydrogen.” Energy is released in this process, mostly in the form of electromagnetic radiation (gamma rays).

The energy of the gamma rays is measured to be 2.224 MeV. Where does this energy come from? It comes from the difference in mass when the hydrogen and neutron combine to form deuterium. The difference between the initial and final masses is

$$\begin{aligned}\Delta m &= m(\text{hydrogen}) + m(\text{neutron}) - m(\text{deuterium}) \\ &= 1.007825 \text{ u} + 1.008665 \text{ u} - 2.014102 \text{ u} = 0.002388 \text{ u}\end{aligned}$$

The initial mass of hydrogen plus neutron is greater than the final mass of deuterium by 0.002388 u. The energy equivalent of this change in mass is

$$\Delta E = (\Delta m)c^2 = 2.224 \text{ MeV}$$

which is equal to the energy released as gamma rays.

Similar experiments have been done to test the $E = mc^2$ relationship by measuring the energy released as gamma rays following the capture of neutrons by atoms of silicon and sulfur, and comparing the gamma-ray energies with the difference between the initial and final masses. These experiments are consistent with $E = mc^2$ to a precision of about 4×10^{-7} . [See S. Rainville et al., *Nature* **438**, 1096 (2006).]

Twin Paradox

Although we cannot perform the experiment to test the twin paradox as we have described it, we can do an equivalent experiment. We take two clocks in our laboratory and synchronize them carefully. We then place one of the clocks in an airplane and fly it around the Earth. When we return the clock to the laboratory and compare the two clocks, we expect to find, if special relativity is correct, that the clock that has left the laboratory is the “younger” one—that is, it will have ticked away fewer seconds and appear to run behind its stationary twin. In this experiment, we must use very precise clocks based on the atomic vibrations of cesium in order to measure the time differences between the clock readings, which amount to only about 10^{-7} s. This experiment is complicated by several factors, all of which can be computed rather precisely: the rotating Earth is *not* an inertial frame (there is a centripetal acceleration), clocks on the surface of the Earth are *already* moving because of the rotation of the Earth, and the *general* theory of relativity predicts that a change in the gravitational field strength, which our moving clock will experience as it changes altitude in its airplane flight, will also change the rate at which the clock runs. In this experiment, as in the others we have discussed, the results are entirely in agreement with the predictions of special relativity. [See J. C. Hafele and R. E. Keating, *Science* **177**, 166 (1972).]

In a similar experiment, a cesium atomic clock carried on the space shuttle was compared with an identical clock on the Earth. The comparison was made through a radio link between the shuttle and the ground station. At an orbital height of about 328 km, the shuttle moves at a speed of about 7712 m/s, or $2.5 \times 10^{-5}c$. A clock moving at this speed runs slower than an identical clock at rest by the time dilation factor. For every second the clock is in orbit, it loses 330 ps relative to the clock on Earth; equivalently, it loses about $1.8\mu\text{s}$ per orbit. These time intervals can be measured with great precision, and the predicted asymmetric aging was verified to a precision of about 0.1%. [See E. Sappl, *Naturwissenschaften* **77**, 325 (1990).]

Chapter Summary

		Section			Section
Galilean relativity	$x' = x - ut, v'_x = v_x - u$	2.1	Lorentz velocity transformation	$v'_x = \frac{v_x - u}{1 - v_x u/c^2},$	2.5
Einstein's postulates	(1) The laws of physics are the same in all inertial frames. (2) The speed of light has the same value c in all inertial frames.	2.3		$v'_y = \frac{v_y \sqrt{1 - u^2/c^2}}{1 - v_x u/c^2},$	
				$v'_z = \frac{v_z \sqrt{1 - u^2/c^2}}{1 - v_x u/c^2}$	
Time dilation	$\Delta t = \frac{\Delta t_0}{\sqrt{1 - u^2/c^2}}$ (Δt_0 = proper time)	2.4	Clock synchronization	$\Delta t' = \frac{uL/c^2}{\sqrt{1 - u^2/c^2}}$	2.5
Length contraction	$L = L_0 \sqrt{1 - u^2/c^2}$ (L_0 = proper length)	2.4	Relativistic momentum	$\vec{p} = \frac{m\vec{v}}{\sqrt{1 - v^2/c^2}}$	2.7
Velocity addition	$v = \frac{v' + u}{1 + v' u/c^2}$	2.4	Relativistic kinetic energy	$K = \frac{mc^2}{\sqrt{1 - v^2/c^2}} - mc^2$	2.7
Doppler effect (source and observer separating)	$f' = f \sqrt{\frac{1 - u/c}{1 + u/c}}$	2.4	Rest energy	$E_0 = mc^2$	2.7
Lorentz transformation	$x' = \frac{x - ut}{\sqrt{1 - u^2/c^2}},$ $y' = y, z' = z,$ $t' = \frac{t - (u/c^2)x}{\sqrt{1 - u^2/c^2}}$	2.5	Relativistic total energy	$E = K + E_0 = \frac{mc^2}{\sqrt{1 - v^2/c^2}}$	2.7
			Momentum–energy relationship	$E = \sqrt{(pc)^2 + (mc^2)^2}$	2.7
			Extreme relativistic approximation	$E \cong pc$	2.7
			Conservation laws	In an isolated system of particles, the total momentum and the relativistic total energy remain constant.	2.8

Questions

- Explain in your own words what is meant by the term “relativity.” Are there different theories of relativity?
- Suppose the two observers and the rock described in the first paragraph of Section 2.1 were isolated in interstellar space. Discuss the two observers’ differing perceptions of the motion of the rock. Is there any experiment they can do to determine whether the rock is moving in any absolute sense?
- Describe the situation of Figure 2.4 as it would appear from the reference frame of O' .
- Does the Michelson–Morley experiment show that the ether does not exist or that it is merely unnecessary?
- Suppose we made a pair of shears in which the cutting blades were many orders of magnitude longer than the handle. Let us in fact make them so long that, when we move the handles at angular velocity ω , a point on the tip of the blade has a tangential velocity $v = \omega r$ that is greater than c . Does this contradict special relativity? Justify your answer.
- Light travels through water at a speed of about 2.25×10^8 m/s. Is it possible for a particle to travel through water at a speed v greater than 2.25×10^8 m/s?
- Is it possible to have particles that travel at the speed of light? What does Eq. 2.36 require of such particles?
- How does relativity combine space and time coordinates into spacetime?
- Einstein developed the relativity theory after trying unsuccessfully to imagine how a light beam would look to an observer traveling with the beam at speed c . Why is this so difficult to imagine?

10. Explain in your own words the terms *time dilation* and *length contraction*.
11. Does the Moon's disk appear to be a different size to a space traveler approaching it at $v = 0.99c$, compared with the view of a person at rest at the same location?
12. According to the time dilation effect, would the life expectancy of someone who lives at the equator be longer or shorter than someone who lives at the North Pole? By how much?
13. Criticize the following argument. "Here is a way to travel faster than light. Suppose a star is 10 light-years away. A radio signal sent from Earth would need 20 years to make the round trip to the star. If I were to travel to the star in my rocket at $v = 0.8c$, to me the distance to the star is contracted by $\sqrt{1 - (0.8)^2}$ to 6 light-years, and at that speed it would take me $6 \text{ light-years}/0.8c = 7.5$ years to travel there. The round trip takes me only 15 years, and therefore I travel faster than light, which takes 20 years."
14. Is it possible to synchronize clocks that are in motion relative to each other? Try to design a method to do so.
- Which observers will believe the clocks to be synchronized?
15. Suppose event A causes event B . To one observer, event A comes before event B . Is it possible that in another frame of reference event B could come before event A ? Discuss.
16. Is mass a conserved quantity in classical physics? In special relativity?
17. "In special relativity, mass and energy are equivalent." Discuss this statement and give examples.
18. Which is more massive, an object at low temperature or the same object at high temperature? A spring at its natural length or the same spring under compression? A container of gas at low pressure or at high pressure? A charged capacitor or an uncharged one?
19. Could a collision be elastic in one frame of reference and inelastic in another?
20. (a) What properties of nature would be different if there were a relativistic transformation law for electric charge? (b) What experiments could be done to prove that electric charge does *not* change with velocity?

Problems

2.1 Classical Relativity

1. You are piloting a small airplane in which you want to reach a destination that is 750 km due north of your starting location. Once you are airborne, you find that (due to a strong but steady wind) to maintain a northerly course you must point the nose of the plane at an angle that is 22° west of true north. From previous flights on this route in the absence of wind, you know that it takes you 3.14 h to make the journey. With the wind blowing, you find that it takes 4.32 h. A fellow pilot calls you to ask about the wind velocity (magnitude and direction). What is your report?
2. A moving sidewalk 95 m in length carries passengers at a speed of 0.53 m/s. One passenger has a normal walking speed of 1.24 m/s. (a) If the passenger stands on the sidewalk without walking, how long does it take her to travel the length of the sidewalk? (b) If she walks at her normal walking speed on the sidewalk, how long does it take to travel the full length? (c) When she reaches the end of the sidewalk, she suddenly realizes that she left a package at the opposite end. She walks rapidly back along the sidewalk at double her normal walking speed to retrieve the package. How long does it take her to reach the package?

2.2 The Michelson–Morley Experiment

3. A shift of one fringe in the Michelson–Morley experiment corresponds to a change in the round-trip travel time along one arm of the interferometer by one period of vibration of light (about 2×10^{-15} s) when the apparatus

is rotated by 90° . Based on the results of Example 2.3, what velocity through the ether would be deduced from a shift of one fringe? (Take the length of the interferometer arm to be 11 m.)

2.4 Consequences of Einstein's Postulates

4. The distance from New York to Los Angeles is about 4000 km and should take about 40 h in a car driving at 100 km/h. (a) How much shorter than 4000 km is the distance according to the car travelers? (b) How much less than 40 h do they age during the trip?
5. How fast must an object move before its length appears to be contracted to one-half its proper length?
6. An astronaut must journey to a distant planet, which is 300 light-years from Earth. What speed will be necessary if the astronaut wishes to age only 12 years during the round trip?
7. The proper lifetime of a certain particle is 120.0 ns. (a) How long does it live in the laboratory if it moves at $v = 0.950c$? (b) How far does it travel in the laboratory during that time? (c) What is the distance traveled in the laboratory according to an observer moving with the particle?
8. High-energy particles are observed in laboratories by photographing the tracks they leave in certain detectors; the length of the track depends on the speed of the particle and its lifetime. A particle moving at $0.993c$ leaves a track 1.15 mm long. What is the proper lifetime of the particle?

9. Carry out the missing steps in the derivation of Equation 2.17.
10. Two spaceships approach the Earth from opposite directions. According to an observer on the Earth, ship A is moving at a speed of $0.743c$ and ship B at a speed of $0.831c$. What is the velocity of ship A as observed from ship B ? Of ship B as observed from ship A ?
11. Rocket A leaves a space station with a speed of $0.811c$. Later, rocket B leaves in the same direction with a speed of $0.665c$. What is the velocity of rocket A as observed from rocket B ?
12. One of the strongest emission lines observed from distant galaxies comes from hydrogen and has a wavelength of 122 nm (in the ultraviolet region). (a) How fast must a galaxy be moving away from us in order for that line to be observed in the visible region at 366 nm ? (b) What would be the wavelength of the line if that galaxy were moving toward us at the same speed?
13. A physics professor claims in court that the reason he went through the red light ($\lambda = 650\text{ nm}$) was that, due to his motion, the red color was Doppler shifted to green ($\lambda = 550\text{ nm}$). How fast was he going?
14. Three rods are joined to form a 45–45–90 triangle, which is placed at rest in the $x'y'$ plane with its hypotenuse along the x' axis. O' moves away from O in the x direction at a speed of $0.92c$. What are the angles of the triangle according to O ?
15. In the Relativistic Heavy Ion Collider (an accelerator at Brookhaven National Laboratory in New York), nuclei of gold atoms are accelerated to a speed of $0.99995c$ and then forced to undergo head-on collisions. The radius of a gold nucleus is about 7.0 fm . Compared to its normal density, what is the density of a gold nucleus traveling at this speed?
20. A rod in the reference frame of observer O makes an angle of 34° with the x axis. According to observer O' , who is in motion in the x direction with velocity u , the rod makes an angle of 52° with the x axis. Find the velocity u .
21. Two events occur at locations separated by a distance of 49.5 m and by a time interval of $0.528\text{ }\mu\text{s}$, according to observer O . Observer O' is in motion away from O with a speed of $0.685c$ in the x direction. According to O' , what are the spatial and time separations of the events?
22. According to observer O , a blue flash occurs at $x_b = 10.4\text{ m}$ when $t_b = 0.124\text{ }\mu\text{s}$, and a red flash occurs at $x_r = 23.6\text{ m}$ when $t_r = 0.138\text{ }\mu\text{s}$. According to observer O' , who is in motion relative to O at velocity u , the two flashes appear to be simultaneous. Find the velocity u .

2.6 The Twin Paradox

2.5 The Lorentz Transformation

16. Derive the Lorentz velocity transformations for v'_x and v'_z .
17. Observer O fires a light beam in the y direction ($v_y = c$). Use the Lorentz velocity transformation to find v'_x and v'_y and show that O' also measures the value c for the speed of light. Assume O' moves relative to O with velocity u in the x direction.
18. A light bulb at point x in the frame of reference of O blinks on and off at intervals $\Delta t = t_2 - t_1$. Observer O' , moving relative to O at speed u , measures the interval to be $\Delta t' = t'_2 - t'_1$. Use the Lorentz transformation expressions to derive the time dilation expression relating Δt and $\Delta t'$.
19. A neutral K meson at rest decays into two π mesons, which travel in opposite directions along the x axis with speeds of $0.815c$. If instead the K meson were moving in the positive x direction with a velocity of $0.453c$, what would be the velocities of the two π mesons?
23. Suppose the speed of light were 1000 mi/h . You are traveling on a flight from Los Angeles to Boston, a distance of 3000 mi . The plane's speed is a constant 600 mi/h . You leave Los Angeles at $10:00\text{ am}$, as indicated by your wristwatch and by a clock in the airport. (a) According to your watch, what time is it when you land in Boston? (b) In the Boston airport is a clock that is synchronized to read exactly the same time as the clock in the Los Angeles airport. What time does that clock read when you land in Boston? (c) The following day when the Boston clock that records Los Angeles time reads $10:00\text{ am}$, you leave Boston to return to Los Angeles on the same airplane. When you land in Los Angeles, what are the times read on your watch and on the airport clock?
24. Suppose rocket traveler Amelia has a clock made on Earth. Every year on her birthday she sends a light signal to brother Casper on Earth. (a) At what rate does Casper receive the signals during Amelia's outward journey? (b) At what rate does he receive the signals during her return journey? (c) How many of Amelia's birthday signals does Casper receive during the journey that he measures to last 20 years?
25. Suppose Amelia traveled at a speed of $0.80c$ to a star that (according to Casper on Earth) is 8.0 light-years away. Casper ages 20 years during Amelia's round trip. How much younger than Casper is Amelia when she returns to Earth?
26. Make a drawing similar to Figure 2.20 showing the worldlines of Casper and Amelia from Casper's frame of reference. Divide the worldline for Amelia's outward journey into 8 equal segments (for the 8 birthdays that Amelia celebrates). For each birthday, draw a line that represents a light signal that Amelia sends to Casper on her birthday. Do the same for Amelia's return journey. (a) According to Casper's time, when does he receive the signal showing Amelia celebrating her 8th birthday after leaving Earth? (b) How long does it take for Casper to receive the signals showing Amelia celebrating birthdays 9 through 16?

27. Two twins make a round-trip journey from Earth to a star that is 12 light-years distant. Alice travels at a speed of $0.6c$. Bob departs 10 years after Alice and travels at a speed of $0.8c$. (a) Show that the two twins arrive back on Earth at the same time. (b) Which twin is the younger when they return?
28. Agnes makes a round trip at a constant speed to a star that is 16 light-years distant from Earth, while twin brother Bert remains on Earth. When Agnes returns to Earth, she reports that she has celebrated 20 birthdays during her journey. (a) What was her speed during her journey? (b) How old is Bert when she returns?

2.7 Relativistic Dynamics

29. (a) Using the relativistically correct final velocities for the collision shown in Figure 2.21a ($v'_{1f} = -0.585c$, $v'_{2f} = +0.294c$), show that relativistic kinetic energy is conserved according to observer O' . (b) Using the relativistically correct final velocities for the collision shown in Figure 2.21b ($v_{1f} = -0.051c$, $v_{2f} = +0.727c$), show that relativistic kinetic energy is conserved according to observer O .
30. Find the momentum, kinetic energy, and total energy of a proton moving at a speed of $0.835c$.
31. An electron is moving with a kinetic energy of 0.923 MeV. What is its speed?
32. The work–energy theorem relates the change in kinetic energy of a particle to the work done on it by an external force: $\Delta K = W = \int F dx$. Writing Newton's second law as $F = dp/dt$, show that $W = \int v dp$ and integrate by parts using the relativistic momentum to obtain Equation 2.34.
33. For what range of velocities of a particle of mass m can we use the classical expression for kinetic energy $\frac{1}{2}mv^2$ to within an accuracy of 1 percent?
34. For what range of velocities of a particle of mass m can we use the extreme relativistic approximation $E = pc$ to within an accuracy of 1 percent?
35. Use Equations 2.32 and 2.36 to derive Equation 2.39.
36. By carrying the binomial expansion one term farther, find the next term after $\frac{1}{2}mv^2$ in the classical approximation of the relativistic kinetic energy. For what value of the speed does this term throw the classical value off by 0.1%?
37. (a) According to observer O , a certain particle has a momentum of 1256 MeV/ c and a total relativistic energy of 1351 MeV. What is the rest energy of this particle? (b) An observer O' in a different frame of reference measures the momentum of this particle to be 857 MeV/ c . What does O' measure for the total relativistic energy of the particle?
38. An electron is moving at a speed of $0.85c$. By how much must its kinetic energy increase to raise its speed to $0.91c$?
39. What is the change in mass when 1 g of copper is heated from 0 to 100°C ? The specific heat capacity of copper is $0.40 \text{ J/g}\cdot\text{K}$.
40. Find the kinetic energy of an electron moving at a speed of (a) $v = 1.00 \times 10^{-4}c$; (b) $v = 1.00 \times 10^{-2}c$; (c) $v = 0.300c$; (d) $v = 0.999c$.
41. An electron and a proton are each accelerated starting from rest through a potential difference of 12.0 million volts. Find the momentum (in MeV/ c) and the kinetic energy (in MeV) of each, and compare with the results of using the classical formulas.
42. In a nuclear reactor, each atom of uranium (of atomic mass 235 u) releases about 210 MeV when it fissions. What is the change in mass when 1.50 kg of uranium-235 is fissioned?
43. (a) Find the energy released as gamma rays when ^2H captures a neutron to form ^3H . Assume all kinetic energies are negligibly small. Atomic masses can be found in Appendix D. (b) Repeat for capture of a neutron by ^3He to form ^4He .

2.8 Conservation Laws in Relativistic Decays and Collisions

44. A π meson of rest energy 139.6 MeV moving at a speed of $0.921c$ collides with and sticks to a proton of rest energy 938.3 MeV that is at rest. (a) Find the total relativistic energy of the resulting composite particle. (b) Find the total linear momentum of the composite particle. (c) Using the results of (a) and (b), find the rest energy of the composite particle.
45. An electron and a positron (an antielectron) make a head-on collision, each moving at $v = 0.99999c$. In the collision, the electrons disappear and are replaced by two muons ($mc^2 = 105.7 \text{ MeV}$), which move off in opposite directions. What is the kinetic energy of each of the muons?
46. It is desired to create a particle of mass $9460 \text{ MeV}/c^2$ in a head-on collision between a proton and an antiproton (each having a mass of $938.3 \text{ MeV}/c^2$) traveling at the same speed. What speed is necessary for this to occur?
47. A particle of rest energy mc^2 is moving with speed v in the positive x direction. The particle decays into two particles, each of rest energy 140 MeV. One particle, with kinetic energy 282 MeV, moves in the positive x direction, and the other particle, with kinetic energy 25 MeV, moves in the negative x direction. Find the rest energy of the original particle and its speed.
48. Let's consider a different approach to Example 2.21, in which two protons collide to form an antiproton. Suppose the two protons collide head-on with equal speeds. (a) In this frame of reference, what is the threshold energy? (b) Find the velocity corresponding to the threshold energy. (c) Now transform to a frame of reference in which one of the initial protons is at rest, and find the speed and the energy of the other proton.

2.9 Experimental Tests of Special Relativity

49. In the muon decay experiment discussed in Section 2.9 as a verification of time dilation, the muons move in the laboratory with a momentum of $3094 \text{ MeV}/c$. Find the dilated lifetime in the laboratory frame. (The proper lifetime is $2.198 \mu\text{s}$.)
50. Derive the relativistic expression $p^2/2K = m + K/2c^2$, which is plotted in Figure 2.28a.

General Problems

51. Suppose we want to send an astronaut on a round trip to visit a star that is 200 light-years distant and at rest with respect to Earth. The life support systems on the spacecraft enable the astronaut to survive at most 20 years. (a) At what speed must the astronaut travel to make the round trip in 20 years of spacecraft time? (b) How much time passes on Earth during the round trip?
52. A “cause” occurs at point 1 (x_1, t_1) and its “effect” occurs at point 2 (x_2, t_2). Use the Lorentz transformation to find $t'_2 - t'_1$, and show that $t'_2 - t'_1 > 0$; that is, O' can never see the “effect” coming before its “cause.”
53. Observer O sees a red flash of light at the origin at $t = 0$ and a blue flash of light at $x = 3.65 \text{ km}$ at a time $t = 8.24 \mu\text{s}$. What are the distance and the time interval between the flashes according to observer O' , who moves relative to O in the direction of increasing x with a speed of $0.534c$? Assume that the origins of the two coordinate systems line up at $t = t' = 0$.
54. Several spacecraft leave a space station at the same time. Relative to an observer on the station, A travels at $0.65c$ in the x direction, B at $0.50c$ in the y direction, C at $0.50c$ in the negative x direction, and D at $0.50c$ at 45° between the y and negative x directions. Find the velocity components, directions, and speeds of B , C , and D as observed from A .
55. Observer O sees a light turn on at $x = 524 \text{ m}$ when $t = 1.52 \mu\text{s}$. Observer O' is in motion at a speed of $0.563c$ in the positive x direction. The two frames of reference are synchronized so that their origins match up ($x = x' = 0$) at $t = t' = 0$. (a) At what time does the light turn on according to O' ? (b) At what location does the light turn on in the reference frame of O' ?
56. Suppose an observer O measures a particle of mass m moving in the x direction to have speed v , energy E , and momentum p . Observer O' , moving at speed u in the x direction, measures v' , E' , and p' for the same object. (a) Use the Lorentz velocity transformation to find E' and p' in terms of m , u , and v . (b) Reduce $E'^2 - (p'c)^2$ to its simplest form and interpret the result.
57. Repeat Problem 56 for the mass moving in the y direction according to O . The velocity u of O' is still along the x direction.
58. Consider again the situation described in Section 2.6. Amelia’s friend Bernice leaves Earth at the same time as Amelia and travels in the same direction at the same speed, but Bernice continues in the original direction when Amelia reaches the planet and turns her ship around. (a) From Bernice’s frame of reference, Casper is moving at a velocity of $-0.60c$. Draw Casper’s worldline in Bernice’s frame of reference. (b) Casper celebrates 20 birthdays during Amelia’s journey. In Bernice’s frame of reference, how long does it take for Casper to celebrate 20 birthdays? (c) In Bernice’s frame of reference, draw a worldline representing Amelia’s outbound journey to the planet. (d) Calculate Amelia’s velocity during her return journey as observed from Bernice’s frame of reference, and draw a worldline showing Amelia’s return journey. Amelia’s and Casper’s worldlines should intersect when Amelia return to Earth. (e) Divide Casper’s worldline into 20 segments, representing his birthdays. He sends a light signal to Amelia on each birthday. Amelia receives a light signal from Casper just as she arrives at the planet. On which birthday did Casper send this signal? (f) Amelia sends Casper a light signal on her 8th birthday. Draw a line on your diagram representing this light signal. When does Casper receive this signal?
59. Electrons are accelerated to high speeds by a two-stage machine. The first stage accelerates the electrons from rest to $v = 0.99c$. The second stage accelerates the electrons from $0.99c$ to $0.999c$. (a) How much energy does the first stage add to the electrons? (b) How much energy does the second stage add in increasing the velocity by only 0.9 percent?
60. A beam of 2.14×10^{11} electrons/s moving at a speed of $0.813c$ strikes a block of copper that is used as a beam stop. The copper block is a cube measuring 2.54 cm on edge. What is the temperature increase of the block after one hour?
61. An electron moving at a speed of $v_i = 0.960c$ in the positive x direction collides with another electron at rest. After the collision, one electron is observed to move with a speed of $v_{if} = 0.956c$ at an angle of $\theta_1 = 9.7^\circ$ with the x axis. (a) Use conservation of momentum to find the velocity (magnitude and direction) of the second electron. (b) Based only on the original data given in the problem, use conservation of energy to find the speed of the second electron.
62. A pion has a rest energy of 135 MeV . It decays into two gamma-ray photons, bursts of electromagnetic radiation that travel at the speed of light. A pion moving through the laboratory at $v = 0.98c$ decays into two gamma-ray photons of equal energies, making equal angles θ with the original direction of motion. Find the angle θ and the energies of the two gamma ray photons.
63. Consider the decay of the K meson from Example 2.19 transformed to a frame of reference in which the K meson is at rest. (a) What is the speed of the K meson in the original reference frame? (b) Using the speed from part (a) to

transform to the reference frame in which the K meson is at rest, find the speed, momentum, and energy of the pi meson in this frame. (c) Find the mass of the unknown particle produced in the decay.

64. An electron and a positron (antielectron) are traveling toward each other, the electron with velocity $+0.834c$ and the positron with velocity $-0.428c$. They collide and stick

together to form a new composite particle. (a) Find the momentum and energy of the new particle. (b) What is the mass of the new particle? (c) Find the change in kinetic energy in the collision. How is the change in kinetic energy related to the mass of the new particle? (d) Which of the above answers would be different according to an observer in a different reference frame?

THE PARTICLE-LIKE PROPERTIES OF ELECTROMAGNETIC RADIATION



Thermal emission, the radiation emitted by all objects due to their temperatures, laid the groundwork for the development of quantum mechanics around the beginning of the 20th century. Today we use thermography for many applications, including the study of heat loss by buildings, medical diagnostics, night vision and other surveillance, and monitoring potential volcanoes. Ted Kinsman / Getty Images

We now turn to a discussion of *wave mechanics*, the second theory on which modern physics is based. One consequence of wave mechanics is the breakdown of the classical distinction between particles and waves. In this chapter, we consider the three early experiments that provided evidence that light, which we usually regard as a wave phenomenon, has properties that we normally associate with particles. Instead of spreading its energy smoothly over a wave front, the energy is delivered in concentrated bundles like particles; a discrete bundle (*quantum*) of electromagnetic energy is known as a *photon*.

Before we begin to discuss the experimental evidence that supports the existence of the photon and the particle-like properties of light, we first review some of the properties of electromagnetic waves.

3.1 REVIEW OF ELECTROMAGNETIC WAVES

A distribution of electric charges sets up an electric field $\vec{E}$, and current-carrying wires set up a magnetic field $\vec{B}$. If the charges do not move and the currents do not change, then $\vec{E}$ and $\vec{B}$ are static fields that vary with location but not with time. However, if the charges are accelerated and the currents vary with time, an electromagnetic wave is produced in which the $\vec{E}$ and $\vec{B}$ fields of the wave vary both with location and time.

The mathematical expression that describes such a wave may have many different forms, depending on the properties of the source of the wave and of the medium through which the wave travels. One special form is the *plane wave*, in which the wave fronts are planes. (A point source, on the other hand, produces spherical waves, in which the wave fronts are spheres.) A plane electromagnetic wave traveling in the positive z direction is described by the expressions

$$\vec{E} = \vec{E}_0 \sin(kz - \omega t), \quad \vec{B} = \vec{B}_0 \sin(kz - \omega t) \quad (3.1)$$

where the *wave number* k is found from the wavelength λ ($k = 2\pi/\lambda$) and the *angular frequency* ω is found from the frequency f ($\omega = 2\pi f$). Because λ and f are related by $c = \lambda f$, k and ω are also related by $c = \omega/k$.

The polarization of the wave is represented by the vector $\vec{E}_0$; the plane of polarization is determined by the direction of $\vec{E}_0$ and the direction of propagation, the z axis in this case. Once we specify the direction of travel and the polarization $\vec{E}_0$, the direction of $\vec{B}_0$ is fixed by the requirements that $\vec{B}$ must be perpendicular to both $\vec{E}$ and the direction of travel and that the vector product $\vec{E} \times \vec{B}$ point in the direction of travel. For example, if $\vec{E}_0$ is in the x direction ($\vec{E}_0 = E_0 \hat{i}$, where $\hat{i}$ is a unit vector in the x direction), then $\vec{B}_0$ must be in the y direction ($\vec{B}_0 = B_0 \hat{j}$). Moreover, the magnitude of $\vec{B}_0$ is determined by

$$B_0 = \frac{E_0}{c} \quad (3.2)$$

where c is the speed of light.

An electromagnetic wave transmits energy from one place to another; the energy flux is specified by the *Poynting vector* $\vec{S}$:

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B} \quad (3.3)$$

For the plane wave, this reduces to

$$\vec{S} = \frac{1}{\mu_0} E_0 B_0 \sin^2(kz - \omega t) \hat{k} \quad (3.4)$$

where $\hat{k}$ is a unit vector in the z direction. The Poynting vector has dimensions of power (energy per unit time) per unit area—for example, J/s/m² or W/m². Figure 3.1 shows the orientation of the vectors $\vec{E}$, $\vec{B}$, and $\vec{S}$ for this special case.

Let us imagine the following experiment. We place a detector of electromagnetic radiation (a radio receiver or a human eye) at some point on the z axis, and we determine the electromagnetic power that this plane wave delivers to the receiver. The receiver is oriented with its sensitive area A perpendicular to the z axis, so that the maximum signal is received; we can therefore drop the vector representation of $\vec{S}$ and work only with its magnitude S . The power P entering the receiver is then

$$P = SA = \frac{1}{\mu_0} E_0 B_0 A \sin^2(kz - \omega t) \quad (3.5)$$

which we can rewrite using Eq. 3.2 as

$$P = \frac{1}{\mu_0 c} E_0^2 A \sin^2(kz - \omega t) \quad (3.6)$$

There are two important features of this expression that you should recognize:

1. The intensity (the average power per unit area) is proportional to E_0^2 . This is a general property of waves: *the intensity is proportional to the square of the amplitude*. We will see later that this same property also characterizes the waves that describe the behavior of material particles.

2. The intensity fluctuates with time, with the frequency $2f = 2(\omega/2\pi)$. We don't usually observe this rapid fluctuation—visible light, for example, has a frequency of about 10^{15} oscillations per second, and because our eye doesn't respond that quickly, we observe the time average of many (perhaps 10^{13}) cycles. If T is the observation time (perhaps 10^{-2} s in the case of the eye), then the average power is

$$P_{\text{av}} = \frac{1}{T} \int_0^T P dt \quad (3.7)$$

and using Eq. 3.6 we obtain the intensity I :

$$I = \frac{P_{\text{av}}}{A} = \frac{1}{2\mu_0 c} E_0^2 \quad (3.8)$$

because the average value of $\sin^2 \theta$ is $1/2$.

Interference and Diffraction

The property that makes waves a unique physical phenomenon is the *principle of superposition*, which, for example, allows two waves to meet at a point, to cause a combined disturbance at the point that might be greater or lesser than the disturbance produced by either wave alone, and finally to emerge from the point of “collision” with all of the properties of each wave totally

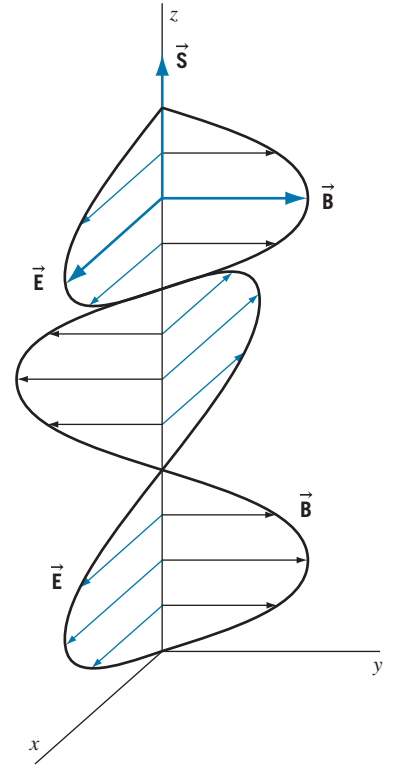


FIGURE 3.1 An electromagnetic wave traveling in the z direction. The electric field $\vec{E}$ lies in the xz plane and the magnetic field $\vec{B}$ lies in the yz plane.

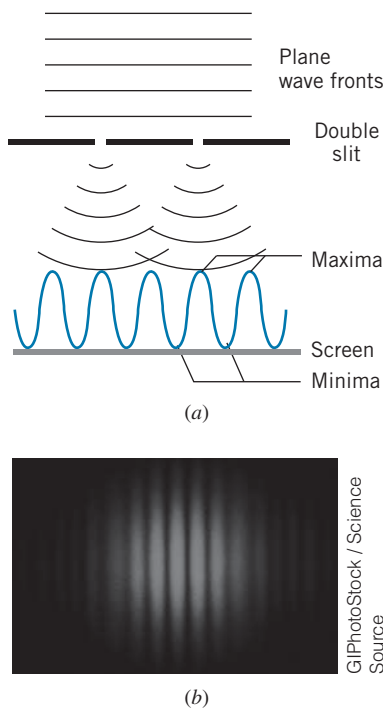


FIGURE 3.2 (a) Young's double-slit experiment. A plane wave front passes through both slits; the wave is diffracted at the slits, and interference occurs where the diffracted waves overlap on the screen. (b) The interference fringes observed on the screen.

unchanged by the collision. To appreciate this important distinction between material objects and waves, imagine trying that trick with two automobiles!

This special property of waves leads to the phenomena of *interference* and *diffraction*. The simplest and best-known example of interference is *Young's double-slit experiment*, in which a monochromatic plane wave is incident on a barrier in which two narrow slits have been cut. (This experiment was first done with light waves, but in fact any wave will do as well, not only other electromagnetic waves, such as microwaves, but also mechanical waves, such as water waves or sound waves. We assume that the experiment is being done with light waves.)

Figure 3.2 illustrates this experimental arrangement. The plane wave is *diffracted* by each of the slits, so that the light passing through each slit covers a much larger area on the screen than the geometric shadow of the slit. This causes the light from the two slits to overlap on the screen, producing the interference. If we move away from the center of the screen just the right distance, we reach a point at which a wave crest passing through one slit arrives at exactly the same time as the previous wave crest that passed through the other slit. When this occurs, the intensity is a maximum, and a bright region appears on the screen. This is *constructive interference*, and it occurs continually at the point on the screen that is exactly one wavelength further from one slit than from the other. That is, if X_1 and X_2 are the distances from the point on the screen to the two slits, then a condition for maximum constructive interference is $|X_1 - X_2| = \lambda$. Constructive interference occurs when any wave crest from one slit arrives simultaneously with another from the other slit, whether it is the next, or the fourth, or the forty-seventh. The general condition for complete constructive interference is that the difference between X_1 and X_2 is an integral number of wavelengths:

$$|X_1 - X_2| = n\lambda \quad n = 0, 1, 2, \dots \quad (3.9)$$

It is also possible for the crest of the wave from one slit to arrive at a point on the screen simultaneously with the trough (valley) of the wave from the other slit. When this happens, the two waves cancel, giving a dark region on the screen. This is known as *destructive interference*. (The existence of destructive interference at intensity minima immediately shows that we must add the electric field vectors $\vec{E}$ of the waves from the two slits and not their powers P , because P can never be negative.) Destructive interference occurs whenever the distances X_1 and X_2 are such that the phase of one wave differs from the other by one-half cycle, or by one and one-half cycles, two and one-half cycles, and so forth:

$$|X_1 - X_2| = \frac{1}{2}\lambda, \frac{3}{2}\lambda, \frac{5}{2}\lambda, \dots = \left(n + \frac{1}{2}\right)\lambda \quad n = 0, 1, 2, \dots \quad (3.10)$$

We can find the locations on the screen where the interference maxima occur in the following way. Let d be the separation of the slits and let D be the distance from the slits to the screen. If y_n is the distance from the center of the screen to the n th maximum, then from the geometry of Figure 3.3 we find (assuming $X_1 > X_2$)

$$X_1^2 = D^2 + \left(\frac{d}{2} + y_n\right)^2 \quad \text{and} \quad X_2^2 = D^2 + \left(\frac{d}{2} - y_n\right)^2 \quad (3.11)$$

Subtracting these equations and solving for y_n , we obtain

$$y_n = \frac{X_1^2 - X_2^2}{2d} = \frac{(X_1 + X_2)(X_1 - X_2)}{2d} \quad (3.12)$$

In experiments with light, D is of the order 1 m, and y_n and d are typically at most 1 mm; thus $X_1 \cong D$ and $X_2 \cong D$, so $X_1 + X_2 \cong 2D$, and to a good approximation

$$y_n = (X_1 - X_2) \frac{D}{d} \quad (3.13)$$

Using Eq. 3.9 for the values of $(X_1 - X_2)$ at the maxima, we find

$$y_n = n \frac{\lambda D}{d} \quad (3.14)$$

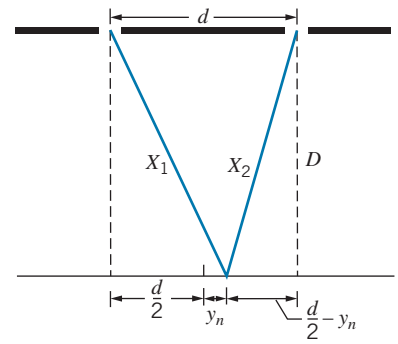


FIGURE 3.3 The geometry of the double-slit experiment.

Crystal Diffraction of X Rays

Another device for observing the interference of light waves is the *diffraction grating*, in which the wave fronts pass through a barrier that has *many* slits (often thousands or tens of thousands) and then recombine. The operation of this device is illustrated in Figure 3.4; interference maxima corresponding to different wavelengths appear at different angles θ , according to

$$d \sin \theta = n\lambda \quad (3.15)$$

where d is the slit spacing and n is the order number of the maximum ($n = 1, 2, 3, \dots$).

The advantage of the diffraction grating is its superior resolution—it enables us to get very good separation of wavelengths that are close to one another, and thus it is a very useful device for measuring wavelengths. Notice, however, that in order to get reasonable values of the angle θ —for example, $\sin \theta$ in the range of 0.3–0.5—we must have d of the order of a few times the wavelength. For visible light, this is not particularly difficult, but for radiations of very short wavelength, mechanical construction of a grating is not possible. For example, for X rays with a wavelength of the order of 0.1 nm, we would need to construct a grating in which the slits were less than 1 nm apart, which is roughly the same as the spacing between the atoms of most materials.

The solution to this problem has been known since the pioneering discoveries of Laue and Bragg*: use the atoms themselves as a diffraction grating! A beam of X rays sees the regular spacings of the atoms in a crystal as a sort of three-dimensional diffraction grating.

Consider the set of atoms shown in Figure 3.5, which represents a small portion of a two-dimensional slice of the crystal. The X rays are reflected from individual atoms in all directions, but in only one direction will the scattered “wavelets” constructively interfere to produce a reflected beam, and in this

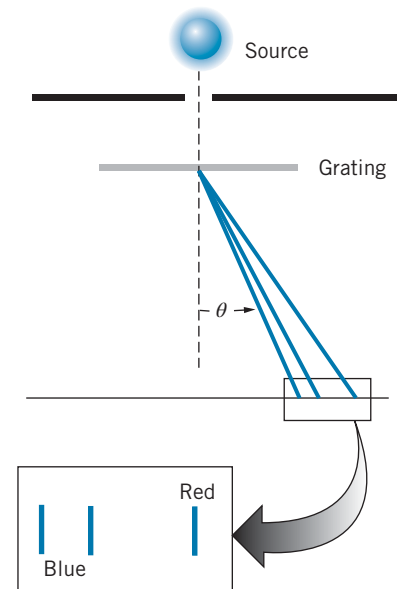


FIGURE 3.4 The use of a diffraction grating to analyze light into its constituent wavelengths.

*Max von Laue (1879–1960, Germany) developed the method of X-ray diffraction for the study of crystal structures, for which he received the 1914 Nobel Prize. Lawrence Bragg (1890–1971, England) developed the Bragg law for X-ray diffraction while he was a student at Cambridge University. He shared the 1915 Nobel Prize with his father, William Bragg, for their research on the use of X rays to determine crystal structures.

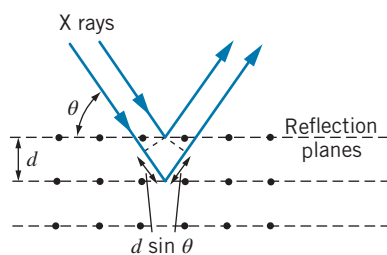


FIGURE 3.5 A beam of X rays reflected from a set of crystal planes of spacing d . The beam reflected from the second plane travels a distance $2d \sin \theta$ greater than the beam reflected from the first plane.

case we can regard the reflection as occurring from a plane drawn through the row of atoms. (This situation is identical with the reflection of light from a mirror—only in one direction will there be a beam of reflected light, and in that direction we can regard the reflection as occurring on a plane with the angle of incidence equal to the angle of reflection.)

Suppose the rows of atoms are a distance d apart in the crystal. Then a portion of the beam is reflected from the front plane, and a portion is reflected from the second plane, and so forth. The wave fronts of the beam reflected from the second plane lag behind those reflected from the front plane, because the wave reflected from the second plane must travel an additional distance of $2d \sin \theta$, where θ is the angle of incidence as *measured from the face of the crystal*. (Note that this is different from the usual procedure in optics, in which angles are defined with respect to the *normal* to the surface.) If this path difference is a whole number of wavelengths, the reflected beams interfere constructively and give an intensity maximum; thus, the basic expression for the interference maxima in X-ray diffraction from a crystal is

$$2d \sin \theta = n\lambda \quad n = 1, 2, 3, \dots \quad (3.16)$$

This result is known as *Bragg's law* for X-ray diffraction. Notice the factor of 2 that appears in Eq. 3.16 but does *not* appear in the otherwise similar expression of Eq. 3.15 for the ordinary diffraction grating.

Example 3.1

A single crystal of table salt (NaCl) is irradiated with a beam of X rays of wavelength 0.250 nm, and the first Bragg reflection is observed at an angle of 26.3° . What is the atomic spacing of NaCl?

Solution

Solving Bragg's law for the spacing d , we have

$$d = \frac{n\lambda}{2 \sin \theta} = \frac{0.250 \text{ nm}}{2 \sin 26.3^\circ} = 0.282 \text{ nm}$$

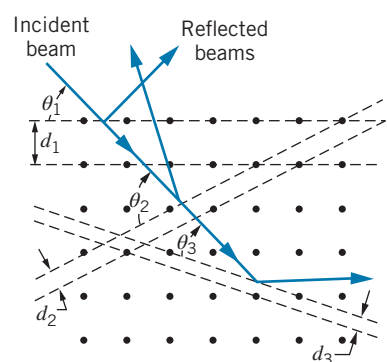


FIGURE 3.6 An incident beam of X rays can be reflected from many different crystal planes.

Our drawing of Figure 3.5 was very arbitrary—we had no basis for choosing which set of atoms to draw the reflecting planes through. Figure 3.6 shows a larger section of the crystal. As you can see, there are many possible reflecting planes, each with a different value of θ and d . (Of course, d_i and θ_i are related and cannot be varied independently.) If we used a beam of X rays of a single wavelength, it might be difficult to find the proper angle and set of planes to observe the interference. However, if we use a beam of X rays of a continuous range of wavelengths, for each d_i and θ_i interference will occur for a certain wavelength λ_i , and so there will be a pattern of interference maxima appearing at different angles of reflection as shown in Figure 3.6. The pattern of interference maxima depends on the spacing and the type of arrangement of the atoms in the crystal.

Figure 3.7 shows sample patterns (called *Laue patterns*) that are obtained from X-ray scattering from two different crystals. The bright dots correspond to interference maxima for wavelengths from the range of incident wavelengths that happen to satisfy Eq. 3.16. The three-dimensional pattern is more complicated than our two-dimensional drawings, but the individual dots have the same interpretation. Figure 3.8 shows the pattern obtained from a sample that consists of many tiny crystals, rather than one single crystal. (It looks like

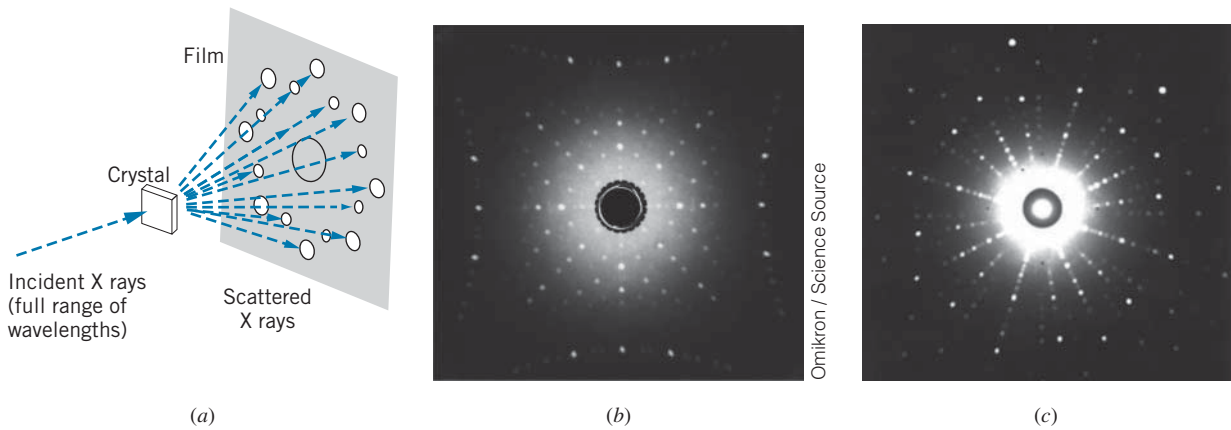


FIGURE 3.7 (a) Apparatus for observing X-ray scattering by a crystal. An interference maximum (dot) appears on the film whenever a set of crystal planes happens to satisfy the Bragg condition for a particular wavelength. (b) Laue pattern of TiO_2 crystal. (c) Laue pattern of a crystal of CuIn_4Se_6 . The differences between the two Laue patterns are due to the differences in the geometric structure of the two crystals. (Source: E. Arushanov *et al.*, Journal of Physics D: Applied Physics **34**, 3480 (2001))

Figure 3.7b or c rotated rapidly about its center.) From such pictures, it is also possible to deduce crystal structures and lattice spacing.

All of the examples we have discussed in this section depend on the wave properties of electromagnetic radiation. However, as we now begin to discuss, there are other experiments that cannot be explained if we regard electromagnetic radiation as waves.

3.2 THE PHOTOELECTRIC EFFECT

We'll now turn to our discussion of the first of three experiments that cannot be explained by the wave theory of light. When a metal surface is illuminated with light, electrons can be emitted from the surface. This phenomenon, known as the *photoelectric effect*, was discovered by Heinrich Hertz in 1887 in the process of his research into electromagnetic radiation. The emitted electrons are called *photoelectrons*.

A sample experimental arrangement for observing the photoelectric effect is illustrated in Figure 3.9. Light falling on a metal surface (the emitter) can release electrons, which travel to the collector. The experiment must be done in an evacuated tube, so that the electrons do not lose energy in collisions with molecules of the air. Among the properties that can be measured are the rate of electron emission and the maximum kinetic energy of the photoelectrons.*

The rate of electron emission can be measured as an electric current i by an ammeter in the external circuit. The maximum kinetic energy of the electrons can be measured by applying a negative potential to the collector that is just enough to repel the most energetic electrons, which then do not have

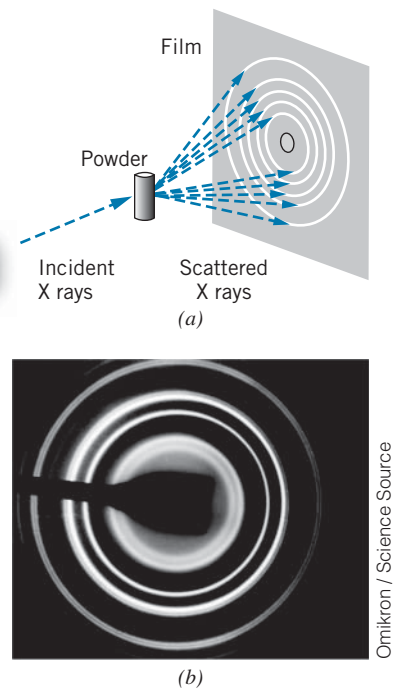


FIGURE 3.8 (a) Apparatus for observing X-ray scattering from a powdered or polycrystalline sample. Because the individual crystals have many different orientations, each scattered ray of Figure 3.7 becomes a cone that forms a circle on the film. (b) Diffraction pattern (known as *Debye-Scherrer* pattern) of polycrystalline gold.

*The electrons can be emitted with many different kinetic energies, depending on how tightly bound they are to the metal. Here we are concerned only with the *maximum* kinetic energy, which depends on the energy needed to remove the least tightly bound electron from the surface of the metal.

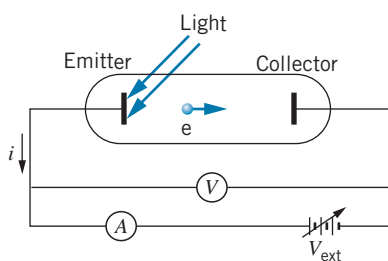


FIGURE 3.9 Apparatus for observing the photoelectric effect. The flow of electrons from the emitter to the collector is measured by the ammeter A as a current i in the external circuit. A variable voltage source V_{ext} establishes a potential difference between the emitter and collector, which is measured by the voltmeter V .

TABLE 3.1 Some Photoelectric Work Functions

Material	ϕ (eV)
Na	2.28
Al	4.08
Co	3.90
Cu	4.70
Zn	4.31
Ag	4.73
Pt	6.35
Pb	4.14

enough energy to “climb” the potential energy hill. That is, if the potential difference between the emitter and the collector is ΔV (a negative quantity), then electrons traveling from the emitter to the collector would gain a potential energy of $\Delta U = q\Delta V = -e\Delta V$ (a positive quantity) and would lose the same amount of kinetic energy. Electrons leaving the emitter with a kinetic energy smaller than this ΔU cannot reach the collector and are pushed back toward the emitter.

As the magnitude of the potential difference is increased, at some point even the most energetic electrons do not have enough kinetic energy to reach the collector. This potential, called the *stopping potential* V_s , is determined by increasing the magnitude of the voltage until the ammeter current drops to zero. At this point, the maximum kinetic energy K_{max} of the electrons as they leave the emitter is just equal to the kinetic energy eV_s lost by the electrons in “climbing” the hill:

$$K_{\text{max}} = eV_s \quad (3.17)$$

where e is the magnitude of the electric charge of the electron. Typical values of V_s are a few volts.*

In the classical picture, the surface of the metal is illuminated by an electromagnetic wave of intensity I . The surface absorbs energy from the wave until the energy exceeds the binding energy of the electron to the metal, at which point the electron is released. The minimum quantity of energy needed to remove an electron is called the *work function* ϕ of the material. Table 3.1 lists some values of the work function of different materials. You can see that the values are typically a few electron-volts.

The Classical Theory of the Photoelectric Effect

What does the classical wave theory predict about the properties of the emitted photoelectrons?

1. *The maximum kinetic energy of the electrons should be proportional to the intensity of the radiation.* As the brightness of the light source is increased, more energy is delivered to the surface (the electric field is greater) and the electrons should be released with greater kinetic energies. Equivalently, increasing the intensity of the light source increases the electric field $\vec{E}$ of the wave, which also increases the force $\vec{F} = -e\vec{E}$ on the electron and its kinetic energy when it eventually leaves the surface.
2. *The photoelectric effect should occur for light of any frequency or wavelength.* According to the wave theory, as long as the light is intense enough to release electrons, the photoelectric effect should occur no matter what the frequency or wavelength.
3. *The first electrons should be emitted in a time interval of the order of seconds after the radiation begins to strike the surface.* In the wave theory, the energy of the wave is uniformly distributed over the wave front. If the electron absorbs energy directly from the wave, the amount of energy delivered to

*The potential difference ΔV read by the voltmeter is not equal to the stopping potential when the emitter and collector are made of different materials. In that case, a correction must be applied to account for the *contact potential difference* between the emitter and the collector.

any electron is determined by how much radiant energy is incident on the surface area in which the electron is confined. Assuming this area is about the size of an atom, a rough calculation leads to an estimate that the time lag between turning on the light and observing the first photoelectrons should be of the order of seconds (see Example 3.2).

Example 3.2

A laser beam with an intensity of 120 W/m^2 (roughly that of a small helium–neon laser) is incident on a surface of sodium. It takes a minimum energy of 2.3 eV to release an electron from sodium (the work function ϕ of sodium). Assuming the electron to be confined to an area of radius equal to that of a sodium atom (0.10 nm), how long will it take for the surface to absorb enough energy to release an electron?

Solution

The average power P_{av} delivered by the wave of intensity I to an area A is IA . An atom on the surface displays a “target area” of $A = \pi r^2 = \pi(0.10 \times 10^{-9} \text{ m})^2 = 3.1 \times 10^{-20} \text{ m}^2$. If the entire electromagnetic power is delivered to the electron, energy is absorbed at the rate

$\Delta E/\Delta t = P_{\text{av}}$. The time interval Δt necessary to absorb an energy $\Delta E = \phi$ can be expressed as

$$\begin{aligned}\Delta t &= \frac{\Delta E}{P_{\text{av}}} = \frac{\phi}{IA} \\ &= \frac{(2.3 \text{ eV})(1.6 \times 10^{-19} \text{ J/eV})}{(120 \text{ W/m}^2)(3.1 \times 10^{-20} \text{ m}^2)} = 0.10 \text{ s}\end{aligned}$$

In reality, electrons in metals are not always bound to individual atoms but instead can be free to roam throughout the metal. However, no matter what reasonable estimate we make for the area over which the energy is absorbed, the characteristic time for photoelectron emission is estimated to have a magnitude of the order of seconds, in a range easily accessible to measurement.

The experimental characteristics of the photoelectric effect were well known by the year 1902. How do the predictions of the classical theory compare with the experimental results?

1. For a fixed value of the wavelength or frequency of the light source, the maximum kinetic energy of the emitted photoelectrons (determined from the stopping potential) is totally independent of the intensity of the light source. Figure 3.10 shows a representation of the experimental results. Doubling the intensity of the source leaves the stopping potential unchanged, indicating no change in the maximum kinetic energy of the electrons. This experimental result disagrees with the wave theory, which predicts that the maximum kinetic energy should depend on the intensity of the light.
2. The photoelectric effect does not occur at all if the frequency of the light source is below a certain value. This value, which is characteristic of the kind of metal surface used in the experiment, is called the *cutoff frequency* f_c . Above f_c , any light source, no matter how weak, will cause the emission of photoelectrons; below f_c , no light source, no matter how strong, will cause the emission of photoelectrons. This experimental result also disagrees with the predictions of the wave theory.
3. The first photoelectrons are emitted virtually instantaneously (within 10^{-9} s) after the light source is turned on. The wave theory predicts a measurable time delay, so this result also disagrees with the wave theory.

These three experimental results all suggest the complete failure of the wave theory to account for the photoelectric effect.

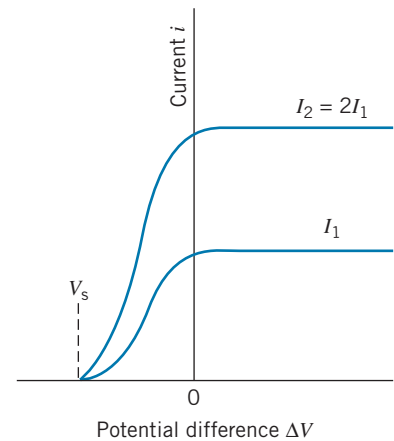


FIGURE 3.10 The photoelectric current i as a function of the potential difference ΔV for two different values of the intensity of the light. When the intensity I is doubled, the current is doubled (twice as many photoelectrons are emitted), but the stopping potential V_s remains the same.

The Quantum Theory of the Photoelectric Effect

A successful theory of the photoelectric effect was developed in 1905 by Albert Einstein. Five years earlier, in 1900, the German physicist Max Planck had developed a theory to explain the wavelength distribution of light emitted by hot, glowing objects (called *thermal radiation*, which is discussed in the next section of this chapter). Based partly on Planck's ideas, Einstein proposed that the energy of electromagnetic radiation is not continuously distributed over the wave front, but instead is concentrated in localized bundles or *quanta* (also known as *photons*). The energy of a photon associated with an electromagnetic wave of frequency f is

$$E = hf \quad (3.18)$$

where h is a proportionality constant known as *Planck's constant*. The photon energy can also be related to the wavelength of the electromagnetic wave by substituting $f = c/\lambda$, which gives

$$E = \frac{hc}{\lambda} \quad (3.19)$$

We often speak about photons as if they were particles, and as concentrated bundles of energy they have particle-like properties. Like the electromagnetic waves, photons travel at the speed of light, and so they must obey the relativistic relationship $p = E/c$. Combining this with Eq. 3.19, we obtain

$$p = \frac{h}{\lambda} \quad (3.20)$$

Photons carry linear momentum as well as energy, and thus they share this characteristic property of particles.

Because a photon travels at the speed of light, it must have zero mass. Otherwise its energy and momentum would be infinite. Similarly, a photon's rest energy $E_0 = mc^2$ must also be zero.

In Einstein's interpretation, a photoelectron is released as a result of an encounter with a *single photon*. The entire energy of the photon is delivered instantaneously to a *single photoelectron*. If the photon energy hf is greater than the work function ϕ of the material, the photoelectron will be released. If the photon energy is smaller than the work function, the photoelectric effect will not occur. This explanation thus accounts for two of the failures of the wave theory: the existence of the cutoff frequency and the lack of any measurable time delay.

If the photon energy hf exceeds the work function, the excess energy appears as the kinetic energy of the electron:

$$K_{\max} = hf - \phi \quad (3.21)$$

The intensity of the light source does not appear in this expression! For a fixed frequency, doubling the intensity of the light means that twice as many photons strike the surface and twice as many photoelectrons are released, but they all have precisely the same maximum kinetic energy.

You can think of Eq. 3.21 as giving a relationship between energy quantities in analogy to making a purchase at a store. The quantity hf represents the payment you hand to the cashier, the quantity ϕ represents the cost of the object, and $K_{\max}$ represents the change you receive. In the photoelectric effect, hf is the amount of energy that is available to “purchase” an electron from the surface, the work function ϕ is the “cost” of removing the least tightly bound electron from the surface, and the difference between the available energy and the removal cost is the leftover energy that appears as the kinetic energy of the emitted electron. (The more tightly bound electrons have a greater “cost” and so emerge with smaller kinetic energies.)

A photon that supplies an energy equal to ϕ , exactly the minimum amount needed to remove an electron, corresponds to light of frequency equal to the cutoff frequency f_c . At this frequency, there is no excess energy for kinetic energy, so Eq. 3.21 becomes $hf_c = \phi$, or

$$f_c = \frac{\phi}{h} \quad (3.22)$$

The corresponding cutoff wavelength $\lambda_c = c/f_c$ is

$$\lambda_c = \frac{hc}{\phi} \quad (3.23)$$

The cutoff wavelength represents the *largest* wavelength for which the photoelectric effect can be observed for a surface with the work function ϕ .

The photon theory appears to explain all of the observed features of the photoelectric effect. The most detailed test of the theory was done by Robert Millikan in 1915. Millikan measured the maximum kinetic energy (stopping potential) for different frequencies of the light and obtained a plot of Eq. 3.21. A sample of his results is shown in Figure 3.11. From the slope of the line, Millikan obtained a value for Planck’s constant of

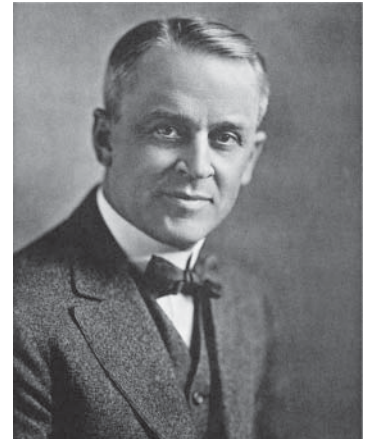
$$h = 6.57 \times 10^{-34} \text{ J} \cdot \text{s}$$

In part for his detailed experiments on the photoelectric effect, Millikan was awarded the 1923 Nobel Prize in physics. Einstein was awarded the 1921 Nobel Prize for his photon theory as applied to the photoelectric effect.

As we discuss in the next section, the wavelength distribution of thermal radiation also yields a value for Planck’s constant, which is in good agreement with Millikan’s value derived from the photoelectric effect. Planck’s constant is one of the fundamental constants of nature; just as c is the characteristic constant of relativity, h is the characteristic constant of quantum mechanics. The value of Planck’s constant has been measured to great precision in a variety of experiments. The presently accepted value is

$$h = 6.6260704 \times 10^{-34} \text{ J} \cdot \text{s}$$

This is an experimentally determined value, with a relative uncertainty of about 1×10^{-8} (± 8 units in the last digit).



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Robert A. Millikan (1868–1953, United States). Perhaps the best experimentalist of his era, his work included the precise determination of Planck’s constant using the photoelectric effect (for which he received the 1923 Nobel Prize) and the measurement of the charge of the electron (using his famous “oil-drop” apparatus).

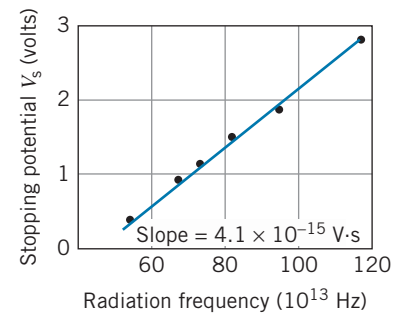


FIGURE 3.11 Millikan’s results for the photoelectric effect in sodium. The slope of the line is h/e ; the experimental determination of the slope gives a way of determining Planck’s constant. The intercept should give the cutoff frequency; however, in Millikan’s time the contact potentials of the electrodes were not known precisely and so the vertical scale is displaced by a few tenths of a volt. The slope is not affected by this correction.

Example 3.3

(a) What are the energy and momentum of a photon of red light of wavelength 650 nm? (b) What is the wavelength of a photon of energy 2.40 eV?

Solution

(a) Using Eq. 3.19 we obtain

$$E = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34} \text{ J} \cdot \text{s}) (3.00 \times 10^8 \text{ m/s})}{650 \times 10^{-9} \text{ m}} = 3.06 \times 10^{-19} \text{ J}$$

Converting to electron-volts, we have

$$E = \frac{3.06 \times 10^{-19} \text{ J}}{1.60 \times 10^{-19} \text{ J/eV}} = 1.91 \text{ eV}$$

This type of problem can be simplified if we express the combination hc in units of $\text{eV} \cdot \text{nm}$:

$$E = \frac{hc}{\lambda} = \frac{1240 \text{ eV} \cdot \text{nm}}{650 \text{ nm}} = 1.91 \text{ eV}$$

The momentum is found in a similar way, using Eq. 3.20

$$p = \frac{h}{\lambda} = \frac{1}{c} \frac{hc}{\lambda} = \frac{1}{c} \left(\frac{1240 \text{ eV} \cdot \text{nm}}{650 \text{ nm}} \right) = 1.91 \text{ eV}/c$$

The momentum could also be found directly from the energy:

$$p = \frac{E}{c} = \frac{1.91 \text{ eV}}{c} = 1.91 \text{ eV}/c$$

(It may be helpful to review the discussion in Example 2.13 about these units of momentum.)

(b) Solving Eq. 3.19 for λ , we find

$$\lambda = \frac{hc}{E} = \frac{1240 \text{ eV} \cdot \text{nm}}{2.40 \text{ eV}} = 517 \text{ nm}$$

Example 3.4

The work function for tungsten metal is 4.52 eV.

(a) What is the cutoff wavelength λ_c for tungsten?

(b) What is the maximum kinetic energy of the electrons when radiation of wavelength 198 nm is used? (c) What is the stopping potential in this case?

Solution

(a) Equation 3.23 gives

$$\lambda_c = \frac{hc}{\phi} = \frac{1240 \text{ eV} \cdot \text{nm}}{4.52 \text{ eV}} = 274 \text{ nm}$$

in the ultraviolet region.

(b) At the shorter wavelength,

$$\begin{aligned} K_{\max} &= hf - \phi = \frac{hc}{\lambda} - \phi \\ &= \frac{1240 \text{ eV} \cdot \text{nm}}{198 \text{ nm}} - 4.52 \text{ eV} \\ &= 1.74 \text{ eV} \end{aligned}$$

(c) The stopping potential is the voltage corresponding to $K_{\max}$:

$$V_s = \frac{K_{\max}}{e} = \frac{1.74 \text{ eV}}{e} = 1.74 \text{ V}$$

Example 3.5

Figure 3.10 shows two curves, corresponding to an original and a doubled intensity of the light source in the photoelectric effect. In comparison with the plot for the original source, how would the curve appear if (a) the frequency of the light source were doubled, keeping the photon emission rate the same, or (b) with the original light source the metal surface were replaced with one having a larger work function?

Solution

(a) If the frequency of the photons increases, the photon energy increases, which, in turn, increases the kinetic energy of the photoelectrons. A larger voltage difference would then be required to stop the current and thus a larger stopping potential. Because there is no change in the photon emission rate from the light source, the magnitude of the current would not

change. So the new curve would have the same plateau in the vertical direction, but the x -intercept (stopping potential) would move to the left (larger negative values).

(b) A larger work function for the same light source means smaller kinetic energy for the photoelectrons, but

with the same emission rate the magnitude of the current would be the same. So the plateau would remain but the x -intercept (stopping potential) would move to smaller negative values, unless the new work function were so large that it exceeded the photon energy (in which case the current would disappear).

3.3 THERMAL RADIATION

The second type of experiment we discuss that cannot be explained by the classical wave theory is *thermal radiation*, which is the electromagnetic radiation emitted by all objects because of their temperature. At room temperature the thermal radiation is mostly in the infrared region of the spectrum, where our eyes are not sensitive. As we heat objects to higher temperatures, they may emit visible light.

A typical experimental arrangement is shown in Figure 3.12. An object is maintained at a temperature T_1 . The radiation emitted by the object is detected by an apparatus that is sensitive to the wavelength of the radiation. For example, a dispersive medium such as a prism can be used so that different wavelengths appear at different angles θ . By moving the radiation detector to different angles θ we can measure the intensity* of the radiation at a specific wavelength. The detector is not a geometrical point (hardly an efficient detector!) but instead subtends a small range of angles $\Delta\theta$, so what we really measure is the amount of radiation in some range $\Delta\theta$ at θ , or, equivalently, in some range $\Delta\lambda$ at λ .

Many experiments were done in the late 19th century to study the wavelength spectrum of thermal radiation. These experiments, as we shall see, gave results that totally disagreed with the predictions of the classical theories of thermodynamics and electromagnetism; instead, the successful analysis of the experiments provided the first evidence of the quantization of energy, which would eventually be seen as the basis for the new quantum theory.

Let's first review the experimental results. The goal of these experiments was to measure the intensity of the radiation emitted by the object as a function of wavelength. Figure 3.13 shows a typical set of experimental results when the object is at a temperature $T_1 = 1000$ K. If we now change the temperature of the object to a different value T_2 , we obtain a different curve, as shown in Figure 3.13 for $T_2 = 1250$ K. If we repeat the measurement for many different temperatures, we obtain systematic results for the radiation intensity that reveal two important characteristics:

1. The total intensity radiated over all wavelengths (that is, the area under each curve) increases as the temperature is increased. This is not a surprising result: we commonly observe that a glowing object glows

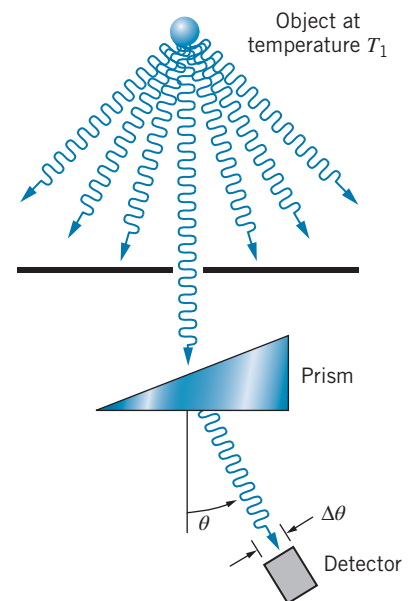


FIGURE 3.12 Measurement of the spectrum of thermal radiation. A device such as a prism is used to separate the wavelengths emitted by the object.

*As always, intensity means energy per unit time per unit area (or power per unit area), as in Eq. 3.8. Previously, "unit area" referred to the wave front, such as would be measured if we recorded the waves with an antenna of a certain area. Here, "unit area" indicates the electromagnetic radiation emitted from each unit area of the surface of the object whose thermal emissions are being observed.

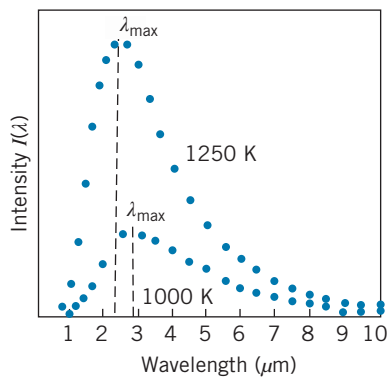


FIGURE 3.13 A possible result of the measurement of the radiation intensity over many different wavelengths. Each different temperature of the emitting body gives a different peak $\lambda_{\max}$.

brighter and thus radiates more energy as we increase its temperature. From careful measurement, we find that the total intensity increases as the fourth power of the absolute or kelvin temperature:

$$I = \sigma T^4 \quad (3.24)$$

where we have introduced the proportionality constant σ . Equation 3.24 is called *Stefan's law* and the constant σ is called the *Stefan–Boltzmann constant*. Its value can be determined from experimental results such as those illustrated in Figure 3.13:

$$\sigma = 5.67037 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$$

2. The wavelength $\lambda_{\max}$ at which the emitted intensity reaches its maximum value decreases as the temperature is increased, in inverse proportion to the temperature: $\lambda_{\max} \propto 1/T$. From results such as those of Figure 3.13, we can determine the proportionality constant, so that

$$\lambda_{\max} T = 2.8978 \times 10^{-3} \text{ m} \cdot \text{K} \quad (3.25)$$

This result is known as *Wien's displacement law*; the term “displacement” refers to the way the peak is moved or displaced as the temperature is varied. Wien's law is qualitatively consistent with our common observation that heated objects first begin to glow with a red color, and at higher temperatures the color becomes more yellow. As the temperature is increased, the wavelength at which most of the radiation is emitted moves from the longer-wavelength (red) part of the visible region toward medium wavelengths. The term “white hot” refers to an object that is hot enough to produce the mixture of all wavelengths in the visible region to make white light.

Example 3.6

(a) At what wavelength does a room-temperature ($T = 20^\circ\text{C}$) object emit the maximum thermal radiation? (b) To what temperature must we heat it until its peak thermal radiation is in the red region of the spectrum ($\lambda = 650 \text{ nm}$)? (c) How many times as much thermal radiation does it emit at the higher temperature?

Solution

(a) Using the absolute temperature, $T_1 = 273 + 20 = 293 \text{ K}$, Wien's displacement law gives

$$\begin{aligned} \lambda_{\max} &= \frac{2.8978 \times 10^{-3} \text{ m} \cdot \text{K}}{T_1} \\ &= \frac{2.8978 \times 10^{-3} \text{ m} \cdot \text{K}}{293 \text{ K}} = 9.89 \mu\text{m} \end{aligned}$$

This is in the infrared region of the electromagnetic spectrum.

(b) For $\lambda_{\max} = 650 \text{ nm}$, we again use Wien's displacement law to find the new temperature T_2 :

$$\begin{aligned} T_2 &= \frac{2.8978 \times 10^{-3} \text{ m} \cdot \text{K}}{\lambda_{\max}} \\ &= \frac{2.8978 \times 10^{-3} \text{ m} \cdot \text{K}}{650 \times 10^{-9} \text{ m}} \\ &= 4460 \text{ K} \end{aligned}$$

(c) The total intensity of radiation is proportional to T^4 , so the ratio of the total thermal emissions will be

$$\begin{aligned} \frac{I_2}{I_1} &= \frac{\sigma T_2^4}{\sigma T_1^4} = \frac{(4460 \text{ K})^4}{(293 \text{ K})^4} \\ &= 5.37 \times 10^4 \end{aligned}$$

Be sure to notice the use of absolute (kelvin) temperatures in this example.

The theoretical analysis of the emission of thermal radiation from an arbitrary object is extremely complicated. It depends on details of the surface properties of the object, and it also depends on how much radiation the object reflects from its surroundings. To simplify our analysis, we consider a special type of object called a *blackbody*, which absorbs all radiation incident on it and reflects none of the incident radiation.

To simplify further, we consider a special type of blackbody: a hole in a hollow metal box whose walls are in thermal equilibrium at temperature T . The box is filled with electromagnetic radiation that is emitted and reflected by the walls. A small hole in one wall of the box allows some of the radiation to escape (Figure 3.14). *It is the hole, and not the box itself, that is the blackbody.* Radiation from outside that is incident on the hole gets lost inside the box and has a negligible chance of reemerging from the hole; thus no reflections occur from the blackbody (the hole). The radiation that emerges from the hole is just a sample of the radiation inside the box, so understanding the nature of the radiation inside the box allows us to understand the radiation that leaves through the hole.

Let's consider the radiation inside the box. It has an energy density (energy per unit volume) per unit wavelength interval $u(\lambda)$. That is, if we could look into the interior of the box and measure the energy density of the electromagnetic radiation with wavelengths between λ and $\lambda + d\lambda$ in a small volume element, the result would be $u(\lambda)d\lambda$. For the radiation in this wavelength interval, what is the corresponding intensity (power per unit area) emerging from the hole? At any particular instant, half of the radiation in the box will be moving away from the hole. The other half of the radiation is moving toward the hole at velocity of magnitude c but directed over a range of angles. Averaging over this range of angles to evaluate the energy flowing perpendicular to the surface of the hole introduces another factor of $1/2$, so the contribution of the radiation in this small wavelength interval to the intensity passing through the hole is

$$I(\lambda) = \frac{c}{4}u(\lambda) \quad (3.26)$$

The quantity $I(\lambda)d\lambda$ is the radiant intensity in the small interval $d\lambda$ at the wavelength λ . This is the quantity whose measurement gives the results displayed in Figure 3.13. Each data point represents a measurement of the intensity in a small wavelength interval. The goal of the theoretical analysis is to find a mathematical function $I(\lambda)$ that gives a smooth fit through the data points of Figure 3.13.

If we wish to find the total intensity emitted in the region between wavelengths λ_1 and λ_2 , we divide the region into narrow intervals $d\lambda$ and add the intensities in each interval, which is equivalent to the integral between those limits:

$$I(\lambda_1:\lambda_2) = \int_{\lambda_1}^{\lambda_2} I(\lambda) d\lambda \quad (3.27)$$

This is similar to Eq. 1.28 for determining the number of molecules with energies between two limits. The total emitted intensity can be found by integrating over all wavelengths:

$$I = \int_0^{\infty} I(\lambda) d\lambda \quad (3.28)$$

This total intensity should work out to be proportional to the 4th power of the temperature, as required by Stefan's law (Eq. 3.24).

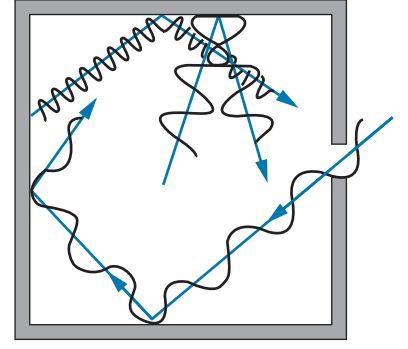


FIGURE 3.14 A cavity filled with electromagnetic radiation in thermal equilibrium with its walls at temperature T . Some radiation escapes through the hole, which represents an ideal blackbody.

Classical Theory of Thermal Radiation

Before discussing the quantum theory of thermal radiation, let's see what the classical theories of electromagnetism and thermodynamics can tell us about the dependence of I on λ . The complete derivation is not given here, only a brief outline of the theory.* The derivation involves first computing the amount of radiation (number of waves) at each wavelength and then finding the contribution of each wave to the total energy in the box.

1. *The box is filled with electromagnetic standing waves.* If the walls of the box are metal, radiation is reflected back and forth with a node of the electric field at each wall (the electric field must vanish inside a conductor). This is the same condition that applies to other standing waves, like those on a stretched string or a column of air in an organ pipe.
2. *The number of standing waves with wavelengths between λ and $\lambda + d\lambda$ is*

$$N(\lambda) d\lambda = \frac{8\pi V}{\lambda^4} d\lambda \quad (3.29)$$

where V is the volume of the box. For one-dimensional standing waves, as on a stretched string of length L , the allowed wavelengths are $\lambda = 2L/n$, ($n = 1, 2, 3, \dots$). The number of possible standing waves with wavelengths between λ_1 and λ_2 is $n_2 - n_1 = 2L(1/\lambda_2 - 1/\lambda_1)$. In the small interval from λ to $\lambda + d\lambda$, the number of standing waves is $N(\lambda)d\lambda = |dn/d\lambda|d\lambda = (2L/\lambda^2)d\lambda$. Equation 3.29 can be obtained by extending this approach to three dimensions.

3. *Each individual wave contributes an average energy of kT to the radiation in the box.* This result follows from an analysis similar to that of Section 1.3 for the statistical mechanics of gas molecules. In this case, we are interested in the statistics of the oscillating atoms in the walls of the cavity, which are responsible for setting up the standing electromagnetic waves in the cavity. For a one-dimensional oscillator, the energies are distributed according to the Maxwell-Boltzmann distribution:†

$$N(E) = \frac{N}{kT} e^{-E/kT} \quad (3.30)$$

Recall from Section 1.3 that $N(E)$ is defined so that the number of oscillators with energies between E and $E + dE$ is $dN = N(E)dE$, and thus the total number of oscillators at all energies is $\int dN = \int_0^\infty N(E)dE$, which (as you should show) works out to N . The average energy per oscillator is then found in the same way as the average energy of a gas molecule (Eq. 1.26):

$$E_{av} = \frac{1}{N} \int_0^\infty E N(E) dE = \frac{1}{kT} \int_0^\infty E e^{-E/kT} dE \quad (3.31)$$

which does indeed work out to $E_{av} = kT$.

*For a more complete derivation, see R. Eisberg and R. Resnick, *Quantum Theory of Atoms, Molecules, Solids, Nuclei, and Particles*, 2nd edition (Wiley, 1985), pp. 9–13.

†The exponential part of this expression is the same as that of Eq. 1.23 for gas molecules, but the rest of the equation is different, because the statistical behavior of one-dimensional oscillators is different from that of gas molecules moving in three dimensions. We'll consider these calculations in greater detail in Chapter 10.

Putting all these ingredients together, we can find the energy density of radiation in the wavelength interval $d\lambda$ inside the cavity: energy density = (number of standing waves per unit volume) $\times$ (average energy per standing wave) or

$$u(\lambda) d\lambda = \frac{N(\lambda) d\lambda}{V} kT = \frac{8\pi}{\lambda^4} kT d\lambda \quad (3.32)$$

The corresponding intensity per unit wavelength interval $d\lambda$ is

$$I(\lambda) = \frac{c}{4} u(\lambda) = \frac{c}{4} \frac{8\pi}{\lambda^4} kT = \frac{2\pi c}{\lambda^4} kT \quad (3.33)$$

This result is known as the *Rayleigh–Jeans formula*; based firmly on the classical theories of electromagnetism and thermodynamics, it represents our best attempt to apply classical physics to understanding the problem of blackbody radiation. In Figure 3.15, the intensity calculated from the Rayleigh–Jeans formula is compared with typical experimental results. The intensity calculated with Eq. 3.33 approaches the data at long wavelengths, but at short wavelengths, the classical theory (which predicts $u \rightarrow \infty$ as $\lambda \rightarrow 0$) fails miserably. The failure of the Rayleigh–Jeans formula at short wavelengths is known as the *ultraviolet catastrophe* and represents a serious problem for classical physics, because the theories of thermodynamics and electromagnetism on which the Rayleigh–Jeans formula is based have been carefully tested in many other circumstances and found to give extremely good agreement with experiment. It is apparent in the case of blackbody radiation that the classical theories do not work and that a new kind of physical theory is needed.

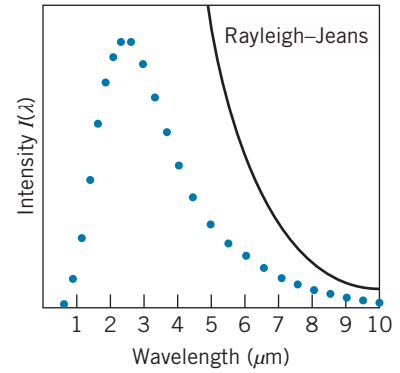


FIGURE 3.15 The failure of the classical Rayleigh–Jeans formula to fit the observed intensity. At long wavelengths the theory approaches the data, but at short wavelengths the classical formula fails miserably.

Quantum Theory of Thermal Radiation

The new physics that gave the correct interpretation of thermal radiation was proposed by the German physicist Max Planck in 1900. The ultraviolet catastrophe occurs because the Rayleigh–Jeans formula predicts too much intensity at short wavelengths (or equivalently at high frequencies). What is needed is a way to make $u \rightarrow 0$ as $\lambda \rightarrow 0$, or as $f \rightarrow \infty$. Again considering the electromagnetic standing waves to result from the oscillations of atoms in the walls of the cavity, Planck tried to find a way to reduce the number of high-frequency standing waves by reducing the number of high-frequency oscillators. He did this by a bold assumption that formed the cornerstone of a new physical theory, *quantum physics*. Associated with this theory is a new version of mechanics, known as *wave mechanics* or *quantum mechanics*. We discuss the methods of wave mechanics in Chapter 5; for now we show how Planck’s theory provided the correct interpretation of the emission spectrum of thermal radiation.

Planck suggested that an oscillating atom can absorb or emit energy only in discrete bundles. This bold suggestion was necessary to keep the average energy of a low-frequency (long-wavelength) oscillator equal to kT (in agreement with the Rayleigh–Jeans law at long wavelength), but it also made the average energy of a high-frequency (low-wavelength) oscillator approach zero. Let’s see how Planck managed this remarkable feat.

In Planck’s theory, each oscillator can emit or absorb energy only in quantities that are integer multiples of a certain basic quantity of energy ϵ

$$E_n = n\epsilon \quad n = 1, 2, 3, \dots \quad (3.34)$$



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Max Planck (1858–1947, Germany). His work on the spectral distribution of radiation, which led to the quantum theory, was honored with the 1918 Nobel Prize. In his later years, he wrote extensively on religious and philosophical topics.

where n is the number of quanta. Furthermore, the energy of each of the quanta is determined by the frequency

$$\epsilon = hf \quad (3.35)$$

where h is the constant of proportionality, now known as Planck's constant. From the mathematical standpoint, the difference between Planck's calculation and the classical calculation using Maxwell–Boltzmann statistics is that the energy of an oscillator at a certain wavelength or frequency is no longer a continuous variable—it is a discrete variable that takes only the values given by Eq. 3.34. The integrals in the classical calculation are then replaced by sums, and the number of oscillators with energy E_n is then

$$N_n = N(1 - e^{-\epsilon/kT})e^{-n\epsilon/kT} \quad (3.36)$$

(Compare this result with Eq. 3.30 for the continuous case.) Here N_n represents the number of oscillators with energy E_n , while N is the total number. You should be able to show that $\sum_{n=0}^{\infty} N_n = N$, again giving the total number of oscillators when summed over all possible energies. Planck's calculation then gives the average energy:

$$E_{\text{av}} = \frac{1}{N} \sum_{n=0}^{\infty} N_n E_n = (1 - e^{-\epsilon/kT}) \sum_{n=0}^{\infty} (n\epsilon) e^{-n\epsilon/kT} \quad (3.37)$$

which gives (see Problem 14)

$$E_{\text{av}} = \frac{\epsilon}{e^{\epsilon/kT} - 1} = \frac{hf}{e^{hf/kT} - 1} = \frac{hc/\lambda}{e^{hc/\lambda kT} - 1} \quad (3.38)$$

Note from this equation that $E_{\text{av}} \cong kT$ at small f (large λ) but that $E_{\text{av}} \rightarrow 0$ at large f (small λ). Thus the small-wavelength oscillators carry a vanishingly small energy, and the ultraviolet catastrophe is solved!

Based on Planck's result, the intensity of the radiation then becomes (using Eqs. 3.26 and 3.29):

$$I(\lambda) = \frac{c}{4} \left(\frac{8\pi}{\lambda^4} \right) \left[\frac{hc/\lambda}{e^{hc/\lambda kT} - 1} \right] = \frac{2\pi hc^2}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1} \quad (3.39)$$

(An alternative approach to deriving this result is given in Section 10.6.) The perfect agreement between experiment and Planck's formula is illustrated in Figure 3.16.

In Problems 15 and 16 at the end of this chapter, you will demonstrate that Planck's formula can be used to deduce Wien's displacement law and Stefan's law. In fact, deducing Stefan's law from Planck's formula results in a relationship between the Stefan–Boltzmann constant and Planck's constant:

$$\sigma = \frac{2\pi^5 k^4}{15c^2 h^3} \quad (3.40)$$

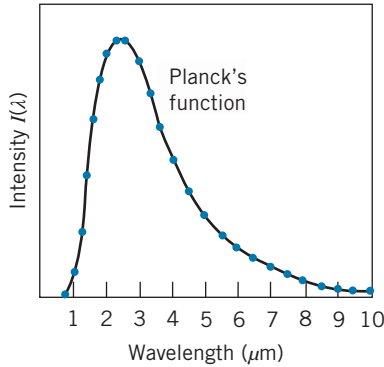


FIGURE 3.16 Planck's function fits the observed data perfectly.

By determining the value of the Stefan–Boltzmann constant from the intensity data available in 1900, Planck was able to determine a value of the constant h :

$$h = 6.56 \times 10^{-34} \text{ J} \cdot \text{s}$$

which agrees very well with the value of h that Millikan deduced 15 years later based on the analysis of data from the photoelectric effect. The good agreement of these two values is remarkable, because they are derived from very different kinds of experiments—one involves the *emission* and the other the *absorption* of electromagnetic radiation. This suggests that the quantization property is not an accident arising from the analysis of one particular experiment, but is instead a property of the electromagnetic field itself. Along with many other scientists of his era, Planck was slow to accept this interpretation. However, later experimental evidence (including the Compton effect) proved to be so compelling that it left no doubt about Einstein's photon theory and the particle-like structure of the electromagnetic field.

Planck's formula still finds important applications today in the measurement of temperature. By measuring the intensity of radiation emitted by an object at a particular wavelength (or, as in actual experiments, in a small interval of wavelengths), Eq. 3.39 can be used to deduce the temperature of the object. Note that only one measurement, at any wavelength, is all that is required to obtain the temperature. A *radiometer* is a device for measuring the intensity of thermal radiation at selected wavelengths, enabling a determination of temperature. Radiometers in orbiting satellites are used to measure the temperature of the land and sea areas of the Earth and of the upper surface of clouds. Other orbiting radiometers have been aimed toward “empty space” to measure the temperature of the radiation from the early history of the universe (Figure 3.17).

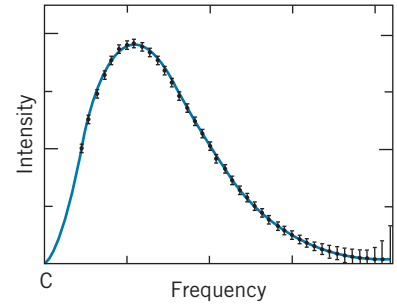


FIGURE 3.17 Data from the COBE satellite, launched in 1989 to determine the temperature of the cosmic microwave background radiation from the early universe. The data points exactly fit the Planck's function corresponding to a temperature of 2.725 K. To appreciate the remarkable precision of this experiment, note that the sizes of the error bars have been increased by a factor of 400 to make them visible! (Source: NASA Office of Space Science)

Example 3.7

You are using a radiometer to observe the thermal radiation from an object that is heated to maintain its temperature at 1278 K. The radiometer records radiation in a wavelength interval of 12.6 nm. By changing the wavelength at which you are measuring, you set the radiometer to record the most intense radiation emission from the object. What is the intensity of the emitted radiation in this interval?

Solution

The wavelength setting for the most intense radiation is determined from Wien's displacement law:

$$\begin{aligned}\lambda_{\max} &= \frac{2.8978 \times 10^{-3} \text{ m} \cdot \text{K}}{T} = \frac{2.8978 \times 10^{-3} \text{ m} \cdot \text{K}}{1278 \text{ K}} \\ &= 2.267 \times 10^{-6} \text{ m} = 2267 \text{ nm}\end{aligned}$$

The given temperature corresponds to $kT = (8.6174 \times 10^{-5} \text{ eV/K})(1278 \text{ K}) = 0.1101 \text{ eV}$. The radiation intensity in this small wavelength interval is

$$\begin{aligned}I(\lambda)d\lambda &= \frac{2\pi hc^2}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1} d\lambda \\ &= 2\pi(6.626 \times 10^{-34} \text{ J} \cdot \text{s})(2.998 \times 10^8 \text{ m/s})^2 \\ &\quad \times (12.6 \times 10^{-9} \text{ m})(2.267 \times 10^{-6} \text{ m})^{-5} \\ &\quad \times (e^{(1240 \text{ eV} \cdot \text{nm})/(2267 \text{ nm})(0.1101 \text{ eV})} - 1)^{-1} \\ &= 552 \text{ W/m}^2\end{aligned}$$

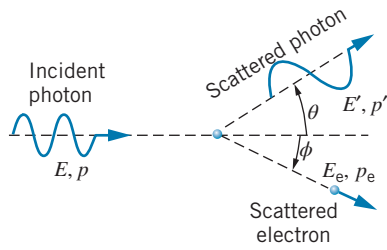


FIGURE 3.18 The geometry of Compton scattering.

3.4 THE COMPTON EFFECT

Another way for radiation to interact with matter is by means of the Compton effect, in which radiation scatters from loosely bound, nearly free electrons. Part of the energy of the radiation is given to the electron; the remainder of the energy is reradiated as electromagnetic radiation. According to the wave picture, the scattered radiation is less energetic than the incident radiation (the difference going into the kinetic energy of the electron) but has the same wavelength. As we will see, the photon concept leads to a very different prediction for the scattered radiation.

The scattering process is analyzed simply as an interaction (a “collision” in the classical sense of particles) between a single photon and an electron, which we assume to be at rest. Figure 3.18 shows the process. Initially, the photon has energy E and linear momentum p given by

$$E = hf = \frac{hc}{\lambda} \quad \text{and} \quad p = \frac{E}{c} \quad (3.41)$$

The electron, initially at rest, has rest energy $m_e c^2$. After the scattering, the photon has energy $E' = hc/\lambda'$ and momentum $p' = E'/c$, and it moves in a direction at an angle θ with respect to the direction of the incident photon. The electron has total final energy E_e and momentum p_e and moves in a direction at an angle ϕ with respect to the initial photon. (To allow for the possibility of high-energy incident photons giving energetic scattered electrons, we use relativistic kinematics for the electron.) The conservation laws for total relativistic energy and momentum are then applied:

$$E_{\text{initial}} = E_{\text{final}}: \quad E + m_e c^2 = E' + E_e \quad (3.42a)$$

$$p_{x,\text{initial}} = p_{x,\text{final}}: \quad p = p_e \cos \phi + p' \cos \theta \quad (3.42b)$$

$$p_{y,\text{initial}} = p_{y,\text{final}}: \quad 0 = p_e \sin \phi - p' \sin \theta \quad (3.42c)$$

We have three equations with four unknowns (θ , ϕ , E_e , E' ; p_e and p' are not independent unknowns) that cannot be solved uniquely, but we can eliminate any two of the four unknowns by solving the equations simultaneously. If we choose to measure the energy and direction of the scattered photon, we eliminate E_e and ϕ . The angle ϕ is eliminated by first rewriting the momentum equations:

$$p_e \cos \phi = p - p' \cos \theta \quad \text{and} \quad p_e \sin \phi = p' \sin \theta \quad (3.43)$$

Squaring these equations and adding the results, we obtain

$$p_e^2 = p^2 - 2pp' \cos \theta + p'^2 \quad (3.44)$$

The relativistic relationship between energy and momentum is, according to Eq. 2.39, $E_e^2 = c^2 p_e^2 + m_e^2 c^4$. Substituting in this equation for E_e from Eq. 3.42a and for p_e^2 from Eq. 3.44, we obtain

$$(E + m_e c^2 - E')^2 = c^2(p^2 - 2pp' \cos \theta + p'^2) + m_e^2 c^4 \quad (3.45)$$

and after a bit of algebra, we find

$$\frac{1}{E'} - \frac{1}{E} = \frac{1}{m_e c^2} (1 - \cos \theta) \quad (3.46)$$

In terms of wavelength, this equation can also be written as

$$\lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta) \quad (3.47)$$

where λ is the wavelength of the incident photon and λ' is the wavelength of the scattered photon. The quantity $h/m_e c$ is known as the *Compton wavelength of the electron* and has a value of 0.002426 nm; however, keep in mind that it is *not* a true wavelength but rather is a *change* of wavelength.

Equations 3.46 and 3.47 give the change in energy or wavelength of the photon, as a function of the *scattering angle* θ . Because the quantity on the right-hand side is never negative, E' is always less than E , so that the scattered photon has less energy than the original incident photon; the difference $E - E'$ is just the kinetic energy given to the electron, $E_e - m_e c^2$. Similarly, λ' is greater than λ , meaning the scattered photon always has a longer wavelength than the incident photon; the change in wavelength ranges from 0 at $\theta = 0^\circ$ to twice the Compton wavelength at $\theta = 180^\circ$. Of course the descriptions in terms of energy and wavelength are equivalent, and the choice of which to use is merely a matter of convenience.

Using $E_e = K_e + m_e c^2$, where K_e is the kinetic energy of the electron, conservation of energy (Eq. 3.42a) can also be written as $E + m_e c^2 = E' + K_e + m_e c^2$. Solving for K_e , we obtain

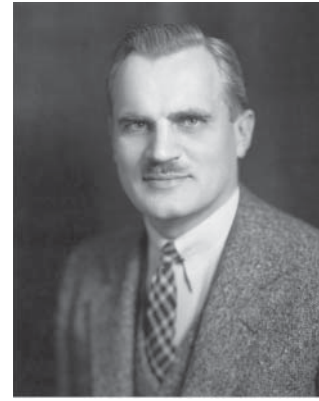
$$K_e = E - E' \quad (3.48)$$

That is, the kinetic energy acquired by the electron is equal to the difference between the initial and final photon energies.

We can also find the direction of the electron's motion by dividing the two momentum relationships in Equation 3.43:

$$\tan \phi = \frac{p_e \sin \phi}{p_e \cos \phi} = \frac{p' \sin \theta}{p - p' \cos \theta} = \frac{E' \sin \theta}{E - E' \cos \theta} \quad (3.49)$$

where the last result comes from using $p = E/c$ and $p' = E'/c$.



Moffett Studio, courtesy AIP Emilio Segre Visual Archives, Weber Collection, W. F. Meggers Gallery of Nobel Laureates / Science Source

Arthur H. Compton (1892–1962, United States). His work on X-ray scattering verified Einstein's photon theory and earned him the 1927 Nobel Prize. He was a pioneer in research with X rays and cosmic rays. During World War II he directed a portion of the U.S. atomic bomb research.

Example 3.8

X rays of wavelength λ are incident on a thin target of carbon. Scattered X rays of wavelength λ' are observed coming from the target at an angle θ relative to the incident X rays, and scattered electrons of kinetic energy K_e are observed at an angle ϕ . If the X ray source is then replaced with another of larger wavelength, what is the effect on (a) the change in wavelength $\Delta\lambda = \lambda' - \lambda$ and (b) the electron kinetic energy if the scattered photon and electron are observed at the original angles?

Solution

(a) According to Eq. 3.47, the change in wavelength of the scattered X ray does not depend on the incident

wavelength, and so $\Delta\lambda$ has the same value with both sources. (b) The electron kinetic energy is *smaller* with the larger incident wavelength. Even though $\Delta\lambda = \lambda' - \lambda$ has the same value, that does *not* imply that $K_e = E - E'$ has the same value. The left side of Eq. 3.46 is $(E - E')/EE'$, and with both E and E' being smaller at the larger incident wavelength, their difference in the numerator must also be smaller.

Example 3.9

X rays of wavelength 0.2400 nm are Compton-scattered, and the scattered beam is observed at an angle of 60.0° relative to the incident beam. Find: (a) the wavelength of the scattered X rays, (b) the energy of the scattered X-ray photons, (c) the kinetic energy of the scattered electrons, and (d) the direction of travel of the scattered electrons.

Solution

(a) λ' can be found immediately from Eq. 3.47:

$$\begin{aligned}\lambda' &= \lambda + \frac{h}{m_e c} (1 - \cos \theta) \\ &= 0.2400 + (0.00243 \text{ nm}) (1 - \cos 60^\circ) \\ &= 0.2412 \text{ nm}\end{aligned}$$

(b) The energy E' can be found directly from λ' :

$$E' = \frac{hc}{\lambda'} = \frac{1240 \text{ eV} \cdot \text{nm}}{0.2412 \text{ nm}} = 5141 \text{ eV}$$

(c) The initial photon energy E is $hc/\lambda = 5167 \text{ eV}$, so

$$K_e = E - E' = 5167 \text{ eV} - 5141 \text{ eV} = 26 \text{ eV}$$

(d) From Eq. 3.49,

$$\begin{aligned}\phi &= \tan^{-1} \frac{E' \sin \theta}{E - E' \cos \theta} \\ &= \tan^{-1} \frac{(5141 \text{ eV}) (\sin 60^\circ)}{(5167 \text{ eV}) - (5141 \text{ eV}) (\cos 60^\circ)} \\ &= 59.7^\circ\end{aligned}$$

The first experimental demonstration of this type of scattering was done by Arthur Compton in 1923. A diagram of his experimental arrangement is shown in Figure 3.19. A beam of X rays of a single wavelength λ is incident on a scattering target, for which Compton used carbon. (Although no scattering target contains actual “free” electrons, the outer or valence electrons in many materials are very weakly attached to the atom and behave like nearly free electrons. The binding energies of these electrons in the atom are so small compared with the energies of the incident X-ray photons that they can be regarded as nearly “free” electrons.) A movable detector measured the energy of the scattered X rays at various angles θ .

Compton’s original results are illustrated in Figure 3.20. At each angle, two peaks appear, corresponding to scattered X-ray photons with two different energies or wavelengths. The wavelength of one peak does not change as the angle is varied; this peak corresponds to scattering that involves “inner”

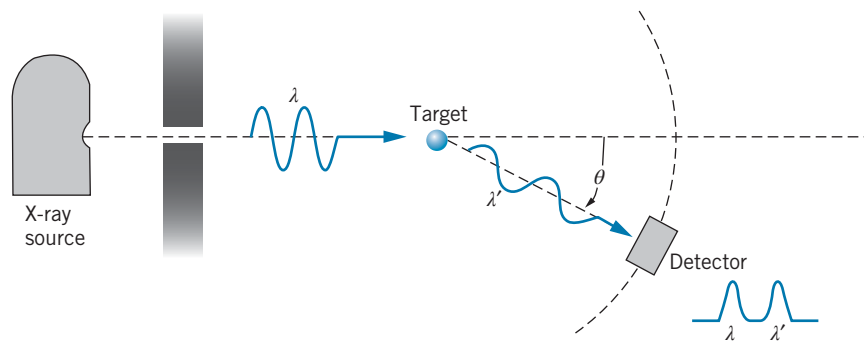


FIGURE 3.19 Schematic diagram of Compton-scattering apparatus. The wavelength λ' of the scattered X rays is measured by the detector, which can be moved to different positions θ . The wavelength difference $\lambda' - \lambda$ varies with θ .

electrons of the atom, which are more tightly bound to the atom so that the photon can scatter with no loss of energy. The wavelength of the other peak, however, varies strongly with angle; as can be seen from Figure 3.21, this variation is exactly as the Compton formula predicts.

Similar results can be obtained for the scattering of gamma rays, which are higher-energy (shorter wavelength) photons emitted in various radioactive decays. Compton also measured the variation in wavelength of scattered gamma rays, as illustrated in Figure 3.22. The change in wavelength in the gamma-ray measurements is identical with the change in wavelength in the X-ray measurements, as Eq. 3.47 predicts—the change in wavelength does not depend on the incident wavelength.

3.5 OTHER PHOTON PROCESSES

Although thermal radiation, the photoelectric effect, and Compton scattering provided the earliest experimental evidence in support of the quantization (particle-like behavior) of electromagnetic radiation, there are numerous other experiments that can also be interpreted correctly only if we assume the existence of photons as discrete quanta of electromagnetic radiation. In this section, we discuss some of these processes, which cannot be understood if we consider only the wave nature of electromagnetic radiation. As you study the descriptions of these processes, note how photons interact with atoms or electrons by delivering energy in discrete bundles, in contrast to the wave interpretation in which the energy can be regarded as arriving continuously.

Interactions of Photons with Atoms

The emission of electromagnetic radiation from atoms takes place in discrete amounts characterized by one or more photons. When an atom emits a photon

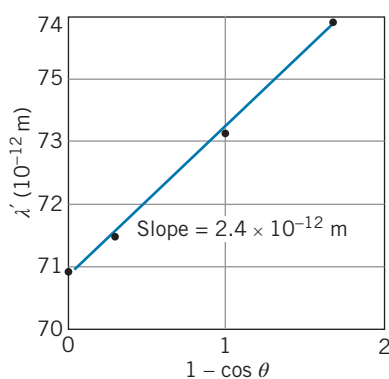


FIGURE 3.21 The scattered X-ray wavelengths λ' , from Figure 3.20, for different scattering angles. The expected slope is $2.43 \times 10^{-12} \text{ m}$, in agreement with the measured slope of Compton's data points.

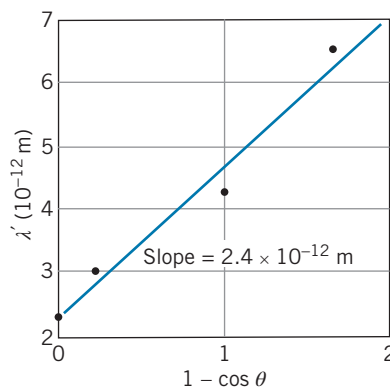


FIGURE 3.22 Compton's results for gamma-ray scattering. The wavelengths are much smaller than for X-rays, but the slope is the same as in Figure 3.21, which the Compton formula, Eq. 3.47, predicts.

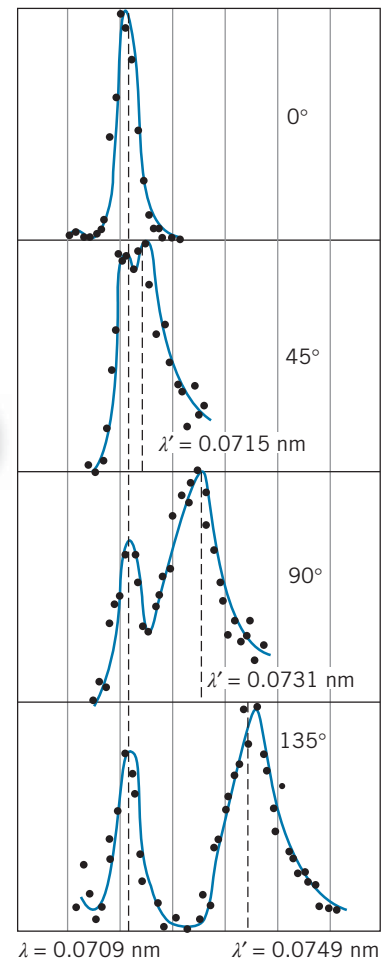


FIGURE 3.20 Compton's original results for X-ray scattering.

of energy E , the atom loses an equivalent amount of energy. Consider an atom at rest that has an initial energy E_i . The atom emits a photon of energy E . After the emission, the atom is left with a final energy E_f , which we will take as the energy associated with the internal structure of the atom. Because of conservation of momentum, the final atom must have a momentum that is equal and opposite to the momentum of the emitted photon, so the atom must also have a “recoil” kinetic energy K . (Normally this kinetic energy is very small.) Conservation of energy then gives

$$E_i = E_f + K + E \quad \text{or} \quad E = (E_i - E_f) - K \quad (3.50)$$

The energy of the emitted photon is equal to the net energy lost by the atom, minus a negligibly small contribution to the recoil kinetic energy of the atom.

In the reverse process, an atom can *absorb* a photon of energy E . If the atom is initially at rest, it must again acquire a small recoil kinetic energy in order to conserve momentum. Now conservation of energy gives

$$E_i + E = E_f + K \quad \text{or} \quad E_f - E_i = E - K \quad (3.51)$$

The energy available to add to the atom’s internal supply of energy is the photon energy, less a recoil kinetic energy that is usually negligible.

Photon emission and absorption experiments are among the most important techniques for acquiring information about the internal structure of atoms, as we discuss in Chapter 6.

Bremsstrahlung and X-Ray Production

When an electric charge, such as an electron, is accelerated or decelerated, it radiates electromagnetic energy; according to the quantum interpretation, we would say that it emits photons. Suppose we have a beam of electrons, which has been accelerated through a potential difference ΔV , so that the electrons experience a loss in potential energy of $-e \Delta V$ and thus acquire a kinetic energy of $K = e \Delta V$ (Figure 3.23). When the electrons strike a target,

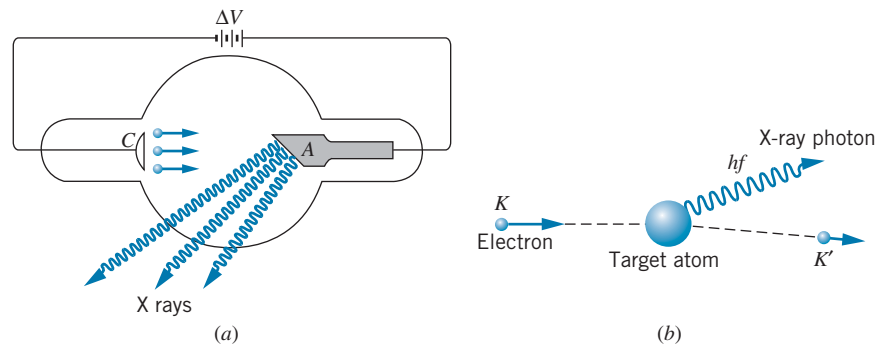


FIGURE 3.23 (a) Apparatus for producing bremsstrahlung. Electrons from a cathode C are accelerated to the anode A through the potential difference ΔV . When an electron encounters a target atom of the anode, it can lose energy, with the accompanying emission of an X-ray photon. (b) A schematic representation of the bremsstrahlung process.

they are slowed down and eventually come to rest, because they make collisions with the atoms of the target material. In such a collision, momentum is transferred to the atom, the electron slows down, and photons are emitted. The recoil kinetic energy of the atom is small (because the atom is so massive) and can safely be neglected. If the electron has a kinetic energy K before the encounter and if it leaves after the collision with a smaller kinetic energy K' , then the photon energy $hf = hc/\lambda$ is

$$hf = \frac{hc}{\lambda} = K - K' \quad (3.52)$$

The amount of energy lost, and therefore the energy and wavelength of the emitted photon, are not uniquely determined, because K is the only known energy in Eq. 3.52. An electron usually will make many collisions, and therefore emit many different photons, before it is brought to rest; the photons then will range all the way from very small energies (large wavelengths) corresponding to small energy losses, up to a maximum photon energy $hf_{\max}$ equal to K , corresponding to an electron that loses all of its kinetic energy K in a single encounter (that is, when $K' = 0$). The smallest emitted wavelength $\lambda_{\min}$ is therefore determined by the maximum possible energy loss

$$\lambda_{\min} = \frac{hc}{K} = \frac{hc}{e \Delta V} \quad (3.53)$$

For typical accelerating voltages in the range of 10,000 V, $\lambda_{\min}$ is in the range of a few tenths of nanometers, which corresponds to the X-ray region of the spectrum. This *continuous* distribution of X rays (which is very different from the *discrete* X-ray energies that are emitted in atomic transitions; more about these in Chapter 8) is called *bremsstrahlung*, which is German for braking, or decelerating, radiation. Some sample bremsstrahlung spectra are illustrated in Figure 3.24.

Symbolically, we can write the bremsstrahlung process as

$$\text{electron} \rightarrow \text{electron} + \text{photon}$$

This is just the reverse process of the photoelectric effect, which is

$$\text{electron} + \text{photon} \rightarrow \text{electron}$$

However, neither process occurs for free electrons. In both cases, there must be a heavy atom in the neighborhood to take care of the recoil momentum.

Pair Production and Annihilation

Another process that can occur when photons encounter atoms is *pair production*, in which the photon loses all its energy and in the process two particles are created: an electron and a positron. (A positron is a particle that is identical in mass to the electron but has a positive electric charge; more about *antiparticles* in Chapter 14.) Here we have an example of the creation of rest energy. The electron did not exist before the encounter of the photon with the atom (it was *not* an electron that was part of the atom). The photon energy hf is converted into the relativistic total energies E_+ and E_- of the positron and electron:

$$hf = E_+ + E_- = (m_e c^2 + K_+) + (m_e c^2 + K_-) \quad (3.54)$$

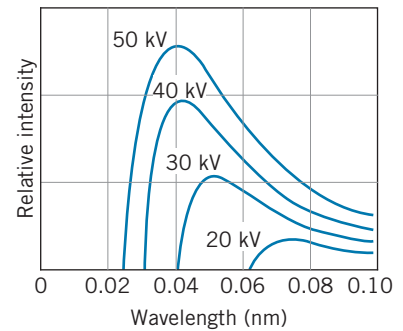


FIGURE 3.24 Some typical bremsstrahlung spectra. Each spectrum is labeled with the value of the accelerating voltage ΔV .

Because K_+ and K_- are always positive, the photon must have an energy of at least $2m_e c^2 = 1.02 \text{ MeV}$ in order for this process to occur; such high-energy photons are in the region of *nuclear gamma rays*. Symbolically,

$$\text{photon} \rightarrow \text{electron} + \text{positron}$$

This process, like bremsstrahlung, will not occur unless there is an atom nearby to supply the necessary recoil momentum. The reverse process

$$\text{electron} + \text{positron} \rightarrow \text{photon}$$

also occurs; this process is known as *electron-positron annihilation* and can occur for free electrons and positrons as long as at least two photons are created. In this process the electron and positron disappear and are replaced by two photons. Conservation of energy requires that

$$(m_e c^2 + K_+) + (m_e c^2 + K_-) = E_1 + E_2 \quad (3.55)$$

where E_1 and E_2 are the photon energies. Usually the kinetic energies K_+ and K_- are negligibly small, so we can assume the positron and electron to be essentially at rest. Momentum conservation then requires the two photons to have equal and opposite momenta and thus equal energies. The two annihilation photons have equal energies of 0.511 MeV ($= m_e c^2$) and move in exactly opposite directions.

3.6 PARTICLES OR WAVES

In some experiments (interference and diffraction, for example), light shows a familiar wave behavior, and it is not possible to understand those experiments in any other way. In other experiments discussed in this chapter (photoelectric effect, thermal radiation, Compton scattering), light behaves as a particle, and the wave picture is not able to explain those experiments.

Unfortunately, “wave” and “particle” are very different kinds of behavior and have no elements in common. A wave distributes its energy over broad wave fronts, while the energy of a particle is concentrated at a single location. The double-slit experiment can be explained only if the wave front goes through both slits, which a particle obviously cannot do. The photoelectric effect, on the other hand, could not occur if the energy of light were distributed over its wave front—the energy of the light must be concentrated in a small bundle to enable an electron to be released.

Light from the same source can be used to demonstrate both double-slit interference and the photoelectric effect. Is the source emitting both particles and waves? Does nature have some sort of secret code by which the kind of experiment we are doing is signaled back to the source so it knows whether to emit light as waves or particles? Yet we can use light from distant galaxies to do both experiments—surely it is not possible to signal to the opposite side of the universe to switch from waves to particles in the time it takes us to replace the double slit with the photoelectric apparatus.

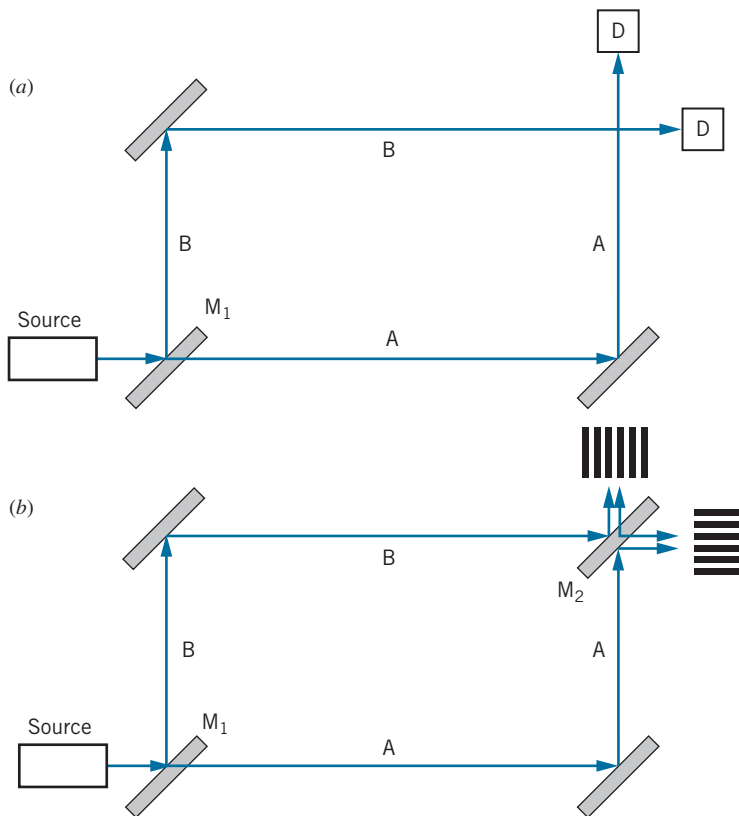


FIGURE 3.25 Apparatus for delayed choice experiment. (a) Light from the source incident on beam-splitting mirror M_1 can travel path A or path B to reach a detector D. (b) With a second beam-splitting mirror M_2 installed, the beams can be recombined to interfere.

Figure 3.25 shows an experiment that tries to distinguish between waves and particles. A light source can be triggered to emit individual photons. Encountering the beam splitter (a half-silvered mirror that reflects half the light and transmits half), the light can travel on either path A or path B, as in Figure 3.25a. Detectors D in line with the paths (perhaps operating through the photoelectric effect) record photons traveling on one or the other path—never are both detectors triggered simultaneously. By inserting another half-silvered mirror into the beam paths, as in Figure 3.25b, the beams can be made to recombine and thus interfere, just as the beams passing through the two slits recombine. So without the final mirror, the light travels *either* path A or path B, never both paths; but with the final mirror inserted the light must travel *both* paths. Another way to say this is that if we know which path the light followed, it is behaving as a particle; but if we lack that knowledge, the possibility is open for the light to exist in both paths like a wave, which results in interference.

Let's emit a bit of light into the apparatus, and after the light has encountered the first half-silvered mirror we use a very fast electronic switch to randomly decide whether or not to insert the second half-silvered mirror. When the decision is made, the light is already traveling in the apparatus, presumably along one path or the other. If the second mirror is inserted, the result suggests that the light has traveled both paths. This type of experiment is called a "delayed choice" experiment, since the choice of whether to examine particle-type or wave-type behavior is made after the light is in the apparatus. It is also known as a "quantum eraser" experiment, because inserting the second half-silvered mirror essentially erases the already fixed information about which path the light is following.

The inescapable conclusion of this kind of experiment is that we cannot separate the wave and particle nature of light. They are both present simultaneously, so that a wave-type experiment reveals one aspect of the nature of light, while a particle-type experiment reveals the other. The terms "wave" and "particle" describe not the composition of light (we don't think of light as containing waves or particles) but rather its interactions with matter. So light might be emitted by electron transitions in atoms as individual photons, then might travel as a wave in encountering a double-slit apparatus, and then might be detected again as photons by triggering individual pixels on a screen.

This situation is often known as *particle-wave duality*. Both are needed for a complete understanding of light, and the two are *complementary* to each other. Yet it took some time for this philosophical view to be accepted by physicists. Einstein proposed his photoelectric effect analysis in 1905 based on Planck's ideas about quantization, but Planck was one of many who tried to hold on to 19th-century ideas about electromagnetic waves. Millikan, whose photoelectric effect experiments verified Einstein's theory, doubted that the theory was fundamentally correct. Einstein's later ideas about the momentum carried by photons were likewise doubted, and even Niels Bohr, one of the founders of the quantum theory, tried (unsuccessfully) to find a wave-based theory to explain Compton scattering.

When photon events are recorded on a screen after the double-slit experiment, what guides the photons to their proper locations on the screen? What steers them toward the interference maxima and away from the interference minima? In the wave interpretation of double-slit interference, the maxima occur where the electric fields of the light passing through the two slits add together, while the minima occur where the electric fields cancel. So there must be some connection between the electric field and the photon build-up. As shown by Eq. 3.8, the light intensity depends on the square of the electric field, so we conclude that

$$\text{probability to observe photons} \propto |\text{electric field amplitude}|^2$$

where the electric field is large, the wave has large intensity and many photons are found. Where the wave has small intensity, few photons are observed. It is this expression that provides the ultimate connection between the wave and particle behaviors. We will see in the next two chapters that a similar relationship connects the wave and particle behaviors of other objects, such as electrons, that were previously believed to behave only as classical particles.

Chapter Summary

		Section			Section
Double-slit maxima	$y_n = n \frac{\lambda D}{d} \quad n = 0, 1, 2, 3, \dots$	3.1	Rayleigh–Jeans formula	$I(\lambda) = \frac{2\pi c}{\lambda^4} kT$	3.3
Bragg’s law for X-ray diffraction	$2d \sin \theta = n\lambda \quad n = 1, 2, 3, \dots$	3.1	Planck’s black-body distribution	$I(\lambda) = \frac{2\pi hc^2}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$	3.3
Energy of photon	$E = hf = hc/\lambda$	3.2	Compton scattering	$\frac{1}{E'} - \frac{1}{E} = \frac{1}{m_e c^2} (1 - \cos \theta),$ $\lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta)$	3.4
Maximum kinetic energy of photoelectrons	$K_{\max} = eV_s = hf - \phi$	3.2	Bremsstrahlung	$\lambda_{\min} = hc/K = hc/e\Delta V$	3.5
Cutoff wavelength	$\lambda_c = hc/\phi$	3.2	Pair production	$hf = E_+ + E_- = (m_e c^2 + K_+) + (m_e c^2 + K_-)$	3.5
Stefan’s law	$I = \sigma T^4$	3.3	Electron–positron annihilation	$(m_e c^2 + K_+) + (m_e c^2 + K_-) = E_1 + E_2$	3.5
Wien’s displacement law	$\lambda_{\max} T = 2.8978 \times 10^{-3} \text{ m} \cdot \text{K}$	3.3			

Questions

- The diameter of an atomic nucleus is about $10 \times 10^{-15} \text{ m}$. Suppose you wanted to study the diffraction of photons by nuclei. What energy of photons would you choose? Why?
- How is the wave nature of light unable to account for the observed properties of the photoelectric effect?
- In the photoelectric effect, why do some electrons have kinetic energies smaller than $K_{\max}$?
- Why doesn’t the photoelectric effect work for free electrons?
- What does the work function tell us about the properties of a metal? Of the metals listed in Table 3.1, which has the least tightly bound electrons? Which has the most tightly bound?
- Electric current is charge flowing per unit time. If we increase the kinetic energy of the photoelectrons (by increasing the energy of the incident photons), shouldn’t the current increase, because the charge flows more rapidly? Why doesn’t it?
- What might be the effects on a photoelectric effect experiment if we were to double the frequency of the incident light? If we were to double the wavelength? If we were to double the intensity?
- In the photoelectric effect, how can a photon moving in one direction eject an electron moving in a different direction? What happens to conservation of momentum?
- In Figure 3.10, why does the photoelectric current rise slowly to its saturation value instead of rapidly, when the potential difference is greater than V_s ? What does this figure indicate about the experimental difficulties that might arise from trying to determine V_s in this way?
- Suppose that the frequency of a certain light source is just above the cutoff frequency of the emitter, so that the photoelectric effect occurs. To an observer in relative motion, the frequency might be Doppler shifted to a lower value that is below the cutoff frequency. Would this moving observer conclude that the photoelectric effect does not occur? Explain.
- Why do cavities that form in a wood fire seem to glow brighter than the burning wood itself? Is the temperature in such cavities hotter than the surface temperature of the exposed burning wood?
- What are the fields of classical physics on which the classical theory of blackbody radiation is based? Why don’t we believe that the “ultraviolet catastrophe” suggests that something is wrong with one of those classical theories?
- In what region of the electromagnetic spectrum do room-temperature objects radiate? What problems would we have if our eyes were sensitive in that region?
- How does the total intensity of thermal radiation vary when the temperature of an object is doubled?
- Compton-scattered photons of wavelength λ' are observed at 90° . In terms of λ' , what is the scattered wavelength observed at 180° ?
- The Compton-scattering formula suggests that objects viewed from different angles should show scattered light of different wavelengths. Why don’t we observe a change in color of objects as we change the viewing angle?

17. You have a monoenergetic source of X rays of energy 84 keV, but for an experiment you need 70 keV X rays. How would you convert the X-ray energy from 84 to 70 keV?
18. TV sets with picture tubes can be significant emitters of X rays. What is the origin of these X rays? Estimate their wavelengths.
19. The X-ray peaks of Figure 3.20 are not sharp but are spread over a range of wavelengths. What reasons might account for that spreading?
20. A beam of photons passes through a block of matter. What are the three ways discussed in this chapter that the photons can lose energy in interacting with the material?
21. Of the photon processes discussed in this chapter (photoelectric effect, thermal radiation, Compton scattering, bremsstrahlung, pair production, electron–positron annihilation), which conserve momentum? Energy? Mass? Number of photons? Number of electrons? Number of electrons minus number of positrons?

Problems

3.1 Review of Electromagnetic Waves

1. A double-slit experiment is performed with sodium light ($\lambda = 589.0$ nm). The slits are separated by 1.25 mm, and the screen is 2.604 m from the slits. Find the separation between adjacent maxima on the screen.
2. In Example 3.1, what angle of incidence will produce the second-order Bragg peak?
3. Monochromatic X rays are incident on a crystal in the geometry of Figure 3.5. The first-order Bragg peak is observed when the angle of incidence is 38.0° . The crystal spacing is known to be 0.327 nm. (a) What is the wavelength of the X rays? (b) Now consider a set of crystal planes that makes an angle of 45° with the surface of the crystal (as shown in Figure 3.6). For X rays of the same wavelength, find the angle of incidence measured from the surface of the crystal that produces the first-order Bragg peak. At what angle from the surface does the emerging beam appear in this case?
4. A certain device for analyzing electromagnetic radiation is based on the Bragg scattering of the radiation from a crystal. For radiation of wavelength 0.149 nm, the first-order Bragg peak appears centered at an angle of 15.15° . The aperture of the analyzer passes radiation in the angular range of 0.015° . What is the corresponding range of wavelengths passing through the analyzer?

3.2 The Photoelectric Effect

5. Find the momentum of (a) a 10.0-MeV gamma ray; (b) a 25-keV X ray; (c) a $1.0\text{-}\mu\text{m}$ infrared photon; (d) a 150-MHz radio-wave photon. Express the momentum in $\text{kg}\cdot\text{m/s}$ and eV/c .
6. What range of photon energies corresponds to radio-wave frequencies of 1 to 100 MHz? Our bodies are continuously bombarded by these photons. Why are they not dangerous to us?
7. (a) What is the wavelength of an X-ray photon of energy 10.0 keV? (b) What is the wavelength of a gamma-ray photon of energy 1.00 MeV? (c) What is the range of

energies of photons of visible light with wavelengths 350–700 nm?

8. What is the cutoff wavelength for the photoelectric effect using an aluminum surface?
9. Light of wavelength 304.2 nm illuminates a metal surface whose cutoff wavelength is 352.8 nm. What is the stopping potential?
10. When light of wavelength λ illuminates a copper surface, the stopping potential is V . In terms of V , what will be the stopping potential if the same wavelength is used to illuminate a sodium surface?
11. The cutoff wavelength for the photoelectric effect in a certain metal is 254 nm. (a) What is the work function for that metal? (b) Will the photoelectric effect be observed for $\lambda > 254$ nm or for $\lambda < 254$ nm?
12. A surface of zinc is illuminated and photoelectrons are observed. (a) What is the largest wavelength that will cause photoelectrons to be emitted? (b) What is the stopping potential when light of wavelength 252.0 nm is used?

3.3 Blackbody Radiation

13. (a) Show that in the classical result for the energy distribution of the cavity wall oscillators (Eq. 3.30), the total number of oscillators at all energies is N . (b) Show that $E_{\text{avg}} = kT$ for the classical oscillators.
14. (a) Writing the discrete Maxwell–Boltzmann distribution for Planck’s cavity wall oscillators as $N_n = Ae^{-E_n/kT}$ (where A is a constant to be determined), show that the condition $\sum_{n=0}^{\infty} N_n = N$ gives $A = N(1 - e^{-\epsilon/kT})$ as in Eq. 3.36. [Hint: Use $\sum_{n=0}^{\infty} e^{nx} = 1/(1 - e^x)$.] (b) By taking the derivative with respect to x of the equation given in the hint, show that $\sum_{n=0}^{\infty} ne^{nx} = e^x/(1 - e^x)^2$. (c) Use this result to derive Eq. 3.38 from Eq. 3.37. (d) Show that $E_{\text{av}} \cong kT$ at large λ and $E_{\text{av}} \rightarrow 0$ for small λ .

15. By differentiating Eq. 3.39 show that $I(\lambda)$ has its maximum as expected according to Wien's displacement law, Eq. 3.25.
 16. Integrate Eq. 3.39 to obtain Eq. 3.24. Use the definite integral $\int_0^\infty x^3 dx/(e^x - 1) = \pi^4/15$ to obtain Eq. 3.40 relating the Stefan-Boltzmann constant to Planck's constant.
 17. Use the numerical value of the Stefan-Boltzmann constant to find the numerical value of Planck's constant from Eq. 3.40.
 18. The surface of the Sun has a temperature of about 6000 K. At what wavelength does the Sun emit its peak intensity? How does this compare with the peak sensitivity of the human eye?
 19. The universe is filled with thermal radiation, which has a blackbody spectrum at an effective temperature of 2.7 K (see Chapter 15). What is the peak wavelength of this radiation? What is the energy (in eV) of quanta at the peak wavelength? In what region of the electromagnetic spectrum is this peak wavelength?
 20. (a) Assuming the human body (skin temperature 34°C) to behave like an ideal thermal radiator, find the wavelength where the intensity from the body is a maximum. In what region of the electromagnetic spectrum is radiation with this wavelength? (b) Making whatever (reasonable) assumptions you may need, estimate the power radiated by a typical person isolated from the surroundings. (c) Estimate the radiation power *absorbed* by a person in a room in which the temperature is 20°C .
 21. A cavity is maintained at a temperature of 1750 K. At what rate does energy escape from the interior of the cavity through a hole in its wall of diameter 1.24 mm?
 22. An analyzer for thermal radiation is set to accept wavelengths in an interval of 1.74 nm. What is the intensity of the radiation in that interval at a wavelength of 932 nm emitted from a glowing object whose temperature is 1546 K?
 23. (a) Assuming the Sun to radiate like an ideal thermal source at a temperature of 6000 K, what is the intensity of the solar radiation emitted in the range 530.0 nm to 532.0 nm? (b) What fraction of the total solar radiation does this represent?
- observed at 90.0° . (a) What is the wavelength of the scattered photons? (b) What is the momentum of the incident photons and the scattered photons? (c) What is the kinetic energy of the scattered electrons? (d) What is the momentum (magnitude and direction) of the scattered electrons?
27. High-energy gamma rays can reach a radiation detector by Compton scattering from the surroundings, as shown in Figure 3.26. This effect is known as back-scattering. Show that, when $E \gg m_e c^2$, the back-scattered photon has an energy of approximately 0.25 MeV, independent of the energy of the original photon, when the scattering angle is nearly 180° .

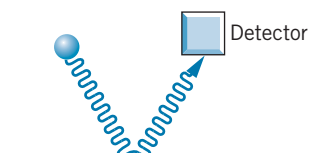


FIGURE 3.26 Problem 27.

3.4 The Compton Effect

24. Show how Eq. 3.46 follows from Eq. 3.45.
 25. Incident photons of energy 11.32 keV are Compton scattered, and the scattered beam is observed at 62.9° relative to the incident beam. (a) What is the energy of the scattered photons at that angle? (b) How much kinetic energy is given to the scattered electron?
 26. X-ray photons of wavelength 0.02218 nm are incident on a target and the Compton-scattered photons are
28. Gamma rays of energy 0.662 MeV are Compton scattered. (a) What is the energy of the scattered photon observed at a scattering angle of 52.2° ? (b) What is the kinetic energy of the scattered electrons?
 29. In Compton's original work he used a slightly different derivation to obtain Eq. 3.47. (a) Conservation of momentum in Figure 3.18 means that the 3 vectors representing the momenta p , p' , and p_e must form a closed triangle such that $\vec{p} = \vec{p}' + \vec{p}_e$. Draw this triangle and apply the law of cosines to the angle θ between $\vec{p}$ and $\vec{p}'$. Express the photon momenta in terms of wavelength and the electron momentum in terms of its speed. (b) Write a second equation for conservation of energy, again expressing the photon energy in terms of wavelength and the electron energy in terms of its speed. (c) Eliminate the electron speed between the 2 equations to obtain Eq. 3.47.
 30. A photon enters a detector and undergoes Compton scattering. The scattered electron is captured within the detector and its kinetic energy is measured. The scattered photon then travels to a second detector where it is captured and its energy measured. In one particular experiment, the electron energy was determined to be 2.302 MeV and the scattered photon energy 0.239 MeV. Determine the energy of the original photon and its direction relative to the scattered photon. This is the process used by the Compton Gamma-Ray Observatory (Figure 3.27) to determine the location in the sky from which energetic gamma rays reach the Earth.

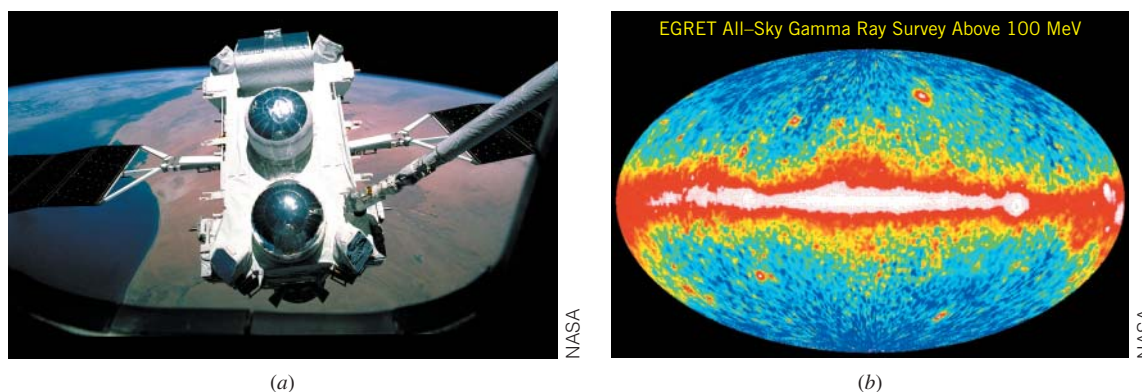


FIGURE 3.27 (a) The NASA Compton Gamma-Ray Observatory at its launch from the Space Shuttle in 1991. It consisted of 4 instruments that used Compton scattering to map gamma rays from the galaxy. (b) An image of gamma rays of energy greater than 100 MeV, showing intense production along the galactic plane and also from the Crab Nebula (at right edge of image).

3.5 Other Photon Processes

31. Suppose an atom of gold at rest emits an X-ray photon of energy 69 keV. Calculate the “recoil” momentum and kinetic energy of the atom. (*Hint:* Do you expect to need classical or relativistic kinetic energy for the atom? Is the kinetic energy likely to be much smaller than the atom’s rest energy?)
32. What is the minimum X-ray wavelength produced in bremsstrahlung by electrons that have been accelerated through 7.5×10^4 V?
33. An atom absorbs a photon of wavelength 425 nm and immediately emits another photon of wavelength 643 nm. What is the net energy absorbed by the atom in this process?
37. A certain cavity has a temperature of 1325 K. (a) At what wavelength will the intensity of the radiation in the cavity have its maximum value? (b) As a fraction of the maximum intensity, what is the intensity at twice the wavelength found in part (a)?
38. In Compton scattering, calculate the maximum kinetic energy given to the scattered electron for a given photon energy.
39. The COBE satellite was launched in 1989 to study the cosmic background radiation and measure its temperature. By measuring at many different wavelengths, researchers were able to show that the background radiation exactly followed the spectral distribution expected for a blackbody. At a wavelength of 0.133 cm, the radiant intensity is 1.440×10^{-7} W/m² in a wavelength interval of 0.00833 cm. What is the temperature of the radiation that would be deduced from these data?

General Problems

34. A certain green light bulb emits at a single wavelength of 550 nm. It consumes 55 W of electrical power and is 75% efficient in converting electrical energy into light. (a) How many photons does the bulb emit in one hour? (b) Assuming the emitted photons to be distributed uniformly in space, how many photons per second strike a 10 cm by 10 cm paper held facing the bulb at a distance of 1.0 m?
35. When sodium metal is illuminated with light of wavelength 4.20×10^2 nm, the stopping potential is found to be 0.65 V; when the wavelength is changed to 3.10×10^2 nm, the stopping potential is 1.69 V. Using *only these data* and the values of the speed of light and the electronic charge, find the work function of sodium and a value of Planck’s constant.
36. A photon of wavelength 157 nm strikes an aluminum surface along a line perpendicular to the surface and releases a photoelectron traveling in the opposite direction. Assume the recoil momentum is taken up by a single aluminum atom on the surface. Calculate the recoil kinetic energy of the atom. Would this recoil energy significantly affect the kinetic energy of the photoelectron?
40. The WMAP satellite launched in 2001 studied the cosmic microwave background radiation and was able to chart small fluctuations in the temperature of different regions of the background radiation. These fluctuations in temperature correspond to regions of large and small density in the early universe. The satellite was able to measure differences in temperature of 2×10^{-5} K at a temperature of 2.7250 K. At the peak wavelength, what is the difference in the radiation intensity per unit wavelength interval between the “hot” and “cold” regions of the background radiation?
41. You have been hired as an engineer on a NASA project to design a microwave spectrometer for an orbital mission to measure the cosmic background radiation, which has a blackbody spectrum with an effective temperature of 2.725 K. (a) The spectrometer is to scan the sky

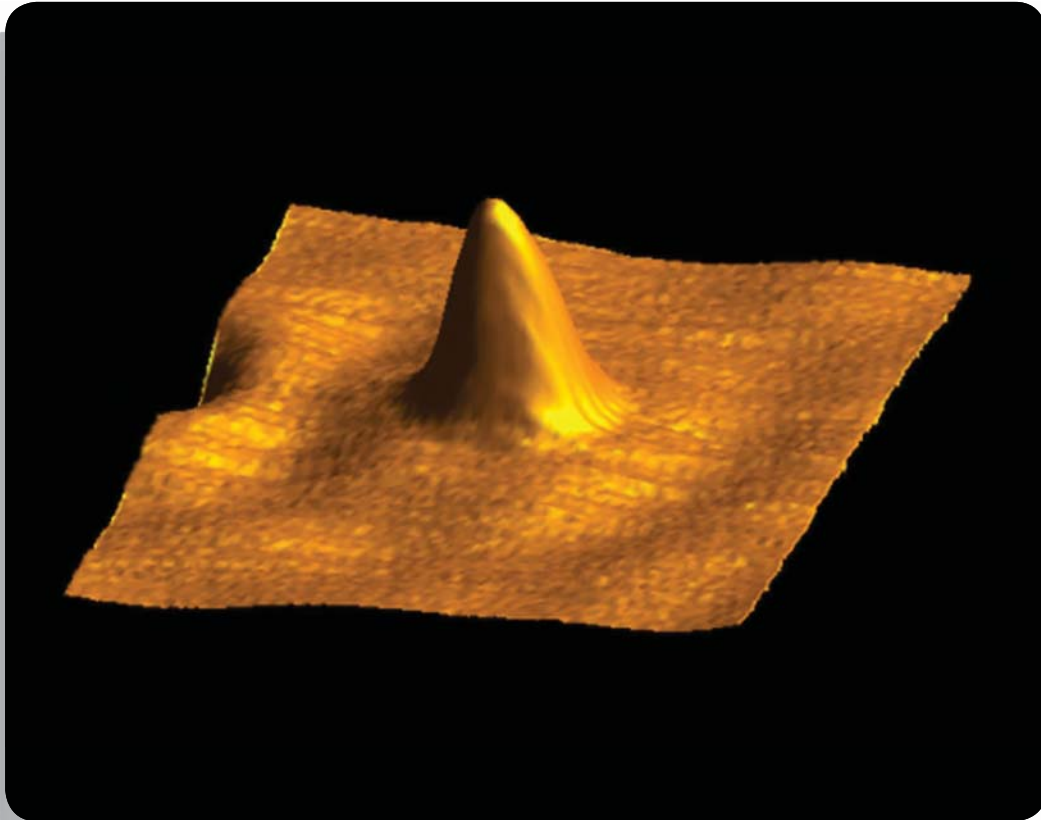
between wavelengths of 0.50 mm and 5.0 mm, and at each wavelength it accepts radiation in a wavelength range of 3.0×10^{-4} mm. What maximum and minimum radiation intensity do you expect to find in this region? (b) The photon detector in the spectrometer is in the form of a disk of diameter 0.86 cm. How many photons per second will the spectrometer record at its maximum and minimum intensities?

42. A photon of wavelength 6.13 pm scatters from a free electron at rest. After the interaction, the electron is observed to be moving in the direction of the original photon. Find the momentum of the electron.
43. A hydrogen atom is moving at a speed of 125.0 m/s. It absorbs a photon of wavelength 97 nm that is moving in the opposite direction. By how much does the speed of the atom change as a result of absorbing the photon?
44. Before a positron and an electron annihilate, they form a sort of “atom” in which each orbits about their common center of mass with identical speeds. As a result of this motion, the photons emitted in the annihilation show a small Doppler shift. In one experiment, the Doppler shift in energy of the photons was observed to be 2.41 keV. (a) What would be the speed of the electron or positron before the annihilation to produce this Doppler shift? (b) The positrons form these atom-like structures with the nearly “free” electrons in a solid. Assuming the positron and electron must have about the same speed to form this structure, find the kinetic energy of the electron. This technique, called “Doppler broadening,” is an important method for learning about the energies of electrons in materials.
45. Prove that it is *not* possible to conserve both momentum and total relativistic energy in the following situation: A free electron moving at velocity $\vec{v}$ emits a photon and then moves at a slower velocity $\vec{v}'$.
46. A photon of energy E interacts with an electron at rest and undergoes pair production, producing a positive electron (positron) and an electron (in addition to the original electron):

$$\text{photon} + e^- \rightarrow e^+ + e^- + e^-$$

The two electrons and the positron move off with identical momenta in the direction of the initial photon. Find the kinetic energy of the three final particles and find the energy E of the photon. (*Hint:* Conserve momentum and total relativistic energy.)
47. You have been hired by NASA to analyze a solar sail that uses the momentum of sunlight for interplanetary travel. A prototype sail of area 1 km² has been developed and is made from a thin lightweight polymer that has a highly reflective aluminum coating on one side. This material has thickness 2 μm and density 0.29 g/cm³. You are limited by the design which requires that the supporting frame and the cargo weigh no more than the film itself. Make any necessary assumptions about the parameters of the calculation, and estimate the travel time of this spacecraft from Earth to Mars.
48. An electron is moving in the $+x$ direction with a speed of $0.46c$. It is struck from behind by a photon of energy 0.172 MeV moving in the same direction. Find the energies of the scattered electron and photon. (*Hint:* Would you expect the scattered photon to be moving in the $+x$ or $-x$ direction?)
49. A certain gamma-ray detector measures photon energies through the Compton interaction: the photon Compton scatters within the detector material, which then absorbs the kinetic energy of the scattered electron. The absorbed energy of the scattered electron is the response of the detector. Suppose photons of energy E are incident on this detector. (a) Find an expression for the maximum energy response E_{max} of this detector and show that E_{max} is less than the original energy of the photon. (b) Evaluate E_{max} for an incident photon energy of 1.5 MeV. (c) Occasionally the detector may report events with energy greater than E_{max} but less than E . What other processes might be responsible for such events? (d) What processes might contribute to the detector reporting the full energy E of the photon?
50. An electron is moving in the negative x direction with a speed of $0.95c$ when it encounters a photon of energy 12.4 keV moving in the positive x direction. Find the energies of the photon and electron after the scattering. This process, in which photons *gain* energy by scattering from very energetic electrons, is called *inverse Compton scattering* and is thought to be the process that produces very high energy gamma rays (1 GeV and above) that reach Earth from space. The Compton Gamma-Ray Observatory was launched by NASA in 1991 and operated until 2000 scanning the sky for gamma rays (Figure 3.27).

THE WAVELIKE PROPERTIES OF PARTICLES



Just as we produce images from light waves that scatter from objects, we can also form images from “particle waves.” The electron microscope produces images from electron waves that enable us to visualize objects on a scale that is much smaller than the wavelength of light. The ability to observe individual human cells and even subcellular objects such as chromosomes has revolutionized our understanding of biological processes. It is even possible to form images of a single atom, such as this cobalt atom on a gold surface. The ripples on the surface are caused by electrons from gold atoms reacting to the presence of the intruder. Drs. Ali Yazdani & Daniel J. Hornbaker / Science Source

In classical physics, the laws describing the behavior of waves and particles are fundamentally different. Projectiles obey particle-type laws, such as Newtonian mechanics. Waves undergo interference and diffraction, which cannot be explained by the Newtonian mechanics associated with particles. The energy carried by a particle is confined to a small region of space; a wave, on the other hand, distributes its energy throughout space in its wavefronts. In describing the behavior of a particle, we often want to specify its location, but this is not so easy to do for a wave. How would you describe the exact location of a sound wave or a water wave?

In contrast to this clear distinction found in classical physics, quantum physics requires that particles sometimes obey the rules that we have previously established for waves, and we shall use some of the language associated with waves to describe particles. The system of mechanics associated with quantum systems is sometimes called “wave mechanics” because it deals with the wavelike behavior of particles. In this chapter, we discuss the experimental evidence in support of this wavelike behavior for particles such as electrons.

As you study this chapter, notice the frequent references to such terms as the *probability* of the outcome of a measurement, the *average* of many repetitions of a measurement, and the *statistical* behavior of a system. These terms are fundamental to quantum mechanics, and you cannot begin to understand quantum behavior until you feel comfortable with discarding such classical notions as fixed trajectories and certainty of outcome, while substituting the quantum mechanical notions of probability and statistically distributed outcomes.

4.1 DE BROGLIE'S HYPOTHESIS

Progress in physics often can be characterized by long periods of experimental and theoretical drudgery punctuated occasionally by flashes of insight that cause profound changes in the way we view the universe. Frequently, the more profound the insight and the bolder the initial step, the simpler it seems in historical perspective, and the more likely we are to sit back and wonder, “Why didn’t I think of that?” Einstein’s special theory of relativity is one example of such insight; the hypothesis of the Frenchman Louis de Broglie is another.*

In the previous chapter, we discussed the double-slit experiment (which can be understood only if light behaves as a wave) and the photoelectric and Compton effects (which can be understood only if light behaves as a particle). Is this dual particle-wave nature a property only of light or of material objects as well? In a bold and daring hypothesis in his 1924 doctoral dissertation, de Broglie chose the latter alternative. Examining Eq. 3.18, $E = hf$, and Eq. 3.20, $p = h/\lambda$, we find some difficulty in applying the first equation in the case of particles, for we cannot be sure whether E should be the kinetic energy, total energy, or total relativistic energy (all, of course, are identical for light). No such difficulties arise from the second relationship. De Broglie suggested, lacking any experimental evidence in support of his hypothesis, that



Meggers Gallery / AIP / Science Source

Louis de Broglie (1892–1987, France). A member of an aristocratic family, his work contributed substantially to the early development of the quantum theory.

*De Broglie’s name should be pronounced “deh-BROY” or “deh-BROY-eh,” but it is often said as “deh-BROH-lee.”

associated with any material particle moving with momentum p there is a wave of wavelength λ , related to p according to

$$\lambda = \frac{h}{p} \quad (4.1)$$

where h is Planck's constant. The wavelength λ of a particle computed according to Eq. 4.1 is called its *de Broglie wavelength*.

Example 4.1

Compute the de Broglie wavelength of the following: Then,

- (a) A 1000-kg automobile traveling at 100 m/s (about 200 mi/h). (b) A 10-g bullet traveling at 500 m/s. (c) A smoke particle of mass 10^{-9} g moving at 1 cm/s. (d) An electron with a kinetic energy of 1 eV. (e) An electron with a kinetic energy of 100 MeV.

$$\begin{aligned} \lambda &= \frac{h}{p} = \frac{6.6 \times 10^{-34} \text{ J} \cdot \text{s}}{5.4 \times 10^{-25} \text{ kg} \cdot \text{m/s}} \\ &= 1.2 \times 10^{-9} \text{ m} = 1.2 \text{ nm} \end{aligned}$$

Solution

(a) Using the classical relation between velocity and momentum,

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{6.6 \times 10^{-34} \text{ J} \cdot \text{s}}{(10^3 \text{ kg})(100 \text{ m/s})} = 6.6 \times 10^{-39} \text{ m}$$

(b) As in part (a),

$$\lambda = \frac{h}{p} = \frac{6.6 \times 10^{-34} \text{ J} \cdot \text{s}}{(10^{-2} \text{ kg})(500 \text{ m/s})} = 1.3 \times 10^{-34} \text{ m}$$

(c)

$$\lambda = \frac{h}{p} = \frac{6.6 \times 10^{-34} \text{ J} \cdot \text{s}}{(10^{-12} \text{ kg})(10^{-2} \text{ m/s})} = 6.6 \times 10^{-20} \text{ m}$$

(d) The rest energy (mc^2) of an electron is 5.1×10^5 eV. Because the kinetic energy (1 eV) is much less than the rest energy, we can use nonrelativistic kinematics.

$$\begin{aligned} p &= \sqrt{2mK} \\ &= \sqrt{2(9.1 \times 10^{-31} \text{ kg})(1 \text{ eV})(1.6 \times 10^{-19} \text{ J/eV})} \\ &= 5.4 \times 10^{-25} \text{ kg} \cdot \text{m/s} \end{aligned}$$

We can also find this solution in the following way, using $p = \sqrt{2mK}$ and $hc = 1240 \text{ eV} \cdot \text{nm}$.

$$\begin{aligned} cp &= c\sqrt{2mK} = \sqrt{2(mc^2)K} \\ &= \sqrt{2(5.1 \times 10^5 \text{ eV})(1 \text{ eV})} = 1.0 \times 10^3 \text{ eV} \\ \lambda &= \frac{h}{p} = \frac{hc}{pc} = \frac{1240 \text{ eV} \cdot \text{nm}}{1.0 \times 10^3 \text{ eV}} = 1.2 \text{ nm} \end{aligned}$$

This method may seem artificial at first, but with practice it becomes quite useful, especially because energies are usually given in electron-volts in atomic and nuclear physics.

(e) In this case, the kinetic energy is much greater than the rest energy, and so we are in the extreme relativistic realm, where $K \cong E \cong pc$, as in Eq. 2.40. The wavelength is

$$\lambda = \frac{hc}{pc} = \frac{1240 \text{ MeV} \cdot \text{fm}}{100 \text{ MeV}} = 12 \text{ fm}$$

Note that the wavelengths computed in parts (a), (b), and (c) are far too small to be observed in the laboratory. Only in the last two cases, in which the wavelength is of the same order as atomic or nuclear sizes, do we have any chance of observing the wavelength. *Because of the smallness of h , only for particles of atomic or nuclear size will the wave behavior be observable.*

Two questions immediately follow. First, just what sort of wave is it that has this de Broglie wavelength? That is, what does the amplitude of the de Broglie

wave measure? We'll discuss the answer to this question later in this chapter. For now, we assume that, associated with the particle as it moves, there is a de Broglie wave of wavelength λ , which shows itself *when a wave-type experiment (such as diffraction) is performed on it*. The outcome of the wave-type experiment depends on this wavelength. The de Broglie wavelength, which characterizes the wave-type behavior of particles, is central to the quantum theory.

The second question then occurs: Why was this wavelength not directly observed before de Broglie's time? As parts (a), (b), and (c) of Example 4.1 showed, for ordinary objects the de Broglie wavelength is very small. Suppose we tried to demonstrate the wave nature of these objects through a double-slit type of experiment. Recall from Eq. 3.14 that the spacing between adjacent fringes in a double-slit experiment is $\Delta y = \lambda D/d$. Putting in reasonable values for the slit separation d and slit-to-screen distance D , you will find that there is no achievable experimental configuration that can produce an observable separation of the fringes (see Problem 9). *There is no experiment that can be done to reveal the wave nature of macroscopic (laboratory-sized) objects.* Experimental verification of de Broglie's hypothesis comes only from experiments with objects on the atomic scale, which are discussed in the next section.

4.2 EXPERIMENTAL EVIDENCE FOR DE BROGLIE WAVES

The indications of wave behavior come mostly from interference and diffraction experiments. Double-slit interference, which was reviewed in Section 3.1, is perhaps the most familiar type of interference experiment, but the experimental difficulties of constructing double slits to do interference experiments with beams of atomic or subatomic particles were not solved until long after the time of de Broglie's hypothesis. We discuss these experiments later in this section. First, we'll discuss diffraction experiments with electrons.

Particle Diffraction Experiments

Diffraction of light waves is discussed in most introductory physics texts and is illustrated in Figure 4.1 for light diffracted by a single slit. For light of wavelength λ incident on a slit of width a , the diffraction minima are located at angles given by

$$a \sin \theta = n\lambda \quad n = 1, 2, 3, \dots \quad (4.2)$$

on either side of the central maximum. Note that most of the light intensity falls in the central maximum.

The experiments that first verified de Broglie's hypothesis involve *electron diffraction*, not through an artificially constructed single slit (as for the diffraction pattern in Figure 4.1) but instead through the atoms of a crystal. The outcomes of these experiments resemble those of the similar X-ray diffraction experiments illustrated in Section 3.1.

In an electron diffraction experiment, a beam of electrons is accelerated from rest through a potential difference ΔV , acquiring a nonrelativistic

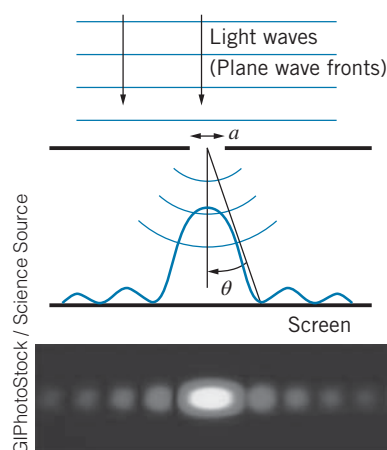


FIGURE 4.1 Light waves (represented as plane wave fronts) are incident on a narrow slit of width a . Diffraction causes the waves to spread after passing through the slit, and the intensity varies along the screen. The photograph shows the resulting intensity pattern.

kinetic energy $K = e \Delta V$ and a momentum $p = \sqrt{2mK}$. Wave mechanics would describe the beam of electrons as a wave of wavelength $\lambda = h/p$. The beam strikes a crystal, and the scattered beam is photographed (Figure 4.2). The similarity between electron diffraction patterns (Figure 4.2) and X-ray diffraction patterns (Figure 3.7) strongly suggests that the electrons are behaving as waves.

The “rings” produced in X-ray diffraction of polycrystalline materials (Figure 3.8b) are also produced in electron diffraction, as shown in Figure 4.3, again providing strong evidence for the similarity in the wave behavior of electrons and X rays. Experiments of the type illustrated in Figure 4.3 were first done in 1927 by G. P. Thomson, who shared the 1937 Nobel Prize for this work. (Thomson’s father, J. J. Thomson, received the 1906 Nobel Prize for his discovery of the electron and measurement of its charge-to-mass ratio. Thus it can be said that Thomson, the father, discovered the particle nature of the electron, while Thomson, the son, discovered its wave nature.)

An electron diffraction experiment gave the first experimental confirmation of the wave nature of electrons (and the quantitative confirmation of the de Broglie relationship $\lambda = h/p$) soon after de Broglie’s original hypothesis. In 1926, at the Bell Telephone Laboratories, Clinton Davisson and Lester Germer were investigating the reflection of electron beams from the surface of nickel crystals. A schematic view of their apparatus is shown in Figure 4.4. A beam of electrons from a heated filament is accelerated through a potential difference ΔV . After passing through a small aperture, the beam strikes a single crystal of nickel. Electrons are scattered in all directions by the atoms of the crystal, some of them striking a detector, which can be moved to any angle ϕ relative to the incident beam and which measures the intensity of the electron beam scattered at that angle.

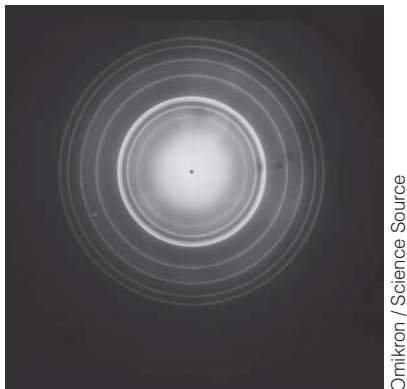


FIGURE 4.3 Electron diffraction of polycrystalline beryllium. Note the similarity between this pattern and the pattern for X-ray diffraction of a polycrystalline material (Figure 3.8b).

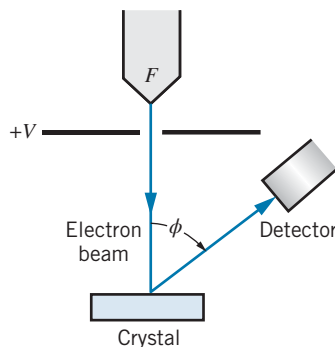
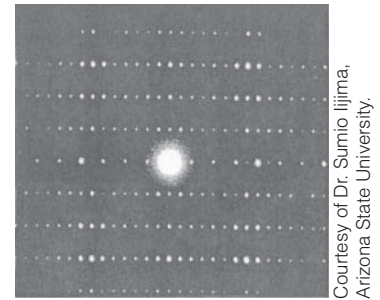
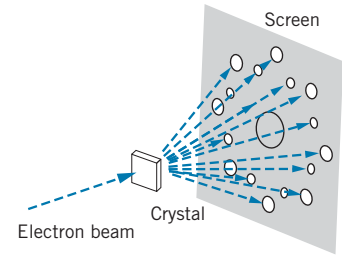


FIGURE 4.4 Apparatus used by Davisson and Germer to study electron diffraction. Electrons leave the filament F and are accelerated by the voltage V . The beam strikes a crystal and the scattered beam is detected at an angle ϕ relative to the incident beam. The detector can be moved in the range of 0 to 90°.



Courtesy of Dr. Sumio Iijima,
Arizona State University.

FIGURE 4.2 (Top) Electron diffraction apparatus. (Bottom) Electron diffraction pattern. Each bright dot is a region of constructive interference, as in the X-ray diffraction patterns of Figure 3.7. The target is a crystal of $\text{Ti}_2\text{Nb}_{10}\text{O}_{29}$.

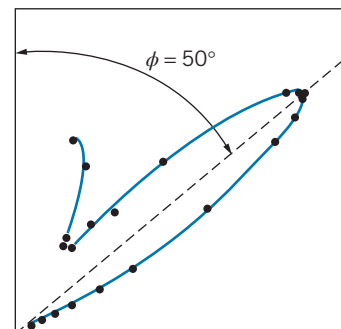


FIGURE 4.5 Results of Davisson and Germer. Each point on the plot represents the relative intensity when the detector in Figure 4.4 is located at the corresponding angle ϕ measured from the vertical axis. Constructive interference causes the intensity of the reflected beam to reach a maximum at $\phi = 50^\circ$ for $V = 54 \text{ V}$.

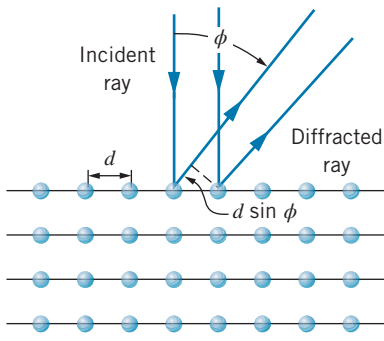


FIGURE 4.6 The crystal surface acts like a diffraction grating with spacing d .

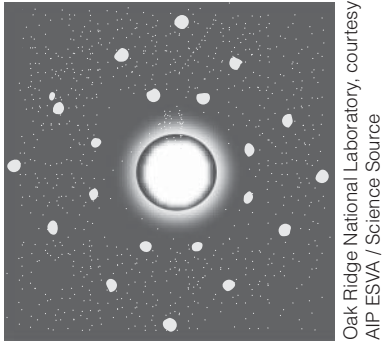


FIGURE 4.7 Diffraction of neutrons by a sodium chloride crystal.

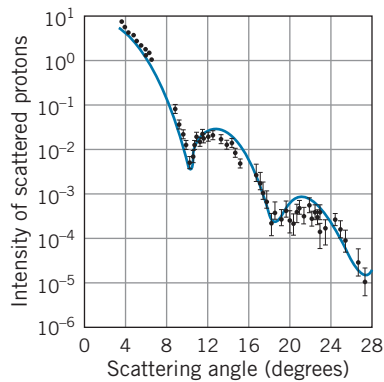


FIGURE 4.8 Diffraction of 1-GeV protons by oxygen nuclei. The pattern of maxima and minima is similar to that of single-slit diffraction of light waves. [Source: H. Palevsky et al., *Physical Review Letters*, **18**, 1200 (1967).]

Figure 4.5 shows the results of one of the experiments of Davisson and Germer. When the accelerating voltage is set at 54 V, there is an intense reflection of the beam at the angle $\phi = 50^\circ$. Let's see how these results give confirmation of the de Broglie wavelength.

Each of the atoms of the crystal can act as a scatterer, so the scattered *electron waves* can interfere, and we have a crystal diffraction grating for the electrons. Figure 4.6 shows a simplified representation of the nickel crystal used in the Davisson–Germer experiment. Because the electrons were of low energy, they did not penetrate very far into the crystal, and it is sufficient to consider the diffraction to take place in the plane of atoms on the surface. The situation is entirely similar to using a reflection-type diffraction grating for light; the spacing d between the rows of atoms on the crystal is analogous to the spacing between the slits in the optical grating. The maxima for a diffraction grating occur at angles ϕ such that the path difference between adjacent rays $d \sin \phi$ is equal to a whole number of wavelengths:

$$d \sin \phi = n\lambda \quad n = 1, 2, 3, \dots \quad (4.3)$$

where n is the order number of the maximum.

From independent data, it is known that the spacing between the rows of atoms in a nickel crystal is $d = 0.215$ nm. The peak at $\phi = 50^\circ$ must be a first-order peak ($n = 1$), because no peaks were observed at smaller angles. If this is indeed an interference maximum, the corresponding wavelength is, from Eq. 4.3,

$$\lambda = d \sin \phi = (0.215 \text{ nm}) (\sin 50^\circ) = 0.165 \text{ nm}$$

We can compare this value with that expected on the basis of the de Broglie theory. An electron accelerated through a potential difference of 54 V has a kinetic energy of 54 eV and therefore a momentum of

$$p = \sqrt{2mK} = \frac{1}{c} \sqrt{2mc^2 K} = \frac{1}{c} \sqrt{2(511,000 \text{ eV})(54 \text{ eV})} = \frac{1}{c}(7430 \text{ eV})$$

The de Broglie wavelength is $\lambda = h/p = hc/pc$. Using $hc = 1240 \text{ eV} \cdot \text{nm}$,

$$\lambda = \frac{hc}{pc} = \frac{1240 \text{ eV} \cdot \text{nm}}{7430 \text{ eV}} = 0.167 \text{ nm}$$

This is in excellent agreement with the value found from the diffraction maximum and provides strong evidence in favor of the de Broglie theory. For this experimental work, Davisson shared the 1937 Nobel Prize with G. P. Thomson.

The wave nature of particles is not exclusive to electrons; *any* particle with momentum p has de Broglie wavelength h/p . Neutrons are produced in nuclear reactors with kinetic energies corresponding to wavelengths of roughly 0.1 nm; these also should be suitable for diffraction by crystals. Figure 4.7 shows that diffraction of neutrons by a salt crystal produces the same characteristic patterns as the diffraction of electrons or X rays. Clifford Shull shared the 1994 Nobel Prize for the development of the neutron diffraction technique.

To study the nuclei of atoms, much smaller wavelengths are needed, of the order of 10^{-15} m. Figure 4.8 shows the diffraction pattern produced by the

scattering of 1-GeV kinetic energy protons by oxygen nuclei. Maxima and minima of the diffracted intensity appear in a pattern similar to the single-slit diffraction shown in Figure 4.1. (The intensity at the minima does not fall to zero because nuclei do not have a sharp boundary. The determination of nuclear sizes from such diffraction patterns is discussed in Chapter 12.)

Example 4.2

A beam of electrons of momentum p passes through a narrow slit of width a . After passing through the slit, the beam strikes a screen, where it produces an image with a bright central band. If the momentum of the electron beam is decreased, does the intensity at the center of the band on the screen *increase, decrease, or remain the same?*

Solution

The width of the central band on the screen (between the two diffraction minima) is determined by the wavelength

of the electron beam and the slit width, according to Eq. 4.2. For smaller momentum, the wavelength would be larger and the angles that limit the width of the central region would be larger. The same amount of beam would be spread over a larger area, so its central intensity (particles per second per unit area) would *decrease*.

Example 4.3

Protons of kinetic energy 1.00 GeV were diffracted by oxygen nuclei, which have a radius of 3.0 fm, to produce the data shown in Figure 4.8. Calculate the expected angles where the first three diffraction minima should appear.

Solution

The total relativistic energy of the protons is $E = K + mc^2 = 1.00 \text{ GeV} + 0.94 \text{ GeV} = 1.94 \text{ GeV}$, so their momentum is

$$p = \frac{1}{c} \sqrt{E^2 - (mc^2)^2} \\ = \frac{1}{c} \sqrt{(1.94 \text{ GeV})^2 - (0.94 \text{ GeV})^2} = 1.70 \text{ GeV}/c$$

The corresponding de Broglie wavelength is

$$\lambda = \frac{h}{p} = \frac{hc}{pc} = \frac{1240 \text{ MeV} \cdot \text{fm}}{1700 \text{ MeV}} = 0.73 \text{ fm}$$

We can represent the oxygen nuclei as circular disks, for which the diffraction formula is a bit different from

Eq. 4.2: $a \sin \theta = 1.22n\lambda$, where a is the diameter of the diffracting object. Based on this formula, the first diffraction minimum ($n = 1$) should appear at the angle

$$\sin \theta = \frac{1.22n\lambda}{a} = \frac{(1.22)(1)(0.73 \text{ fm})}{6.0 \text{ fm}} = 0.148$$

or $\theta = 8.5^\circ$. Because the sine of the diffraction angle is proportional to the index n , the $n = 2$ minimum should appear at the angle where $\sin \theta = 2 \times 0.148 = 0.296$ ($\theta = 17.2^\circ$), and the $n = 3$ minimum where $\sin \theta = 3 \times 0.148 = 0.444$ ($\theta = 26.4^\circ$).

From the data in Figure 4.8, we see the first diffraction minimum at an angle of about 10° , the second at about 18° , and the third at about 27° , all in very good agreement with the expected values. The data don't exactly follow the formula for diffraction by a disk, because nuclei don't behave quite like disks. In particular, they have diffuse rather than sharp edges, which prevents the intensity at the diffraction minima from falling to zero and also alters slightly the locations of the minima.

Double-Slit Experiments with Particles

The definitive evidence for the wave nature of *light* was deduced from the double-slit experiment performed by Thomas Young in 1801 (discussed in

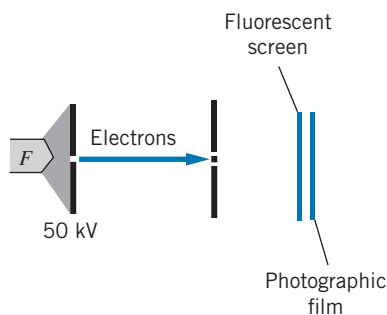


FIGURE 4.9 Double-slit apparatus for electrons. Electrons from the filament F are accelerated through 50 kV and pass through the double slit. They produce a visible pattern when they strike a fluorescent screen (like a TV screen), and the resulting pattern is photographed. A photograph is shown in Figure 4.10. [See C. Jönsson, *American Journal of Physics* **42**, 4 (1974).]

Section 3.1). In principle, it should be possible to do double-slit experiments with *particles* and thereby directly observe their wavelike behavior. However, the technological difficulties of producing double slits for particles are formidable, and such experiments did not become possible until long after the time of de Broglie. The first double-slit experiment with electrons was done in 1961. A diagram of the apparatus is shown in Figure 4.9. The electrons from a hot filament were accelerated through 50 kV (corresponding to $\lambda = 5.4$ pm) and then passed through a double slit of separation $2.0\ \mu\text{m}$ and width $0.5\ \mu\text{m}$. A photograph of the resulting intensity pattern is shown in Figure 4.10. The similarity with the double-slit pattern for light (Figure 3.2) is striking.

A similar experiment can be done for neutrons. A beam of neutrons from a nuclear reactor can be slowed to a room-temperature “thermal” energy distribution (average $K \approx kT \approx 0.025$ eV), and a specific wavelength can be selected by a scattering process similar to Bragg diffraction (see Eq. 3.16 and Problem 35 at the end of the present chapter). In one experiment, neutrons of kinetic energy 0.00024 eV and de Broglie wavelength 1.85 nm passed through a gap of diameter $148\ \mu\text{m}$ in a material that absorbs virtually all of the neutrons incident on it (Figure 4.11). In the center of the gap was a boron wire (also highly absorptive for neutrons) of diameter $104\ \mu\text{m}$. The neutrons could pass on either side of the wire through slits of width $22\ \mu\text{m}$. The intensity of neutrons that pass through this double slit was observed by sliding another slit across the beam and measuring the intensity of neutrons passing through this “scanning slit.” Figure 4.12 shows the resulting pattern of intensity maxima and minima, which leaves no doubt that interference is occurring and that the neutrons have a corresponding wave nature. The wavelength can be deduced from the slit separation using Eq. 3.14 to obtain the spacing between adjacent maxima, $\Delta y = y_{n+1} - y_n$. Estimating the spacing Δy from Figure 4.12 to be about $75\ \mu\text{m}$, we obtain

$$\lambda = \frac{d\Delta y}{D} = \frac{(126\ \mu\text{m})(75\ \mu\text{m})}{5\ \text{m}} = 1.89\ \text{nm}$$

This result agrees very well with the de Broglie wavelength of 1.85 nm selected for the neutron beam.

It is also possible to do a similar experiment with atoms. In this case, a source of helium atoms formed a beam (of velocity corresponding to a kinetic energy of 0.020 eV) that passed through a double slit of separation $8\ \mu\text{m}$ and

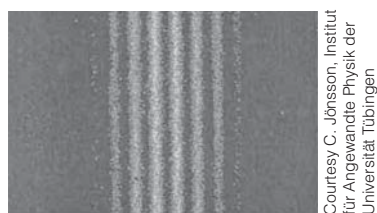


FIGURE 4.10 Double-slit interference pattern for electrons.

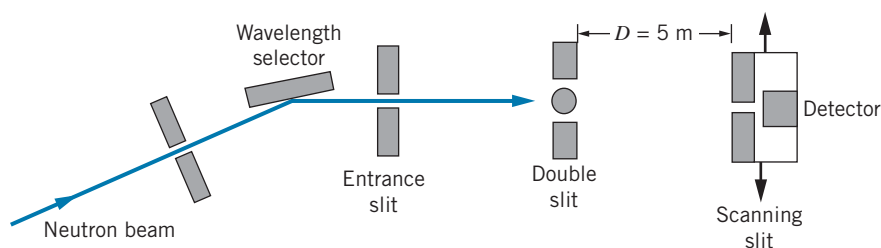


FIGURE 4.11 Double-slit apparatus for neutrons. Thermal neutrons from a reactor are incident on a crystal; scattering through a particular angle selects the energy of the neutrons. After passing through the double slit, the neutrons are counted by the scanning slit assembly, which moves laterally.

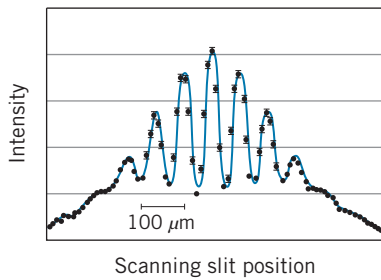


FIGURE 4.12 Intensity pattern observed for double-slit interference with neutrons. The spacing between the maxima is about $75\ \mu\text{m}$. [Source: R. Gahler and A. Zeilinger, *American Journal of Physics* **59**, 316 (1991).]

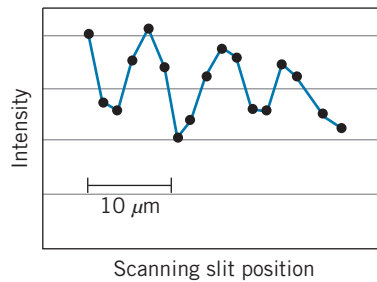


FIGURE 4.13 Intensity pattern observed for double-slit interference with helium atoms. [Source: O. Carnal and J. Mylnek, *Physical Review Letters* **66**, 2689 (1991).]

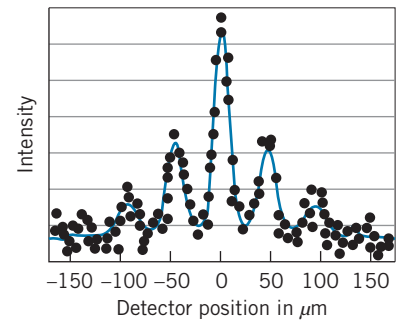


FIGURE 4.14 Diffraction grating pattern produced by C_{60} molecules. [Source: O. Nairz, M. Arndt, and A. Zeilinger, *American Journal of Physics* **71**, 319 (2003).]

width $1\ \mu\text{m}$. Again a scanning slit was used to measure the intensity of the beam passing through the double slit. Figure 4.13 shows the resulting intensity pattern. Although the results are not as dramatic as those for electrons and neutrons, there is clear evidence of interference maxima and minima, and the separation of the maxima gives a wavelength that is consistent with the de Broglie wavelength (see Problem 8).

Diffraction can be observed with even larger objects. Figure 4.14 shows the pattern produced by fullerene molecules (C_{60}) in passing through a diffraction grating with a spacing of $d = 100\ \text{nm}$. The diffraction pattern was observed at a distance of $1.2\ \text{m}$ from the grating. Estimating the separation of the maxima in Figure 4.14 as $50\ \mu\text{m}$, we get the angular separation of the maxima to be $\theta \approx \tan \theta = (50\ \mu\text{m})/(1.2\ \text{m}) = 4.2 \times 10^{-5}\ \text{rad}$, and thus $\lambda = d \sin \theta = 4.2\ \text{pm}$. For C_{60} molecules with a speed of $117\ \text{m/s}$ used in this experiment, the expected de Broglie wavelength is $4.7\ \text{pm}$, in good agreement with our estimate from the diffraction pattern.

In this chapter, we have discussed several interference and diffraction experiments using different particles—electrons, protons, neutrons, atoms, and molecules. These experiments are not restricted to any particular type of particle or to any particular type of observation. They are examples of a *general* phenomenon, the wave nature of particles, that was unobserved before 1920 because the necessary experiments had not yet been done. Today this wave nature is used as a basic tool by scientists. For example, neutron diffraction gives detailed information on the structure of solid crystals and of complex molecules (Figure 4.15). The electron microscope uses electron waves to illuminate and form an image of objects; because the wavelength can be made thousands of times smaller than that of visible light, it is possible to resolve and observe small details that are not observable with visible light (Figure 4.16).

Through Which Slit Does the Particle Pass?

When we do a double-slit experiment with particles such as electrons, it is tempting to try to determine through which slit the particle passes. For

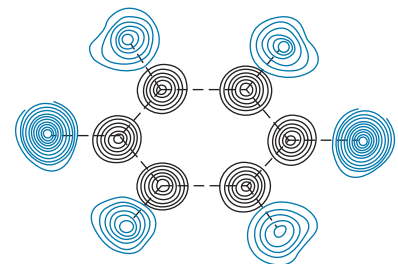
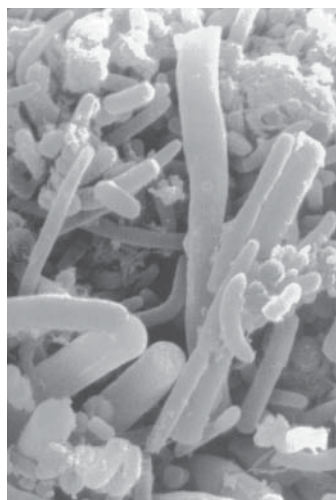


FIGURE 4.15 The atomic structure of solid benzene as deduced from neutron diffraction. The circles indicate contours of constant density. The black circles show the locations of the six carbon atoms that form the familiar benzene ring. The blue circles show the locations of the hydrogen atoms.



SCIMAT / Science Source

FIGURE 4.16 Electron microscope image of bacteria on the surface of a human tongue. The magnification here is about a factor of 5000.

example, we could surround each slit with an electromagnetic loop that causes a meter to deflect whenever a charged particle or perhaps a particle with a magnetic moment passes through the loop (Figure 4.17). If we fired the particles through the slits at a slow enough rate, we could track each particle as it passed through one slit or the other and then appeared on the screen.

If we performed this imaginary experiment, the result would no longer be an interference pattern on the screen. Instead, we would observe a pattern similar to that shown in Figure 4.17, with “hits” in front of each slit, but no interference fringes. No matter what sort of device we use to determine through which slit the particle passes, the interference pattern will be destroyed. The classical *particle* must pass through one slit or the other; only a *wave* can reveal interference, which depends on parts of the wavefront passing through *both* slits and then recombining.

When we ask through which slit the particle passed, we are investigating only the *particle* aspects of its behavior, and we cannot observe its wave nature (the interference pattern). Conversely, when we study the wave nature, we cannot simultaneously observe the particle nature. The electron will behave as a particle *or* a wave, but we cannot observe *both* aspects of its behavior simultaneously. This curious aspect of quantum mechanics was also discussed for photons in Section 3.6, where we discovered that experiments can reveal either the particle nature of the photon or its wave nature, but not both aspects simultaneously.

This is the basis for the *principle of complementarity*, which asserts that the complete description of a photon or a particle such as an electron cannot be made in terms of only particle properties or only wave properties, but that both aspects of its behavior must be considered. Moreover, the particle and wave natures cannot be observed simultaneously, and the type of behavior that we

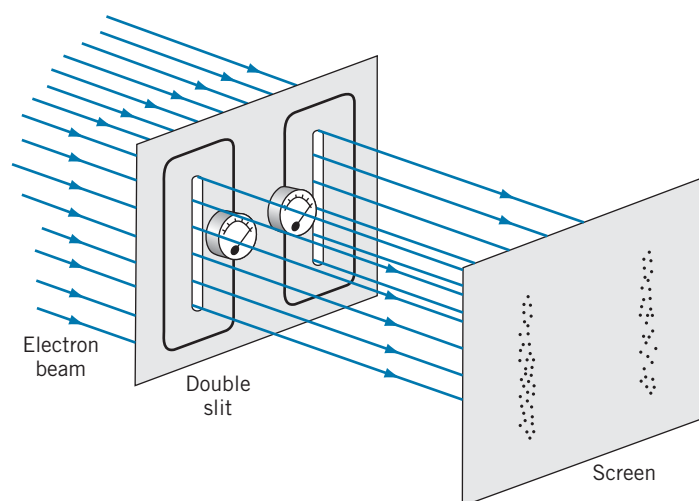


FIGURE 4.17 Apparatus to record passage of electrons through slits. Each slit is surrounded by a loop with a meter that signals the passage of an electron through the slit. No interference fringes are seen on the screen.

observe depends on the kind of experiment we are doing: a particle-type experiment shows only particle like behavior, and a wave-type experiment shows only wavelike behavior.

4.3 UNCERTAINTY RELATIONSHIPS FOR CLASSICAL WAVES

In quantum mechanics, we want to use de Broglie waves to describe particles. In particular, the amplitude of the wave will tell us something about the location of the particle. Clearly a pure sinusoidal wave, as in Figure 4.18a, is not much use in locating a particle—the wave extends from $-\infty$ to $+\infty$, so the particle might be found anywhere in that region. On the other hand, a narrow wave pulse like Figure 4.18b does a pretty good job of locating the particle in a small region of space, but this wave does not have an easily identifiable wavelength. In the first case, we know the wavelength exactly but have no knowledge of the location of the particle, while in the second case we have a good idea of the location of the particle but a poor knowledge of its wavelength. Because wavelength is associated with momentum by the de Broglie relationship (Eq. 4.1), a poor knowledge of the wavelength is associated with a poor knowledge of the particle's momentum. For a classical particle, we would like to know both its location and its momentum as precisely as possible. For a quantum particle, we are going to have to make some compromises—the better we know its momentum (or wavelength), the less we know about its location. We can improve our knowledge of its location only at the expense of our knowledge of its momentum.

This competition between knowledge of location and knowledge of wavelength is not restricted to de Broglie waves—classical waves show the same effect. All real waves can be represented as *wave packets*—disturbances that are localized to a finite region of space. We will discuss more about constructing wave packets in Section 4.5. In this section, we will examine this competition between specifying the location and the wavelength of classical waves more closely.

Figure 4.19a shows a very small wave packet. The disturbance is well localized to a small region of space of length Δx . (Imagine listening to a very short burst of sound, of such brief duration that it is hard for you to recognize the pitch or frequency of the wave.) Let's try to measure the wavelength of this wave packet. Placing a measuring stick along the wave, we have some difficulty defining exactly where the wave starts and where it ends. Our measurement of the wavelength is therefore subject to a small *uncertainty* $\Delta\lambda$. Let's represent this uncertainty as a fraction ϵ of the wavelength λ , so that $\Delta\lambda \sim \epsilon\lambda$. The fraction ϵ is certainly less than 1, but it is probably greater than 0.01, so we estimate that $\epsilon \sim 0.1$ to within an order of magnitude. (In our discussion of uncertainty, we use the $\sim$ symbol to indicate a rough order-of-magnitude estimate.) That is, the uncertainty in our measurement of the wavelength might be roughly 10% of the wavelength.

The size of this wave disturbance is roughly one wavelength, so $\Delta x \approx \lambda$. For this discussion, we want to examine the product of the size of the wave

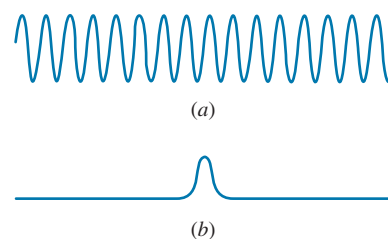


FIGURE 4.18 (a) A pure sine wave, which extends from $-\infty$ to $+\infty$. (b) A narrow wave pulse.

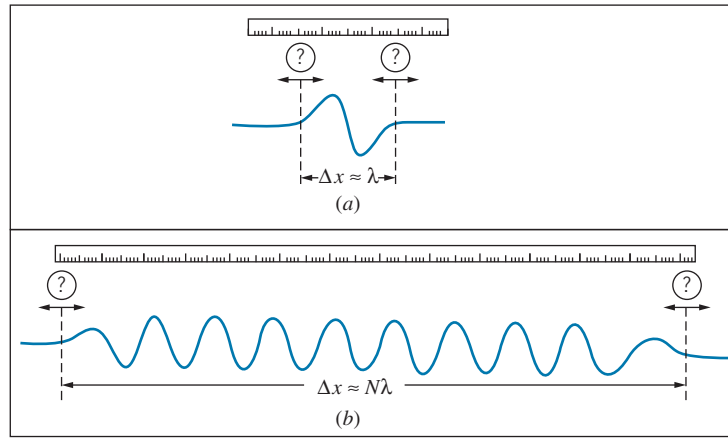


FIGURE 4.19 (a) Measuring the wavelength of a wave represented by a small wave packet of length roughly one wavelength. (b) Measuring the wavelength of a wave represented by a large wave packet consisting of N waves.

packet and the uncertainty in the wavelength, Δx times $\Delta\lambda$ with $\Delta x \approx \lambda$ and $\Delta\lambda \sim \epsilon\lambda$:

$$\Delta x \Delta\lambda \sim \epsilon\lambda^2 \quad (4.4)$$

This expression shows the inverse relationship between the size of the wave packet and the uncertainty in the wavelength: for a given wavelength, the smaller the size of the wave packet, the greater the uncertainty in our knowledge of the wavelength. That is, as Δx gets smaller, $\Delta\lambda$ must become larger.

Making a larger wave packet doesn't help us at all. Figure 4.19b shows a larger wave packet with the same wavelength. Suppose this larger wave packet contains N cycles of the wave, so that $\Delta x \approx N\lambda$. Again using our measuring stick, we try to measure the size of N wavelengths, and dividing this distance by N we can then determine the wavelength. We still have the same uncertainty of $\epsilon\lambda$ in locating the start and end of this wave packet, but when we divide by N to find the wavelength, the uncertainty in one wavelength becomes $\Delta\lambda \sim \epsilon\lambda/N$. For this larger wave packet, the product of Δx and $\Delta\lambda$ is $\Delta x \Delta\lambda \sim (N\lambda)(\epsilon\lambda/N) = \epsilon\lambda^2$, exactly the same as in the case of the smaller wave packet. Equation 4.4 is a fundamental property of classical waves, independent of the type of wave or the method used to measure its wavelength. This is the first of the *uncertainty relationships* for classical waves.

Example 4.4

In a measurement of the wavelength of water waves, 10 wave cycles are counted in a distance of 196 cm. Estimate the minimum uncertainty in the wavelength that might be obtained from this experiment.

Solution

With 10 wave crests in a distance of 196 cm, the wavelength is about $(196 \text{ cm})/10 = 19.6 \text{ cm}$. We can take $\epsilon \sim 0.1$ as a good order-of-magnitude estimate of the

typical precision that might be obtained. From Eq. 4.4, we can find the uncertainty in wavelength:

$$\Delta\lambda \sim \frac{\epsilon\lambda^2}{\Delta x} = \frac{(0.1)(19.6 \text{ cm})^2}{196 \text{ cm}} = 0.2 \text{ cm}$$

With an uncertainty of 0.2 cm, the “true” wavelength might range from 19.5 cm to 19.7 cm, so we might express this result as $19.6 \pm 0.1 \text{ cm}$.

The Frequency-Time Uncertainty Relationship

We can take a different approach to uncertainty for classical waves by imagining a measurement of the period rather than the wavelength of the wave that comprises our wave packet. Suppose we have a timing device that we use to measure the duration of the wave packet, as in Figure 4.20. Here we are plotting the wave disturbance as a function of time rather than location. The “size” of the wave packet is now its duration in time, which is roughly one period T for this wave packet, so that $\Delta t \approx T$. Whatever measuring device we use, we have some difficulty locating exactly the start and end of one cycle, so we have an uncertainty ΔT in measuring the period. As before, we’ll assume this uncertainty is some small fraction of the period: $\Delta T \sim \epsilon T$. To examine the competition between the duration of the wave packet and our ability to measure its period, we calculate the product of Δt and ΔT :

$$\Delta t \Delta T \sim \epsilon T^2 \quad (4.5)$$

This is the second of our uncertainty relationships for classical waves. It shows that for a wave of a given period, the smaller the duration of the wave packet, the larger is the uncertainty in our measurement of the period. Note the similarity between Eqs. 4.4 and 4.5, one representing relationships in space and the other in time.

It will turn out to be more useful if we write Eq. 4.5 in terms of frequency instead of period. Given that period T and frequency f are related by $f = 1/T$, how is Δf related to ΔT ? The correct relationship is certainly *not* $\Delta f = 1/\Delta T$, which would imply that a very small uncertainty in the period would lead to a very large uncertainty in the frequency. Instead, they should be directly related—the better we know the period, the better we know the frequency. Here is how we obtain the relationship: Beginning with $f = 1/T$, we take differentials on both sides:

$$df = -\frac{1}{T^2} dT$$

Next, we convert the infinitesimal differentials to finite intervals, and because we are interested only in the magnitude of the uncertainties we can ignore the minus sign:

$$\Delta f = \frac{1}{T^2} \Delta T \quad (4.6)$$

Combining Eqs. 4.5 and 4.6, we obtain

$$\Delta f \Delta t \sim \epsilon \quad (4.7)$$

Equation 4.7 shows that the longer the duration of the wave packet, the more precisely we can measure its frequency.

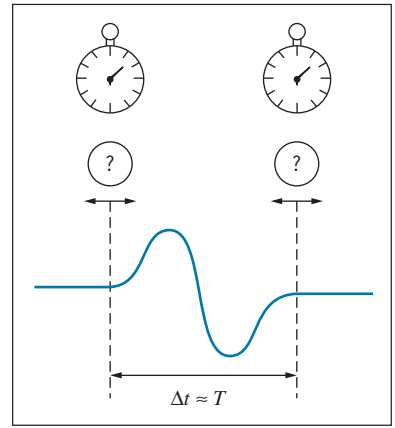


FIGURE 4.20 Measuring the period of a wave represented by a small wave packet of duration roughly one period.

Example 4.5

An electronics salesman offers to sell you a frequency-measuring device. When hooked up to a sinusoidal signal, it automatically displays the frequency of the signal, and to account for frequency variations, the frequency is remeasured once each second and the display is updated. The salesman

claims the device to be accurate to 0.01 Hz. Is this claim valid?

in a time $\Delta t = 1\text{ s}$ must have an associated uncertainty of about

$$\Delta f \sim \frac{\epsilon}{\Delta t} = \frac{0.1}{1\text{ s}} = 0.1\text{ Hz}$$

Solution

Based on Eq. 4.7, and again estimating ϵ to be about 0.1, we know that a measurement of frequency

It appears that the salesman may be exaggerating the precision of this device.

4.4 HEISENBERG UNCERTAINTY RELATIONSHIPS

The uncertainty relationships discussed in the previous section apply to *all* waves, and we should therefore apply them to de Broglie waves. We can use the basic de Broglie relationship $p = h/\lambda$ to relate the uncertainty in the momentum Δp to the uncertainty in wavelength $\Delta\lambda$, using the same procedure that we used to obtain Eq. 4.6. Starting with $p = h/\lambda$, we take differentials on both sides and obtain $dp = (-h/\lambda^2)d\lambda$. Now we change the differentials into differences, ignoring the minus sign:

$$\Delta p = \frac{h}{\lambda^2} \Delta\lambda \quad (4.8)$$

An uncertainty in the momentum of the particle is directly related to the uncertainty in the wavelength associated with the particle's de Broglie wave packet.

Combining Eq. 4.8 with Eq. 4.4, we obtain

$$\Delta x \Delta p \sim \epsilon h \quad (4.9)$$

Just like Eq. 4.4, this equation suggests an inverse relationship between Δx and Δp . The smaller the size of the wave packet of the particle, the larger is the uncertainty in its momentum (and thus in its velocity).

Quantum mechanics provides a formal procedure for calculating Δx and Δp for wave packets corresponding to different physical situations and for different schemes for confining a particle. One outcome of these calculations gives the wave packet with the smallest possible value of the product $\Delta x \Delta p$, which turns out to be $h/4\pi$, as we will discuss in the next chapter. Thus, $\epsilon = 1/4\pi$ in this case. All other wave packets will have larger values for $\Delta x \Delta p$.

The combination $h/2\pi$ occurs frequently in quantum mechanics and is given the special symbol $\hbar$ ("h-bar")

$$\hbar = \frac{h}{2\pi} = 1.05 \times 10^{-34}\text{ J}\cdot\text{s} = 6.58 \times 10^{-16}\text{ eV}\cdot\text{s}$$

In terms of $\hbar$, we can write the uncertainty relationship as

$$\Delta x \Delta p_x \geq \frac{1}{2} \hbar \quad (4.10)$$

The x subscript has been added to the momentum to remind us that Eq. 4.10 applies to motion in a given direction and relates the uncertainties in position and momentum in that direction only. Similar and independent relationships



Bettmann / Getty Images

Werner Heisenberg (1901–1976, Germany). Best known for the uncertainty principle, he also developed a complete formulation of the quantum theory based on matrices.

can be applied in the other directions as necessary; thus, $\Delta y \Delta p_y \geq \hbar/2$ or $\Delta z \Delta p_z \geq \hbar/2$.

Equation 4.10 is the first of the *Heisenberg uncertainty relationships*. It sets the limit of the best we can possibly do in an experiment to measure *simultaneously* the location and the momentum of a particle. Another way of interpreting this equation is to say that the more we try to confine a particle, the less we know about its momentum.

Because the limit of $\hbar/2$ represents the minimum value of the product $\Delta x \Delta p_x$, in most cases we will do worse than this limit. It is therefore quite acceptable to take

$$\Delta x \Delta p_x \sim \hbar \quad (4.11)$$

as a rough estimate of the relationship between the uncertainties in location and the momentum.

As an example, let's consider a beam of electrons incident on a single slit, as in Figure 4.21. We know this experiment as single-slit diffraction, which produces the characteristic diffraction pattern illustrated in Figure 4.1. We'll assume that the particles are initially moving in the y direction and that we know their momentum in that direction as precisely as possible. If the electrons initially have no component of their momentum in the x direction, we know p_x exactly (it is exactly zero), so that $\Delta p_x = 0$; thus, we know nothing about the x coordinates of the electrons ($\Delta x = \infty$). This situation represents a very wide beam of electrons, only a small fraction of which pass through the slit.

At the instant that some of the electrons pass through the slit, we know quite a bit more about their x location. In order to pass through the slit, the uncertainty in their x location is no larger than a , the width of the slit; thus, $\Delta x = a$. This improvement in our knowledge of the electron's location comes at the expense of our knowledge of its momentum, however. According to Eq. 4.11, the uncertainty in the x component of its momentum is now $\Delta p_x \sim \hbar/a$. Measurements beyond the slit no longer show the particle moving precisely in the y direction (for which $p_x = 0$); the momentum now has a small x component as well, with values distributed about zero but now with a range of roughly $\pm \hbar/a$. In passing through the slit, a particle acquires on the average an x component of momentum of roughly $\hbar/a$, according to the uncertainty principle.

Let us now find the angle θ that specifies where a particle with this value of p_x lands on the screen. For small angles, $\sin \theta \approx \tan \theta$ and so

$$\sin \theta \approx \tan \theta = \frac{p_x}{p_y} = \frac{\hbar/a}{p_y} = \frac{\lambda}{2\pi a}$$

using $\lambda = h/p_y$ for the de Broglie wavelength of the electrons. The first minimum of the diffraction pattern of a single slit is located at $\sin \theta = \lambda/a$, which is larger than the spread of angles into which most of the particles are diffracted. The calculation shows that the distribution of transverse momentum given by the uncertainty principle is roughly equivalent to the spreading of the beam into the central diffraction peak, and it illustrates again the close connection between wave behavior and uncertainty in particle location.

The diffraction (spreading) of a beam following passage through a slit is just the effect of the uncertainty principle on our attempt to specify the location of

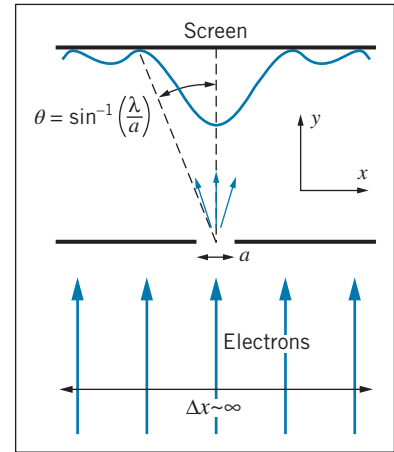


FIGURE 4.21 Single-slit diffraction of electrons. A wide beam of electrons is incident on a narrow slit. The electrons that pass through the slit acquire a component of momentum in the x direction.

the particle. As we make the slit narrower, p_x increases and the beam spreads even more. In trying to obtain more precise knowledge of the location of the particle by making the slit narrower, we have lost knowledge of the direction of its travel. This trade-off between observations of position and momentum is the essence of the Heisenberg uncertainty principle.

We can also apply the second of our classical uncertainty relationships (Eq. 4.7) to de Broglie waves. If we assume the energy–frequency relationship for light, $E = hf$, can be applied to particles, then we immediately obtain $\Delta E = h\Delta f$. Combining this with Eq. 4.7, we obtain

$$\Delta E \Delta t \sim \epsilon h \quad (4.12)$$

Once again, the minimum uncertainty wave packet gives $\epsilon = 1/4\pi$, and so

$$\Delta E \Delta t \geq \frac{1}{2} \hbar \quad (4.13)$$

This is the second of the Heisenberg uncertainty relationships. It tells us that the more precisely we try to determine the time coordinate of a particle, the less precisely we know its energy. For example, if a particle has a very short lifetime between its creation and decay ($\Delta t \rightarrow 0$), a measurement of its rest energy (and thus its mass) will be very imprecise ($\Delta E \rightarrow \infty$). Conversely, the rest energy of a stable particle (one with an infinite lifetime, so that $\Delta t = \infty$) can in principle be measured with unlimited precision ($\Delta E = 0$).

As in the case of the first Heisenberg relationship, we can take

$$\Delta E \Delta t \sim \hbar \quad (4.14)$$

as a reasonable estimate for most wave packets.

The Heisenberg uncertainty relationships are the mathematical representations of the *Heisenberg uncertainty principle*, which states:

It is not possible to make a simultaneous determination of the position and the momentum of a particle with unlimited precision,

and

It is not possible to make a simultaneous determination of the energy and the time coordinate of a particle with unlimited precision.

These relationships give an estimate of the minimum uncertainty that can result from any experiment; measurement of the position and momentum of a particle will give a spread of values of widths Δx and Δp_x . We may, for other reasons, do much worse than Eqs. 4.10 and 4.13, but *we can do no better*.

These relationships have a profound impact on our view of nature. It is quite acceptable to say that there is an uncertainty in locating the position of a water wave. It is quite another matter to make the same statement about a de Broglie wave, because *there is an implied corresponding uncertainty in the position of the particle*. Equations 4.10 and 4.13 say that *nature imposes a limit on the accuracy with which we can do experiments*. To emphasize this point, the Heisenberg relationships are sometimes called “indeterminacy” rather than “uncertainty” principles, because the idea of uncertainty may suggest an experimental limit

that can be reduced by using better equipment or technique. In actuality, these coordinates are *indeterminate* to the limits provided by Eqs. 4.10 and 4.13—no matter how hard we try, it is simply not possible to measure more precisely.

Example 4.6

A beam of electrons moving in the x direction with momentum p_x passes through a narrow opening of width a that is transverse to the direction of the beam. How does passage through the slit affect our knowledge of p_x ?

Solution

The Heisenberg uncertainty principle can be expressed as $\Delta x \Delta p_x \sim \hbar$ and $\Delta y \Delta p_y \sim \hbar$. The slit defines a region

$\Delta y = a$ that increases our knowledge of the y location of the electrons and in the process introduces a corresponding uncertainty Δp_y into their motion (before passage through the slit, p_y was 0). Here the x and y motions are independent, so application of the uncertainty principle to the y motion does not affect the motion in the x direction. Even though the electrons can acquire a momentum component in the y direction, their x -momentum remains at the same value p_x before and after the slit.

Example 4.7

An electron moves in the x direction with a speed of 3.6×10^6 m/s. We can measure its speed to a precision of 1%. With what precision can we simultaneously measure its x coordinate?

Solution

The electron's momentum is

$$\begin{aligned} p_x &= mv_x = (9.11 \times 10^{-31} \text{ kg})(3.6 \times 10^6 \text{ m/s}) \\ &= 3.3 \times 10^{-24} \text{ kg} \cdot \text{m/s} \end{aligned}$$

The uncertainty Δp_x is 1% of this value, or 3.3×10^{-26} kg · m/s. The uncertainty in position is then

$$\Delta x \sim \frac{\hbar}{\Delta p_x} = \frac{1.05 \times 10^{-34} \text{ J} \cdot \text{s}}{3.3 \times 10^{-26} \text{ kg} \cdot \text{m/s}} = 3.2 \text{ nm}$$

which is roughly 10 atomic diameters.

Example 4.8

Repeat the calculations of the previous example in the case of a pitched baseball ($m = 0.145$ kg) moving at a speed of 95 mi/h (42.5 m/s). Again assume that its speed can be measured to a precision of 1%.

Solution

The baseball's momentum is

$$p_x = mv_x = (0.145 \text{ kg})(42.5 \text{ m/s}) = 6.16 \text{ kg} \cdot \text{m/s}$$

The uncertainty in momentum is 6.16×10^{-2} kg · m/s, and the corresponding uncertainty in position is

$$\Delta x \sim \frac{\hbar}{\Delta p_x} = \frac{1.05 \times 10^{-34} \text{ J} \cdot \text{s}}{6.16 \times 10^{-2} \text{ kg} \cdot \text{m/s}} = 1.7 \times 10^{-33} \text{ m}$$

This uncertainty is 19 orders of magnitude smaller than the size of an atomic nucleus. The uncertainty principle cannot be blamed for the batter missing the pitch! Once again we see that, because of the small magnitude of Planck's constant, quantum effects are not observable for ordinary objects.

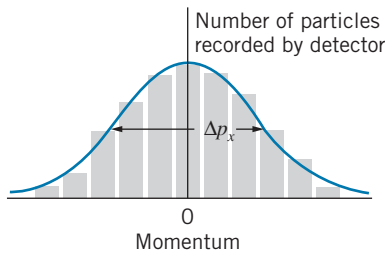


FIGURE 4.22 Results that might be obtained from measuring the number of electrons in a given time interval at different locations on the screen of Figure 4.21. The distribution is centered around $p_x = 0$ and has a width that is characterized by Δp_x .

A Statistical Interpretation of Uncertainty

A diffraction pattern, such as that shown in Figure 4.21, is the result of the passage of many particles or photons through the slit. So far, we have been discussing the behavior of only one particle. Let's imagine that we do an experiment in which a large number of particles pass (one at a time) through the slit, and we measure the transverse (x component) momentum of each particle after it passes through the slit. We can do this experiment simply by placing a detector at different locations on the screen where we observe the diffraction pattern. Because the detector actually accepts particles over a finite region on the screen, it measures in a range of deflection angles or equivalently in a range of transverse momentum. The result of the experiment might look something like Figure 4.22. The vertical scale shows the number of particles with momentum in each interval corresponding to different locations of the detector on the screen. The values are symmetrically arranged about zero, which indicates that the mean or average value of p_x is zero. The *width* of the distribution is characterized by Δp_x .

Figure 4.22 resembles a statistical distribution, and in fact the precise definition of Δp_x is similar to that of the standard deviation σ_A of a quantity A that has a mean or average value A_{av} :

$$\sigma_A = \sqrt{(A^2)_{av} - (A_{av})^2}$$

If there are N individual measurements of A , then $A_{av} = N^{-1} \sum A_i$ and $(A^2)_{av} = N^{-1} \sum A_i^2$.

By analogy, we can make a rigorous definition of the uncertainty in momentum as

$$\Delta p_x = \sqrt{(p_x^2)_{av} - (p_{x,av})^2} \quad (4.15)$$

The average value of the transverse momentum for the situation shown in Figure 4.22 is zero, so

$$\Delta p_x = \sqrt{(p_x^2)_{av}} \quad (4.16)$$

which gives in effect a root-mean-square value of p_x . This can be taken to be a rough measure of the magnitude of p_x . Thus, it is often said that Δp_x gives a measure of the magnitude of the momentum of the particle. As you can see from Figure 4.22, this is indeed true.*

Example 4.9

In nuclear beta decay, electrons are observed to be ejected from the atomic nucleus. Suppose we assume that electrons are somehow trapped within the nucleus, and that occasionally one escapes and is observed in the laboratory. Take the diameter of a typical nucleus to be 1.0×10^{-14} m, and use the uncertainty principle to estimate the range of kinetic energies that such an electron must have.

Solution

If the electron were trapped in a region of width $\Delta x \approx 10^{-14}$ m, the corresponding uncertainty in its momentum would be

$$\Delta p_x \sim \frac{\hbar}{\Delta x} = \frac{1}{c} \frac{\hbar c}{\Delta x} = \frac{1}{c} \frac{197 \text{ MeV} \cdot \text{fm}}{10 \text{ fm}} = 19.7 \text{ MeV}/c$$

*The relationship between the value of Δp_x calculated from Eq. 4.16 and the width of the distribution shown in Figure 4.22 depends on the exact shape of the distribution. You should consider the value from Eq. 4.16 as a rough order-of-magnitude estimate of the width of the distribution.

Note the use of $\hbar c = 197 \text{ MeV} \cdot \text{fm}$ in this calculation. This momentum is clearly in the relativistic regime for electrons, so we must use the relativistic formula to find the kinetic energy for a particle of momentum $19.7 \text{ MeV}/c$:

$$K = \sqrt{p^2 c^2 + (mc^2)^2} - mc^2 \\ = \sqrt{(19.7 \text{ MeV})^2 + (0.5 \text{ MeV})^2} - 0.5 \text{ MeV} = 19 \text{ MeV}$$

where we have used Eq. 4.16 to relate Δp_x to p_x^2 . This result gives the spread of kinetic energies corresponding to a spread in momentum of $19.7 \text{ MeV}/c$.

Electrons emitted from the nucleus in nuclear beta decay typically have kinetic energies of about 1 MeV , much smaller than the typical spread in energy required by the uncertainty principle for electrons confined inside the nucleus. This suggests that beta-decay electrons of such low energies cannot be confined in a region of the size of the nucleus, and that another explanation must be found for the electrons observed in nuclear beta decay. (As we discuss in Chapter 12, these electrons cannot pre-exist in the nucleus, which would violate the uncertainty principle, but are “manufactured” by the nucleus at the instant of the decay.)

Example 4.10

(a) A charged pi meson has a rest energy of 140 MeV and a lifetime of 26 ns . Find the energy uncertainty of the pi meson, expressed in MeV and also as a fraction of its rest energy. (b) Repeat for the uncharged pi meson with a rest energy of 135 MeV and a lifetime of $8.3 \times 10^{-17} \text{ s}$. (c) Repeat for the rho meson with a rest energy of 765 MeV and a lifetime of $4.4 \times 10^{-24} \text{ s}$.

Solution

(a) If the pi meson lives for 26 ns , we have only that much time in which to measure its rest energy, and Eq. 4.8 tells us that *any* energy measurement done in a time Δt is uncertain by an amount of at least

$$\Delta E = \frac{\hbar}{\Delta t} = \frac{6.58 \times 10^{-16} \text{ eV} \cdot \text{s}}{26 \times 10^{-9} \text{ s}} \\ = 2.5 \times 10^{-8} \text{ eV} \\ = 2.5 \times 10^{-14} \text{ MeV} \\ \frac{\Delta E}{E} = \frac{2.5 \times 10^{-14} \text{ MeV}}{140 \text{ MeV}} \\ = 1.8 \times 10^{-16}$$

(b) In a similar way,

$$\Delta E = \frac{\hbar}{\Delta t} = \frac{6.58 \times 10^{-16} \text{ eV} \cdot \text{s}}{8.3 \times 10^{-17} \text{ s}} = 7.9 \text{ eV} \\ = 7.9 \times 10^{-6} \text{ MeV} \\ \frac{\Delta E}{E} = \frac{7.9 \times 10^{-6} \text{ MeV}}{135 \text{ MeV}} = 5.9 \times 10^{-8}$$

(c) For the rho meson,

$$\Delta E = \frac{\hbar}{\Delta t} = \frac{6.58 \times 10^{-16} \text{ eV} \cdot \text{s}}{4.4 \times 10^{-24} \text{ s}} = 1.5 \times 10^8 \text{ eV} \\ = 150 \text{ MeV} \\ \frac{\Delta E}{E} = \frac{150 \text{ MeV}}{765 \text{ MeV}} = 0.20$$

In the first case, the uncertainty principle does not give a large enough effect to be measured—particle masses cannot be measured to a precision of 10^{-16} (about 10^{-6} is the best precision that we can obtain). In the second example, the uncertainty principle contributes at about the level of 10^{-7} , which approaches the limit of our measuring instruments and therefore might be observable in the laboratory. In the third example, we see that the uncertainty principle can contribute substantially to the precision of our knowledge of the rest energy of the rho meson; measurements of its rest energy will show a statistical distribution centered about 765 MeV with a spread of 150 MeV , and no matter how precise an instrument we use to measure the rest energy, we can never reduce that spread.

The lifetime of a very short-lived particle such as the rho meson cannot be measured directly. In practice, we reverse the procedure of the calculation of this example—we measure the rest energy, which gives a distribution similar to Figure 4.22, and from the “width” ΔE of the distribution we deduce the lifetime using Eq. 4.8. This procedure is discussed in Chapter 14.

Example 4.11

Estimate the minimum velocity that would be measured for a billiard ball ($m \approx 100$ g) confined to a billiard table of dimension 1 m.

Solution

For $\Delta x \approx 1$ m, we have

$$\Delta p_x \sim \frac{\hbar}{\Delta x} = \frac{1.05 \times 10^{-34} \text{ J} \cdot \text{s}}{1 \text{ m}} = 1 \times 10^{-34} \text{ kg} \cdot \text{m/s}$$

so

$$\Delta v_x = \frac{\Delta p_x}{m} = \frac{1 \times 10^{-34} \text{ kg} \cdot \text{m/s}}{0.1 \text{ kg}} = 1 \times 10^{-33} \text{ m/s}$$

Thus, quantum effects might result in motion of the billiard ball with a speed distribution having a spread of about 1×10^{-33} m/s. At this speed, the ball would move a distance of 1% of the diameter of an atomic nucleus in a time equal to the age of the universe! Once again, we see that quantum effects are not observable with macroscopic objects.

4.5 WAVE PACKETS

In Section 4.3, we described measurements of the wavelength or frequency of a wave packet, which we consider to be a finite group of oscillations of a wave. That is, the wave amplitude is large over a finite region of space or time and is very small outside that region.

Before we begin our discussion, it is necessary to keep in mind that we are discussing *traveling waves*, which we imagine as moving in one direction with a uniform speed. (We'll discuss the speed of the wave packet later.) As the wave packet moves, individual locations in space will oscillate with the frequency or wavelength that characterizes the wave packet. When we show a static picture of a wave packet, it doesn't matter that some points within the packet appear to have positive displacement, some have negative displacement, and some may even have zero displacement. As the wave travels, those locations are in the process of oscillating, and our drawings may "freeze" that oscillation. What is important is the locations in space where the overall wave packet has a large oscillation amplitude and where it has a very small amplitude.*

In this section, we discuss how to build a wave packet by adding waves together. A pure sinusoidal wave is of no use in representing a particle—the wave extends from $-\infty$ to $+\infty$, so the particle could be found anywhere. We would like the particle to be represented by a wave packet that describes how the particle is *localized* to a particular region of space, such as an atom or a nucleus.

The key to the process of building a wave packet involves adding together waves of different wavelength. We represent our waves as $A \cos kx$, where k is

*By analogy, think of a radio wave traveling from the station to your receiver. At a particular instant of time, some points in space may have instantaneous electromagnetic field values of zero, but that doesn't affect your reception of the signal. What is important is the overall amplitude of the traveling wave.

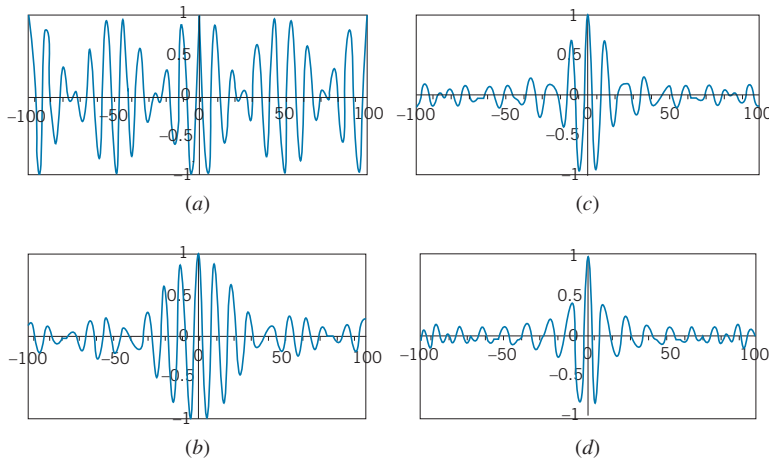


FIGURE 4.23 (a) Adding two waves of wavelengths 9 and 11 gives beats. (b) Adding 5 waves with wavelengths ranging from 9 to 11. (c) Adding 9 waves with wavelengths ranging from 8 to 12. (d) Adding 13 waves with wavelengths ranging from 7 to 13. All of the patterns repeat from $-\infty$ to $+\infty$.

the wave number ($k = 2\pi/\lambda$) and A is the amplitude. For example, let's add together two waves:

$$y(x) = A_1 \cos k_1 x + A_2 \cos k_2 x = A_1 \cos(2\pi x/\lambda_1) + A_2 \cos(2\pi x/\lambda_2) \quad (4.17)$$

This sum is illustrated in Figure 4.23a for the case $A_1 = A_2$ and $\lambda_1 = 9, \lambda_2 = 11$. This combined wave shows the phenomenon known as *beats* in the case of sound waves. So far we don't have a result that looks anything like the wave packet we are after, but you can see that by adding together two different waves we have reduced the amplitude of the wave packet at some locations. This pattern repeats endlessly from $-\infty$ to $+\infty$, so the particle is still not localized.

Let's try a more detailed sum. Figure 4.23b shows the result of adding 5 waves with wavelengths 9, 9.5, 10, 10.5, 11. Here we have been a bit more successful in restricting the amplitude of the wave packet in some regions. By adding even more waves with a larger range of wavelengths, we can obtain still narrower regions of large amplitude: Figure 4.23c shows the result of adding 9 waves with wavelengths 8, 8.5, 9, ..., 12, and Figure 4.23d shows the result of adding 13 waves of wavelengths 7, 7.5, 8, ..., 13. Unfortunately, all of these patterns (including the regions of large amplitude) repeat endlessly from $-\infty$ to $+\infty$, so even though we have obtained increasingly large regions where the wave packet has small amplitude, we haven't yet created a wave packet that might represent a particle localized to a particular region. If these wave packets did represent particles, then the particle would not be confined to any finite region.

The regions of large amplitude in Figures 4.23b,c,d do show how adding more waves of a greater range of wavelengths helps to restrict the size of the wave packet. The region of large amplitude in Figure 4.23b ranges from about $x = -40$ to $+40$, while in Figure 4.23c it is from about $x = -20$ to $+20$ and in Figure 4.23d from about $x = -15$ to $+15$. This shows again the inverse relationship between Δx and $\Delta \lambda$ expected for wave packets given by Eq. 4.4: as

the range of wavelengths increases from 2 to 4 to 6, the size of the “allowed” regions decreases from about 80 to 40 to 30. Once again we find that to restrict the size of the wave packet we must sacrifice the precise knowledge of the wavelength.

Note that for all four of these wave patterns, the disturbance seems to have a wavelength of about 10, equal to the central wavelength of the range of values of the functions we constructed. We can therefore regard these functions as a cosine wave with a wavelength of 10 that is shaped or *modulated* by the other cosine waves included in the function. For example, for the case of $A_1 = A_2 = A$, Eq. 4.17 can be rewritten after a bit of trigonometric manipulation as

$$y(x) = 2A \cos\left(\frac{\pi x}{\lambda_1} - \frac{\pi x}{\lambda_2}\right) \cos\left(\frac{\pi x}{\lambda_1} + \frac{\pi x}{\lambda_2}\right) \quad (4.18)$$

If λ_1 and λ_2 are close together (that is, if $\Delta\lambda = \lambda_2 - \lambda_1 \ll \lambda_1, \lambda_2$), this can be approximated as

$$y(x) = 2A \cos\left(\frac{\Delta\lambda\pi x}{\lambda_{av}^2}\right) \cos\left(\frac{2\pi x}{\lambda_{av}}\right) \quad (4.19)$$

where $\lambda_{av} = (\lambda_1 + \lambda_2)/2 \approx \lambda_1$ or λ_2 . The second cosine term represents a wave with a wavelength of 10, and the first cosine term provides the shaping envelope that produces the beats.

Any finite combination of waves with discrete wavelengths will produce patterns that repeat between $-\infty$ and $+\infty$, so this method of adding waves will not work in constructing a finite wave packet. To construct a wave packet with a finite width, we must replace the first cosine term in Eq. 4.19 with a function that is large in the region where we want to confine the particle but that falls to zero as $x \rightarrow \pm\infty$. For example, the simplest function that has this property is $1/x$, so we might imagine a wave packet whose mathematical form is

$$y(x) = \frac{2A}{x} \sin\left(\frac{\Delta\lambda\pi x}{\lambda_0^2}\right) \cos\left(\frac{2\pi x}{\lambda_0}\right) \quad (4.20)$$

Here λ_0 represents the central wavelength, replacing λ_{av} . (In going from Eq. 4.19 to Eq. 4.20, the cosine modulating term has been changed to a sine; otherwise, the function would blow up at $x = 0$.) This function is plotted in Figure 4.24a. It looks more like the kind of function we are seeking—it has large amplitude only in a small region of space, and the amplitude drops rapidly to zero outside that region. Another function that has this property is the *Gaussian* modulating function:

$$y(x) = Ae^{-2(\Delta\lambda\pi x/\lambda_0^2)^2} \cos\left(\frac{2\pi x}{\lambda_0}\right) \quad (4.21)$$

which is shown in Figure 4.24b.

Both of these functions show the characteristic inverse relationship between an arbitrarily defined size of the wave packet Δx and the wavelength range parameter $\Delta\lambda$ that is used in constructing the wave packet. For example, consider the wave packet shown in Figure 4.24a. Let's arbitrarily define the width of the wave packet as the distance over which the amplitude of the central

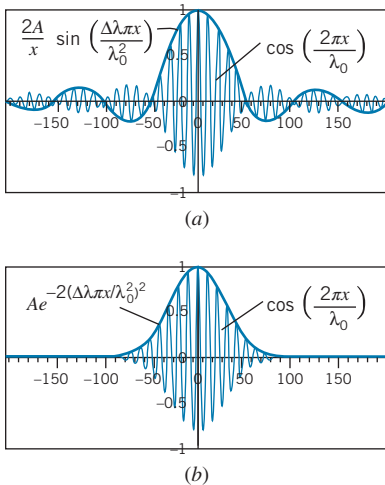


FIGURE 4.24 (a) A wave packet in which the modulation envelope decreases in amplitude like $1/x$. (b) A wave packet with a Gaussian modulating function. Both curves are drawn for $\lambda_0 = 10$ and $\Delta\lambda = 0.58$, which corresponds approximately to Figure 4.23b.

region falls by $1/2$. That occurs roughly where the argument of the sine has the value $\pm\pi/2$, which gives $\Delta x \Delta \lambda \sim \lambda_0^2$, consistent with our classical uncertainty estimate.

These wave packets can also be constructed by adding together waves of differing amplitude and wavelength, but the wavelengths form a continuous rather than a discrete set. It is a bit easier to illustrate this if we work with wave number $k = 2\pi/\lambda$ rather than wavelength. So far we have been adding waves in the form of $A \cos kx$, so that

$$y(x) = \sum A_i \cos k_i x \quad (4.22)$$

where $k_i = 2\pi/\lambda_i$. The waves plotted in Figure 4.23 represent applications of the general formula of Eq. 4.22 carried out over different numbers of discrete waves. If we have a continuous set of wave numbers, the sum in Eq. 4.22 becomes an integral:

$$y(x) = \int A(k) \cos kx \, dk \quad (4.23)$$

where the integral is carried out over whatever range of wave numbers is permitted (possibly infinite).

For example, suppose we have a range of wave numbers from $k_0 - \Delta k/2$ to $k_0 + \Delta k/2$, which is a continuous distribution of wave numbers of width Δk centered at k_0 . If all of the waves have the same amplitude A_0 , then from Eq. 4.23 the form of the wave packet can be shown to be (see Problem 27 at the end of the chapter)

$$y(x) = \frac{2A_0}{x} \sin\left(\frac{\Delta k}{2}x\right) \cos k_0 x \quad (4.24)$$

This is identical with Eq. 4.20 with $k_0 = 2\pi/\lambda_0$ and $\Delta k = 2\pi\Delta\lambda/\lambda_0^2$. This relationship between Δk and $\Delta\lambda$ follows from a procedure similar to what was used to obtain Eq. 4.6. With $k = 2\pi/\lambda$, taking differentials gives $dk = -(2\pi/\lambda^2)d\lambda$. Replacing the differentials with differences and ignoring the minus sign gives the relationship between Δk and $\Delta\lambda$.

A better approximation of the shape of the wave packet can be found by letting $A(k)$ vary according to a Gaussian distribution $A(k) = A_0 e^{-(k-k_0)^2/2(\Delta k)^2}$. This gives a range of wave numbers that has its largest contribution at the central wave number k_0 and falls off to zero for larger or smaller wave numbers with a characteristic width of Δk . Applying Eq. 4.23 to this case, with k ranging from $-\infty$ to $+\infty$ gives (see Problem 28)

$$y(x) = A_0 \Delta k \sqrt{2\pi} e^{-(\Delta k x)^2/2} \cos k_0 x \quad (4.25)$$

which shows how the form of Eq. 4.21 originates.

By specifying the distribution of wavelengths, we can construct a wave packet of any desired shape. A wave packet that restricts the particle to a region in space of width Δx will have a distribution of wavelengths characterized by a width $\Delta\lambda$. The smaller we try to make Δx , the larger will be the spread $\Delta\lambda$ of the wavelength distribution. The mathematics of this process gives a result that is consistent with the uncertainty relationship for classical waves (Eq. 4.4).

4.6 THE MOTION OF A WAVE PACKET

Let's consider again the “beats” wave packet represented by Eq. 4.17 and illustrated in Figure 4.23a. We now want to turn our “static” waves into traveling waves. It will again be more convenient for this discussion to work with the wave number k instead of the wavelength. To turn a static wave $y(x) = A \cos kx$ into a traveling wave moving in the positive x direction, we replace kx with $kx - \omega t$, so that the traveling wave is written as $y(x, t) = A \cos(kx - \omega t)$. (For motion in the negative x direction, we would replace kx with $kx + \omega t$.) Here ω is the *circular frequency* of the wave: $\omega = 2\pi f$. The combined traveling wave then would be represented as

$$y(x, t) = A_1 \cos(k_1 x - \omega_1 t) + A_2 \cos(k_2 x - \omega_2 t) \quad (4.26)$$

For any individual wave, the wave speed is related to its frequency and wavelength according to $v = \lambda f$. In terms of the wave number and circular frequency, we can write this as $v = (2\pi/k)(\omega/2\pi)$, so $v = \omega/k$. This quantity is sometimes called the *phase speed* and represents the speed of one particular phase or component of the wave packet. In general, each individual component may have a different phase speed. As a result, the shape of the wave packet may change with time.

For Figure 4.23a, we chose $A_1 = A_2$ and $\lambda_1 = 9$, $\lambda_2 = 11$. Let's choose $v_1 = 6$ units/s and $v_2 = 4$ units/s. Figure 4.25 shows the waveform at a time of $t = 1$ s. In that time, wave 1 will have moved a distance of 6 units in the positive x direction, and wave 2 will have moved a distance of 4 units in the positive x direction. However, the combined wave moves a much greater distance in that time: the center of the beat that was formerly at $x = 0$ has moved to $x = 15$ units. How is it possible that the combined waveform moves faster than either of its component waves?

To produce the peak at $x = 0$ and $t = 0$, the two component waves were exactly in phase—their two maxima lined up exactly to produce the combined maximum of the wave. At $x = 15$ units and $t = 1$ s, two individual maxima line up once again to produce a combined maximum. They are *not* the same two maxima that lined up to produce the maximum at $t = 0$, but it happens that

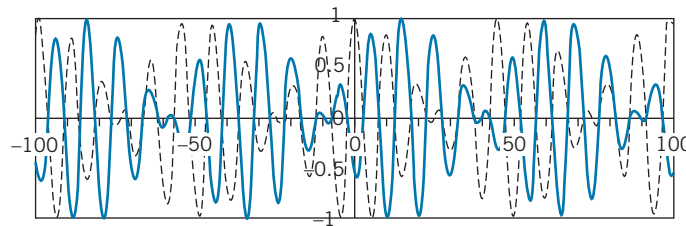


FIGURE 4.25 The solid line shows the waveform of Figure 4.23a at $t = 1$ s, and the dashed line shows the same waveform at $t = 0$. Note that the peak that was originally at $x = 0$ has moved to $x = 15$ at $t = 1$ s.

two other maxima are in phase at $x = 15$ units and $t = 1$ s to produce the combined maximum. If we were to watch an animation of the wave, we would see the maximum originally at $x = 0$ move gradually to $x = 15$ between $t = 0$ and $t = 1$ s.

We can understand how this occurs by writing Eq. 4.26 in a form similar to Eq. 4.18 using trigonometric identities. The result is (again assuming $A_1 = A_2 = A$, as we did in Eq. 4.18):

$$y(x, t) = 2A \cos\left(\frac{\Delta k}{2}x - \frac{\Delta\omega}{2}t\right) \cos\left(\frac{k_1 + k_2}{2}x - \frac{\omega_1 + \omega_2}{2}t\right) \quad (4.27)$$

As in Eq. 4.18, the second term in Eq. 4.27 represents the rapid variation of the wave within the envelope given by the first term. It is the first term that dictates the overall shape of the waveform, so it is this term that determines the speed of travel of the waveform. For a wave that is written as $\cos(kx - \omega t)$, the speed is ω/k . For this wave envelope, the speed is $(\Delta\omega/2)/(\Delta k/2) = \Delta\omega/\Delta k$. This speed is called the *group speed* of the wave packet. As we have seen, the group speed of the wave packet can be very different from the phase speed of the component waves. For more complicated situations than the two-component “beat” waveform, the group speed can be generalized by turning the differences into differentials:

$$v_{\text{group}} = \frac{d\omega}{dk} \quad (4.28)$$

The group speed depends on the relationship between frequency and wavelength for the component waves. If the phase speed of all component waves is the same and is independent of frequency or wavelength (as, for example, light waves in empty space), then the group speed is identical to the phase speed and the wave packet keeps its original shape as it travels. In general, the propagation of a component wave depends on the properties of the medium, and different component waves will travel with different speeds. Light waves in glass or sound waves in most solids travel with a speed that varies with frequency or wavelength, and so their wave packets change shape as they travel. De Broglie waves in general have different phase speeds, so that their wave packets expand as they travel.

Example 4.12

Certain ocean waves travel with a phase velocity $v_{\text{phase}} = \sqrt{g\lambda/2\pi}$, where g is the acceleration due to gravity. What is the group velocity of a “wave packet” of these waves?

Solution

With $k = 2\pi/\lambda$, we can write the phase velocity as a function of k as

$$v_{\text{phase}} = \sqrt{g/k}$$

But with $v_{\text{phase}} = \omega/k$, we have $\omega/k = \sqrt{g/k}$, so $\omega = \sqrt{gk}$ and Eq. 4.28 gives

$$v_{\text{group}} = \frac{d\omega}{dk} = \frac{d}{dk} \sqrt{gk} = \frac{1}{2} \sqrt{\frac{g}{k}} = \frac{1}{2} \sqrt{\frac{g\lambda}{2\pi}}$$

Note that the group speed of the wave packet increases as the wavelength increases.

The Group Speed of de Broglie Waves

Suppose that we have a localized particle, represented by a group of de Broglie waves. For each component wave, the energy of the particle is related to the frequency of the de Broglie wave by $E = hf = \hbar\omega$, and so $dE = \hbar d\omega$. Similarly, the momentum of the particle is related to the wavelength of the de Broglie wave by $p = h/\lambda = \hbar k$, so $dp = \hbar dk$. The group speed of the de Broglie wave then can be expressed as

$$v_{\text{group}} = \frac{d\omega}{dk} = \frac{dE/\hbar}{dp/\hbar} = \frac{dE}{dp} \quad (4.29)$$

For a classical particle having only kinetic energy $E = K = p^2/2m$, we can find dE/dp as

$$\frac{dE}{dp} = \frac{d}{dp} \left(\frac{p^2}{2m} \right) = \frac{p}{m} = v \quad (4.30)$$

which is the velocity of the particle.

Combining Eqs. 4.29 and 4.30 we obtain an important result:

$$v_{\text{group}} = v_{\text{particle}} \quad (4.31)$$

The speed of a particle is equal to the group speed of the corresponding wave packet. The wave packet and the particle move together—wherever the particle goes, its de Broglie wave packet moves along with it like a shadow. If we do a wave-type experiment on the particle, the de Broglie wave packet is always there to reveal the wave behavior of the particle. A particle can never escape its wave nature!

The Spreading of a Moving Wave Packet

Suppose we have a wave packet that represents a confined particle at $t = 0$. For example, the particle might have passed through a single-slit apparatus. Its initial uncertainty in position is Δx_0 , and its initial uncertainty in momentum is Δp_{x0} . The wave packet moves in the x direction with velocity v_x , but that velocity is not precisely known—the uncertainty in its momentum gives a corresponding uncertainty in velocity: $\Delta v_{x0} = \Delta p_{x0}/m$. Because there is an uncertainty in the velocity of the wave packet, we can't be sure where it will be located at time t . That is, its location at time t is $x = v_x t$, with velocity $v_x = v_{x0} \pm \Delta v_{x0}$. Thus there are two contributions to the uncertainty in its location at time t : the initial uncertainty Δx_0 and an additional amount equal to $\Delta v_{x0} t$ that represents the spreading of the wave packet. We'll assume that these two contributions add quadratically, like experimental uncertainties, so that the total uncertainty in the location of the particle is

$$\Delta x = \sqrt{(\Delta x_0)^2 + (\Delta v_{x0} t)^2} = \sqrt{(\Delta x_0)^2 + (\Delta p_{x0} t/m)^2} \quad (4.32)$$

Taking $\Delta p_{x0} = \hbar/\Delta x_0$ according to the uncertainty principle, we have

$$\Delta x = \sqrt{(\Delta x_0)^2 + (\hbar t/m\Delta x_0)^2} \quad (4.33)$$

If we try to make the wave packet very small at $t = 0$ (Δx_0 is small), then the second term under the square root makes the wave packet expand rapidly, because Δx_0 appears in the denominator of that term. *The more successful we are at confining a wave packet, the more quickly it spreads.* This reminds us of the single-slit experiment discussed in Section 4.4: the narrower we make the slit, the more the waves diverge after passing through the slit. Figure 4.26 shows how the size of the wave packet expands with time for three different initial sizes, and you can see that the smaller initial wave packet grows more rapidly than the larger initial packet.

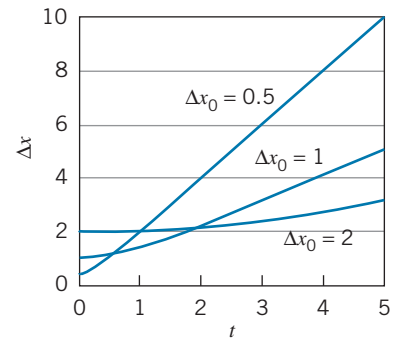


FIGURE 4.26 The smaller the initial wave packet, the more quickly it grows.

4.7 PROBABILITY AND RANDOMNESS

Any single measurement of the position or momentum of a particle can be made with as much precision as our experimental skill permits. How then does the wavelike behavior of a particle become observable? How does the uncertainty in position or momentum affect our experiment?

Suppose that we prepare an atom by attaching an electron to a nucleus. (For this example, we regard the nucleus as being fixed in space.) Some time after preparing our atom, we measure the position of the electron. We then repeat the procedure, preparing the atom *in an identical way*, and find that a remeasurement of the position of the electron yields a value different from that found in our first measurement. In fact, each time we repeat the measurement, we may obtain a different outcome. If we repeat the measurement a large number of times, we find ourselves led to a conclusion that runs counter to a basic notion of classical physics—*systems that are prepared in identical ways do not show identical subsequent behavior*. What hope do we then have of constructing a mathematical theory that has any usefulness at all in predicting the outcome of a measurement, if that outcome is completely random?

The solution to this dilemma lies in the consideration of the *probability* of obtaining any given result from an experiment whose possible results are subject to the laws of statistics. We cannot predict the outcome of a single flip of a coin or roll of the dice, because any *single result* is as likely as any other *single result*. We can, however, predict the *distribution* of a large number of individual measurements. For example, on a single flip of a coin, we cannot predict whether the outcome will be “heads” or “tails”; the two are equally likely. If we make a large number of trials, we expect that approximately 50% will turn up “heads” and 50% will yield “tails”; even though we cannot predict the result of any single toss of the coin, we can predict reasonably well the result of a large number of tosses.

Our study of systems governed by the laws of quantum physics leads us to a similar situation. We cannot predict the outcome of any *single* measurement of the position of the electron in the atom we prepared, but if we do a large number of measurements, we ought to find a statistical distribution of results. We cannot develop a mathematical theory that predicts the result of a single measurement, but we do have a mathematical theory that predicts the statistical

behavior of a system (or of a large number of identical systems). The quantum theory provides this mathematical procedure, which enables us to calculate the average or probable outcome of measurements and the distribution of individual outcomes about the average. This is not such a disadvantage as it may seem, for in the realm of quantum physics, we seldom do measurements with, for example, a single atom. If we were studying the emission of light by a radiant system or the properties of a solid or the scattering of nuclear particles, we would be dealing with a large number of atoms, and so our concept of statistical averages is very useful.

For example, if we say that the electron in a hydrogen atom has a 50% probability of circulating in a clockwise direction, we mean that in observing a large collection of similarly prepared atoms we find 50% to be circulating clockwise. Of course a *single measurement* shows *either* clockwise or counterclockwise circulation.

It could be argued that the flip of a coin or the roll of the dice is not a random process, but that the apparently random nature of the outcome simply reflects our lack of knowledge of the state of the system. For example, if we knew exactly how the dice were thrown (magnitude and direction of initial velocity, initial orientation, rotational speed) and precisely what the laws are that govern their bouncing on the table, we should be able to predict exactly how they would land. When we instead analyze the outcomes in terms of probabilities, we are really admitting our inability to do the analysis exactly. There is a school of thought that asserts that the same situation exists in quantum physics. According to this interpretation, we could predict *exactly* the behavior of the electron in our atom if only we knew the nature of a set of so-called “hidden variables” that determine its motion. However, experimental evidence disagrees with this theory, and so we must conclude that the random behavior of a system governed by the laws of quantum physics is a fundamental aspect of nature and not a result of our limited knowledge of the properties of the system.

The Probability Amplitude

What does the amplitude of the de Broglie wave represent? In any wave phenomenon, a physical quantity such as displacement or pressure varies with location and time. What is the physical property that varies as the de Broglie wave propagates?

A localized particle is represented by a wave packet. If a particle is confined to a region of space of dimension Δx , its wave packet has large amplitude only in a region of space of dimension Δx and has small amplitude elsewhere. That is, the amplitude is large where the particle is likely to be found and small where the particle is less likely to be found. *The probability of finding the particle at any point depends on the amplitude of its de Broglie wave at that point.* In analogy with classical physics, in which the intensity of any wave is proportional to the square of its amplitude, we have

$$\text{probability to observe particles} \propto |\text{de Broglie wave amplitude}|^2$$

Compare this with the similar relationship for photons discussed in Section 3.6:

$$\text{probability to observe photons} \propto |\text{electric field amplitude}|^2$$

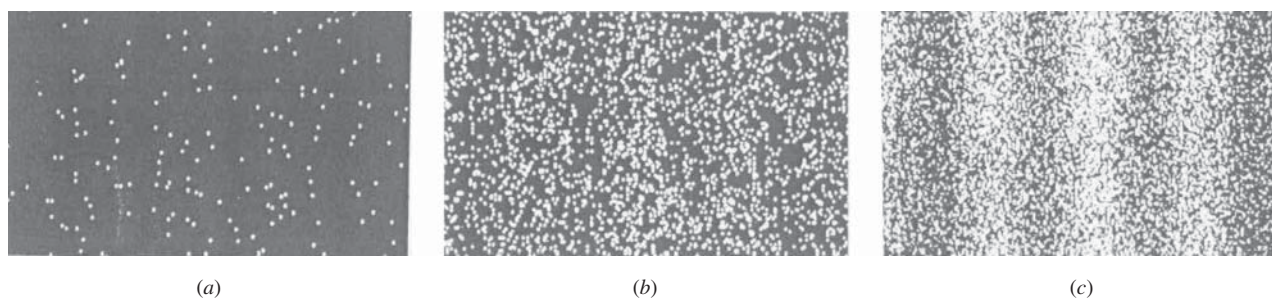


FIGURE 4.27 The buildup of an electron interference pattern as increasing numbers of electrons are detected (a) 100 electrons; (b) 3000 electrons; (c) 70,000 electrons. Reprinted with permission from A. Tonomura, J. Endo, T. Matsuda, and T. Kawasaki, Demonstration of single-electron buildup of an interference pattern, *American Journal of Physics* **57**, 117 (1989). Reproduced with permission from American Association of Physics Teachers publications.

Just as the electric field amplitude of an electromagnetic wave indicates regions of high and low probability for observing photons, the de Broglie wave performs the same function for particles. Figure 4.27 illustrates this effect, as individual electrons in a double-slit type of experiment eventually produce the characteristic interference fringes. The path of each electron is guided by its de Broglie wave toward the allowed regions of high probability. This statistical effect is not apparent for a small number of electrons, but it becomes quite apparent when a large number of electrons has been detected.

In the next chapter, we discuss the mathematical framework for computing the wave amplitudes for a particle in various situations, and we also develop a more rigorous mathematical definition of the probability.

Chapter Summary

Section			Section		
De Broglie wavelength	$\lambda = h/p$	4.1	Heisenberg energy–time uncertainty	$\Delta E \Delta t \sim \hbar$	4.4
Single-slit diffraction	$a \sin \theta = n\lambda$ $n = 1, 2, 3, \dots$	4.2	Statistical momentum uncertainty	$\Delta p_x = \sqrt{(p_x^2)_{\text{av}} - (p_{x,\text{av}})^2}$	4.4
Classical position–wavelength uncertainty	$\Delta x \Delta \lambda \sim \epsilon \lambda^2$	4.3	Wave packet (discrete k)	$y(x) = \sum A_i \cos k_i x$	4.5
Classical frequency–time uncertainty	$\Delta f \Delta t \sim \epsilon$	4.3	Wave packet (continuous k)	$y(x) = \int A(k) \cos kx \, dk$	4.5
Heisenberg position–momentum uncertainty	$\Delta x \Delta p_x \sim \hbar$	4.4	Group speed of wave packet	$v_{\text{group}} = \frac{d\omega}{dk}$	4.6

Questions

- When an electron moves with a certain de Broglie wavelength, does any aspect of the electron's motion vary with that wavelength?
- Imagine a different world in which the laws of quantum physics still apply, but which has $h = 1 \text{ J} \cdot \text{s}$. What might be some of the difficulties of life in such a world? (See

- Mr. Tompkins in Paperback* by George Gamow for an imaginary account of such a world.)
- Suppose that we try to measure an unknown frequency f by listening for beats between f and a known (and controllable) frequency f' . (We assume f' is known to arbitrarily small uncertainty.) The beat frequency is $|f' - f|$. If we hear no beats, then we conclude that $f = f'$. (a) How long must we listen to hear “no” beats? (b) If we hear no beats in 1 s, how accurately have we determined f ? (c) If we hear no beats in 10 s, how accurately? In 100 s? (d) How is this experiment related to Eq. 4.7?
 - What difficulties does the uncertainty principle cause in trying to pick up an electron with a pair of forceps?
 - Does the uncertainty principle apply to nature itself or only to the results of experiments? That is, is it the position and momentum that are *really* uncertain, or merely our knowledge of them? What is the difference between these two interpretations?
 - The uncertainty principle states in effect that the more we try to confine an object, the faster we are likely to find it moving. Is this why you can’t seem to keep money in your pocket for long? Make a numerical estimate.
 - Consider a collection of gas molecules trapped in a container. As we move the walls of the container closer together (compressing the gas) the molecules move faster (the temperature increases). Does the gas behave this way because of the uncertainty principle? Justify your answer with some numerical estimates.
 - Many nuclei are unstable and undergo radioactive decay to other nuclei. The lifetimes for these decays are typically of the order of days to years. Do you expect that the uncertainty principle will cause a measurable effect in the precision to which we can measure the masses of atoms of these nuclei?
 - Just as the classical limit of relativity can be achieved by letting $c \rightarrow \infty$, the classical limit of quantum behavior is achieved by letting $\hbar \rightarrow 0$. Consider the following in the $\hbar \rightarrow 0$ limit and explain how they behave classically: the size of the energy quantum of an electromagnetic wave, the de Broglie wavelength of an electron, the Heisenberg uncertainty relationships.
 - Assume the electron beam in a television tube is accelerated through a potential difference of 25 kV and then passes through a deflecting capacitor of interior width 1 cm. Are diffraction effects important in this case? Justify your answer with a calculation.
 - The structure of crystals can be revealed by X-ray diffraction (Figures 3.7 and 3.8), electron diffraction (Figure 4.2), and neutron diffraction (Figure 4.7). In what ways do these experiments reveal similar structure? In what ways are they different?
 - Often it happens in physics that great discoveries are made inadvertently. What would have happened if Davisson and Germer had their accelerating voltage set below 32 V?
 - Suppose that we cover one slit in the two-slit electron experiment with a very thin sheet of fluorescent material that emits a photon of light whenever an electron passes through. We then fire electrons one at a time at the double slit; whether or not we see a flash of light tells us which slit the electron went through. What effect does this have on the interference pattern? Why?
 - In another attempt to determine through which slit the electron passes, we suspend the double slit itself from a very fine spring balance and measure the “recoil” momentum of the slit as a result of the passage of the electron. Electrons that strike the screen near the center must cause recoils in opposite directions depending on which slit they pass through. Sketch such an apparatus and describe its effect on the interference pattern. (*Hint:* Consider the uncertainty principle $\Delta x \Delta p_x \sim \hbar$ as applied to the motion of the slits suspended from the spring. How precisely do we know the position of the slit?)
 - Is it possible for v_{phase} to be greater than c ? Can v_{group} be greater than c ?
 - In a nondispersive medium, $v_{\text{group}} = v_{\text{phase}}$; this is another way of saying that all waves travel with the same phase velocity, no matter what their wavelengths. Is this true for (a) de Broglie waves? (b) Light waves in glass? (c) Light waves in vacuum? (d) Sound waves in air? What difficulties would be encountered in attempting communication (by speech or by radio signals, for example) in a strongly dispersive medium?

Problems

4.1 De Broglie’s Hypothesis

- Find the de Broglie wavelength of (a) a 7-MeV proton; (b) a 45-GeV electron; (c) an electron moving at $v = 1.35 \times 10^6$ m/s.
- The neutrons produced in a reactor are known as *thermal neutrons*, because their kinetic energies have been reduced (by collisions) until $K = \frac{3}{2}kT$ where T is room

temperature (293 K). (a) What is the kinetic energy of such neutrons? (b) What is their de Broglie wavelength? Because this wavelength is of the same order as the lattice spacing of the atoms of a solid, neutron diffraction (like X-ray and electron diffraction) is a useful means of studying solid lattices.

3. By doing a nuclear diffraction experiment, you measure the de Broglie wavelength of a proton to be 8.29 fm. (a) What is the speed of the proton? (b) Through what potential difference must it be accelerated to achieve that speed?
4. A proton is accelerated from rest through a potential difference of -3.26×10^5 V. What is its de Broglie wavelength?

4.2 Experimental Evidence for De Broglie Waves

5. Through what potential difference must electrons be accelerated to resolve the following particles? (a) A virus of diameter 15 nm; (b) an atom of diameter 0.096 nm; (c) a proton of diameter 1.2 fm.
6. In an electron microscope, we wish to study particles of diameter about $0.10 \mu\text{m}$ (about 1000 times the size of a single atom). (a) What should be the de Broglie wavelength of the electrons? (b) Through what potential difference should the electrons be accelerated to have that de Broglie wavelength?
7. In order to study the atomic nucleus, we would like to observe the diffraction of particles whose de Broglie wavelength is about the same size as the nuclear diameter, about 14 fm for a heavy nucleus such as lead. What kinetic energy should we use if the diffracted particles are (a) electrons, (b) neutrons, and (c) alpha particles ($m = 4 \text{ u}$)?
8. In the double-slit interference pattern for helium atoms (Figure 4.13), the beam of atoms had a kinetic energy of 0.020 eV. (a) What is the de Broglie wavelength of a helium atom with this kinetic energy? (b) Estimate the de Broglie wavelength of the atoms from the fringe spacing in Figure 4.13 and compare your estimate with the value obtained in part (a). The distance from the double slit to the scanning slit is 64 cm.
9. Suppose that we wish to do a double-slit experiment with a beam of the smoke particles of Example 4.1c. Assume that we can construct a double slit whose separation is about the same size as the particles. Estimate the separation between the fringes if the double slit and the screen were on opposite coasts of the United States.
10. In the Davisson–Germer experiment using a Ni crystal, a second-order beam is observed at an angle of 55° . For what accelerating voltage does this occur?
11. A certain crystal is cut so that the rows of atoms on its surface are separated by a distance of 0.352 nm. A beam of electrons is accelerated through a potential difference of 175 V and is incident normally on the surface. If all possible diffraction orders could be observed, at what angles (relative to the incident beam) would the diffracted beams be found?
12. To appreciate the difficulty of observing the double-slit interference pattern for electrons, calculate the spacing of the interference fringes shown in Figure 4.10 as obtained with the apparatus shown in Figure 4.9. The distance between the double slit and the screen is 35 cm.

4.3 Uncertainty Relationships for Classical Waves

13. Suppose that a traveling wave has a speed v (where $v = \lambda f$). Instead of measuring waves over a distance Δx , we stay in one place and count the number of wave crests which pass in a time Δt . Show that Eq. 4.7 is equivalent to Eq. 4.4 for this case.
14. Sound waves travel through air at a speed of 330 m/s. A whistle blast at a frequency of about 1.2 kHz lasts for 2.5 s. (a) Over what distance in space does the “wave train” representing the sound extend? (b) What is the wavelength of the sound? (c) Estimate the precision with which an observer could measure the wavelength. (d) Estimate the precision with which an observer could measure the frequency.
15. A stone tossed into a body of water creates a disturbance at the point of impact which lasts for 5.2 s. The wave speed is 29 cm/s. (a) Over what distance on the surface of the water does the group of waves extend? (b) An observer counts 12 wave crests in the group. Estimate the precision with which the wavelength can be determined.
16. A radar transmitter emits a pulse of electromagnetic radiation with wavelength 0.275 m. The pulses have a duration of $1.27 \mu\text{s}$. The receiver is set to accept a range of frequencies about the central frequency. To what range of frequencies should the receiver be set?
17. Estimate the signal processing time that would be necessary if you want to design a device to measure frequencies to a precision of no worse than 10,000 Hz.
18. A wave train of 36 vibrations covers a length of 148 m. What is the minimum uncertainty that can be measured for the wavelength?

4.4 Heisenberg Uncertainty Relationships

19. The speed of an electron is measured to within an uncertainty of 2.8×10^4 m/s. What is the size of the smallest region of space in which the electron can be confined?
20. An electron is confined to a region of space of the size of an atom (0.1 nm). (a) What is the uncertainty in the momentum of the electron? (b) What is the kinetic energy of an electron with a momentum equal to Δp ? (c) Does this give a reasonable value for the kinetic energy of an electron in an atom?
21. The Σ^* particle has a rest energy of 1385 MeV and a lifetime of 2.0×10^{-23} s. What would be a typical range of outcomes of measurements of the Σ^* rest energy?
22. A pi meson (pion) and a proton can briefly join together to form a Δ particle. A measurement of the energy of the πp system (Figure 4.28) shows a peak at 1236 MeV, corresponding to the rest energy of the Δ particle, with an experimental spread of 120 MeV. What is the lifetime of the Δ ?

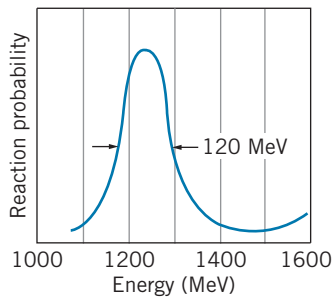


FIGURE 4.28 Problem 22.

23. A nucleus emits a gamma ray of energy 1.2 MeV from a state that has a lifetime of 2.1 ns. What is the uncertainty in the energy of the gamma ray? The best gamma-ray detectors can measure gamma-ray energies to a precision of no better than a few eV. Will this uncertainty be directly measurable?
24. In special conditions (see Section 12.9), it is possible to measure the energy of a gamma-ray photon to 1 part in 10^{15} . For a photon energy of 75 keV, estimate the maximum lifetime that could be determined by a direct measurement of the spread of the photon energy.
25. Alpha particles are emitted in nuclear decay processes with typical energies of 5 MeV. In analogy with Example 4.9, determine the range of kinetic energies of an alpha particle trapped in a nucleus-sized region and deduce whether the alpha particle can exist inside the nucleus.
26. The nucleus of a gold atom has a radius of 7.0 fm. Estimate the kinetic energy of a proton or neutron confined to a gold nucleus.

4.5 Wave Packets

27. Use a distribution of wave numbers of constant amplitude in a range Δk about k_0 :

$$A(k) = A_0 \quad k_0 - \frac{\Delta k}{2} \leq k \leq k_0 + \frac{\Delta k}{2}$$

$$= 0 \quad \text{otherwise}$$

and obtain Eq. 4.24 from Eq. 4.23.

28. Use the distribution of wave numbers $A(k) = A_0 e^{-(k-k_0)^2/2(\Delta k)^2}$ for $k = -\infty$ to $+\infty$ to derive Eq. 4.25.
29. Do the trigonometric manipulation necessary to obtain Eq. 4.18.

4.6 The Motion of a Wave Packet

30. Show that the data used in Figure 4.25 are consistent with Eq. 4.27, that is, use $\lambda_1 = 9$ and $\lambda_2 = 11$, $v_1 = 6$ and $v_2 = 4$ to show $v_{\text{group}} = 15$.
31. (a) Show that the group velocity and phase velocity are related by

$$v_{\text{group}} = v_{\text{phase}} - \lambda \frac{dv_{\text{phase}}}{d\lambda}$$

(b) When white light travels through glass, the phase velocity of each wavelength depends on the wavelength. (This is the origin of dispersion and the breaking up of white light into its component colors—different wavelengths travel at different speeds and have different indices of refraction.) How does v_{phase} depend on λ ? Is $dv_{\text{phase}}/d\lambda$ positive or negative? Therefore, is $v_{\text{group}} > v_{\text{phase}}$ or $< v_{\text{phase}}$?

32. Certain surface waves in a fluid travel with phase velocity $\sqrt{b/\lambda}$, where b is a constant. Find the group velocity of a packet of surface waves, in terms of the phase velocity.
33. By a calculation similar to that of Eq. 4.30, show that $dE/dp = v$ remains valid when E represents the relativistic kinetic energy of the particle.

General Problems

34. A free electron bounces elastically back and forth in one dimension between two walls that are $L = 0.50$ nm apart. (a) Assuming that the electron is represented by a de Broglie standing wave with a node at each wall, show that the permitted de Broglie wavelengths are $\lambda_n = 2L/n$ ($n = 1, 2, 3, \dots$). (b) Find the values of the kinetic energy of the electron for $n = 1, 2$, and 3.
35. A beam of thermal neutrons (see Problem 2) emerges from a nuclear reactor and is incident on a crystal as shown in Figure 4.29. The beam is Bragg scattered, as in Figure 3.5, from a crystal whose scattering planes are separated by 0.247 nm. From the continuous energy spectrum of the beam we wish to select neutrons of energy 0.0105 eV. Find the Bragg scattering angle that results in a scattered beam of this energy. Will other energies also be present in the scattered beam at that angle?

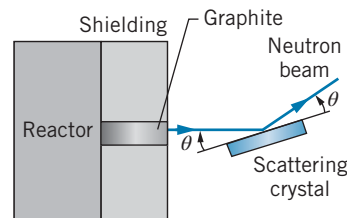


FIGURE 4.29 Problem 35.

36. (a) Find the de Broglie wavelength of a nitrogen molecule in air at room temperature (293 K). (b) The density of air at room temperature and atmospheric pressure is 1.292 kg/m^3 . Find the average distance between air molecules at this temperature and compare with the de Broglie wavelength. What do you conclude about the importance of quantum effects in air at room temperature? (c) Estimate the temperature at which quantum effects might become important.

37. In designing an experiment, you want a beam of photons and a beam of electrons with the same wavelength of 0.281 nm, equal to the separation of the Na and Cl ions in a crystal of NaCl. Find the energy of the photons and the kinetic energy of the electrons.
38. A nucleus of helium with mass 5 u breaks up from rest into a nucleus of ordinary helium (mass = 4 u) plus a neutron (mass = 1 u). The rest energy liberated in the break-up is 0.89 MeV, which is shared (*not* equally) by the products. (a) Using energy and momentum conservation, find the kinetic energy of the neutron. (b) The lifetime of the original nucleus is 1.0×10^{-21} s. What range of values of the neutron kinetic energy might we measure in the laboratory as a result of the uncertainty relationship?
39. In a metal, the conduction electrons are not attached to any one atom, but are relatively free to move throughout the entire metal. Consider a cube of copper measuring 1.0 cm on each edge. (a) What is the uncertainty in any one component of the momentum of an electron confined to the metal? (b) Estimate the average kinetic energy of an electron in the metal. (Assume $\Delta p = [(\Delta p_x)^2 + (\Delta p_y)^2 + (\Delta p_z)^2]^{1/2}$.) (c) Assuming the heat capacity of copper to be 24.5 J/mole · K, would the contribution of this motion to the internal energy of the copper be important at room temperature? What do you conclude from this? (See also Problem 41.)
40. A proton or a neutron can sometimes “violate” conservation of energy by emitting and then reabsorbing a pi meson, which has a mass of 135 MeV/ c^2 . This is possible as long as the pi meson is reabsorbed within a short enough time Δt consistent with the uncertainty principle, (a) Consider $p \rightarrow p + \pi$. By what amount ΔE is energy conservation violated? (Ignore any kinetic energies.) (b) For how long a time Δt can the pi meson exist? (c) Assuming that the pi meson travels at very nearly the speed of light, how far from the proton can it go? (This procedure, as we discuss in Chapter 12, gives us an estimate of the *range* of the nuclear force, because protons and neutrons are held together in the nucleus by exchanging pi mesons.)
41. In a crystal, the atoms are a distance L apart; that is, each atom must be localized to within a distance of at most L . (a) What is the minimum uncertainty in the momentum of the atoms of a solid that are 0.20 nm apart? (b) What is the average kinetic energy of such an atom of mass 65 u? (c) What would a collection of such atoms contribute to the internal energy of a typical solid, such as copper? Is this contribution important at room temperature? (See also Problem 39.)
42. An apparatus is used to prepare an atomic beam by heating a collection of atoms to a temperature T and allowing the beam to emerge through a hole of diameter d in one side of the oven. The beam then travels through a straight path of length L . Show that the uncertainty principle causes the diameter of the beam at the end of the path to be larger than d by an amount of order $Lh/d\sqrt{3mkT}$, where m is the mass of an atom. Make a numerical estimate for typical values of $T = 1500$ K, $m = 7$ u (lithium atoms), $d = 3$ mm, and $L = 2$ m.
43. An electron is trapped in a rectangular region of length $x = 1.25$ nm and width $y = 2.76$ nm. What is the minimum value that might be measured for its kinetic energy?
44. It is possible to construct oscillatory wave packets without using trigonometric functions. Consider the function $y(x) = (64x^6 - 240x^4 + 180x^2 - 15)e^{-x^2}$. Wave packets using polynomials occur in quantum mechanics as solutions to the simple harmonic oscillator and the hydrogen atom, as we discuss later in this text. (a) Sketch this function in the region where it has reasonably large amplitude. (b) What is the width of this wave packet? Make a rough estimate from your sketch. (c) Estimate the average wavelength. (d) Estimate the uncertainty in the wavelength.

THE SCHRÖDINGER EQUATION



Quantum mechanics provides a mathematical framework in which the description of a process often includes different and possibly contradictory outcomes. A favorite illustration of that situation is the case of Schrödinger's cat. The cat is confined in a chamber with a radioactive atom, the decay of which will trigger the release of poison from a vial. Because we don't know exactly when that decay will occur, until an observation of the condition of the cat is made the quantum-mechanical description of the cat must include both "cat alive" and "cat dead" components. Mehau Kulyk / Science Source

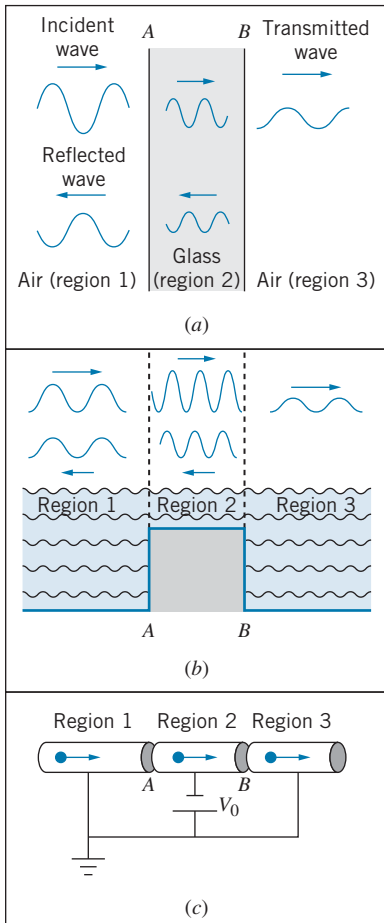


FIGURE 5.1 (a) A light wave in air is incident on a slab of glass, showing transmitted and reflected waves at the two boundaries (A and B). (b) A surface wave in water incident on a region of smaller depth similarly has transmitted and reflected waves. (c) The de Broglie waves of electrons moving from a region of constant zero potential to a region of constant negative potential V_0 also have transmitted and reflected components.

The future behavior of a particle in a classical (nonrelativistic, nonquantum) situation may be predicted with absolute certainty using Newton's laws. If a particle interacts with its environment through a known force $\vec{F}$ (which might be associated with a potential energy U), we can do the mathematics necessary to solve Newton's second law, $\vec{F} = d\vec{p}/dt$ (a second-order, linear differential equation), and find the particle's location $\vec{r}(t)$ and velocity $\vec{v}(t)$ at all future times t . The mathematics may be difficult, and in fact it may not be possible to solve the equations in closed form (in which case an approximate solution can be obtained with the help of a computer). Aside from any such mathematical difficulties, the physics of the problem consists of writing down the original equation $\vec{F} = d\vec{p}/dt$ and interpreting its solutions $\vec{r}(t)$ and $\vec{v}(t)$. For example, a satellite or planet moving under the influence of a $1/r^2$ gravitational force can be shown, after the equations have been solved, to follow exactly an elliptical path.

In the case of nonrelativistic quantum physics, the basic equation to be solved is a second-order differential equation known as the *Schrödinger equation*. Like Newton's laws, the Schrödinger equation is written for a particle interacting with its environment, although we describe the interaction in terms of the potential energy rather than the force. Unlike Newton's laws, the Schrödinger equation does not give the trajectory of the particle; instead, its solution gives the *wave function* of the particle, which carries information about the particle's wavelike behavior. In this chapter, we introduce the Schrödinger equation, obtain some of its solutions for certain potential energies, and learn how to interpret those solutions.

5.1 BEHAVIOR OF A WAVE AT A BOUNDARY

In studying wave motion, we often must analyze what occurs when a wave moves from one region or medium to a different region or medium in which the properties of the wave may change. For example, when a light wave moves from air into glass, its wavelength and the amplitude of its electric field both decrease. At every such boundary, a portion of the incident wave intensity is transmitted into the second medium and a portion is reflected back into the first medium.

Let's consider the case of a light wave incident on a glass plate, as in Figure 5.1a. At boundary A , the light wave moves from air (region 1) into glass (region 2), while at B the light wave moves from glass into air (region 3). The wavelength in air in region 3 is the same as the original wavelength of the incident wave in region 1, but the amplitude in region 3 is less than the amplitude in region 1, because some of the intensity is reflected at A and at B .

Other types of waves show similar behavior. For example, Figure 5.1b shows a surface water wave that moves into a region of shallower depth. In that region, its wavelength is smaller (but its amplitude is larger) compared with the original incident wave. When the wave enters region 3, in which the depth is the same as in region 1, the wavelength returns to its original value, but the amplitude of the wave is smaller in region 3 than in region 1 because some of the intensity was reflected at the two boundaries.

The same type of behavior occurs for de Broglie waves that characterize particles. Consider, for example, the apparatus shown in Figure 5.1c. Electrons are incident from the left and move inside a narrow metal tube that is at ground

potential ($V = 0$). Another narrow tube in region 2 is connected to the negative terminal of a battery, which maintains it at a uniform potential of $-V_0$. Region 3 is connected to region 1 at ground potential. The gaps between the tubes can in principle be made so small that we can regard the changes in potential at A and B as occurring suddenly. In region 1, the electrons have kinetic energy K , momentum $p = \sqrt{2mK}$, and de Broglie wavelength $\lambda = h/p$. In region 2, the potential energy for the electrons is $U = qV = (-e)(-V_0) = +eV_0$. We assume that the original kinetic energy of the electrons in region 1 is greater than eV_0 , so that the electrons move into region 2 with a smaller kinetic energy (equal to $K - eV_0$), a smaller momentum, and thus a greater wavelength. When the electrons move from region 2 into region 3, they gain back the lost kinetic energy and move with their original kinetic energy K and thus with their original wavelength. As in the case of the light wave or the water wave, the amplitude of the de Broglie wave in region 3 is smaller than in region 1, meaning that the current of electrons in region 3 is smaller than the incident current, because some of the electrons are reflected at the boundaries at A and B .

We can thus identify a total of 5 waves moving in the three regions: (1) a wave moving to the right in region 1 (the incident wave); (2) a wave moving to the left in region 1 (representing the net combination of waves reflected from boundary A plus waves reflected from boundary B and then transmitted through boundary A back into region 1); (3) a wave moving to the right in region 2 (representing waves transmitted through boundary A plus waves reflected at B and then reflected again at A); (4) a wave moving to the left in region 2 (waves reflected at B); and (5) a wave moving to the right in region 3 (the transmitted waves at boundary B). Because we are assuming that waves are incident from region 1, it is not possible to have a wave moving to the left in region 3.

Penetration of the Reflected Wave

Another property of classical waves that carries over into quantum waves is penetration of a totally reflected wave into a forbidden region. When a light wave is completely reflected from a boundary, an exponentially decreasing wave called the *evanescent wave* penetrates into the second medium. Because 100% of the light wave intensity is reflected, the evanescent wave carries no energy and so cannot be directly observed in the second medium. But if we make the second medium very thin (perhaps equal to a few wavelengths of light), the light wave can emerge on the opposite side of the second medium. We'll discuss this phenomenon in more detail at the end of this chapter.

The same effect occurs with de Broglie waves. Suppose we increase the battery voltage in Figure 5.1c so that the potential energy in region 2 (equal to eV_0) is greater than the initial kinetic energy in region 1. The electrons do not have enough energy to enter region 2 (they would have negative kinetic there) and so all electrons are reflected back into region 1.

Like light waves, de Broglie waves can also penetrate into the forbidden region with exponentially decreasing amplitudes. However, because de Broglie waves are associated with the motion of electrons, that means that electrons must also penetrate a short distance into the forbidden region. The electrons cannot be directly observed in that region, because they have negative kinetic energy there. Nor can we do any experiment that would reveal their "real"

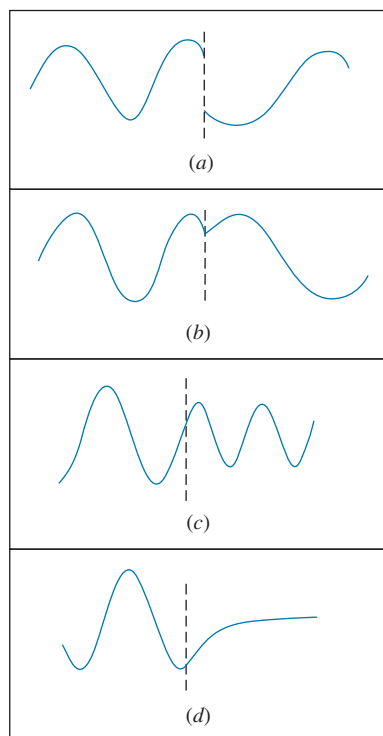


FIGURE 5.2 (a) A discontinuous wave. (b) A continuous wave with a discontinuous slope. (c) Two sine waves join smoothly. (d) A sine wave and an exponential join smoothly.

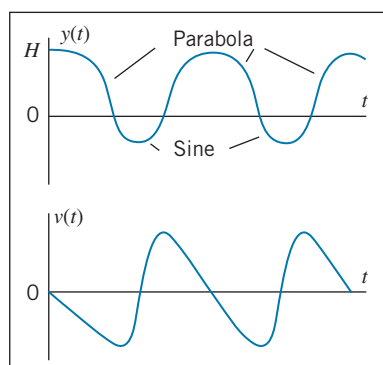


FIGURE 5.3 The position and velocity of a ball dropped from a height H above a springlike rubber sheet at $y = 0$.

existence in the forbidden region, such as measuring the speed of their passage through that region or detecting the magnetic field that their motion might produce.

One explanation for the penetration of the electrons into the forbidden region relies on the uncertainty principle—because we can't know exactly the energy of the incident electrons, we can't say with certainty that they don't have enough kinetic energy to penetrate into the forbidden region. For short enough time Δt , the energy uncertainty $\Delta E \sim \hbar/\Delta t$ might allow the electron to travel a short distance into the forbidden region, but this extra energy does not "belong to" the electron in any permanent sense. Later in this chapter we'll discuss a more mathematical approach to this explanation of penetration into the forbidden region.

Continuity at the Boundaries

When a wave such as a light wave or a water wave crosses a boundary as in Figure 5.1, the mathematical function that describes the wave must have two properties at each boundary:

1. The wave function must be continuous.
2. The slope of the wave function must be continuous, except when the boundary height is infinite.

Figure 5.2a shows a discontinuous wave function; the wave displacement changes suddenly at a single location. This type of behavior is not allowed. Figure 5.2b shows a continuous wave function (there are no gaps) with a discontinuous slope. This type of behavior is also not allowed, unless the boundary is of infinite height. Figures 5.2c,d show how two sine curves and an exponential and a sine can be joined so that both the function and the slope are continuous.

Across any noninfinite boundary, the wave must be smooth—no gaps in the function and no sharp changes in slope. When we solve for the mathematical form of a wave function, there are usually undetermined parameters, such as the amplitude and phase of the wave. In order to make the wave smooth at the boundary, we obtain the values of those coefficients by applying the two *boundary conditions* to make the function and its slope continuous. For example, at boundary A in Figure 5.1, we first evaluate the total wave function in region 1 at A and set it equal to the wave function in region 2 at A . This guarantees that the total wave function is continuous at A . We then take the derivative of the wave function in region 1, evaluate it at A , and set that equal to the derivative of the wave function in region 2 evaluated at A . This step makes the slope in region 1 match the slope in region 2 at boundary A . These two steps give us two equations relating the parameters of the waves and allow us to find relationships between the amplitudes and phases of the waves in regions 1 and 2. The process must be repeated at every boundary, such as at B in Figure 5.1 to match the waves in regions 2 and 3.

We can understand the exception to the continuity of the slope for infinite boundaries with an example from classical physics. Imagine a ball dropped from a height $y = H$ above a stretched rubber sheet at $y = 0$. The ball falls freely under gravity until it strikes the sheet, which we assume behaves like an elastic spring. The sheet stretches as the ball is brought to rest, after which the restoring force propels the ball upward. The motion of the ball might be represented by Figure 5.3. Above the sheet ($y > 0$) the motion is represented

by parabolas, and while the ball is in contact with the sheet ($y < 0$) the motion is described by sine curves. Note how the curves join smoothly at $y = 0$, and note how both $y(t)$ and its derivative $v(t)$ are continuous.

On the other hand, imagine a ball hitting a steel surface, which we assume to be perfectly rigid. The ball rebounds elastically, and at the instant it is in contact with the surface its velocity reverses direction. The motion of the ball is represented in Figure 5.4. At the points of contact with the surface, there is a sudden change in the velocity, corresponding to an infinite acceleration and thus to an infinite force. The function $y(t)$ is continuous, but its slope is not—the function has no gaps, but it does have sharp “points” where the slope changes suddenly.

The assumption of the perfectly rigid surface is an idealization that we make to help us understand the situation and also to help simplify the mathematics. In reality, the steel surface will flex slightly and ultimately behave somewhat like a much stiffer version of the rubber sheet. In quantum mechanics, we will also sometimes use an assumption of a perfectly rigid or impenetrable boundary to help us understand and simplify the analysis of a more complicated physical situation.

In this section, we have established several properties of classical waves that also apply to quantum waves:

1. When a wave crosses a boundary between two regions, part of the wave intensity is reflected and part is transmitted.
2. When a wave encounters a boundary to a region from which it is forbidden, the wave will penetrate perhaps by a few wavelengths before reflecting.
3. At a finite boundary, the wave and its slope are continuous. At an infinite boundary, the wave is continuous but its slope is discontinuous.

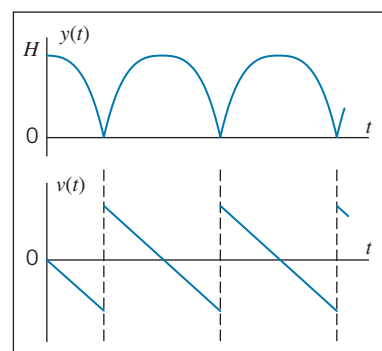


FIGURE 5.4 The position and velocity of a ball dropped from a height H above a rigid surface.

Example 5.1

In the geometry of Figure 5.1, the wave in region 1 is given by $y_1(x) = C_1 \sin(2\pi x/\lambda_1 - \phi_1)$, where $C_1 = 11.5$, $\lambda_1 = 4.97$ cm, and $\phi_1 = -65.3^\circ$. In region 2, the wavelength is $\lambda_2 = 10.5$ cm. The boundary A is located at $x = 0$, and the boundary B is located at $x = L$, where $L = 20.0$ cm. Find the wave functions in regions 2 and 3.

Solution

The general form of the wave in region 2 can be represented in a form similar to that of the wave in region 1: $y_2(x) = C_2 \sin(2\pi x/\lambda_2 - \phi_2)$. To find the complete wave function in region 2, we must find the amplitude C_2 and the phase ϕ_2 by applying the boundary conditions on the function and its slope at boundary A ($x = 0$). Setting $y_1(x = 0) = y_2(x = 0)$ gives

$$-C_1 \sin \phi_1 = -C_2 \sin \phi_2$$

The slopes can be found from the derivative of the general form $dy/dx = (2\pi/\lambda)C \cos(2\pi x/\lambda - \phi)$ evaluated at $x = 0$:

$$\frac{2\pi}{\lambda_1} C_1 \cos \phi_1 = \frac{2\pi}{\lambda_2} C_2 \cos \phi_2$$

Dividing the first equation by the second eliminates C_2 and allows us to solve for ϕ_2 :

$$\begin{aligned} \phi_2 &= \tan^{-1} \left(\frac{\lambda_1}{\lambda_2} \tan \phi_1 \right) \\ &= \tan^{-1} \left(\frac{4.97 \text{ cm}}{10.5 \text{ cm}} \tan(-65.3^\circ) \right) \\ &= -45.8^\circ \end{aligned}$$

We can solve for C_2 using the result from applying the first boundary condition:

$$C_2 = C_1 \frac{\sin \phi_1}{\sin \phi_2} = 11.5 \frac{\sin(-65.3^\circ)}{\sin(-45.8^\circ)} = 14.6$$

To find the wave function in region 3, which we assume to have the same form $y_3(x) = C_3 \sin(2\pi x/\lambda_1 - \phi_3)$, we must apply the boundary conditions on y_2 and y_3 at

$x = L$. Applying the two boundary conditions in the same way we did at $x = 0$, we obtain

$$C_2 \sin\left(\frac{2\pi L}{\lambda_2} - \phi_2\right) = C_3 \sin\left(\frac{2\pi L}{\lambda_1} - \phi_3\right)$$

$$\frac{2\pi}{\lambda_2} C_2 \cos\left(\frac{2\pi L}{\lambda_2} - \phi_2\right) = \frac{2\pi}{\lambda_1} C_3 \cos\left(\frac{2\pi L}{\lambda_1} - \phi_3\right)$$

Proceeding as we did before, we divide these two equations to find $\phi_3 = 60.9^\circ$, and then from either equation obtain $C_3 = 7.36$. Our two solutions are then $y_2(x) = 14.6 \sin(2\pi x/10.5 + 45.8^\circ)$ and $y_3(x) = 7.36 \sin(2\pi x/4.97 + 14.6^\circ)$, with x measured in centimeters. Figure 5.5 shows the wave in all three regions. Note how the waves join smoothly at the boundaries.

How is it possible that the amplitude of y_2 can be greater than the amplitude of y_1 ? Keep in mind that y_1

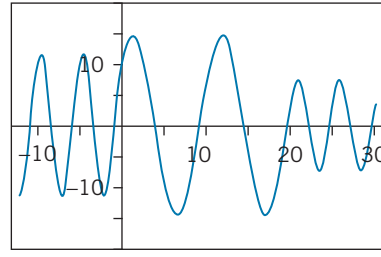


FIGURE 5.5 Example 5.1.

represents the total wave in region 1, which includes the incident wave and the reflected wave. Depending on the phase difference between them, when the incident and reflected waves are added to obtain y_1 , the amplitude of the resultant can be smaller than the amplitude of either wave.

5.2 CONFINING A PARTICLE

A free particle (that is, a particle on which no forces act anywhere) is by definition not confined, so it can be located anywhere. It has, as we discussed in Chapter 4, a definite wavelength, momentum, and energy (for which we can choose any value).

A confined particle, on the other hand, is represented by a wave packet that makes it likely to be found only in a region of space of size Δx . We can construct such a wave packet by adding together different sine or cosine waves to obtain the desired mathematical shape.

In quantum mechanics, we often want to analyze the behavior of confined particles, for example, an electron that is attached to a specific atom or molecule. We'll consider the properties of atomic electrons beginning in Chapter 6, but for now let's look at a simpler problem: an electron moving in one dimension and confined by a series of electric fields. Figure 5.6 shows how the apparatus of Figure 5.1c might be modified for this purpose. The center section is grounded (so that $V = 0$) and the two side sections are connected to batteries so that they are at potentials of $-V_0$ relative to the center section. As before, we assume that the gaps between the center section and the side sections can be made as narrow as possible, so we can regard the potential energy as changing instantaneously at the boundaries A and B . This arrangement is often called a *potential energy well*.

The potential energy of an electron in this situation is then 0 in the center section and $U_0 = qV = (-e)(-V_0) = +eV_0$ in the two side sections as shown in Figure 5.6. To confine the electron, we want to consider cases in which it moves in the center section with a kinetic energy K that is less than U_0 . For example, the electron might have a kinetic energy of 5 eV in the center section, and the side sections might have potential energies of 10 eV. The electron thus does not have enough energy to “climb” the potential energy hill between the

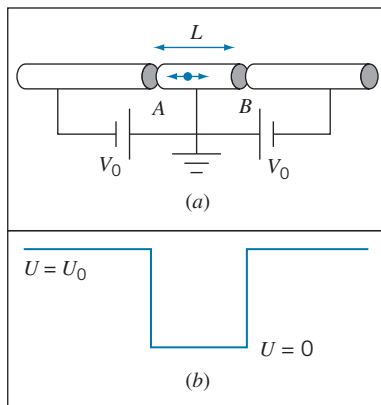


FIGURE 5.6 (a) Apparatus for confining an electron to the center region of length L . (b) The potential energy of an electron in this apparatus.

center section and the side sections, and (at least from the classical point of view) the electron is confined to the center section.

We'll discuss the full solution to this problem later in this chapter, but for now let's simplify even further and consider the case of an infinitely high potential energy barrier at A and B . This is a good approximation to the situation in which the kinetic energy of the electron in the center section is much smaller than the potential energy supplied by the batteries. In this case, the penetration into the forbidden region, which we discussed in Section 5.1, cannot occur. The probability to find the electron in either of the side regions is therefore precisely zero everywhere in those regions, and thus the wave amplitude is zero everywhere in those regions, including at the boundaries (locations A and B). For the wave function to be continuous, the wave function in the center section must have values of zero at A and B .

Of all the possible waves that might be used to describe the particle in this center section, the continuity condition restricts us to waves that have zero amplitude at the boundaries. Some of those waves are illustrated in Figure 5.7. Note that the wave function is continuous, but its slope is not (there are sharp points in the function at locations A and B). This is an example of the exception to the second boundary condition—the slope may be discontinuous at an infinite barrier.

In contrast to the free particle for which the wavelength could have any value, *only certain values of the wavelength are allowed*. The de Broglie relationship then tells us that only certain values of the momentum are allowed, and consequently only certain values of the energy are allowed. The energy is not a continuous variable, free to take on any arbitrary value; instead, the energy is a discrete variable that is restricted to a certain set of values. This is known as *quantization of energy*.

You can see directly from Figure 5.7 that the allowed wavelengths are $2L, L, 2L/3, \dots$, where L is the length of the center section. We can write these wavelengths as

$$\lambda_n = \frac{2L}{n} \quad n = 1, 2, 3, \dots \quad (5.1)$$

This set of wavelengths is identical to the wavelengths of the classical problem of standing waves on a string stretched between two points. From the de Broglie relationship $\lambda = h/p$ we obtain

$$p_n = n \frac{h}{2L} \quad (5.2)$$

The energy of the particle in the center section is only kinetic energy $p^2/2m$, and so

$$E_n = n^2 \frac{h^2}{8mL^2} \quad (5.3)$$

These are the allowed or quantized values of the energy of the electron.

A wave packet describing the electron in this region must be a combination of waves with the allowed values of the wavelengths. However, it is not necessary to construct a wave packet from a combination of waves to describe this confined particle. Even a single one of these waves represents the confined particle, because the wave function must be zero in the forbidden regions. So the waveforms shown in Figure 5.7 can represent wave packets of this confined electron, each wave packet consisting of only a single wave.

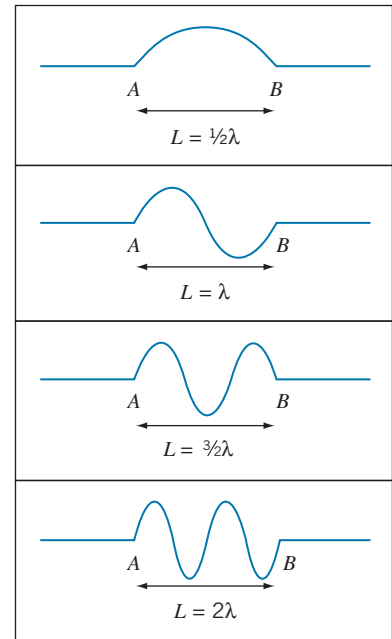


FIGURE 5.7 Some possible waves that might be used to describe an electron confined by an infinite potential energy barrier to a region of length L .

The appearance of energy quantization accompanies every attempt to confine a particle to a finite region of space. Quantization of energy is one of the principal features of the quantum theory, and studying the quantized energy levels of systems (such as by observing the energies of emitted photons) is an important technique of experimental physics that gives us information about the properties of atoms and nuclei.

Applying the Uncertainty Principle to a Confined Particle

In Chapter 4, we constructed wave packets and showed how the uncertainty principle related the size of the wave packet to the range of wavelengths that was used in its construction. Let's now see how the Heisenberg uncertainty relationships apply in the case of a confined particle.

In the arrangement of Figure 5.6 (with infinitely high barriers on each side), the particle is known to be somewhere in the center section of the apparatus, and thus $\Delta x \sim L$ is a reasonable estimate of the uncertainty in its location. To find the uncertainty in its momentum, we use the rigorous definition of uncertainty given in Eq. 4.15: $\Delta p_x = \sqrt{(p_x^2)_{\text{av}} - (p_{x,\text{av}})^2}$. The particle moving in the center section can be considered to be moving to the left or to the right with equal probability (just as the classical standing-wave problem can be analyzed as the superposition of identical waves moving to the left and to the right). Thus $p_{x,\text{av}} = 0$. If the particle is moving with a momentum given by Eq. 5.2, $p_x^2 = (nh/L)^2$ and so $\Delta p_x = nh/L$. Combining the uncertainties in position and momentum, we have

$$\Delta x \Delta p_x \sim L \frac{nh}{L} = nh \quad (5.4)$$

The product of the uncertainties is certainly greater than $\hbar/2$, and so the result of confining the particle is entirely consistent with the Heisenberg uncertainty relationship. Note that even the smallest possible value of the product of the uncertainties (which is obtained for $n = 1$) is still much larger than the minimum value given by the uncertainty principle.

Later in this chapter, we will use a more rigorous way to evaluate the uncertainty in position using a formula similar to Eq. 4.15 to find the uncertainty in position, and we will find that the result does not differ very much from the estimate of Eq. 5.4.

5.3 THE SCHRÖDINGER EQUATION

The differential equation whose solution gives us the wave behavior of particles is called the *Schrödinger equation*. It was developed in 1926 by Austrian physicist Erwin Schrödinger. The equation cannot be derived from any previous laws or postulates; like Newton's equations of motion or Maxwell's equations of electromagnetism, it is a new and independent result whose correctness can be determined only by comparing its predictions with experimental results. For nonrelativistic motion, the Schrödinger equation gives results that correctly account for observations at the atomic and subatomic level.

We can justify the form of the Schrödinger equation by examining the solution expected for the free particle, which should give a wave whose shape at



Erwin Schrödinger (1887–1961, Austria). Although he disagreed with the probabilistic interpretation that was later given to his work, he developed the mathematical theory of wave mechanics that for the first time permitted the wave behavior of physical systems to be calculated.

any particular time, specified by the *wave function* $\psi(x)$, is that of a simple de Broglie wave, such as $\psi(x) = A \sin kx$, where A is the amplitude of the wave and $k = 2\pi/\lambda$. If we are looking for a differential equation, then we need to take some derivatives:

$$\frac{d\psi}{dx} = kA \cos kx, \quad \frac{d^2\psi}{dx^2} = -k^2 A \sin kx = -k^2 \psi(x)$$

Note that the second derivative gives the original function again. With the kinetic energy $K = p^2/2m = (\hbar/\lambda)^2/2m = \hbar^2 k^2/2m$, we can then write

$$\frac{d^2\psi}{dx^2} = -k^2 \psi(x) = -\frac{2m}{\hbar^2} K \psi(x) = -\frac{2m}{\hbar^2} (E - U) \psi(x)$$

where $E = K + U$ is the nonrelativistic total energy of the particle. For a free particle, $U = 0$ so $E = K$; however, we are using the free particle solution to try to extend to the more general case in which there is a potential energy $U(x)$. The equation then becomes

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + U(x)\psi(x) = E\psi(x) \quad (5.5)$$

Equation 5.5 is the *time-independent Schrödinger equation* for one-dimensional motion.

The solution to Eq. 5.5 gives the shape of the wave at time $t = 0$. The mathematical function that describes a one-dimensional *traveling* wave must involve both x and t . This wave is represented by the function $\Psi(x, t)$:

$$\Psi(x, t) = \psi(x)e^{-i\omega t} \quad (5.6)$$

The time dependence is given by the complex exponential function $e^{-i\omega t}$ with $\omega = E/\hbar$. (You can find a few useful formulas involving complex numbers in Appendix B.) We'll discuss the time-dependent part later in this chapter. For now, we'll concentrate on the time-independent function $\psi(x)$.

We assume that we know the potential energy $U(x)$, and we wish to obtain the wave function $\psi(x)$ and the energy E for that potential energy. This is a general example of a type of problem known as an *eigenvalue* problem; we find that it is possible to obtain solutions to the equation only for particular values of E , which are known as the *energy eigenvalues*.

The general procedure for solving the Schrödinger equation is as follows:

1. Begin by writing Eq. 5.5 with the appropriate $U(x)$. Note that if the potential energy changes discontinuously [$U(x)$ may be represented by a discontinuous function; $\psi(x)$ may *not*], we may need to write different equations for different regions of space. Examples of this sort are given in Section 5.4.
2. Using general mathematical techniques suited to the form of the equation, find a mathematical function $\psi(x)$ that is a solution to the differential equation. Because there is no one specific technique for solving differential equations, we will study several examples to learn how to find solutions.
3. In general, several solutions may be found. By applying boundary conditions, some of these may be eliminated and some arbitrary constants may be determined. It is generally the application of the boundary conditions that selects out the allowed energies.

4. If you are seeking solutions for a potential energy that changes discontinuously, you must apply the continuity conditions on $\psi(x)$ (and usually on $d\psi/dx$) at the boundary between different regions.

Because the Schrödinger equation is linear, any constant multiplying a solution is also a solution. The method to determine the amplitude of the wave function is discussed in the next section.

Probabilities and Normalization

The remaining steps in the procedure for applying the Schrödinger recipe depend on the physical interpretation of the solution to the differential equation. Our original goal in solving the Schrödinger equation was to obtain the wave properties of the particle. What does the amplitude of $\psi(x)$ represent, and what is the physical variable that is waving? It is certainly not a displacement, as in the case of a water wave or a wave on a stretched piano wire, nor is it a pressure wave, as in the case of sound. *It is a very different kind of wave, whose squared absolute amplitude gives the probability for finding the particle in a given region of space.*

If we define $P(x)$ as the *probability density* (probability per unit length, in one dimension), then according to the Schrödinger recipe

$$P(x) dx = |\psi(x)|^2 dx \quad (5.7)$$

as indicated in Figure 5.8. In Eq. 5.7, $|\psi(x)|^2 dx$ gives the probability to find the particle in the interval dx at x (that is, between x and $x + dx$).^{*} Because the wave function $\psi(x)$ might be a complex function, it is necessary to square its absolute magnitude to make sure that the probability is a positive real number.

The squared magnitude of the general time-dependent wave function (Eq. 5.6) is

$$|\Psi(x, t)|^2 = |\psi(x)|^2 |e^{-i\omega t}|^2 = |\psi(x)|^2 \quad (5.8)$$

where the last step can be taken because the magnitude of the time-dependent factor is 1. For this reason, the probability density associated with a solution to the Schrödinger equation (for any allowed value of E) is independent of time. These special quantum states are called *stationary states*.

This interpretation of $|\psi(x)|^2$ helps us to understand the continuity condition of $\psi(x)$. We must not allow the probability to change discontinuously, but, like any well-behaved wave, the probability to locate the particle varies smoothly and continuously.

This interpretation of $\psi(x)$ now permits us to complete the Schrödinger recipe and to illustrate how to use the wave function to calculate quantities that we can measure in the laboratory. Steps 1 through 4 were given previously; the recipe continues:

5. For a wave function describing a single particle, the probability summed over all locations must give 100%—that is, the particle must be located *somewhere* between $x = -\infty$ and $x = +\infty$. The probability to find the

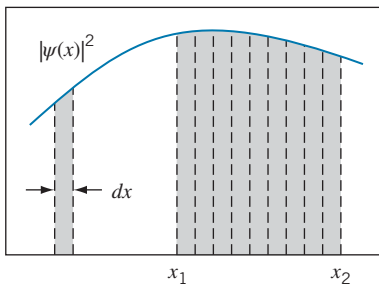


FIGURE 5.8 The probability to find the particle in a small region of width dx is equal to the area of the strip under the $|\psi(x)|^2$ curve. The total probability to find the particle between x_1 and x_2 is the sum of the areas of the strips, equal to the integral between those limits.

^{*}It is not correct to speak of “the probability to find the particle at the point x .” A single point is a mathematical abstraction with no physical dimension. The probability of finding a particle *at a point* is zero, but there can be a nonzero probability of finding the particle *in an interval*.

particle in a small interval was given in Eq. 5.7. The total probability to find the particle in all such intervals must be exactly 1:

$$\int_{-\infty}^{+\infty} |\psi(x)|^2 dx = 1 \quad (5.9)$$

The Schrödinger equation is linear, which means that if $\psi(x)$ is a solution then any constant times $\psi(x)$ is also a solution. For the probability to be a meaningful concept, this constant must be chosen so that Eq. 5.9 is satisfied. A wave function with its multiplicative constant chosen in this way is said to be *normalized*, and Eq. 5.9 is known as the *normalization condition*.

6. Because the solution to the Schrödinger equation represents a probability, any solution that becomes infinite must be discarded—it makes no sense to have an infinite probability to find a particle in any interval. In practice, we “discard” a solution by setting its multiplicative constant equal to zero. For example, if the mathematical solution to the differential equation yields $\psi(x) = Ae^{kx} + Be^{-kx}$ for the *entire* region $x > 0$, then we must require $A = 0$ for the solution to be physically meaningful; otherwise $|\psi(x)|^2$ would become infinite as x goes to infinity. On the other hand, if this solution is to be valid in the *entire* region $x < 0$, then we must set $B = 0$. However, if the solution is to be valid only in a small portion of the range of x —say, $0 < x < L$ —then we cannot set either $A = 0$ or $B = 0$.
7. Suppose the interval between two points x_1 and x_2 is divided into a series of infinitesimal intervals of width dx (Figure 5.8). To find the total probability for the particle to be located between x_1 and x_2 , which we represent as $P(x_1 : x_2)$, we calculate the sum of all the probabilities $P(x) dx$ in each interval dx . This sum can be expressed as an integral:

$$P(x_1 : x_2) = \int_{x_1}^{x_2} P(x) dx = \int_{x_1}^{x_2} |\psi(x)|^2 dx \quad (5.10)$$

If the wave function has been properly normalized, Eq. 5.10 will always yield a probability that lies between 0 and 1.

8. Because we can no longer speak with certainty about the position of the particle, we can no longer guarantee the outcome of a single measurement of any physical quantity that depends on its position. Instead, we can find the *average* outcome of a large number of measurements. For example, suppose we wish to find the average location of a particle by measuring its coordinate x . From a large number of measurements, we find the value x_1 a certain number of times n_1 , x_2 a number of times n_2 , etc., and in the usual way we can find the average value

$$x_{av} = \frac{n_1 x_1 + n_2 x_2 + \cdots}{n_1 + n_2 + \cdots} = \frac{\sum n_i x_i}{\sum n_i} \quad (5.11)$$

The number of times n_i that we measure each x_i is proportional to the probability $P(x_i)dx$ to find the particle in the interval dx at x_i . Making this substitution and changing the sums to integrals, we have

$$x_{av} = \frac{\int_{-\infty}^{+\infty} P(x)x dx}{\int_{-\infty}^{+\infty} P(x) dx} = \frac{\int_{-\infty}^{+\infty} |\psi(x)|^2 x dx}{\int_{-\infty}^{+\infty} |\psi(x)|^2 dx} \quad (5.12)$$

where the last step can be made if the wave function is normalized, in which case the denominator of Eq. 5.12 is equal to 1.

By analogy, the average value of any function of x can be found:

$$[f(x)]_{\text{av}} = \int_{-\infty}^{+\infty} P(x)f(x) dx = \int_{-\infty}^{+\infty} |\psi(x)|^2 f(x) dx \quad (5.13)$$

Average values calculated according to Eq. 5.12 or 5.13 are known as *expectation values*.

5.4 APPLICATIONS OF THE SCHRÖDINGER EQUATION

Solutions for Constant Potential Energy

First let's examine the solutions to the Schrödinger equation for the special case of a constant potential energy, equal to U_0 . Then Eq. 5.5 becomes

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + U_0\psi(x) = E\psi(x) \quad (5.14)$$

or (assuming for now that $E > U_0$)

$$\frac{d^2\psi}{dx^2} = -k^2\psi(x) \quad \text{with} \quad k = \sqrt{\frac{2m(E - U_0)}{\hbar^2}} \quad (5.15)$$

The parameter k in this equation is equal to the wave number $2\pi/\lambda$.

The solution to Eq. 5.15 is a function of x that, when differentiated twice, gives back the original function multiplied by the negative constant $-k^2$. The function that has this property is $\sin kx$ or $\cos kx$. The most general solution to the equation is

$$\psi(x) = A \sin kx + B \cos kx \quad (5.16)$$

The constants A and B must be determined by applying the continuity and normalization requirements. We can demonstrate that Eq. 5.16 satisfies Eq. 5.15 by taking two derivatives:

$$\begin{aligned} \frac{d\psi}{dx} &= kA \cos kx - kB \sin kx \\ \frac{d^2\psi}{dx^2} &= -k^2 A \sin kx - k^2 B \cos kx = -k^2(A \sin kx + B \cos kx) = -k^2\psi(x) \end{aligned}$$

so the original equation is indeed satisfied.

To analyze the penetration of a particle into a forbidden region, we must consider the case in which the energy E of the particle is smaller than the potential energy U_0 . For this case we can rewrite Eq. 5.14 as

$$\frac{d^2\psi}{dx^2} = k'^2\psi(x) \quad \text{with} \quad k' = \sqrt{\frac{2m(U_0 - E)}{\hbar^2}} \quad (5.17)$$

In this case, the general solution in the forbidden region is

$$\psi(x) = Ae^{k'x} + Be^{-k'x} \quad (5.18)$$

Once again, we can demonstrate that Eq. 5.18 is a solution of Eq. 5.17 by taking two derivatives:

$$\begin{aligned}\frac{d\psi}{dx} &= k' A e^{k'x} - k' B e^{-k'x} \\ \frac{d^2\psi}{dx^2} &= k'^2 A e^{k'x} + k'^2 B e^{-k'x} = k'^2 (A e^{k'x} + B e^{-k'x}) = k'^2 \psi(x)\end{aligned}$$

We will use Eqs. 5.16 and 5.18 as our solutions to the Schrödinger equation for constant potential energy in the allowed ($E > U_0$) and forbidden ($E < U_0$) regions.

The Free Particle

For a free particle, the force is zero and so the potential energy is constant. We may choose any value for that constant, so for convenience we'll choose $U_0 = 0$. The solution is given by Eq. 5.16, $\psi(x) = A \sin kx + B \cos kx$. The energy of the particle is

$$E = \frac{\hbar^2 k^2}{2m} \quad (5.19)$$

Our solution has placed no restrictions on k , so the energy is permitted to have any value (in the language of quantum physics, we say that the energy is *not* quantized). We note that Eq. 5.19 is the kinetic energy of a particle with momentum $p = \hbar k$ or, equivalently, $p = h/\lambda$. This is as we would have expected, because the free particle can be represented by a de Broglie wave with any wavelength.

Solving for A and B presents some difficulties because the normalization integral, Eq. 5.9, cannot be evaluated from $-\infty$ to $+\infty$ for this wave function. We therefore cannot determine probabilities for the free particle from the wave function of Eq. 5.16.

It is also instructive to write the wave function in terms of complex exponentials, using $\sin kx = (e^{ikx} - e^{-ikx})/2i$ and $\cos kx = (e^{ikx} + e^{-ikx})/2$:

$$\psi(x) = A \left(\frac{e^{ikx} - e^{-ikx}}{2i} \right) + B \left(\frac{e^{ikx} + e^{-ikx}}{2} \right) = A' e^{ikx} + B' e^{-ikx} \quad (5.20)$$

where $A' = A/2i + B/2$ and $B' = -A/2i + B/2$. To interpret this solution in terms of waves, we form the complete time-dependent wave function using Eq. 5.6:

$$\Psi(x,t) = (A' e^{ikx} + B' e^{-ikx}) e^{-i\omega t} = A' e^{i(kx - \omega t)} + B' e^{-i(kx + \omega t)} \quad (5.21)$$

The dependence of the first term on $kx - \omega t$ identifies this term as representing a wave moving to the right (in the positive x direction) with amplitude A' , and the second term involving $kx + \omega t$ represents a wave moving to the left (in the negative x direction) with amplitude B' .

If we want the wave to represent a beam of particles moving in the $+x$ direction, then we must set $B' = 0$. The probability density associated with this wave is then, according to Eq. 5.7,

$$P(x) = |\psi(x)|^2 = |A'|^2 e^{ikx} e^{-ikx} = |A'|^2 \quad (5.22)$$

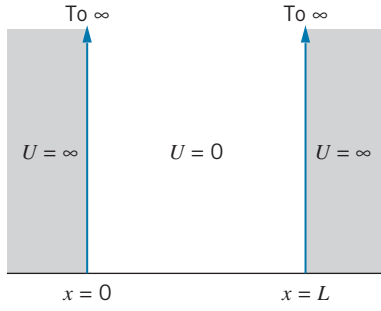


FIGURE 5.9 The potential energy of a particle that moves freely ($U = 0$) in the region $0 \leq x \leq L$ but is completely excluded ($U = \infty$) from the regions $x < 0$ and $x > L$.

The probability density is constant, meaning the particles are equally likely to be found anywhere along the x axis. This is consistent with our discussion of the free-particle de Broglie wave in Chapter 4—a wave of precisely defined wavelength extends from $x = -\infty$ to $x = +\infty$ and thus gives a completely unlocalized particle.

Infinite Potential Energy Well

Now we'll consider the formal solution to the problem we discussed in Section 5.2: a particle is trapped in the region between $x = 0$ and $x = L$ by infinitely high potential energy barriers. Imagine an apparatus like that of Figure 5.6, in which the particle moves freely in this region and makes elastic collisions with the perfectly rigid barriers that confine it. This problem is sometimes called “a particle in a box.” For now we'll assume that the particle moves in only one dimension; later we'll expand to two and three dimensions.

The potential energy may be expressed as

$$U(x) = \begin{cases} 0 & 0 \leq x \leq L \\ \infty & x < 0, x > L \end{cases} \quad (5.23)$$

The potential energy is shown in Figure 5.9. We are free to choose any constant value for U in the region $0 \leq x \leq L$; we choose it to be zero for convenience.

Because the potential energy is different in the regions inside and outside the well, we must find separate solutions in each region. We can analyze the outside region in either of two ways. If we examine Eq. 5.5 for the region outside the well, we find that the only way to keep the equation from becoming meaningless when $U \rightarrow \infty$ is to require $\psi = 0$, so that $U\psi$ will not become infinite. Alternatively, we can go back to the original statement of the problem. If the walls at the boundaries of the well are perfectly rigid, the particle must always be in the well, and the probability for finding it outside must be zero. To make the probability zero everywhere outside the well, we must make $\psi = 0$ everywhere outside. Thus we have

$$\psi(x) = 0 \quad x < 0, x > L \quad (5.24)$$

The Schrödinger equation for $0 \leq x \leq L$, when $U(x) = 0$, is identical with Eq. 5.14 with $U_0 = 0$ and has the same solution:

$$\psi(x) = A \sin kx + B \cos kx \quad 0 \leq x \leq L \quad (5.25)$$

with

$$k = \sqrt{\frac{2mE}{\hbar^2}} \quad (5.26)$$

Our solution is not yet complete, for we have not evaluated A or B , nor have we found the allowed values of the energy E . To do this, we must apply the requirement that $\psi(x)$ is continuous across any boundary. In this case, we require that our solutions for $x < 0$ and $x > 0$ match up at $x = 0$; similarly, the solutions for $x > L$ and $x < L$ must match at $x = L$.

Let us begin at $x = 0$. At $x < 0$, we have found that $\psi = 0$, and so we must set $\psi(x)$ of Eq. 5.25 to zero at $x = 0$.

$$\psi(0) = A \sin 0 + B \cos 0 = 0 \quad (5.27)$$

which gives $B = 0$. Because $\psi = 0$ for $x > L$, the second boundary condition is $\psi(L) = 0$, so

$$\psi(L) = A \sin kL + B \cos kL = 0 \quad (5.28)$$

We have already found $B = 0$, so we must now have $A \sin kL = 0$. Either $A = 0$, in which case $\psi = 0$ everywhere, $\psi^2 = 0$ everywhere, and there is no particle (a meaningless solution) or else $\sin kL = 0$, which is true only when $kL = \pi, 2\pi, 3\pi, \dots$, or

$$kL = n\pi \quad n = 1, 2, 3, \dots \quad (5.29)$$

With $k = 2\pi/\lambda$, we have $\lambda = 2L/n$; this is identical with the result obtained in introductory mechanics for the wavelengths of the standing waves in a string of length L fixed at both ends, which we already obtained in Section 5.2 (Eq. 5.1). *Thus the solution to the Schrödinger equation for a particle trapped in a linear region of length L is a series of standing de Broglie waves!* Not all wavelengths are permitted; only certain values, determined from Eq. 5.29, may occur.

From Eq. 5.26 we find that, because only certain values of k are permitted by Eq. 5.29, only certain values of E may occur—the energy is quantized! Solving Eq. 5.29 for k and substituting into Eq. 5.26, we obtain

$$E_n = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2 \pi^2 n^2}{2mL^2} = \frac{h^2 n^2}{8mL^2} \quad n = 1, 2, 3, \dots \quad (5.30)$$

For convenience, let $E_0 = \hbar^2 \pi^2 / 2mL^2 = h^2 / 8mL^2$; this unit of energy is determined by the mass of the particle and the width of the well. Then $E_n = n^2 E_0$, and the only allowed energies for the particle are $E_0, 4E_0, 9E_0, 16E_0$, etc. All intermediate values, such as $3E_0$ or $6.2E_0$, are forbidden. Figure 5.10 shows the allowed energy levels. The lowest energy state, for which $n = 1$, is known as the *ground state*, and the states with higher energies ($n > 1$) are known as *excited states*.

Because the energy is purely kinetic in this case, our result means that only certain speeds are permitted for the particle. This is very different from the case of the classical trapped particle, in which the particle can be given any initial velocity and will move forever, back and forth, at the same speed. In the quantum case, this is not possible; only certain initial speeds can result in sustained states of motion; these special conditions represent the “stationary states.” Average values calculated according to Eq. 5.13 likewise do not change with time.

From one energy state, the particle can make jumps or transitions to another energy state by absorbing or releasing an amount of energy equal to the energy difference between the two states. By absorbing energy the particle will move to a higher energy state, and by releasing energy it moves to a lower energy state. A similar effect occurs for electrons in atoms, in which the absorbed or released energy is usually in the form of a photon of visible light or other electromagnetic radiation. For example, from the state with $n = 3$ ($E_3 = 9E_0$), the particle might absorb an energy of $\Delta E = 7E_0$ and jump upward to the $n = 4$ state ($E_4 = 16E_0$) or might release energy of $\Delta E = 5E_0$ and jump downward to the $n = 2$ state ($E_2 = 4E_0$).

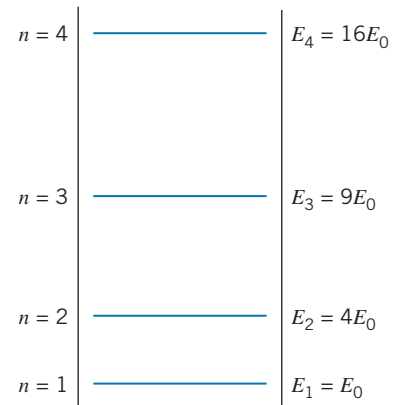


FIGURE 5.10 The first four energy levels in a one-dimensional infinite potential energy well.

Example 5.2

An electron is trapped in a one-dimensional region of length 1.00×10^{-10} m (a typical atomic diameter). (a) Find the energies of the ground state and first two excited states. (b) How much energy must be supplied to excite the electron from the ground state to the second excited state? (c) From the second excited state, the electron drops down to the first excited state. How much energy is released in this process?

Solution

(a) The basic quantity of energy needed for this calculation is

$$E_0 = \frac{h^2}{8mL^2} = \frac{(hc)^2}{8mc^2L^2}$$

$$= \frac{(1240 \text{ eV} \cdot \text{nm})^2}{8(511,000 \text{ eV})(0.100 \text{ nm})^2} = 37.6 \text{ eV}$$

With $E_n = n^2 E_0$, we can find the energy of the states:

$$\begin{aligned} n = 1: & \quad E_1 = E_0 = 37.6 \text{ eV} \\ n = 2: & \quad E_2 = 4E_0 = 150.4 \text{ eV} \\ n = 3: & \quad E_3 = 9E_0 = 338.4 \text{ eV} \end{aligned}$$

(b) The energy difference between the ground state and the second excited state is

$$\Delta E = E_3 - E_1 = 338.4 \text{ eV} - 37.6 \text{ eV} = 300.8 \text{ eV}$$

This is the energy that must be absorbed for the electron to make this jump.

(c) The energy difference between the second and first excited states is

$$\Delta E = E_3 - E_2 = 338.4 \text{ eV} - 150.4 \text{ eV} = 188.0 \text{ eV}$$

This is the energy that is released when the electron makes this jump.

To complete the solution for $\psi(x)$, we must determine the constant A by using the normalization condition given in Eq. 5.9, $\int_{-\infty}^{+\infty} |\psi(x)|^2 dx = 1$. The integrand is zero in the regions $-\infty < x \leq 0$ and $L \leq x < +\infty$, so all that remains is

$$\int_0^L A^2 \sin^2 \frac{n\pi x}{L} dx = 1 \quad (5.31)$$

from which we find $A = \sqrt{2/L}$. The complete wave function for $0 \leq x \leq L$ is then

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} \quad n = 1, 2, 3, \dots \quad (5.32)$$

In Figure 5.11, the wave functions and probability densities ψ^2 are illustrated for the lowest several states.

In the ground state, the particle has the greatest probability to be found near the middle of the well ($x = L/2$), and the probability falls off to zero between the center and the sides of the well. This is very different from the behavior of a classical particle—a classical particle moving at constant speed would be found with equal probability at every location inside the well. The quantum particle also has constant speed but yet is still found with differing probability at various locations in the well. It is the wave nature of the quantum particle that is responsible for this very nonclassical behavior.

Another example of nonclassical behavior occurs for the first excited state. The probability density has a maximum at $x = L/4$ and another maximum at $x = 3L/4$. Between the two maxima, there is zero probability to find the particle in the center of the well at $x = L/2$. How can the particle travel from $x = L/4$ to $x = 3L/4$ without ever being at $x = L/2$? Of course, no classical

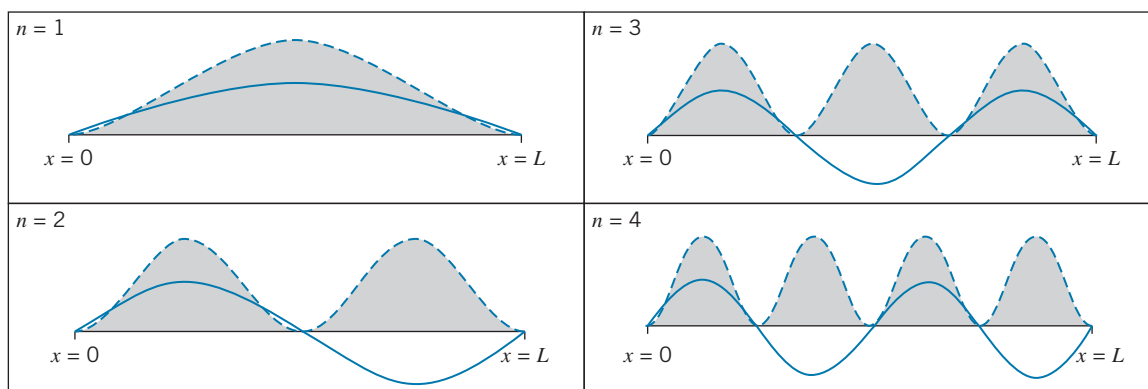


FIGURE 5.11 The wave functions (solid lines) and probability densities (shaded regions) of the first four states in the one-dimensional infinite potential energy well.

particle could behave in such a way, but it is a common behavior for waves. For example, the first overtone of a vibrating string has a node at its midpoint and antinodes (vibrational maxima) at the $1/4$ and $3/4$ locations.

Example 5.3

A particle is in the ground state of an infinite potential energy well of width L . One of the walls of the well is *slowly* moved from $x = L$ to $x = 2L$, after which the particle is found to be in the ground state of the new potential energy well. Suppose instead the wall is *suddenly* moved from $x = L$ to $x = 2L$. (Here “suddenly” means in a time short compared with the time it takes the particle to travel across the well.) In what state of the new well would the particle be found?

Solution

The particle will try to keep the same pattern of nodes and antinodes in the new well. The locations of the nodes

and antinodes in the ground state of the original well match up with those of the first excited state in the new well (for example, antinodes at $x = L/2$ and $x = 3L/2$), so that is where the particle will go. Note that the energy of the $n = 1$ (ground) state of the original well is the same as the energy of the first excited ($n = 2$) state in the new well, so the energy of the particle does not change.

A similar effect can be observed in a plucked guitar string with a finger fretting it at its center. If the finger slides slowly down the string, the pitch of the tone gradually decreases, but if the finger is removed suddenly, the pitch remains the same.

Example 5.4

Consider again an electron trapped in a one-dimensional region of length $1.00 \times 10^{-10} \text{ m} = 0.100 \text{ nm}$. (a) In the ground state, what is the probability of finding the electron in the region from $x = 0.0090 \text{ nm}$ to 0.0110 nm ? (b) In the first excited state, what is the probability of finding the electron between $x = 0$ and $x = 0.025 \text{ nm}$?

Solution

(a) When the interval is small, it is often simpler to use Eq. 5.7 to find the probability, instead of using the integration method. The width of the small interval is

$dx = 0.0110 \text{ nm} - 0.0090 \text{ nm} = 0.0020 \text{ nm}$. Evaluating the wave function at the midpoint of the interval ($x = 0.0100 \text{ nm}$), we can use the $n = 1$ wave function with Eq. 5.7 to find

$$\begin{aligned} P(x) dx &= |\psi_1(x)|^2 dx = \frac{2}{L} \sin^2 \frac{\pi x}{L} dx \\ &= \frac{2}{0.100 \text{ nm}} \sin^2 \frac{\pi(0.0100 \text{ nm})}{0.100 \text{ nm}} (0.002 \text{ nm}) \\ &= 0.0038 = 0.38\% \end{aligned}$$

(b) For this wide interval, we must use the integration method to find the probability:

$$\begin{aligned} P(x_1 : x_2) &= \int_{x_1}^{x_2} |\psi_2(x)|^2 dx \\ &= \frac{2}{L} \int_{x_1}^{x_2} \sin^2 \frac{2\pi x}{L} dx \\ &= \left(\frac{x}{L} - \frac{1}{4\pi} \sin \frac{4\pi x}{L} \right) \Big|_{x_1}^{x_2} \end{aligned}$$

Evaluating this expression using the limits $x_1 = 0$ and $x_2 = 0.025$ nm gives a probability of 0.25% or 25%. This result is of course what we would expect by inspection of the graph of ψ^2 for $n = 2$ in Figure 5.11. The interval from $x = 0$ to $x = L/4$ contains 25% of the total area under the ψ^2 curve.

Example 5.5

Show that the average value of x is $L/2$, independent of the quantum state.

Solution

We use Eq. 5.12; because $\psi = 0$ except for $0 \leq x \leq L$, the limits of integration are 0 and L :

$$x_{\text{av}} = \int_0^L |\psi(x)|^2 x dx = \frac{2}{L} \int_0^L \sin^2 \frac{n\pi x}{L} x dx$$

This can be integrated by parts or found in integral tables; the result is

$$x_{\text{av}} = \frac{L}{2}$$

Note that, as required, this result is independent of n . Thus a measurement of the average position of the particle yields no information about its quantum state.

Let's now look at how the uncertainty principle applies to the motion of this trapped particle. By solving Problems 38 and 39, you will find that the uncertainties in position and momentum for a particle in an infinite potential well are $\Delta x = L\sqrt{1/12 - 1/2\pi^2 n^2}$ and $\Delta p = hn/2L$. The product of the uncertainties is

$$\Delta x \Delta p = \frac{hn}{2} \sqrt{\frac{1}{12} - \frac{1}{2\pi^2 n^2}} = \frac{h}{2} \sqrt{\frac{n^2}{12} - \frac{1}{2\pi^2}}$$

Clearly, the product of the uncertainties grows as n grows. The minimum value occurs for $n = 1$, in which case $\Delta x \Delta p = 0.090h = 0.57\hbar$. The ground state represents a fairly “compact” wave packet, but it is somewhat less compact than the minimum possible limit of $0.50\hbar$ (Eq. 4.10). You can see from Figure 5.11 how the wave becomes less compact (spreads out more) as n increases. Even for $n = 2$, the product of the uncertainties grows quickly to $1.67\hbar$.

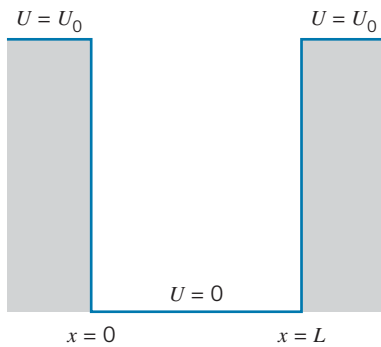


FIGURE 5.12 The potential energy of a particle that is confined to the region $0 \leq x \leq L$ by finite barriers U_0 at $x = 0$ and $x = L$.

Finite Potential Energy Well

Because the infinite potential energy well is an idealization of a technique for confining a particle, we should examine the solution when the barriers at the sides of the well are finite rather than infinite. The potential energy well can be described by

$$\begin{aligned} U(x) &= 0 & 0 \leq x \leq L \\ &= U_0 & x < 0, x > L \end{aligned} \quad (5.33)$$

and is sketched in Figure 5.12. We look for solutions in which the particle is confined to this well, and thus the energies that we deduce for the particle must be less than U_0 .

The solution in the center region (between $x = 0$ and $x = L$) is exactly the same as it was for the infinite well (Eq. 5.25):

$$\psi(x) = A \sin kx + B \cos kx \quad (0 \leq x \leq L) \quad (5.34)$$

although the values that we deduced previously for the coefficients A and B are not valid in this calculation. The region $x < 0$ is an example of a situation in which the energy E of the particle is less than the potential energy U_0 , and so we must use the solution in the form of Eq. 5.18, $\psi(x) = Ce^{k'x} + De^{-k'x}$ with k' given in Eq. 5.17. This region includes $x = -\infty$, for which the term with the coefficient D becomes infinite. Because we cannot allow the probability to become infinite, we must discard this term by setting $D = 0$. The solution for $x < 0$ is then

$$\psi(x) = Ce^{k'x} \quad (x < 0) \quad (5.35)$$

In the region $x > L$, the energy E is once again smaller than U_0 , and so the solution is also in the form of Eq. 5.18, $\psi(x) = Fe^{k'x} + Ge^{-k'x}$. Here the region now includes $x = +\infty$, for which the term with the coefficient F would become infinite. We must prevent that possibility by setting $F = 0$, so the solution in this region is

$$\psi(x) = Ge^{-k'x} \quad (x > L) \quad (5.36)$$

We now have four coefficients to determine (A, B, C, G) along with the energy E . For this determination, we have four equations from the boundary conditions (the continuity of both ψ and $d\psi/dx$ at both $x = 0$ and $x = L$) and one equation from the normalization condition. As you might imagine, solving five equations in five unknowns presents a straightforward but very tedious algebraic challenge. Moreover, the resulting solution for the energy values cannot be obtained in terms of a direct equation such as Eq. 5.30, but instead must be found numerically by solving a transcendental equation. The result is a series of increasing energy values, but the number of energy values is finite rather than infinite, because the energy cannot be allowed to exceed the value of U_0 .

As we did for the infinite potential energy well in Example 5.2, let's consider a well of width $L = 0.100$ nm. We'll choose the depth of the well to be $U_0 = 400$ eV. Applying the boundary conditions at $x = 0$ and $x = L$, we can eliminate all of the coefficients and find an equation that involves only k and k' (both of which depend on the energy E). Solving that equation numerically, we find four possible values of the energy: $E_1 = 26$ eV, $E_2 = 104$ eV, $E_3 = 227$ eV, $E_4 = 375$ eV. Here the subscript just numbers the energy values, starting at the ground state; there is no simple functional dependence of the energies on the quantum number n as there was for the infinite well. The allowed energy levels are shown in Figure 5.13.

The probability densities (square of the wave functions) for these four states are shown in Figure 5.14. In some ways, they are similar to the probability densities in the infinite well—note that each state has n maxima in its probability density, just like the infinite well (see Figure 5.11). Unlike the infinite well, these probability densities show the property of penetration into the classically forbidden region. Look carefully at the continuity of the wave function and its slope at $x = 0$ and $x = L$; see how smoothly the sine and cosine function inside the well joins the exponentials in the forbidden regions.

The energy levels of the finite well are smaller than those of the infinite well of the same width (38 eV, 150 eV, 338 eV, 602 eV), and the differences increase

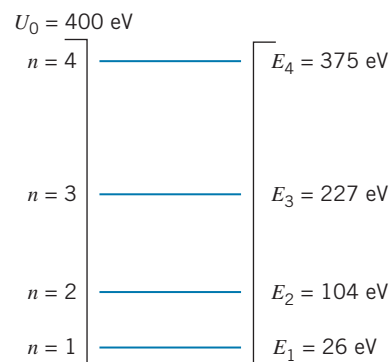


FIGURE 5.13 The energy levels in a potential energy well of depth 400 eV. There are only four energy states in this well.

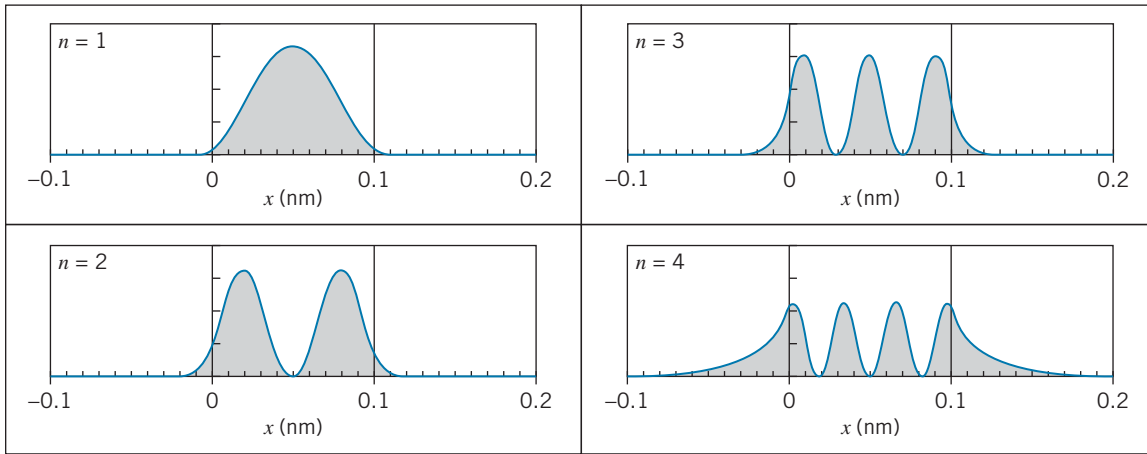


FIGURE 5.14 The probability densities of the four states in the one-dimensional potential energy well of width 0.100 nm and depth 400 eV.

as we go to higher states. This is consistent with the uncertainty principle—because of the penetration into the forbidden region, Δx is larger for the finite well and thus Δp_x must be smaller. As a result, the kinetic energies are smaller for the finite well. From Figure 5.14, we see that the penetration distance increases as we go up in energy, so the difference between Δx for the finite well and the infinite well increases and the energy discrepancy also increases.

For an energy close to the top of the well such as E_4 , a smaller uncertainty ΔE is necessary to reach the top of the well, giving a larger $\Delta t \sim \hbar/\Delta E$ and thus a larger penetration distance. At the bottom of the well, the state E_1 requires much more energy to reach the top of the well and thus needs a much larger ΔE ; the smaller resulting Δt gives a smaller penetration distance into the forbidden region.

Two-Dimensional Infinite Potential Energy Well

When we extend the previous calculation to two and three dimensions, the principal features of the solution remain the same, but an important new feature is introduced. In this section, we show how this occurs; this new feature, known as *degeneracy*, will turn out to be very important in our study of atomic physics.

To begin with, we need a Schrödinger equation that is valid in more than one dimension; our previous version, Eq. 5.5, included only one spatial dimension. If the potential energy is a function of x and y , we expect that ψ also depends on both x and y , and the derivatives with respect to x must be replaced by derivatives with respect to x and y . In two dimensions, we then have*

$$-\frac{\hbar^2}{2m} \left(\frac{\partial^2 \psi(x, y)}{\partial x^2} + \frac{\partial^2 \psi(x, y)}{\partial y^2} \right) + U(x, y)\psi(x, y) = E\psi(x, y) \quad (5.37)$$

*The first two terms on the left side of this equation require *partial* derivatives; for well-behaved functions, these involve taking the derivative with respect to one variable while keeping the other constant. Thus if $f(x, y) = x^2 + xy + y^2$, then $\partial f/\partial x = 2x + y$ and $\partial f/\partial y = x + 2y$.

The two-dimensional potential energy well is

$$U(x, y) = \begin{cases} 0 & 0 \leq x \leq L; 0 \leq y \leq L \\ \infty & \text{otherwise} \end{cases} \quad (5.38)$$

The particle is confined by infinitely high barriers to the square region with the vertices $(x, y) = (0, 0), (L, 0), (L, L), (0, L)$, as shown in Figure 5.15. A classical analog might be a small disk sliding without friction on a tabletop and colliding elastically with walls at $x = 0, x = L, y = 0$, and $y = L$. (For simplicity, we have made the allowed region square; we could have made it rectangular by setting $U = 0$ when $0 \leq x \leq a$ and $0 \leq y \leq b$.)

Solving *partial* differential equations requires a technique more involved than we need to consider, so we will not give the details of the solution. We suspect that, as in the previous case, $\psi(x, y) = 0$ outside the allowed region, in order to make the probability zero there. Inside the well, we consider solutions that are *separable*; that is, our function of x and y can be expressed as the product of one function that depends only on x and another that depends only on y :

$$\psi(x, y) = f(x)g(y) \quad (5.39)$$

where the functions f and g are similar to Eq. 5.16:

$$f(x) = A \sin k_x x + B \cos k_x x, \quad g(y) = C \sin k_y y + D \cos k_y y \quad (5.40)$$

The wave number k of the one-dimensional problem has become the separate wave numbers k_x for $f(x)$ and k_y for $g(y)$. We show later how these are related. (See also Problem 20 at the end of this chapter.)

The continuity condition on $\psi(x, y)$ requires that the solutions inside and outside match at the boundary. Because $\psi = 0$ everywhere outside, the continuity condition then requires that $\psi = 0$ everywhere on the boundary. That is,

$$\begin{aligned} \psi(0, y) = 0 & \quad \text{and} \quad \psi(L, y) = 0 & \quad \text{for all } y \\ \psi(x, 0) = 0 & \quad \text{and} \quad \psi(x, L) = 0 & \quad \text{for all } x \end{aligned}$$

In analogy with the one-dimensional problem, the condition at $x = 0$ gives $f(0) = 0$, which requires $B = 0$ in Eq. 5.40. Similarly, the condition at $y = 0$ gives $g(0) = 0$, which requires $D = 0$. The condition $f(L) = 0$ requires that $\sin k_x L = 0$, and thus that $k_x L$ be an integer multiple of π ; the condition $g(L) = 0$ similarly requires that $k_y L$ be an integer multiple of π . These two integers do not necessarily need to be the same, so we call them n_x and n_y . Making all these substitutions into Eq. 5.39, we obtain

$$\psi(x, y) = A' \sin \frac{n_x \pi x}{L} \sin \frac{n_y \pi y}{L} \quad (5.41)$$

where we have combined A and C into A' . The coefficient A' is once again found by the normalization condition, which in two dimensions becomes

$$\int \int \psi^2 dx dy = 1 \quad (5.42)$$

For our case this gives

$$\int_0^L dy \int_0^L A'^2 \sin^2 \frac{n_x \pi x}{L} \sin^2 \frac{n_y \pi y}{L} dx = 1 \quad (5.43)$$

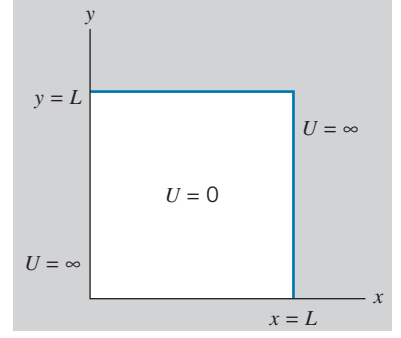


FIGURE 5.15 A particle moves freely in the two-dimensional region $0 < x < L, 0 < y < L$, but encounters infinite barriers beyond that region.

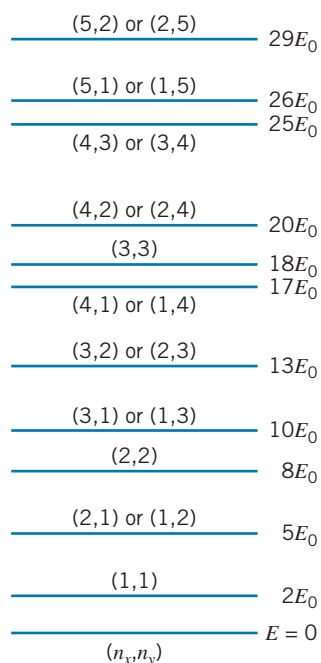


FIGURE 5.16 The lower permitted energy levels of a particle confined to an infinite two-dimensional potential energy well.

from which follows

$$A' = \frac{2}{L} \quad (5.44)$$

(The solutions to this problem, which are standing de Broglie waves on a two-dimensional surface, are similar to the solutions of the classical problem of the vibrations of a stretched membrane such as a drumhead.)

Finally, we can substitute our solution for $\psi(x, y)$ back into Eq. 5.41 to find the energy:

$$E = \frac{\hbar^2 \pi^2}{2mL^2} (n_x^2 + n_y^2) = \frac{\hbar^2}{8mL^2} (n_x^2 + n_y^2) \quad (5.45)$$

Compare this result with Eq. 5.30. Once again we let $E_0 = \hbar^2 \pi^2 / 2mL^2 = \hbar^2 / 8mL^2$ so that $E = E_0(n_x^2 + n_y^2)$. In Figure 5.16, the energies of the excited states are shown. You can see how different the energies are from those of the one-dimensional case shown in Figure 5.10.

Figure 5.17 shows the probability density ψ^2 for several different combinations of the *quantum numbers* n_x and n_y . The probability has maxima and minima, just like the probability in the one-dimensional problem. For example, if we gave the particle an energy of $8E_0$ and then made a large number of measurements of its position, we would expect to find it most often near the four points $(x, y) = (L/4, L/4), (L/4, 3L/4), (3L/4, L/4),$ and $(3L/4, 3L/4)$; we expect *never* to find it near $x = L/2$ or $y = L/2$. The *shape* of the probability density tells us something about the quantum numbers and therefore about the energy. Thus if we measured the probability density and

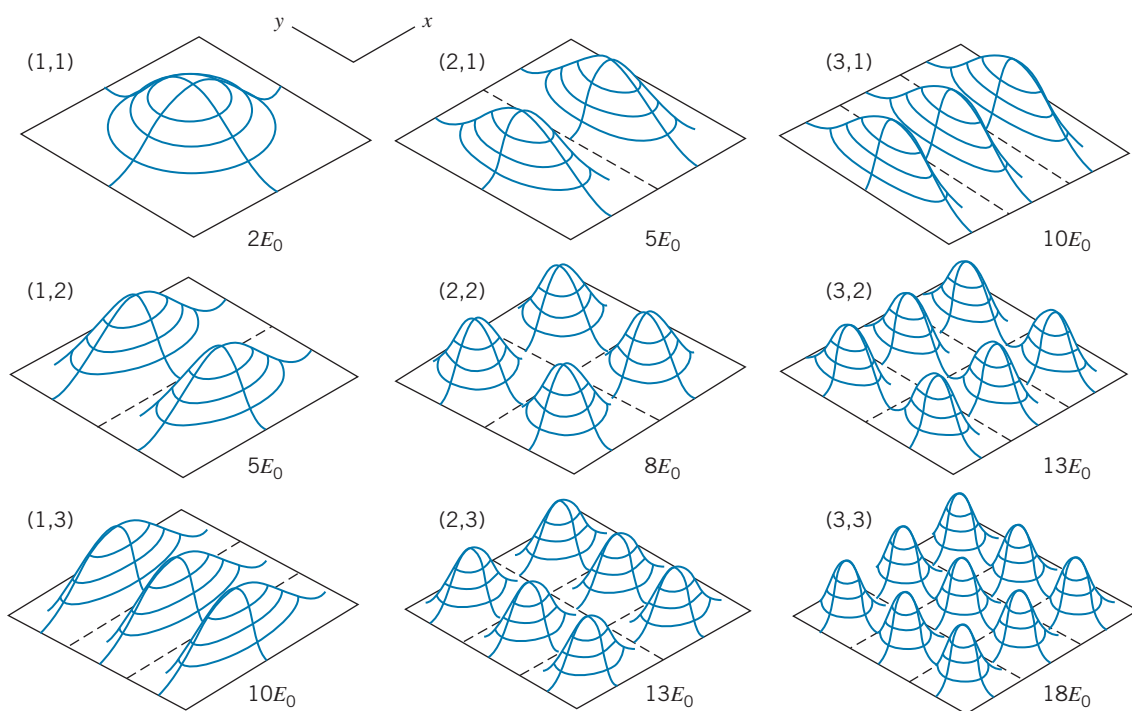


FIGURE 5.17 The probability density for some of the lower energy levels of a particle confined to the infinite two-dimensional potential energy well. The individual plots are labeled with the quantum numbers (n_x, n_y) and with the value of the energy E .

found six maxima, as shown in Figure 5.17, we would deduce that the particle had an energy of $13E_0$ with $n_x = 2$ and $n_y = 3$, or else $n_x = 3, n_y = 2$.

Recently, it has become possible to photograph the probability densities of electrons confined in a two-dimensional region. The tip of an electron microscope was used to place 48 individual iron atoms on a metal surface in a ring or “corral” of radius 7.13 nm that formed the walls of a potential well, as shown in Figure 5.18. Inside the ring, the waves of probability density for electrons trapped in the potential well are clearly visible. The potential well is circular, rather than square, but otherwise the analysis follows the procedures described in this section; when the Schrödinger equation is solved in cylindrical polar coordinates with the potential energy for a circular well, the calculated probability density gives a close match with the observed one. These beautiful results are a dramatic confirmation of the wave functions obtained for the two-dimensional potential energy well.

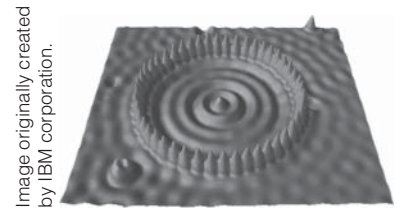


FIGURE 5.18 A ring of iron atoms on a copper surface forms a “corral” within which the probability density of trapped electrons is clearly visible. This image was obtained with a scanning tunneling electron microscope.

Degeneracy Occasionally, it happens that two different sets of quantum numbers n_x and n_y have exactly the same energy. This situation is known as *degeneracy*, and the energy levels are said to be *degenerate*. For example, the energy level at $E = 13E_0$ is degenerate, because both $n_x = 2, n_y = 3$ and $n_x = 3, n_y = 2$ have $E = 13E_0$. This degeneracy arises from interchanging n_x and n_y (which is the same as interchanging the x and y axes), so the probability distributions in the two cases are not very different. However, consider the state with $E = 50E_0$, for which there are three sets of quantum numbers: $n_x = 7, n_y = 1$; $n_x = 1, n_y = 7$; and $n_x = 5, n_y = 5$. The first two sets of quantum numbers result from the interchange of n_x and n_y and so have similar probability distributions, but the third represents a *very* different state of motion, as shown in Figure 5.19. The level at $E = 13E_0$ is said to be *two-fold* degenerate, while the level at $E = 50E_0$ is *three-fold* degenerate; we could also say that one level has a degeneracy of 2, while the other has a degeneracy of 3.

Degeneracy occurs in general whenever a system is labeled by two or more quantum numbers; as we have seen in the above calculation, different combinations of quantum numbers often can give the same value of the energy. The number of different quantum numbers required by a given physical problem turns out to be exactly equal to the number of dimensions in which the problem is being solved—one-dimensional problems need only one quantum number, two-dimensional problems need two, and so forth. When we get to three dimensions, as in Problem 21 at the end of this chapter

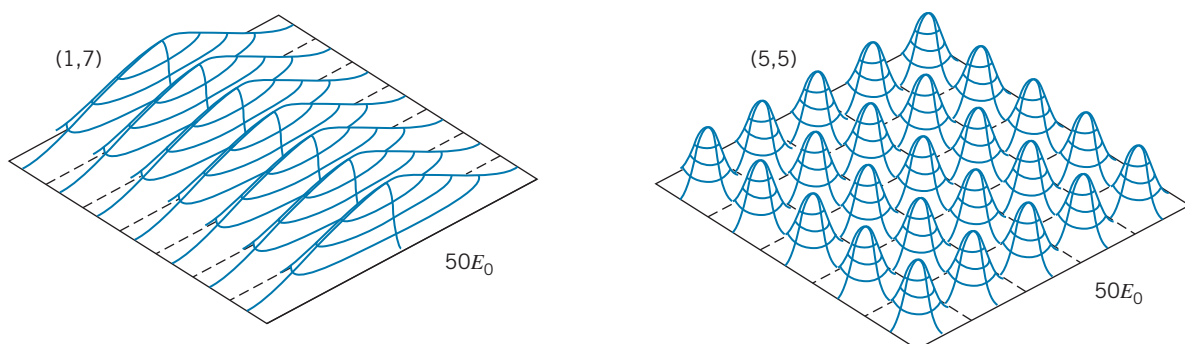


FIGURE 5.19 Two very different probability densities with exactly the same energy.

and especially in the hydrogen atom in Chapter 7, we find that the effects of degeneracy become more significant; in the case of atomic physics, the degeneracy is a major contributor to the structure and properties of atoms.

5.5 THE SIMPLE HARMONIC OSCILLATOR

Another situation that can be analyzed using the Schrödinger equation is the one-dimensional simple harmonic oscillator. The classical oscillator is an object of mass m attached to a spring of force constant k . The spring exerts a restoring force $F = -kx$ on the object, where x is the displacement from its equilibrium position. Using Newton's laws, we can analyze the oscillator and show that it has a (circular or angular) frequency $\omega_0 = \sqrt{k/m}$ and a period $T = 2\pi\sqrt{m/k}$. The maximum distance of the oscillating object from its equilibrium position is x_0 , the amplitude of the oscillation. The oscillator has its maximum kinetic energy at $x = 0$; its kinetic energy vanishes at the *turning points* $x = \pm x_0$. At the turning points the oscillator comes to rest for an instant and then reverses its direction of motion. The motion is, of course, confined to the region $-x_0 \leq x \leq +x_0$.

Why analyze the motion of such a system using quantum mechanics? Although we never find in nature an example of a one-dimensional quantum oscillator, there are systems that behave approximately as one—a vibrating diatomic molecule, for example. In fact, any system in a smoothly varying potential energy well near its minimum behaves approximately like a simple harmonic oscillator.

A force $F = -kx$ has the associated potential energy $U = \frac{1}{2}kx^2$, and so we have the Schrödinger equation:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + \frac{1}{2}kx^2\psi = E\psi \quad (5.46)$$

(Because we are working in one dimension, U and ψ are functions only of x .) There are no boundaries between different regions of potential energy here, so the wave function must fall to zero for both $x \rightarrow +\infty$ and $x \rightarrow -\infty$. The simplest function that satisfies these conditions, which turns out to be the correct ground state wave function, is $\psi(x) = Ae^{-ax^2}$. The constant a and the energy E can be found by substituting this function into Eq. 5.46. We begin by evaluating $d^2\psi/dx^2$.

$$\begin{aligned} \frac{d\psi}{dx} &= -2ax(Ae^{-ax^2}) \\ \frac{d^2\psi}{dx^2} &= -2a(Ae^{-ax^2}) - 2ax(-2ax)Ae^{-ax^2} = (-2a + 4a^2x^2)Ae^{-ax^2} \end{aligned}$$

Substituting into Eq. 5.46 and canceling the common factor Ae^{-ax^2} yields

$$\frac{\hbar^2 a}{m} - \frac{2a^2 \hbar^2}{m} x^2 + \frac{1}{2} k x^2 = E \quad (5.47)$$

Equation 5.47 is *not* an equation to be solved for x , because we are looking for a solution that is valid for *any* x , not just for one specific value. In order

for this to hold for *any* x , the coefficients of x^2 must cancel and the remaining constants must be equal. (That is, consider the equation $bx^2 = c$. It will be true for *any* and *all* x only if both $b = 0$ and $c = 0$.) Thus

$$-\frac{2a^2\hbar^2}{m} + \frac{1}{2}k = 0 \quad \text{and} \quad \frac{\hbar^2 a}{m} = E \quad (5.48)$$

which yield

$$a = \frac{\sqrt{km}}{2\hbar} \quad \text{and} \quad E = \frac{1}{2}\hbar\sqrt{k/m} \quad (5.49)$$

We can also write the energy in terms of the classical frequency $\omega_0 = \sqrt{k/m}$ as

$$E = \frac{1}{2}\hbar\omega_0 \quad (5.50)$$

The coefficient A is found from the normalization condition (see Problem 22 at the end of the chapter). The result, which is valid *only* for this ground-state wave function, is $A = (m\omega_0/\hbar\pi)^{1/4}$. The complete wave function of the ground state is then

$$\psi(x) = \left(\frac{m\omega_0}{\hbar\pi}\right)^{1/4} e^{-(\sqrt{km}/2\hbar)x^2} \quad (5.51)$$

The probability density for this wave function is illustrated in Figure 5.20. Note that, as in the case of the finite potential energy well, the probability density can penetrate into the forbidden region beyond the classical turning points at $x = \pm x_0$ (in this region the potential energy is greater than E).

The solution we have found corresponds only to the *ground state* of the oscillator. The general solution is of the form $\psi_n(x) = Af_n(x)e^{-ax^2}$, where $f_n(x)$ is a polynomial in which the highest power of x is x^n . The corresponding energies are

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega_0 \quad n = 0, 1, 2, \dots \quad (5.52)$$

These levels are shown in Figure 5.21. Note that they are *uniformly spaced*, in contrast to the one-dimensional infinite potential energy well. Probability densities are shown in Figure 5.22. All of the solutions have the property of penetration of probability density into the forbidden region beyond the classical turning points. The probability density oscillates, somewhat like a sine wave, between the turning points, and decreases like e^{-ax^2} to zero beyond the turning points. Note the great similarity between the probability densities for the quantum oscillator and those of the finite potential energy well (Figure 5.14).

A sequence of vibrational excited states similar to Figure 5.21 is commonly found in diatomic molecules such as HCl (see Chapter 9). The spacing between the states is typically 0.1–1 eV; the states are observed when photons (in the infrared region of the spectrum) are emitted or absorbed as the molecule jumps from one state to another. A similar sequence is observed in nuclei, where the spacing is 0.1–1 MeV and the radiations are in the gamma-ray region of the spectrum.

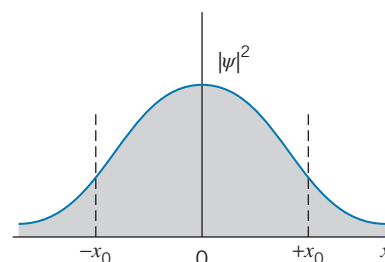


FIGURE 5.20 The probability density for the ground state of the simple harmonic oscillator. The classical turning points are at $x = \pm x_0$.

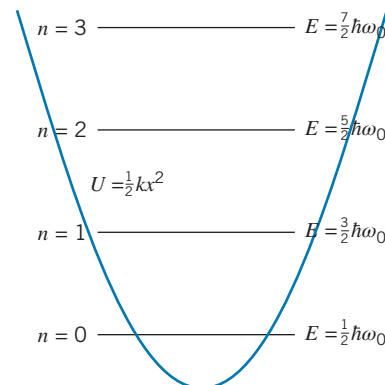


FIGURE 5.21 Energy levels of the simple harmonic oscillator. Note that the levels have equal spacings and that the distance between the classical turning points increases with energy.

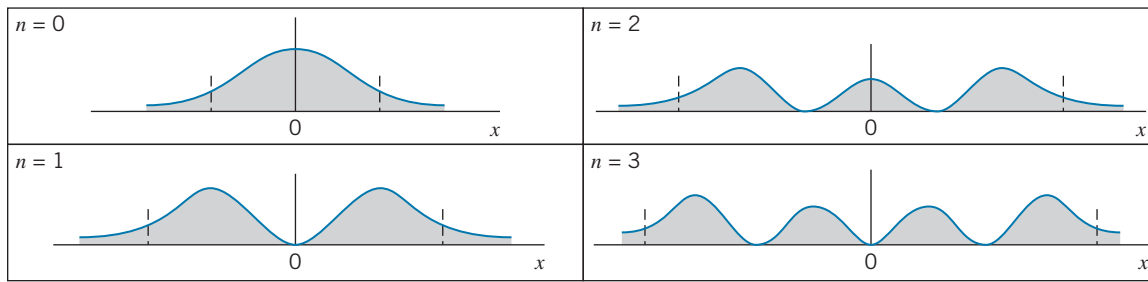


FIGURE 5.22 Probability densities for the simple harmonic oscillator. Note how the distance between the classical turning points (marked by the short vertical lines) increases with energy. Compare with the probability densities for the finite potential energy well (Figure 5.14).

Example 5.6

Consider the “half oscillator” that is restricted to the region $x > 0$ by the potential energy

$$\begin{aligned} U(x) &= \infty & x &\leq 0 \\ U(x) &= \frac{1}{2}kx^2 & x &> 0 \end{aligned}$$

What are the energies of the ground state and first excited state?

Solution

Because $\psi(x) = 0$ for $x \leq 0$, the continuity condition on $\psi(x)$ at $x = 0$ requires that the solutions for $x > 0$ must also have $\psi(x) = 0$. From the harmonic oscillator probability distributions shown in Figure 5.22, it is apparent that only the $n = 1$ and $n = 3$ wave functions satisfy this condition. So the ground state ($n = 1$) energy is $\frac{3}{2}\hbar\omega_0$ and the first excited state ($n = 3$) energy is $\frac{7}{2}\hbar\omega_0$.

Example 5.7

An electron is bound to a region of space by a springlike force with an effective spring constant of $k = 95.7 \text{ eV/nm}^2$. (a) What is its ground-state energy? (b) How much energy must be absorbed for the electron to jump from the ground state to the second excited state?

Solution

(a) The ground-state energy is

$$E = \frac{1}{2}\hbar\omega_0 = \frac{1}{2}\hbar\sqrt{\frac{k}{m}} = \frac{1}{2}\hbar c\sqrt{\frac{k}{mc^2}}$$

$$\begin{aligned} &= \frac{1}{2} (197 \text{ eV} \cdot \text{nm}) \sqrt{\frac{95.7 \text{ eV/nm}^2}{0.511 \times 10^6 \text{ eV}}} \\ &= 1.35 \text{ eV} \end{aligned}$$

(b) The difference between adjacent energy levels is $\hbar\omega_0 = 2.70 \text{ eV}$ for all energy levels, so the energy that must be absorbed to go from the ground state to the second excited state is $\Delta E = 2 \times 2.70 \text{ eV} = 5.40 \text{ eV}$.

Example 5.8

For the electron of Example 5.7 in its ground state, what is the probability to find it in a narrow interval of width 0.004 nm located halfway between the equilibrium position and the classical turning point?

Solution

First we need to find the location of the turning point. At the classical turning points $x = \pm x_0$, the kinetic energy is zero and so the total energy is all potential.

Thus $E = \frac{1}{2}kx_0^2$, and so

$$x_0 = \sqrt{\frac{2E}{k}} = \sqrt{\frac{2(1.35 \text{ eV})}{95.7 \text{ eV/nm}^2}} = 0.168 \text{ nm}$$

Evaluating the parameters of the wave function Ae^{-ax^2} (the normalization constant A and the exponential coefficient a), we have

$$\begin{aligned} A &= \left(\frac{m\omega_0}{\hbar\pi} \right)^{1/4} = \left(\frac{mc^2\hbar\omega_0}{\hbar^2c^2\pi} \right)^{1/4} \\ &= \left(\frac{(0.511 \times 10^6 \text{ eV})(2.70 \text{ eV})}{(197 \text{ eV} \cdot \text{nm})^2\pi} \right)^{1/4} = 1.83 \text{ nm}^{-1/2} \end{aligned}$$

$$\begin{aligned} a &= \frac{\sqrt{km}}{2\hbar} = \frac{\sqrt{kmc^2}}{2\hbar c} \\ &= \frac{\sqrt{(95.7 \text{ eV/nm}^2)(0.511 \times 10^6 \text{ eV})}}{2(197 \text{ eV} \cdot \text{nm})} \\ &= 17.74 \text{ nm}^{-2} \end{aligned}$$

The probability in the interval $dx = 0.004 \text{ nm}$ at $x = x_0/2 = 0.084 \text{ nm}$ is then

$$\begin{aligned} P(x) dx &= |\psi(x)|^2 dx = A^2 e^{-2ax^2} dx \\ &= (1.83 \text{ nm}^{-1/2})^2 e^{-2(17.74 \text{ nm}^{-2})(0.084 \text{ nm})^2} (0.004 \text{ nm}) \\ &= 0.0104 = 1.04\% \end{aligned}$$

As we did in the case of the infinite potential energy well, let's look at the application of the uncertainty principle to the wave packet represented by the harmonic oscillator. Using the results of Problems 24 and 25 for the uncertainties in position and momentum for the ground state of the oscillator, $\Delta x = \sqrt{\hbar/2m\omega_0}$ and $\Delta p = \sqrt{\hbar\omega_0 m/2}$, the product of the uncertainties is $\Delta x \Delta p = \hbar/2$. This is the minimum possible value for this product, according to Eq. 4.10. The ground state of the oscillator thus represents the most "compact" wave packet in which the product of the uncertainties has its smallest value. You can see from Figure 5.22 that the excited states of the oscillator are much less compact (more spread out) than the ground state.

5.6 STEPS AND BARRIERS

In this general type of problem, we analyze what happens when a particle moving (again in one dimension) in a region of constant potential energy suddenly moves into a region of different, but also constant, potential energy. We will not discuss in detail the solutions to these problems, but the methods of solution of each are so similar that we can outline the steps to take in the solution. In this discussion, we let E be the (fixed) total energy of the particle and U_0 will be the value of the constant potential energy. In these calculations, the particle is *not* confined, so the energy is not quantized—we are free to choose any value for the particle energy.

Potential Energy Step, $E > U_0$

Consider the potential energy step shown in Figure 5.23:

$$\begin{aligned} U(x) &= 0 & x < 0 \\ &= U_0 & x \geq 0 \end{aligned} \quad (5.53)$$

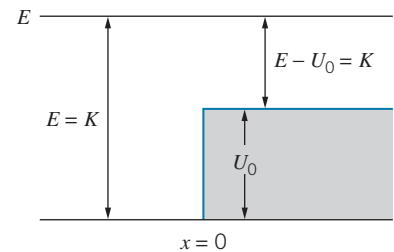


FIGURE 5.23 A step of height U_0 . Particles are incident from the left with energy E . The kinetic energy is equal to E in the region $x < 0$ and is reduced to $E - U_0$ in the region $x > 0$.

If the total energy E of the particle is greater than U_0 , then we can write the solutions to the Schrödinger equation in the two regions based on the general form of Eq. 5.16:

$$\psi_0(x) = A \sin k_0 x + B \cos k_0 x \quad k_0 = \sqrt{\frac{2mE}{\hbar^2}} \quad x < 0 \quad (5.54a)$$

$$\psi_1(x) = C \sin k_1 x + D \cos k_1 x \quad k_1 = \sqrt{\frac{2m}{\hbar^2}(E - U_0)} \quad x > 0 \quad (5.54b)$$

Relationships among the four coefficients, A , B , C , and D , may be found by applying the condition that $\psi(x)$ and $\psi'(x) = d\psi/dx$ must be continuous at the boundary; thus $\psi_0(0) = \psi_1(0)$ and $\psi'_0(0) = \psi'_1(0)$. A typical solution might look like Figure 5.24. Note the smooth transition between the solutions at $x = 0$, which results from applying the continuity conditions.

The coefficients A , B , C , and D are in general complex, so to visualize the complete wave we need both the real and imaginary parts of ψ . We can use the equation $e^{i\theta} = \cos \theta + i \sin \theta$ to transform these solutions from sines and cosines to complex exponentials:

$$\psi_0(x) = A' e^{ik_0 x} + B' e^{-ik_0 x} \quad x < 0 \quad (5.55a)$$

$$\psi_1(x) = C' e^{ik_1 x} + D' e^{-ik_1 x} \quad x > 0 \quad (5.55b)$$

The coefficients A' , B' , C' , D' can be found from the coefficients A , B , C , D . The time dependent wave functions are obtained by multiplying each term by $e^{-i\omega t}$, which gives

$$\Psi_0(x, t) = A' e^{i(k_0 x - \omega t)} + B' e^{-i(k_0 x + \omega t)} \quad (5.56a)$$

$$\Psi_1(x, t) = C' e^{i(k_1 x - \omega t)} + D' e^{-i(k_1 x + \omega t)} \quad (5.56b)$$

We can then make the following identification of the component waves, recalling that $(kx - \omega t)$ is the phase of a wave moving in the positive x direction, while $(kx + \omega t)$ is the phase of a wave moving in the negative x direction, and assuming that *the squared magnitude of each coefficient gives the intensity of the corresponding component wave*. In the region $x < 0$, Eq. 5.56a describes the superposition of a wave $e^{i(k_0 x - \omega t)}$ of intensity $|A'|^2$ moving in the positive x direction (from $-\infty$ to 0) and a wave $e^{-i(k_0 x + \omega t)}$ of intensity $|B'|^2$ moving in the

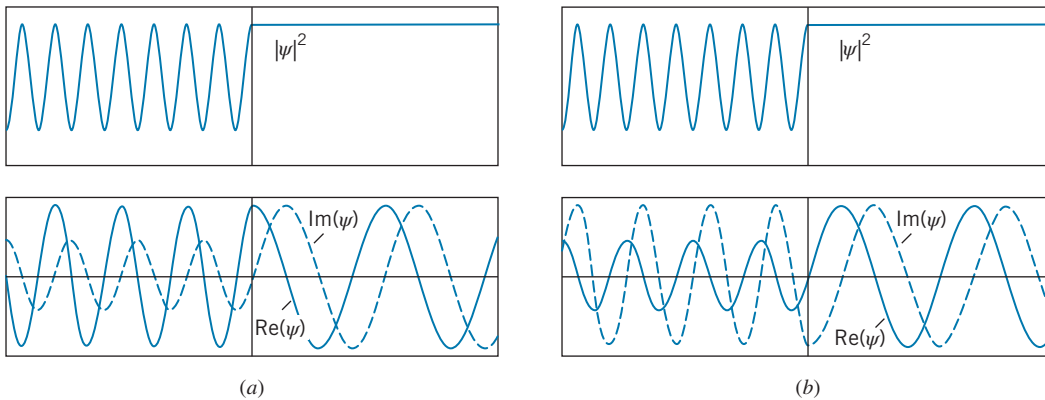


FIGURE 5.24 Wave function for electrons incident from the left on a potential energy step for $E > U_0$. The probability density and the real and imaginary parts of the wavefunction are shown for (a) $t = 0$ and (b) $t = 1/4$ period. The vertical line marks the location of the step.

negative x direction. Suppose we had intended our solution to describe particles that are incident from the left on this step. Then $|A'|^2$ gives the intensity of the incident wave (more exactly, the de Broglie wave describing the incident beam of particles) and $|B'|^2$ gives the intensity of the reflected wave. The ratio $|B'|^2/|A'|^2$ tells us the reflected fraction of the incident wave intensity.

In the region $x > 0$, Eq. 5.56b describes the transmitted wave $e^{i(k_1x - \omega t)}$ of intensity $|C'|^2$ moving to the right and a wave $e^{-i(k_1x + \omega t)}$ of intensity $|D'|^2$ moving to the left. If particles are incident from $-\infty$, it is not possible to have particles in the region $x > 0$ moving to the left, so in this particular experimental situation we are justified in setting D' to zero.

Figure 5.24a shows that the probability density has the same value everywhere in the region $x > 0$. You can see this immediately from Eq. 5.56b with $D' = 0$; taking the squared magnitude of the remaining term gives a constant result, independent of x and t . This is consistent with what we expect for the de Broglie wave of free particles; the particles can be found anywhere in the region $x > 0$ with equal probability.

In the region $x < 0$, the incident and reflected waves combine to produce a standing wave, for which the probability density has fixed maxima and minima. The probability density in this region does not vary with time, as suggested by the plots for the two different times ($t = 0$ and $t = 1/4$ period) shown in Figure 5.24.

To illustrate the propagation of the de Broglie wave, it is instructive also to plot the real and imaginary parts of the wave function, which are shown in Figure 5.24. Here you can see the change in wavelength (corresponding to the change in kinetic energy or momentum) in crossing the step. You can also see something of the time dependence—the wave propagates in both regions, but it does so in a way that the real and imaginary parts combine to give a probability density that remains unchanged in time.

Potential Energy Step, $E < U_0$

If the energy of the particle is less than the height of the potential energy step, then the solution in the region $x > 0$ is of the form of Eq. 5.18:

$$\psi_0(x) = A \sin k_0x + B \cos k_0x \quad k_0 = \sqrt{\frac{2mE}{\hbar^2}} \quad x < 0 \quad (5.57a)$$

$$\psi_1(x) = Ce^{k_1x} + De^{-k_1x} \quad k_1 = \sqrt{\frac{2m}{\hbar^2}(U_0 - E)} \quad x > 0 \quad (5.57b)$$

We set $C = 0$ to keep $\psi_1(x)$ from becoming infinite as $x \rightarrow \infty$, and we apply the boundary conditions on $\psi(x)$ and $\psi'(x)$ at $x = 0$. The resulting solution is shown in Figure 5.25. The probability density for $x > 0$ shows penetration into the classically forbidden region. All particles are reflected from the barrier; that is, if we write $\Psi_0(x, t)$ in the form of Eq. 5.56a, then we must have $|A'| = |B'|$, indicating that the waves moving to the right (the incident wave) and to the left (the reflected wave) in the region $x < 0$ have equal amplitudes.

Figure 5.25 shows the probability density at two different times, illustrating that the probability density does not change with time. In the region $x < 0$, we again have standing waves with fixed maxima and minima. Viewing the real and imaginary parts at two different times ($t = 0$ and $t = 1/4$ period) shows that the wave is propagating, even though the probability density does not change with time.

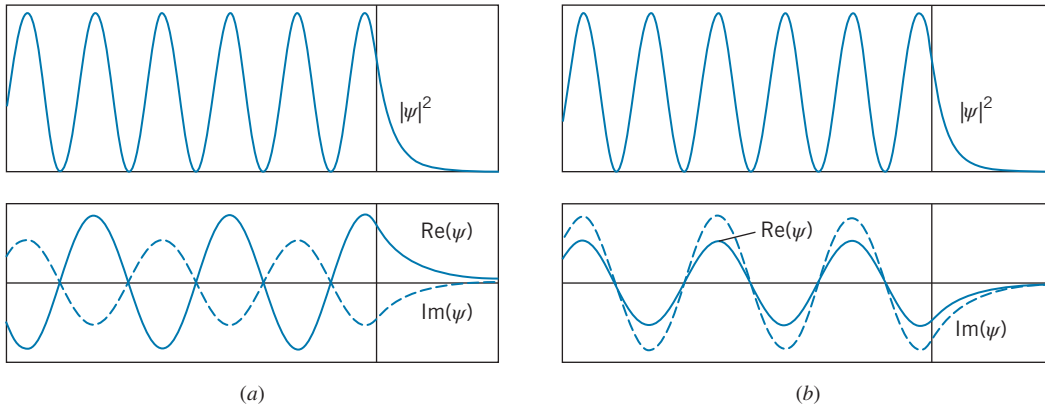


FIGURE 5.25 Wave function for electrons incident from the left on a potential energy step for $E < U_0$. The probability density and the real and imaginary parts of the wavefunction are shown for (a) $t = 0$ and (b) $t = 1/4$ period. The vertical line marks the location of the step.

Penetration into the forbidden region is associated with the wave nature of the particle and also with the uncertainty in the particle's energy or location. The probability density in the $x > 0$ region is $|\psi|^2$, which according to Eq. 5.57b is proportional to e^{-2k_1x} . If we define a representative penetration distance Δx to be the distance over which the probability drops by $1/e$, then $e^{-2k_1\Delta x} = e^{-1}$ and so

$$\Delta x = \frac{1}{2k_1} = \frac{1}{2} \frac{\hbar}{\sqrt{2m(U_0 - E)}} \quad (5.58)$$

To be able to enter the region with $x > 0$, the *particle* must gain an energy of at least $U_0 - E$ in order to get over the potential energy step; it must in addition gain some kinetic energy if it is to move in the region $x > 0$. Of course, it is a violation of conservation of energy for the particle to spontaneously gain *any* amount of energy, but according to the uncertainty relationship $\Delta E \Delta t \sim \hbar$ conservation of energy does not apply at times smaller than Δt except to within an amount $\Delta E \sim \hbar/\Delta t$. That is, if the particle “borrows” an amount of energy ΔE and “returns” the borrowed energy within a time $\Delta t \sim \hbar/\Delta E$, we observers will still believe energy is conserved. Suppose the particle borrows an energy sufficient to give it a kinetic energy of K in the forbidden region. How far into the forbidden region does the particle penetrate?

The “borrowed” energy is $(U_0 - E) + K$; the energy $(U_0 - E)$ gets the particle to the top of the step, and the extra kinetic energy K gives it its motion. The energy must be returned within a time

$$\Delta t = \frac{\hbar}{U_0 - E + K} \quad (5.59)$$

The particle moves with speed $v = \sqrt{2K/m}$, and so the distance it can travel is

$$\Delta x = \frac{1}{2} v \Delta t = \frac{1}{2} \sqrt{\frac{2K}{m}} \frac{\hbar}{U_0 - E + K} \quad (5.60)$$

(The factor of $1/2$ is present because in the time Δt the particle must penetrate the distance Δx into the forbidden region and return through that same distance to the allowed region.)

In the limit $K \rightarrow 0$, the penetration distance Δx goes to 0 according to Eq. 5.60 because the particle has zero velocity; similarly, $\Delta x \rightarrow 0$ in the limit $K \rightarrow \infty$, because it moves for a vanishing time interval Δt . In between those limits, there must be a maximum value of Δx for some particular K . Differentiating Eq. 5.60 with respect to K , we can find the maximum value

$$\Delta x_{\max} = \frac{1}{2} \frac{\hbar}{\sqrt{2m(U_0 - E)}} \quad (5.61)$$

This value of Δx is identical with Eq. 5.58! This demonstrates that the penetration into the forbidden region given by the solution to the Schrödinger equation is entirely consistent with the uncertainty relationship. (The agreement between Eqs. 5.58 and 5.61 is really somewhat accidental, because the factor $1/e$ used to obtain Eq. 5.58 was chosen arbitrarily. What we have really demonstrated is that the estimates of uncertainty given by the Heisenberg relationships are consistent with the wave properties of the particle obtained from the Schrödinger equation. This should not be surprising, because the uncertainty principle can be derived as a consequence of the Schrödinger equation.)

Potential Energy Barrier

Consider now the potential energy barrier shown in Figure 5.26:

$$\begin{aligned} U(x) &= 0 & x < 0 \\ &= U_0 & 0 \leq x \leq L \\ &= 0 & x > L \end{aligned} \quad (5.62)$$

Particles with energy E less than U_0 are incident from the left. Our experience then leads us to expect solutions of the form shown in Figure 5.27—sinusoidal oscillation in the region $x < 0$ (an incident wave and a reflected wave), exponentials in the region $0 \leq x \leq L$, and sinusoidal oscillations in the region $x > L$ (the transmitted wave). Note that the intensity of the transmitted wave ($x > L$) is much smaller than the intensity of the incident + reflected waves ($x < 0$), which means that most of the particles are reflected and few are transmitted through the barrier. Also note that the wavelengths are the same on either side of the barrier (because the kinetic energies are the same).

The intensity of the transmitted wave, which can be found by application of the continuity conditions, depends on the energy of the particle and on the height and thickness of the barrier. Classically, the particles should never appear at $x > L$, because they do not have sufficient energy to overcome the barrier. This situation is an example of *barrier penetration*, sometimes called quantum mechanical *tunneling*. Particles cannot be *observed* while they are in the classically forbidden region $0 \leq x \leq L$, but can “tunnel” *through* that region and be observed at $x > L$.

Every particle incident on the barrier of Figure 5.26 is either reflected or transmitted; the number of incident particles is equal to the number reflected back to $x < 0$ plus the number transmitted to $x > L$. None are “trapped” or ever seen in the forbidden region $0 < x < L$. How can the incident *particle* get from $x < 0$ to $x > L$? As a classical particle, *it can't!* However, the wave representing the particle *can* penetrate through the barrier, which allows the particle to be observed in the classically *allowed* region $x > L$.

This phenomenon of penetration of a forbidden region is a well-known property of classical waves. Quantum physics provides a new aspect to this

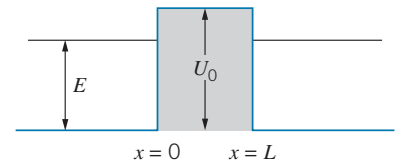


FIGURE 5.26 A barrier of height U_0 and width L .

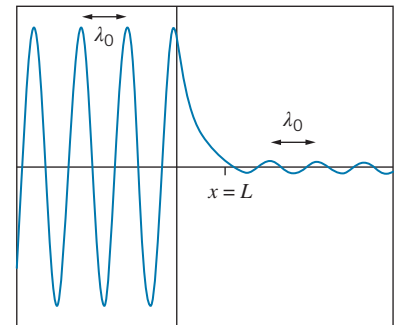


FIGURE 5.27 The real part of the wave function of a particle of energy $E < U_0$ encountering a barrier (the particle is incident from the left in the figure). The wavelength λ_0 is the same on both sides of the barrier, but the amplitude beyond the barrier is much less than the original amplitude.

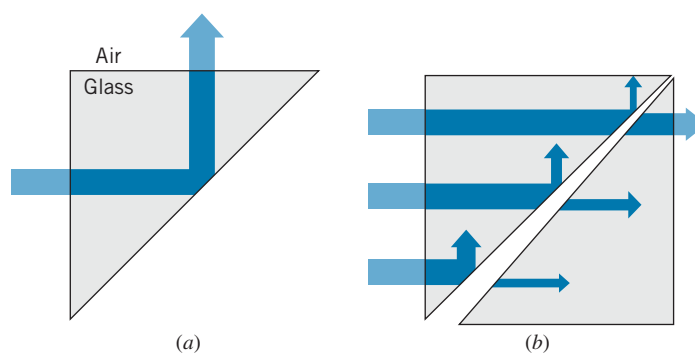


FIGURE 5.28 (a) Total internal reflection of light waves at a glass–air boundary. (b) Frustrated total internal reflection. The thicker the air gap, the smaller the probability to penetrate. Note that the light beam does not appear *in* the gap.

phenomenon by associating a particle with the wave, and thus allowing a particle to pass through a classically forbidden region. An example of the penetration effect for classical waves occurs for total internal reflection* of light waves. Figure 5.28a shows a light beam in glass incident on a boundary with air. The beam is totally reflected in the glass. However, if a second piece of glass is brought close to the first, as in Figure 5.28b, the beam can appear in the second piece of glass. This effect is called *frustrated total internal reflection*. The intensity of the beam in the second piece, represented by the widths of the arrows in Figure 5.28b, decreases rapidly as the thickness of the gap increases.

Just like our unobservable quantum wave, which penetrates a few wavelengths into the forbidden region, an unobservable light wave of exponentially decreasing amplitude, the *evanescent wave*, penetrates into the air even when the light wave undergoes total reflection in the glass. The evanescent wave carries no energy away from the interface, so it cannot be directly observed in the air, but it can be observed in another medium such as a second piece of glass placed close to the first. Evanescent waves have applications in microscopy, where they enable the production of images of individual molecules.

Although the potential energy barrier of Figure 5.29 is rather artificial, there are many practical examples of quantum tunneling:

1. **Alpha Decay.** An atomic nucleus consists of protons and neutrons in a constant state of motion; occasionally these particles form themselves into an aggregate of two protons and two neutrons, called an alpha particle. In one form of radioactive decay, the nucleus can emit an alpha particle, which can be detected in the laboratory. However, in order to escape from the nucleus the alpha particle must penetrate a barrier of the form shown in Figure 5.29. The probability for the alpha particle to penetrate the barrier, and be detected in the laboratory, can be computed based on the energy of the alpha particle and the height and thickness of the barrier. The decay

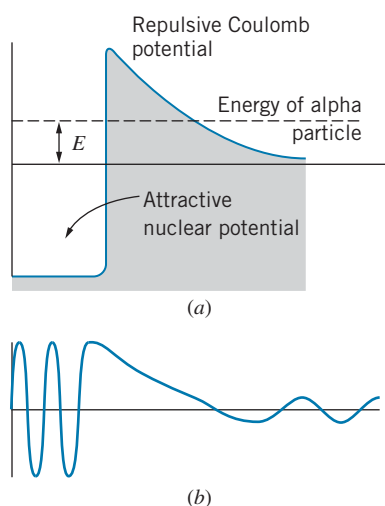


FIGURE 5.29 (a) A nuclear potential energy barrier is penetrated by an alpha particle of energy E . (b) A representation of the real part of the wave function of the alpha particle. The probability to penetrate the barrier depends strongly on the energy of the alpha particle.

*Total internal reflection occurs when a light beam is incident on a boundary between two substances, such as glass and air, from the side with the higher index of refraction. If the angle of incidence inside the glass exceeds a certain critical value, the light beam is totally reflected back into the glass.

probability can be measured in the laboratory, and it is found to be in excellent agreement with the value obtained from a quantum-mechanical calculation based on barrier penetration.

2. **Nuclear Fusion in Stars.** Stars produce their radiated energy by nuclear fusion, in which two nuclear particles collide and fuse to form a heavier nucleus. This is in many ways a reverse of the situation shown in Figure 5.29a—here we have the particle incident on the barrier from outside rather than from inside. The barrier now acts to prevent the two particles from fusing. In stars, the thermal energy of particles is typically in the keV region, which is insufficient to overcome the MeV barriers due to the repulsive Coulomb force of the two positively charged particles. Were it not for tunneling, stellar fusion would not occur and stars would not shine.
3. **Ammonia Inversion.** Figure 5.30 is a representation of the ammonia molecule NH_3 . If we were to try to move the nitrogen atom along the axis of the molecule, toward the plane of the hydrogen atoms, we find repulsion caused by the three hydrogen atoms, which produces a potential energy of the form shown in Figure 5.31. According to classical mechanics, unless we give the nitrogen atom sufficient energy, it should not be able to surmount the barrier and appear on the other side of the plane of hydrogens. According to quantum mechanics, the nitrogen can tunnel through the barrier and appear on the other side of the molecule. In fact, the nitrogen atom actually tunnels back and forth with a frequency in excess of 10^{10} oscillations per second.

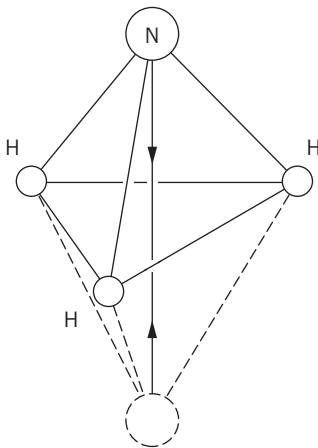


FIGURE 5.30 A schematic diagram of the ammonia molecule. The Coulomb repulsion of the three hydrogens establishes a barrier against the nitrogen atom moving to a symmetric position (shown in dashed lines) on the opposite side of the plane of hydrogens.

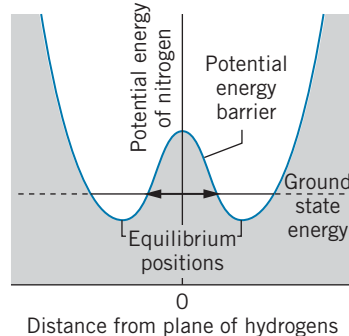


FIGURE 5.31 The potential energy seen by the nitrogen atom in an ammonia molecule. The nitrogen can penetrate the barrier and move from one equilibrium position to another.

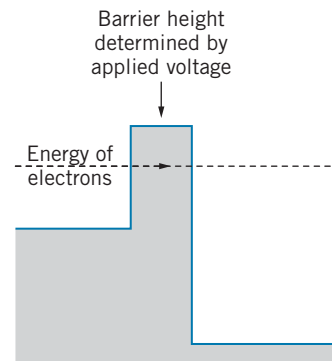


FIGURE 5.32 The potential energy barrier seen by an electron in a tunnel diode. The conductivity of the device is determined by the electron's probability to penetrate the barrier, which depends on the height of the barrier.

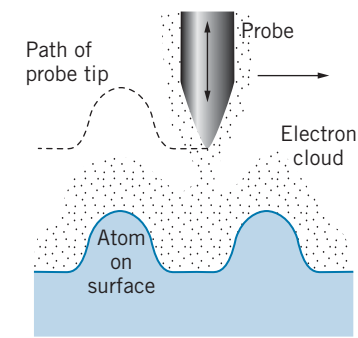


FIGURE 5.33 In a scanning tunneling microscope, a needlelike probe is scanned over a surface. The probe is moved vertically so that the distance between the probe and the surface remains constant as the probe scans laterally.

4. **The Tunnel Diode.** A tunnel diode is an electronic device that uses the phenomenon of tunneling. Schematically, the potential energy of an electron in a tunnel diode can be represented by Figure 5.32. The current that flows through the device is produced by electrons tunneling through the barrier. The rate of tunneling, and therefore the current, can be regulated merely by changing the height of the barrier, which can be done with an applied voltage. This can be done rapidly, so that switching frequencies in excess of 10^9 Hz can be obtained. Ordinary semiconductor diodes depend on the diffusion of electrons across a junction, and therefore operate on much longer time scales (that is, at lower frequencies).
5. **The Josephson Junction.** In a superconductor, linked pairs of electrons can travel without electrical resistance. If a thin insulating layer is sandwiched between two superconductors, the electron pairs can tunnel through the insulator from one superconductor to the other. Applications of the Josephson junction include sensitive measurements of very weak magnetic fields (such as those due to brain activity), the definition of the international standard of the volt, and computers based on quantum effects. More details about the Josephson junction can be found in Section 11.5.
6. **The Scanning Tunneling Microscope.** Images of individual atoms on the surface of materials (such as Figure 5.18) can be made with the scanning tunneling microscope. Electrons are trapped in a surface by a potential energy barrier (the work function of the material). When a needlelike probe is placed within about 1 nm of the surface (Figure 5.33), electrons can tunnel through the barrier between the surface and the probe and produce a current that can be recorded in an external circuit. The current is very sensitive to the width of the barrier (the distance from the probe to the surface). In practice, a feedback mechanism keeps the current constant by moving the tip up and down. The motion of the tip gives a map of the surface that reveals details smaller than 0.01 nm, about 1/100 the diameter of an atom! For the development of the scanning tunneling microscope, Gerd Binnig and Heinrich Rohrer were awarded the 1986 Nobel Prize in physics.

Chapter Summary

	Section		Section
Time-independent Schrödinger equation	$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + U(x)\psi(x) = E\psi(x)$ 5.3	Probability in interval x_1 to x_2	$P(x_1 : x_2) = \int_{x_1}^{x_2} \psi(x) ^2 dx$ 5.3
Time-dependent Schrödinger equation	$\Psi(x, t) = \psi(x)e^{-i\omega t}$ 5.3	Average or expectation value of $f(x)$	$[f(x)]_{av} = \int_{-\infty}^{+\infty} \psi(x) ^2 f(x) dx$ 5.3
Probability density	$P(x) = \psi(x) ^2$ 5.3	Constant potential energy, $E > U_0$	$\psi(x) = A \sin kx + B \cos kx,$ $k = \sqrt{2m(E - U_0)}/\hbar$ 5.4
Normalization condition	$\int_{-\infty}^{+\infty} \psi(x) ^2 dx = 1$ 5.3		

		Section			Section
Constant potential energy, $E < U_0$	$\psi(x) = Ae^{k'x} + Be^{-k'x}$, $k' = \sqrt{2m(U_0 - E)/\hbar^2}$	5.4	Simple harmonic oscillator ground state	$\psi(x) = (m\omega_0/\hbar\pi)^{1/4} e^{-(\sqrt{km}/2\hbar)x^2}$	5.5
Infinite potential energy well	$\psi_n(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$, $E_n = \frac{h^2 n^2}{8mL^2} \quad (n = 1, 2, 3, \dots)$	5.4	Simple harmonic oscillator energies	$E_n = \left(n + \frac{1}{2}\right) \hbar\omega_0 \quad (n = 0, 1, 2, \dots)$	5.5
Two-dimensional infinite well	$\psi(x, y) = \frac{2}{L} \sin \frac{n_x \pi x}{L} \sin \frac{n_y \pi y}{L}$ $E = \frac{h^2}{8mL^2} (n_x^2 + n_y^2)$	5.4	Potential energy step, $E > U_0$	$\psi_0(x < 0) = A \sin k_0 x + B \cos k_0 x$ $\psi_1(x > 0) = C \sin k_1 x + D \cos k_1 x$	5.6
			Potential energy step, $E < U_0$	$\psi_0(x < 0) = A \sin k_0 x + B \cos k_0 x$ $\psi_1(x > 0) = Ce^{k_1 x} + De^{-k_1 x}$	5.6

Questions

- Newton's laws can be solved to give the future behavior of a particle. In what sense does the Schrödinger equation also do this? In what sense does it not?
- Why is it important for a wave function to be normalized? Is an unnormalized wave function a solution to the Schrödinger equation?
- What is the physical meaning of $\int_{-\infty}^{+\infty} |\psi|^2 dx = 1$?
- What are the dimensions of $\psi(x)$? Of $\psi(x, y)$?
- None of the following are permitted as solutions of the Schrödinger equation. Give the reasons in each case.
 - $\psi(x) = A \cos kx \quad x < 0$
 $\psi(x) = B \sin kx \quad x > 0$
 - $\psi(x) = Ax^{-1}e^{-kx} \quad -L \leq x \leq L$
 - $\psi(x) = A \sin^{-1} kx$
 - $\psi(x) = A \tan kx \quad x > 0$
- What happens to the probability density in the infinite well when $n \rightarrow \infty$? Is this consistent with classical physics?
- How would the solution to the infinite potential energy well be different if the well extended from $x = x_0$ to $x = x_0 + L$, where x_0 is a nonzero value of x ? Would any of the measurable properties be different?
- How would the solution to the one-dimensional infinite potential energy well be different if the potential energy were not zero for $0 \leq x \leq L$ but instead had a constant value U_0 ? What would be the energies of the excited states? What would be the wavelengths of the standing de Broglie waves? Sketch the behavior of the lowest two wave functions.
- Assuming a pendulum to behave like a quantum oscillator, what are the energy differences between the quantum states of a pendulum of length 1 m? Are such differences observable?
- For the potential energy barrier (Figure 5.26), is the wavelength for $x > L$ the same as the wavelength for $x < 0$? Is the amplitude the same?
- Suppose particles were incident on the potential energy step from the *positive* x direction. Which of the four coefficients of Eq. 5.56 would be set to zero? Why?
- The energies of the excited states of the systems we have discussed in this chapter have been exact—there is no energy uncertainty. What does this suggest about the lifetime of particles in those excited states? Left on its own, will a particle ever make transitions from one state to another?
- Explain how the behavior of a particle in a one-dimensional infinite well can be considered in terms of standing de Broglie waves.
- How would you design an experiment to observe barrier penetration with sound waves? What range of thicknesses would you choose for the barrier?
- If U_0 were negative in Figure 5.26, how would the wave functions appear for $E > 0$?
- Does Eq. 5.2 imply that we know the momentum of the particle exactly? If so, what does the uncertainty principle indicate about our knowledge of its location? How can you reconcile this with our knowledge that the particle *must* be in the well?
- Do sharp boundaries and discontinuous jumps of potential energy occur in nature? If not, how would our analysis of potential energy steps and barriers be different?

Problems

5.1 Behavior of a Wave at a Boundary

1. A ball falls from rest at a height H above a lake. Let $y = 0$ at the surface of the lake. As it falls, it experiences a gravitational force $-mg$. When it enters the water, it experiences a buoyant force B so the net force in the water is $B - mg$. (a) Write expressions for $v(t)$ and $y(t)$ while the ball is falling in air. (b) In the water, let $v_2(t) = at + b$ and $y_2(t) = \frac{1}{2}at^2 + bt + c$, where $a = (B - mg)/m$. Use the continuity conditions at the surface of the water to find the constants b and c .
2. A wave has the form $y = A \cos(2\pi x/\lambda + \pi/3)$ when $x < 0$. For $x > 0$, the wavelength is $\lambda/2$. By applying continuity conditions at $x = 0$, find the amplitude (in terms of A) and phase of the wave in the region $x > 0$. Sketch the wave, showing both $x < 0$ and $x > 0$.

5.2 Confining a Particle

3. The lowest energy of a particle in an infinite one-dimensional well is 5.6 eV. If the width of the well is doubled, what is its lowest energy?
4. What are the three longest wavelengths of the de Broglie waves that describe an electron that is confined in an infinite well of width 0.144 nm?
5. An electron is trapped in a one-dimensional region of width 0.062 nm. Find the three smallest possible values allowed for the energy of the electron.
6. What is the minimum energy of a neutron ($mc^2 = 940$ MeV) confined to a region of space of nuclear dimensions (1.2×10^{-14} m)?

5.3 The Schrödinger Equation

7. In the region $0 \leq x \leq a$, a particle is described by the wave function $\psi_1(x) = -b(x^2 - a^2)$. In the region $a \leq x \leq w$, its wave function is $\psi_2(x) = (x - d)^2 - c$. For $x \geq w$, $\psi_3(x) = 0$. (a) By applying the continuity conditions at $x = a$, find c and d in terms of a and b . (b) Find w in terms of a and b .
8. A particle is described by the wave function $\psi(x) = b(a^2 - x^2)$ for $-a \leq x \leq +a$ and $\psi(x) = 0$ for $x \leq -a$ and $x \geq +a$, where a and b are positive real constants. (a) Using the normalization condition, find b in terms of a . (b) What is the probability to find the particle at $x = +a/2$ in a small interval of width $0.010a$? (c) What is the probability for the particle to be found between $x = +a/2$ and $x = +a$?
9. In a certain region of space, a particle is described by the wave function $\psi(x) = Cxe^{-bx}$ where C and b are real constants. By substituting into the Schrödinger equation, find the potential energy in this region and also find the energy of the particle. (Hint: Your solution must give an

energy that is a constant everywhere in this region, independent of x .)

10. A particle is represented by the following wave function:

$$\begin{aligned}\psi(x) &= 0 & x < -L/2 \\ &= C(2x/L + 1) & -L/2 < x < 0 \\ &= C(-2x/L + 1) & 0 < x < +L/2 \\ &= 0 & x > +L/2\end{aligned}$$

- (a) Use the normalization condition to find C . (b) Evaluate the probability to find the particle in an interval of width $0.010L$ at $x = L/4$ (that is, between $x = 0.245L$ and $x = 0.255L$. (No integral is necessary for this calculation.) (c) Evaluate the probability to find the particle between $x = 0$ and $x = +L/4$. (d) Find the average value of x and the rms value of x : $x_{\text{rms}} = \sqrt{(x^2)_{\text{av}}}$.
11. (a) Find the normalization constant A for the wave function $\psi(x) = Axe^{-bx}$. (b) Find the normalization constant for the wave function $\psi(x) = Ae^{-b|x|}$.
12. The wave function of a particle is $\psi(x) = Ae^{-bx}$ for $x > 0$ and $\psi(x) = Ae^{bx}$ for $x < 0$. Find the corresponding potential energy and energy eigenvalue.

5.4 Applications of the Schrödinger Equation

13. An electron is trapped in an infinitely deep one-dimensional well of width 0.285 nm. Initially, the electron occupies the $n = 4$ state. (a) Suppose the electron jumps to the ground state with the accompanying emission of a photon. What is the energy of the photon? (b) Find the energies of other photons that might be emitted if the electron takes other paths between the $n = 4$ state and the ground state.
14. A particle in an infinite well is in the ground state with energy 1.54 eV. How much energy must be added to the particle to reach the second excited state ($n = 3$)? The third excited state ($n = 4$)?
15. Show that Eq. 5.31 gives the value $A = \sqrt{2/L}$.
16. A particle is trapped in an infinite one-dimensional well of width L . If the particle is in its ground state, evaluate the probability to find the particle (a) between $x = 0$ and $x = L/3$; (b) between $x = L/3$ and $x = 2L/3$; (c) between $x = 2L/3$ and $x = L$.
17. A particle is confined between rigid walls separated by a distance $L = 0.189$ nm. The particle is in the second excited state ($n = 3$). Evaluate the probability to find the particle in an interval of width 1.00 pm located at (a) $x = 0.188$ nm; (b) $x = 0.031$ nm; (c) $x = 0.079$ nm. (Hint: No integrations are required for this problem; use Eq. 5.7 directly.) (d) What would be the corresponding results for a classical particle?

18. What is the next level (above $E = 50E_0$) of the two-dimensional particle in a box in which the degeneracy is greater than 2?
19. A particle is confined to a two-dimensional box of length L and width $2L$. The energy values are $E = (\hbar^2 \pi^2 / 2mL^2)(n_x^2 + n_y^2/4)$. Find the two lowest degenerate levels.
20. Show by direct substitution that Eq. 5.39 gives a solution to the two-dimensional Schrödinger equation, Eq. 5.37. Find the relationship between k_x , k_y , and E .
21. A particle is confined to a three-dimensional region of space of dimensions L by L by L . The energy levels are $(\hbar^2 \pi^2 / 2mL^2)(n_x^2 + n_y^2 + n_z^2)$, where n_x , n_y , and n_z are integers ≥ 1 . Sketch an energy level diagram, showing the energies, quantum numbers, and degeneracies for the lowest 10 energy levels.

5.5 The Simple Harmonic Oscillator

22. Using the normalization condition, show that the constant A has the value $(m\omega_0/\hbar\pi)^{1/4}$ for the one-dimensional simple harmonic oscillator in its ground state.
23. (a) At the classical turning points $\pm x_0$ of the simple harmonic oscillator, $K = 0$ and so $E = U$. From this relationship, show that $x_0 = (\hbar\omega_0/k)^{1/2}$ for an oscillator in its ground state. (b) Find the turning points in the first and second excited states.
24. Use the ground-state wave function of the simple harmonic oscillator to find x_{av} , $(x^2)_{av}$, and Δx . Use the normalization constant $A = (m\omega_0/\hbar\pi)^{1/4}$.
25. (a) Using a symmetry argument rather than a calculation, determine the value of p_{av} for a simple harmonic oscillator. (b) Conservation of energy for the harmonic oscillator can be used to relate p^2 to x^2 . Use this relation, along with the value of $(x^2)_{av}$ from Problem 24, to find $(p^2)_{av}$ for the oscillator in its ground state. (c) Using the results of (a) and (b), show that $\Delta p = \sqrt{\hbar\omega_0 m/2}$.
26. The ground-state energy of an oscillating electron is 1.24 eV. How much energy must be added to the electron to move it to the second excited state? The fourth excited state?
27. Compare the probabilities for an oscillating particle in its ground state to be found in a small interval of width dx at the center of the well and at the classical turning points.
28. There is a minimum energy of $\frac{1}{2}\hbar\omega_0$ in any vibrating system; this energy is sometimes known as the *zero-point motion*. (a) Use an argument based on the uncertainty principle to explain why the vibrating system can never have $E = 0$. (b) The hydrogen molecule H_2 can be treated as a vibrating system, with an effective force constant $k = 3.5 \times 10^3$ eV/nm². Compute the zero-point energy of one of the protons in H_2 . How does it compare with the molecular binding energy of 4.5 eV? (c) Compute the amplitude of the zero-point motion and compare with the atomic spacing of 0.074 nm.
29. Find the locations where the probability density has its maximum values for the wave function $\psi(x) = A(2ax^2 - 1)e^{-ax^2/2}$, which represents the second excited state of the simple harmonic oscillator.

5.6 Steps and Barriers

30. Find the value of K at which Eq. 5.60 has its maximum value, and show that Eq. 5.61 is the maximum value of Δx .
31. For a particle with energy $E < U_0$ incident on the potential energy step, use ψ_0 and ψ_1 from Eqs. 5.57, and evaluate the constants B and D in terms of A by applying the boundary conditions at $x = 0$.
32. Using the wave functions of Eq. 5.55 for the potential energy step, apply the boundary conditions of ψ and $d\psi/dx$ to find B' and C' in terms of A' , for the potential step when particles are incident from the negative x direction. Evaluate the ratios $|B'|^2/|A'|^2$ and $|C'|^2/|A'|^2$ and interpret.
33. (a) Write down the wave functions for the three regions of the potential energy barrier (Figure 5.25) for $E < U_0$. You will need six coefficients in all. Use complex exponential notation. (b) Use the boundary conditions at $x = 0$ and at $x = L$ to find four relationships among the six coefficients. (Do not try to solve these relationships.) (c) Suppose particles are incident on the barrier from the left. Which coefficient should be set to zero? Why?
34. Repeat Problem 33 for the potential energy barrier when $E > U_0$, and sketch a representative probability density that shows several cycles of the wave function. In your sketch, make sure the amplitude and wavelength in each region accurately describe the situation.

General Problems

35. An electron is trapped in a one-dimensional well of width 0.132 nm. The electron is in the $n = 10$ state. (a) What is the energy of the electron? (b) What is the uncertainty in its momentum? (*Hint*: Use Eq. 4.10.) (c) What is the uncertainty in its position? How do these results change as $n \rightarrow \infty$? Is this consistent with classical behavior?
36. Sketch the form of a possible solution to the Schrödinger equation for each of the potential energies shown in Figure 5.34. The potential energies go to infinity at the boundaries. In each case show several cycles of the wave function. In your sketches, pay attention to the continuity conditions (where applicable) and to changes in the wavelength and amplitude.
37. Show that the average value of x^2 in the one-dimensional infinite potential energy well is $L^2(1/3 - 1/2n^2\pi^2)$.

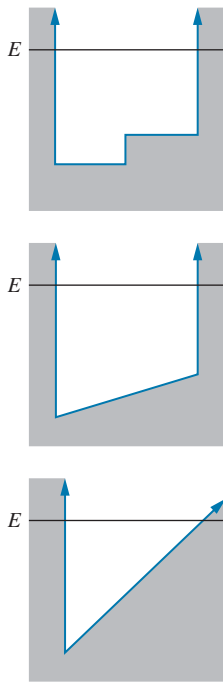


FIGURE 5.34 Problem 36.

38. Use the result of Problem 37 to show that, for the infinite one-dimensional well, defining $\Delta x = \sqrt{(x^2)_{\text{av}} - (x_{\text{av}})^2}$ gives $\Delta x = L\sqrt{1/12 - 1/2\pi^2 n^2}$.
39. (a) In the infinite one-dimensional well, what is p_{av} ? (Use a symmetry argument.) (b) What is $(p^2)_{\text{av}}$? [Hint: What is $(p^2/2m)_{\text{av}}$?] (c) Defining $\Delta p = \sqrt{(p^2)_{\text{av}} - (p_{\text{av}})^2}$, show that $\Delta p = \hbar n/2L$.
40. The first excited state of the harmonic oscillator has a wave function of the form $\psi(x) = Axe^{-ax^2}$. (a) Follow the

method outlined in Section 5.5 to find a and the energy E . (b) Use the normalization condition to show that $A = \pi^{-1/4} \sqrt{2(m\omega_0/\hbar)^{3/4}}$.

41. Using the normalization constant A from Problem 22 and the value of a from Eq. 5.49, evaluate the probability to find an oscillator in the ground state beyond the classical turning points $\pm x_0$. This problem cannot be solved in closed, analytic form. Develop an approximate, numerical method using a graph, calculator, or computer. Assume an electron bound to an atomic-sized region ($x_0 = 0.1 \text{ nm}$) with an effective force constant of 1.0 eV/nm^2 .
42. A two-dimensional harmonic oscillator has energy $E = \hbar\omega_0(n_x + n_y + 1)$, where n_x and n_y are integers beginning with zero. (a) Justify this result based on the energy of the one-dimensional oscillator. (b) Sketch an energy-level diagram similar to Figure 5.20, showing the lowest four energy levels. For each level, show the value of E (in units of $\hbar\omega_0$), the quantum numbers n_x and n_y , and the degeneracy. (c) Show that the degeneracy of each level is equal to $n_x + n_y + 1$.
43. Consider the normalized wave function $\psi(x) = b^{-1/2}e^{-|x|/b}$. (a) Evaluate Δx for this wave function. (b) Estimate Δx from the width of the probability distribution where it falls to half its maximum value.
44. (a) Using the normalized wave function for the first excited state of the harmonic oscillator (see Problem 40 and Figure 5.22), find the locations of the maxima in the probability density. (b) For either maximum, find the locations where the probability density drops to half of its maximum value. Use a numerical or graphical method. (c) Use the result of part (b) to estimate the width of the region where the particle is most probably located. (d) Compare the result of part (c) with the calculated value of Δx .

THE RUTHERFORD-BOHR MODEL OF THE ATOM



This model of the atom, based on the work of Rutherford and Bohr, shows electrons circulating about the nucleus like planets circulating about the Sun. It can be a useful model for some purposes, but it does not represent even approximately the structure of real atoms. In Chapters 7 and 8, we will learn more about the behavior and properties of electrons in atoms. Michael Dunning / Science Source

Our goal in this chapter is to understand some of the details of atomic structure that can be learned from experimental studies of atoms. In particular, we discuss two types of experiments that are important in the development of our theory of atomic structure: the scattering of charged particles by atoms, which tells us about the distribution of electric charge in atoms, and the emission or absorption of radiation by atoms, which tells us about their excited states.

We use the information obtained from these experiments to develop an *atomic model*, which helps us understand and explain the properties of atoms. A model is usually an oversimplified picture of a more complex system, which provides some insight into its operation but may not be sufficiently detailed to explain *all* of its properties.

In this chapter, we discuss the experiments that led to the *Rutherford-Bohr model* (also known simply as the Bohr model), which is based on the familiar “planetary” structure in which the electrons orbit about the nucleus like planets about the Sun. Even though this model is not strictly valid from the standpoint of wave mechanics, it does help us understand many atomic properties, especially the excited states of the simplest atom, hydrogen. In Chapter 7, we show how wave mechanics changes our picture of the hydrogen atom, and in Chapter 8 we consider the structure of more complicated atoms.

6.1 BASIC PROPERTIES OF ATOMS

Before we begin to construct a model of the atom, it is helpful to summarize some of the basic properties of atoms.

1. *Atoms are very small*, about 0.1 nm (0.1×10^{-9} m) in radius. Thus any effort to “see” an atom using visible light ($\lambda = 500$ nm) is hopeless owing to diffraction effects. We can make a crude estimate of the *maximum* size of an atom in the following way. Consider a cube of elemental matter—for example, iron. Iron has a density of about 8 g/cm³ and a molar mass of 56 g. One mole of iron (56 g) contains Avogadro’s number of atoms, about 6×10^{23} . Thus 6×10^{23} atoms occupy about 7 cm³ and so 1 atom occupies about 10^{-23} cm³. If we assume that the atoms of a solid are packed together in the most efficient possible way, like hard spheres in contact, then the diameter of one atom is about $\sqrt[3]{10^{-23} \text{ cm}^3} = 2 \times 10^{-8} \text{ cm} = 0.2 \text{ nm}$.
2. *Atoms are stable*—they do not spontaneously break apart into smaller pieces or collapse; therefore the internal forces that hold the atom together must be in equilibrium. This immediately tells us that the forces that pull the parts of an atom together must be opposed in some way; otherwise atoms would collapse.
3. *Atoms contain negatively charged electrons, but are electrically neutral*. If we disturb an atom or collection of atoms with sufficient force, electrons are emitted. We learn this fact from studying the Compton effect and the photoelectric effect. We also learned in Chapter 4 that even though electrons are emitted from the nuclei of atoms in certain radioactive decay processes, they don’t “exist” in those nuclei but are manufactured there by some process. Electrons were excluded from the nucleus based on the uncertainty principle, which forbids emitted electrons of the energies observed in the

laboratory from existing in the nucleus (see Example 4.9). The uncertainty principle places no such restriction on the existence of electrons in a volume as large as an atom (see Problem 1).

We can also easily observe that bulk matter is electrically neutral, and we assume that this is likewise a property of the atoms. Experiments with beams of individual atoms support this assumption. From these experimental facts, we deduce that an atom with Z negatively charged electrons must also contain a net positive charge of Ze .

4. *Atoms emit and absorb electromagnetic radiation.* This radiation may take many forms—visible light ($\lambda \sim 500$ nm), X rays ($\lambda \sim 1$ nm), ultraviolet rays ($\lambda \sim 10$ nm), infrared rays ($\lambda \sim 0.1$ μm), and so forth. In fact it is from observation of these emitted and absorbed radiations, which can be measured with great precision, that we learn most of what we know about atoms. In a typical emission measurement, an electric current is passed through a glass tube containing a small sample of the gas phase of the element under study, and radiation is emitted when an excited atom returns to its ground state. The absorption wavelengths can be measured by passing a beam of white light through a sample of the gas and noting which colors are removed from the white light by absorption in the gas. One particularly curious feature of the atomic radiations is that atoms don't always emit and absorb radiations at the same wavelengths—some wavelengths present in the *emission* experiment do not also appear in the *absorption* experiment. Any successful theory of atomic structure must be able to account for these emission and absorption wavelengths.

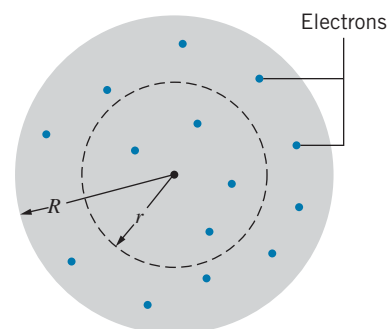


FIGURE 6.1 The Thomson model of the atom. Z electrons are imbedded in a uniform sphere of positive charge Ze and radius R . An imaginary spherical surface of radius r contains a fraction r^3/R^3 of the positive charge.

6.2 SCATTERING EXPERIMENTS AND THE THOMSON MODEL

An early model of the structure of the atom was proposed (in 1904) by J. J. Thomson, who was known for his previous identification of the electron and measurement of its charge-to-mass ratio e/m . The Thomson model incorporates many of the known properties of atoms: size, mass, number of electrons, and electrical neutrality. In this model, an atom contains Z electrons that are embedded in a uniform sphere of positive charge (Figure 6.1). The total positive charge of the sphere is Ze , the mass of the sphere is essentially the mass of the atom (the electrons don't contribute significantly to the total mass), and the radius R of the sphere is the radius of the atom. (This model is sometimes known as the “plum-pudding” model, because the electrons are distributed throughout the atom like raisins in a plum pudding.) As we will see, the Thomson model gives predictions that disagree with experiment, and so it is not the correct way of understanding the structure of atoms.

One way of studying atoms is by probing the distribution of electric charge in their interior, which we can do by bombarding the atom with charged particles and observing the angle by which particles are deflected from their original direction. This type of experiment is called a *scattering experiment*. Ideally we would like to do this experiment with a single atom, such as is represented in Figure 6.2. The scattering angle θ depends on the *impact parameter* b , which measures the distance from the center of the atom that a projectile would pass

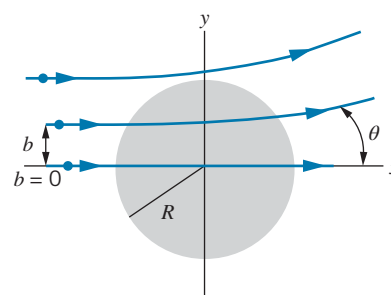


FIGURE 6.2 A positively charged particle is deflected by an angle θ as it passes through a positively charged sphere, representing a Thomson model atom. The scattering angle depends on the value of the impact parameter b , which varies from 0 to R .

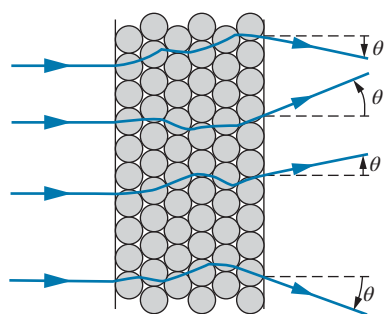
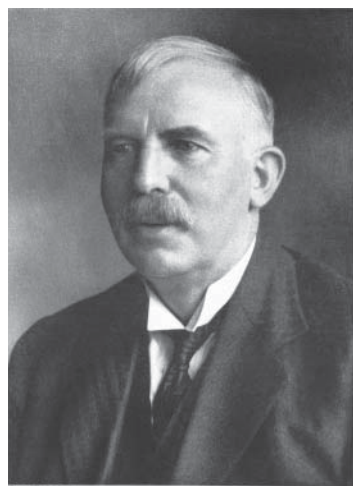


FIGURE 6.3 Scattering by a thin foil. Some individual scatterings tend to increase θ , while others tend to decrease θ .



Ernest Rutherford (1871–1937, England). Founder of nuclear physics, he is known for his pioneering work on alpha-particle scattering and radioactive decays. His inspiring leadership influenced a generation of British nuclear and atomic scientists.

if it were not deflected. Each different value of the impact parameter results in a different value of the scattering angle.

The particle is deflected from its original trajectory by the electrical forces exerted on the particle by the atom. For a positively charged particle, these forces are (1) a repulsive force due to the positive charge of the atom and (2) an attractive force due to the negatively charged electrons. We assume that the mass of the deflected particle is much greater than the mass of an electron but also much less than the mass of the atom. In the encounter between the projectile and an electron, the forces exerted on each by the other are equal and opposite (by Newton's third law), and so the principal victim of the encounter is the much less massive electron; the effect on the projectile is negligible. (Imagine rolling a bowling ball through a field of Ping-Pong balls!) We thus need consider only the positively charged atom as a cause of the deflection of the particle. By the same argument, we neglect any possible motion of the more massive atom caused by the passage of the projectile. The basic experiment, then, is the scattering of a positively charged projectile by the stationary positively charged massive part of the atom.

In practice we cannot do the experiment with one atom. Instead, we bombard a thin foil, as in Figure 6.3. The scattering angle θ that we observe in the laboratory is the result of scattering by many atoms, with impact parameters that we do not know and cannot control. Let's assume that for a single atom the average scattering angle is θ_{av} , which represents an average over all possible impact parameters from zero up to the atomic radius R . For a typical foil thickness of $1\mu\text{m}$ (10^{-6} m), the projectile is scattered by about 10^4 atoms.

The total scattering angle θ is determined by statistical considerations, because some of the individual scatterings move the projectile toward larger scattering angles and some toward smaller angles, as represented in Figure 6.3. This is an example of a “random walk” problem—for N scatterings, the most likely observed net scattering angle θ is related to the average individual scattering angle by

$$\theta \simeq \sqrt{N}\theta_{av} \quad (6.1)$$

According to the Thomson model, the average scattering angle for a single atom is on the order of 0.01° , and for a foil that is 10^4 atoms thick the net scattering angle should be about 1° . This is consistent with experimental observations.

The most critical test of the Thomson model, which it fails completely, occurs when we examine the probability for scattering at large angles. If each individual scattering deflects the projectile through an angle of around 0.01° , then to observe projectiles scattered through a total angle greater than 90° , we must have about 10^4 successive scatterings, *all* of which push the projectile toward larger angles. Because the probabilities of individual scatterings toward either larger or smaller angles are equal, the probability of having 10^4 successive scatterings toward larger angles, like the probability of finding 10^4 successive heads in tossing a coin, is about $(1/2)^{10,000} = 10^{-3000}$.

An experiment to observe this scattering was performed by Hans Geiger and Ernest Marsden in the laboratory of Ernest Rutherford at Manchester University in 1910. For projectiles they used alpha particles, which are nuclei of helium (of charge $+2e$) emitted in radioactive decay. Their results showed that the probability of an alpha-particle scattering at angles greater than 90°

was about 10^{-4} . This remarkable discrepancy between the expected value based on the Thomson model (10^{-3000}) and the observed value (10^{-4}) was described by Rutherford in this way:

It was quite the most incredible event that ever happened to me in my life. It was as incredible as if you fired a 15-inch shell at a piece of tissue paper and it came back and hit you.

The analysis of the results of such scattering experiments led Rutherford to propose that the mass and positive charge of the atom are not distributed uniformly over the volume of the atom, but instead are concentrated in an extremely small region, about 10^{-14} m in diameter, at the center of the atom. In Section 6.3, we will see how this proposal is consistent with the large-angle scattering results.

Scattering in the Thomson Model (Optional)

Let's assume that a projectile of positive charge ze is incident on an atom of radius R that we represent according to the Thomson model as a uniform sphere of positive charge Ze . The force on the projectile when it is a distance r from the center of the atom can be computed using Gauss's law (see Problem 2):

$$F = \frac{zZe^2}{4\pi\epsilon_0 R^3} r \quad (6.2)$$

Before discussing the scattering, we should note that this equation can also describe (if we put $z = 1$) the force on an electron embedded in the Thomson atom at a distance r from its center. This force can be written $F = kr$ with $k = Ze^2/4\pi\epsilon_0 R^3$. This linear restoring force permits the electrons to oscillate about their equilibrium positions just like a mass on a spring subject to the linear restoring force $F = kx$. We therefore expect the electrons in the Thomson atom to oscillate about their equilibrium positions with a frequency $f = (2\pi)^{-1} \sqrt{k/m}$, where k is the force constant. Because an oscillating electric charge radiates electromagnetic waves whose frequency is identical to the oscillation frequency, we might expect, based on the Thomson model, that the radiation emitted by atoms would show this characteristic frequency. This turns out *not* to be true (see Problem 3); the calculated frequencies do not correspond to the frequencies observed for radiation emitted by atoms.

The exact calculation of the scattering angle for different values of the impact parameter in the Thomson model of the atom is fairly complicated, but for our purposes we want only an estimate of the average value of the angle. As we will find out later, it's not very important if our estimate is off by a small factor.

Initially the projectile moves in the x direction in the geometry of Figure 6.2, but the atom exerts a force in the y direction that produces a small component of momentum p_y in that direction. Using Newton's second law we can find the momentum from the *impulse* received by the projectile due to the electrostatic force:

$$p_y = \int F_y dt \quad (6.3)$$

Rather than carry out this complicated integral for a force that is changing in magnitude and direction as the projectile travels, we'll estimate the average

scattering angle by choosing an average value for the impact parameter, namely $b = R/2$ (representing the middle trajectory of Figure 6.2), and we'll assume the force acts in the y direction for a time Δt determined by the projectile's flight along a line of length roughly equal to R . This underestimates the amount of time during which the force acts but overestimates the effect of the force (which doesn't act purely in the y direction along the entire trajectory), so to some extent these two effects should cancel each other.

Making these approximations, we obtain

$$p_y \cong F \Delta t \cong \frac{zZe^2(R/2)}{4\pi\epsilon_0 R^3} \frac{R}{v} = \frac{zZe^2}{8\pi\epsilon_0 Rv} \quad (6.4)$$

The angle θ is small, so we can make the approximation $\tan \theta \cong \theta$, and we can assume that p_x changes very little from its initial value mv , and so the average scattering angle is

$$\theta_{av} \cong \tan \theta_{av} = \frac{p_y}{p_x} = \frac{p_y}{mv} = \frac{zZe^2}{8\pi\epsilon_0 Rv} \frac{1}{mv} = \frac{zZe^2}{16\pi\epsilon_0 RK} \quad (6.5)$$

using the nonrelativistic kinetic energy $K = \frac{1}{2}mv^2$. This gives an estimate of the scattering angle when the impact parameter b is equal to half the radius R . Smaller values of b will give smaller deflection angles, and larger values of b will give larger angles, so this is a reasonable estimate for the average scattering angle for a Thomson model atom.

Example 6.1

Using the Thomson model, estimate the average scattering angle when alpha particles ($z = 2$) with kinetic energy 3 MeV are scattered from gold ($Z = 79$). The atomic radius of gold is 0.179 nm.

Solution

Using $e^2/4\pi\epsilon_0 = 1.44 \text{ eV} \cdot \text{nm}$, we have

$$\begin{aligned} \theta_{av} &\cong \frac{zZe^2}{16\pi\epsilon_0 RK} = \frac{1}{4} \frac{e^2}{4\pi\epsilon_0} \frac{zZ}{RK} \\ &= \frac{0.25(1.44 \text{ eV} \cdot \text{nm})(2)(79)}{(0.179 \text{ nm})(3 \times 10^6 \text{ eV})} \\ &= 1 \times 10^{-4} \text{ rad} = 0.01^\circ \end{aligned}$$

Even though this result represents a rough estimate of the average scattering angle in the Thomson model of the atom, its accuracy does not affect our conclusions about the failure of the model. Even if our estimate were too small by as much as a factor of 10 (which is highly unlikely), we would be comparing an expected probability of 10^{-300} (instead of 10^{-3000}) with the observed 10^{-4} , still a spectacular disagreement. Any reasonable estimate shows the complete failure of the Thomson model to account for these scattering experiments.

6.3 THE RUTHERFORD NUCLEAR ATOM

In analyzing the scattering of alpha particles, Rutherford concluded that the most likely way an alpha particle ($m = 4 \text{ u}$) can be deflected through large angles is by a *single* collision with a more massive object. Rutherford therefore proposed that the charge and mass of the atom are concentrated at its center, in a region called the *nucleus*. Figure 6.4 illustrates the scattering

geometry in this case. The projectile, of charge ze , experiences a repulsive force due to the positively charged nucleus:

$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1||q_2|}{r^2} = \frac{(ze)(Ze)}{4\pi\epsilon_0 r^2} \quad (6.6)$$

(Compare this with Eq. 6.2, which describes a projectile that is inside the sphere of charge Ze and so feels only a portion of the positive charge. We assume now that the projectile is always outside the nucleus, so it feels the full nuclear charge Ze .) The atomic electrons, with their small mass, do not appreciably affect the path of the projectile and we neglect their effect on the scattering. We also assume that the nucleus is so much more massive than the projectile that it does not move during the scattering process; because no recoil motion is given to the nucleus, the initial and final kinetic energies K of the projectile are equal.

As Figure 6.4 shows, for each impact parameter b , there is a certain scattering angle θ , and we need the relationship between b and θ . The projectile can be shown* to follow a hyperbolic path; in polar coordinates r and ϕ , the equation of the hyperbola is

$$\frac{1}{r} = \frac{1}{b} \sin \phi + \frac{zZe^2}{8\pi\epsilon_0 b^2 K} (\cos \phi - 1) \quad (6.7)$$

As shown in Figure 6.5, the initial position of the particle is $\phi = 0, r \rightarrow \infty$, and the final position is $\phi = \pi - \theta, r \rightarrow \infty$. Using the coordinates at the final position, Eq. 6.7 reduces to

$$b = \frac{zZe^2}{8\pi\epsilon_0 K} \cot \frac{1}{2}\theta = \frac{zZ}{2K} \frac{e^2}{4\pi\epsilon_0} \cot \frac{1}{2}\theta \quad (6.8)$$

(This result is written in this form so that $e^2/4\pi\epsilon_0 = 1.44 \text{ eV} \cdot \text{nm}$ or $\text{MeV} \cdot \text{fm}$ can be easily inserted.) A projectile that approaches the nucleus with impact parameter b will be scattered at an angle θ ; projectiles approaching with smaller values of b will be scattered through larger angles, as shown in Figure 6.4.

We divide our study of the scattering of charged projectiles by nuclei (which is commonly called *Rutherford scattering*) into three parts: (1) calculation of the fraction of projectiles scattered at angles greater than some value of θ , (2) the Rutherford scattering formula and its experimental verification, and (3) the closest approach of a projectile to the nucleus.

1. The Fraction of Projectiles Scattered at Angles Greater than θ . From Figure 6.4 we see immediately that every projectile with impact parameters less than a given value of b will be scattered at angles greater than its corresponding θ . What is the chance of a projectile having an impact parameter less than a given value of b ? Suppose the foil were one atom thick—a single layer of atoms packed tightly together, as in Figure 6.6. Each atom is represented by a circular disc of area πR^2 . If the foil contains N atoms, its total area is $N\pi R^2$. For scattering at angles greater than θ , the impact parameter must fall between zero and b —that is, the projectile must approach the atom within a circular disc of area πb^2 . If the projectiles are spread uniformly over

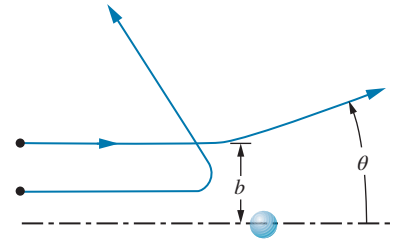


FIGURE 6.4 Scattering by a nuclear atom. The path of the scattered particle is a hyperbola. Smaller impact parameters give larger scattering angles.

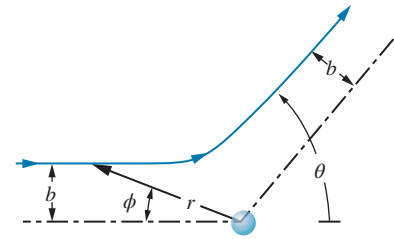


FIGURE 6.5 The hyperbolic trajectory of a scattered particle.

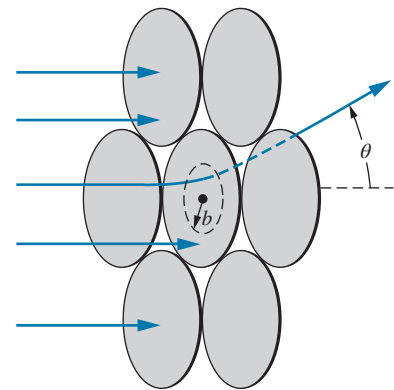


FIGURE 6.6 Scattering geometry for many atoms. For impact parameter b , the scattering angle is θ . If the particle enters the atom within the disc of area πb^2 , its scattering angle will be larger than θ .

*See, for example, R. M. Eisberg and R. Resnick, *Quantum Physics of Atoms, Molecules, Solids, Nuclei, and Particles*, 2nd ed. (New York, Wiley, 1985).

the area of the disc, then the fraction of projectiles that fall within that area is just $\pi b^2/\pi R^2$.

A real scattering foil may be thousands or tens of thousands of atoms thick. Let t be the thickness of the foil and A its area, and let ρ and M be the density and molar mass of the material of which the foil is made. The volume of the foil is then At , its mass is ρAt , the number of moles is $\rho At/M$, and the number of atoms or nuclei per unit volume is

$$n = N_A \frac{\rho At}{M} \frac{1}{At} = \frac{N_A \rho}{M} \quad (6.9)$$

where N_A is Avogadro's number (the number of atoms per mole). As seen by an incident projectile, the number of nuclei per unit area is $nt = N_A \rho t/M$; that is, on average, each nucleus contributes an area $(N_A \rho t/M)^{-1}$ to the field of view of the projectile. For scattering at angles greater than θ , it must once again be true that the projectile must fall within an area πb^2 of the center of an atom; the fraction scattered at angles greater than θ is just the fraction that approaches an atom within the area πb^2 :

$$f_{>\theta} = f_{>\theta} = nt\pi b^2 \quad (6.10)$$

assuming that the incident particles are spread uniformly over the area of the foil.

Example 6.2

A gold foil ($\rho = 19.3 \text{ g/cm}^3$, $M = 197 \text{ g/mole}$) has a thickness of $2.0 \times 10^{-4} \text{ cm}$. It is used to scatter alpha particles of kinetic energy 8.0 MeV . (a) What fraction of the alpha particles is scattered at angles greater than 90° ? (b) What fraction of the alpha particles is scattered at angles between 90° and 45° ?

Solution

(a) For this case the number of nuclei per unit volume can be computed as

$$\begin{aligned} n &= \frac{N_A \rho}{M} = \frac{(6.02 \times 10^{23} \text{ atoms/mole})(19.3 \text{ g/cm}^3)}{(197 \text{ g/mole})(1 \text{ m/}10^2 \text{ cm})^3} \\ &= 5.9 \times 10^{28} \text{ m}^{-3} \end{aligned}$$

For scattering at 90° , the impact parameter b can be found from Eq. 6.8 as

$$\begin{aligned} b &= \frac{zZ}{2K} \frac{e^2}{4\pi\epsilon_0} \cot \frac{1}{2}\theta \\ &= \frac{(2)(79)}{2(8.0 \text{ MeV})} (1.44 \text{ MeV} \cdot \text{fm}) \cot 45^\circ \\ &= 14 \text{ fm} = 1.4 \times 10^{-14} \text{ m} \end{aligned}$$

and using Eq. 6.10 we then have

$$\begin{aligned} f_{>90^\circ} &= nt\pi b^2 \\ &= (5.9 \times 10^{28} \text{ m}^{-3})(2.0 \times 10^{-6} \text{ m})\pi(1.4 \times 10^{-14} \text{ m})^2 \\ &= 7.5 \times 10^{-5} \end{aligned}$$

(b) Repeating the calculation for $\theta = 45^\circ$, we find

$$\begin{aligned} b &= \frac{zZ}{2K} \frac{e^2}{4\pi\epsilon_0} \cot \frac{1}{2}\theta \\ &= \frac{(2)(79)}{2(8.0 \text{ MeV})} (1.44 \text{ MeV} \cdot \text{fm}) \cot 22.5^\circ \\ &= 34 \text{ fm} = 3.4 \times 10^{-14} \text{ m} \\ f_{>45^\circ} &= nt\pi b^2 \\ &= (5.9 \times 10^{28} \text{ m}^{-3})(2.0 \times 10^{-6} \text{ m})\pi(3.4 \times 10^{-14} \text{ m})^2 \\ &= 4.4 \times 10^{-4} \end{aligned}$$

If a total fraction of 4.4×10^{-4} is scattered at angles greater than 45° , and of that, 7.5×10^{-5} is scattered at angles greater than 90° , the fraction scattered *between* 45° and 90° must be

$$4.4 \times 10^{-4} - 7.5 \times 10^{-5} = 3.6 \times 10^{-4}$$

2. The Rutherford Scattering Formula and Its Experimental Verification. In order to find the probability that a projectile will be scattered into a small angular range at θ (between θ and $\theta + d\theta$), we require that the impact parameter lie within a small range of values db at b (see Figure 6.7). The fraction, df , is then

$$df = nt(2\pi b db) \quad (6.11)$$

from Eq. 6.10. Differentiating Eq. 6.8 we find db in terms of $d\theta$:

$$db = \frac{zZ}{2K} \frac{e^2}{4\pi\epsilon_0} \left(-\csc^2 \frac{1}{2}\theta \right) \left(\frac{1}{2} d\theta \right) \quad (6.12)$$

and so

$$|df| = \pi nt \left(\frac{zZ}{2K} \right)^2 \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \csc^2 \frac{1}{2}\theta \cot \frac{1}{2}\theta d\theta \quad (6.13)$$

(This minus sign in Eq. 6.12 is not important—it just tells us that θ increases as b decreases.) Suppose we place a detector for the scattered projectiles at the angle θ a distance r from the nucleus. The probability for a projectile to be scattered into the detector depends on df , which gives the probability for scattered particles to pass through the ring of radius $r \sin \theta$ and width $r d\theta$. The area of the ring is $dA = (2\pi r \sin \theta) r d\theta$. In order to calculate the rate at which projectiles are scattered *into the detector* we must know the probability *per unit area* for scattering into the ring. This is $|df|/dA$, which we call $N(\theta)$, and, after some manipulation, we find:

$$N(\theta) = \frac{nt}{4r^2} \left(\frac{zZ}{2K} \right)^2 \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \frac{1}{\sin^4 \frac{1}{2}\theta} \quad (6.14)$$

This is the *Rutherford scattering formula*.

In Rutherford's laboratory, Hans Geiger and Ernest Marsden tested the predictions of this formula in a remarkable series of experiments involving the scattering of alpha particles ($z = 2$) from a variety of thin metal foils. In those days before electronic recording and processing equipment was available, Geiger and Marsden observed and recorded the alpha particles by counting the scintillations (flashes of light) produced when the alpha particles struck a zinc sulfide screen. A schematic view of their apparatus is shown in Figure 6.8. Alpha particles from a shielded radioactive source were scattered by a foil F into a narrow angular range $d\theta$. Each particle that struck the scintillator S produced a tiny flash of light, which was observed with a microscope M that could move to different angular positions θ . In all, four predictions of the Rutherford scattering formula were tested:

(a) $N(\theta) \propto t$. With a source of 8-MeV alpha particles from radioactive decay, Geiger and Marsden used scattering foils of varying thicknesses t while keeping the scattering angle θ fixed at about 25° . Their results are summarized in Figure 6.9, and the linear dependence of $N(\theta)$ on t is apparent. This is also evidence that, even at this moderate scattering angle, *single* scattering is much more important than *multiple* scattering. (In a random statistical theory of multiple scattering, the probability for scattering at a large angle would be proportional to the square root of the number of single scatterings, and we would expect $N(\theta) \propto t^{1/2}$. Figure 6.9 shows clearly that this is not true.)

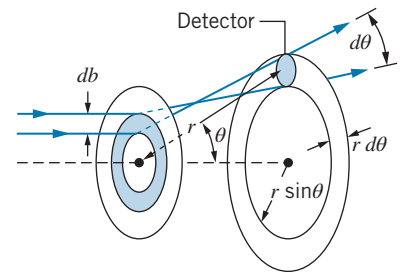


FIGURE 6.7 Particles entering the ring between b and $b + db$ are distributed uniformly along a ring of angular width $d\theta$. A detector is at a distance r from the scattering foil.

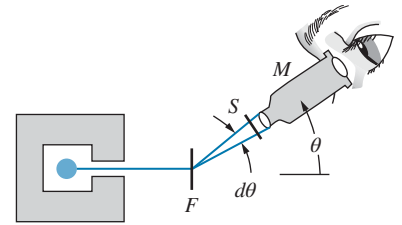


FIGURE 6.8 Schematic diagram of Geiger–Marsden alpha-particle scattering experiment. Alpha particles from the shielded source strike the foil F and are observed on the scintillator S .

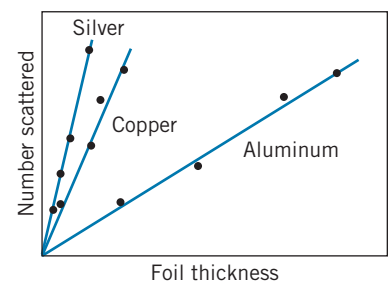


FIGURE 6.9 The dependence of scattering rate on foil thickness for three different scattering foils.

This result emphasizes a significant difference between scattering by a Thomson model atom and a Rutherford nuclear atom: In the Thomson model, the projectile is scattered by *every* atom along its path as it passes through the foil (see Figure 6.3), while in the Rutherford nuclear model the nucleus is so tiny that the chance of even a single significant encounter is small and the chance of encountering more than one nucleus is negligible.

(b) $N(\theta) \propto Z^2$. In this experiment, Geiger and Marsden used a variety of different scattering materials, of approximately (*but not exactly*) the same thickness. This proportionality is therefore much more difficult to test than the previous one, since it involves the comparison of *different thicknesses* of *different materials*. However, as shown in Figure 6.10, the results are consistent with the proportionality of $N(\theta)$ to Z^2 .

(c) $N(\theta) \propto K^{-2}$. In order to test this prediction of the Rutherford scattering formula, Geiger and Marsden kept the thickness of the scattering foil constant and varied the speed of the alpha particles. They accomplished this by slowing down the alpha particles emitted from the radioactive source by passing them through thin sheets of mica. From independent measurements they knew the effect of different thicknesses of mica on the velocity of the alpha particles. The results of the experiment are shown in Figure 6.11; once again we see excellent agreement with the expected relationship.

(d) $N(\theta) \propto \sin^{-4} \frac{1}{2} \theta$. This dependence of N on θ is perhaps the most important and distinctive feature of the Rutherford scattering formula. It also produces the largest variation in N over the range accessible by experiment. In the tests discussed so far, N varied by perhaps an order of magnitude; in this case N varies by about *five* orders of magnitude from the smaller to the larger angles. Geiger and Marsden used a gold foil and varied θ from 5 to 150°, to obtain the relationship between N and θ plotted in Figure 6.12. The agreement with the Rutherford formula is again very good.

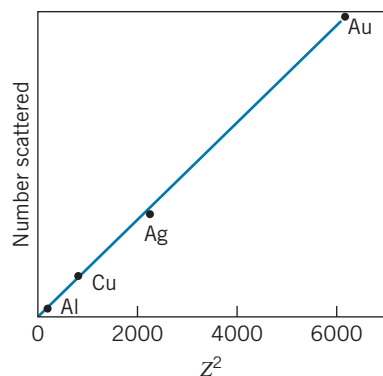


FIGURE 6.10 The dependence of scattering rate on the nuclear charge Z for foils of different materials. The data are plotted against Z^2 .

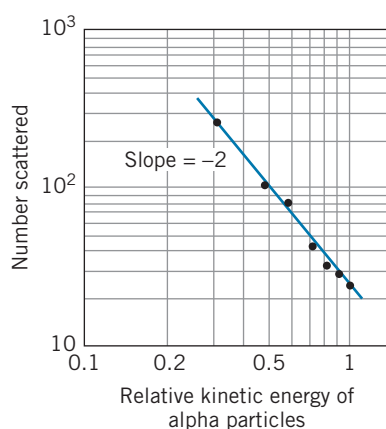


FIGURE 6.11 The dependence of scattering rate on the kinetic energy of the incident alpha particles for scattering by a single foil. The slope of -2 on the log-log scale shows that $N \propto K^{-2}$, as expected from the Rutherford formula.

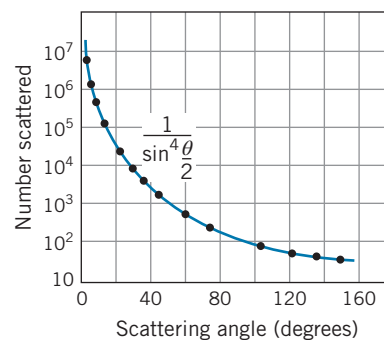


FIGURE 6.12 The dependence of scattering rate on the scattering angle θ , using a gold foil. The $\sin^{-4}(\theta/2)$ dependence is exactly as predicted by the Rutherford formula.

Thus all predictions of the Rutherford scattering formula were confirmed by experiment, and the “nuclear atom” was verified.

Example 6.3

Alpha particles are scattered from a thin foil that is an alloy consisting of 25% gold atoms and 75% copper atoms. What fraction of the alpha particles observed at 35° is due to scattering by gold?

Solution

The scattering rate depends on the number density n and on Z^2 , so the ratio of gold to copper scattering

is $(1/3)(79^2/29^2) = 2.47$, independent of the scattering angle. The fraction due to gold (at any angle) is $2.47/(2.47 + 1) = 0.71$ or 71%.

3. The Closest Approach of a Projectile to the Nucleus. A positively charged projectile slows down as it approaches a nucleus, exchanging part of its initial kinetic energy for the electrostatic potential energy due to the nuclear repulsion. The closer the projectile gets to the nucleus, the more potential energy it gains, because

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r} = \frac{1}{4\pi\epsilon_0} \frac{zZe^2}{r} \quad (6.15)$$

The maximum potential energy, and thus the minimum kinetic energy, occurs at the minimum value of r . We assume that $U = 0$ when the projectile is far from the nucleus, where it has total energy $E = K = \frac{1}{2}mv^2$. As the projectile approaches the nucleus, K decreases and U increases, but $U + K$ remains constant. At the distance $r_{\min}$, the speed is $v_{\min}$ and

$$E = \frac{1}{2}mv_{\min}^2 + \frac{1}{4\pi\epsilon_0} \frac{zZe^2}{r_{\min}} = \frac{1}{2}mv^2 \quad (6.16)$$

(See Figure 6.13.)

Angular momentum is also conserved. Far from the nucleus, the angular momentum L is mvb , and at $r_{\min}$, the angular momentum is $mv_{\min}r_{\min}$, so

$$mvb = mv_{\min}r_{\min} \quad (6.17)$$

which gives $v_{\min} = bv/r_{\min}$. Substituting this result into Eq. 6.16, we find

$$\frac{1}{2}mv^2 = \frac{1}{2}m \left(\frac{b^2 v^2}{r_{\min}^2} \right) + \frac{1}{4\pi\epsilon_0} \frac{zZe^2}{r_{\min}} \quad (6.18)$$

This expression can be solved for the value of $r_{\min}$.

Notice that the kinetic energy of the projectile is not zero at $r_{\min}$, *unless* $b = 0$. (See Figure 6.13.) In this case, the projectile would lose all of its kinetic energy, and thus get closest to the nucleus. At this point its distance from the nucleus is d , the *distance of closest approach*. We find this distance by solving Eq. 6.18 for $r_{\min}$ when $b = 0$ and obtain

$$d = \frac{1}{4\pi\epsilon_0} \frac{zZe^2}{K} \quad (6.19)$$

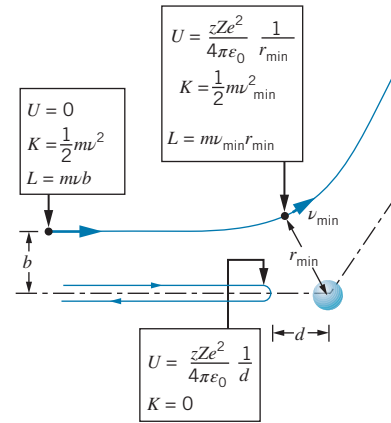


FIGURE 6.13 Closest approach of the projectile to the nucleus.

Example 6.4

Find the distance of closest approach of an 8.0-MeV alpha particle incident on a gold foil.

Solution

$$d = \frac{zZe^2}{4\pi\epsilon_0 K} \frac{1}{K} = (2)(79)(1.44 \text{ MeV} \cdot \text{fm}) \frac{1}{8.0 \text{ MeV}} = 28 \text{ fm}$$

Although a distance of 28 fm is very small (much less than an atomic radius, for example), it is larger than the nuclear radius of gold (about 7 fm). Thus the projectile is always *outside* of the nuclear charge distribution, and the Rutherford scattering law, which was derived assuming the projectile to remain outside the nucleus, correctly describes the scattering. If we increase the kinetic energy of the projectile, or decrease the electrostatic repulsion by using a target nucleus with low Z , this may not be the case. Under certain circumstances, the distance of closest approach can be less than the nuclear radius. When this happens, the projectile no longer feels the full nuclear charge, and the Rutherford scattering law no longer holds. In fact, as we discuss in Chapter 12, this gives us a convenient way of measuring the size of the nucleus.

6.4 LINE SPECTRA

The radiation from atoms can be classified into continuous spectra and discrete or line spectra. In a continuous spectrum, all wavelengths from some minimum, perhaps 0, to some maximum, perhaps approaching ∞ , are emitted. The radiation from a hot, glowing object is an example of this category. White light is a mixture of all of the different colors of visible light; an object that glows white hot is emitting light at all wavelengths of the visible spectrum. If, on the other hand, we force an electric discharge in a tube containing a small amount of the gas or vapor of a certain element, such as mercury, sodium, or neon, light is emitted at a few discrete wavelengths and not at any others. Examples of such emission “line” spectra are shown in Figure 6.14. The strong 436 nm (blue) and 546 nm (green) lines in the mercury emission

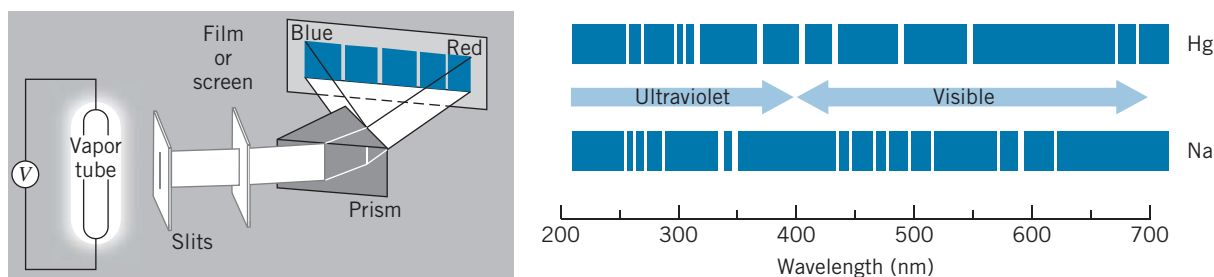


FIGURE 6.14 Apparatus for observing emission spectra. Light is emitted when an electric discharge is created in a tube containing a vapor of an element. The light passes through a dispersive medium, such as a prism or a diffraction grating, which displays the individual component wavelengths at different locations on a film or screen. Sample line spectra are shown for mercury and sodium in the visible and near ultraviolet.

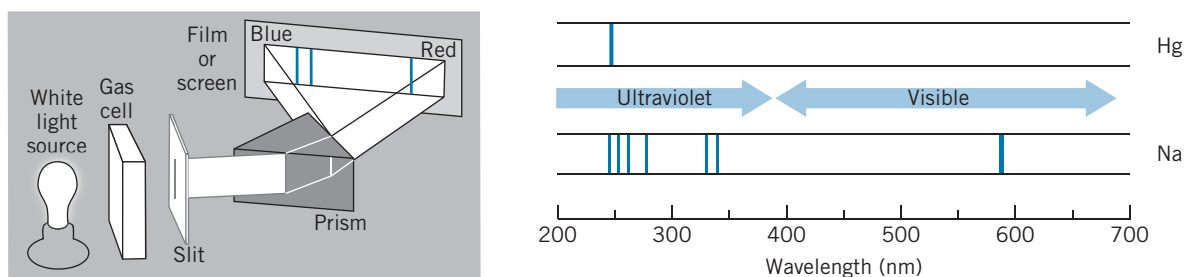


FIGURE 6.15 Apparatus for observing absorption spectra. A light source produces a continuous range of wavelengths, some of which are absorbed by a gaseous element. The light is dispersed, as in Figure 6.14. The result is a continuous “rainbow” spectrum, with dark lines at wavelengths where the light was absorbed by the gas.

spectrum give mercury-vapor street lights their blue-green tint; the strong yellow line at 590 nm in the sodium spectrum (which is actually a *doublet*—two very closely spaced lines) gives sodium-vapor street lights a softer, yellowish color. The intense red lines of neon are responsible for the red color of “neon signs.”

Another possible experiment is to pass a beam of white light, containing all wavelengths, through a sample of a gas. When we do so, we find that certain wavelengths have been absorbed from the light, and again a line spectrum results. In this case there are dark lines, superimposed on the bright continuous spectrum, at the wavelengths where the absorption occurred. These wavelengths correspond to many (*but not all*) of the wavelengths seen in the emission spectrum. Examples of absorption spectra are shown in Figure 6.15.

In general, the interpretation of line spectra is very difficult in complex atoms, and so we will deal for now with the line spectra of the simplest atom, hydrogen. Regularities appear in both the emission and the absorption spectra, as shown in Figure 6.16. Notice that, as with the mercury and sodium spectra, some lines present in the emission spectrum are missing from the absorption spectrum.

In 1885, Johann Balmer, a Swiss mathematics teacher, noticed (mostly by trial and error) that the wavelengths of the group of emission lines of hydrogen in the visible region could be calculated very accurately from the formula

$$\lambda = (364.5 \text{ nm}) \frac{n^2}{n^2 - 4} \quad (n = 3, 4, 5, \dots) \quad (6.20)$$

For example, for $n = 3$, the formula gives $\lambda = 656.1 \text{ nm}$, which corresponds exactly to the longest wavelength of the series of hydrogen lines in the visible region (see Figure 6.16). This formula is now known as the *Balmer formula* and the series of lines that it fits is called the *Balmer series*. The wavelength 364.5 nm, corresponding to $n \rightarrow \infty$, is called the *series limit* (which is shown as the dashed line at the left end of the Balmer series in Figure 6.16).

It was soon discovered that all of the groupings of lines in the hydrogen spectrum could be fit with a similar formula of the form

$$\lambda = \lambda_{\text{limit}} \frac{n^2}{n^2 - n_0^2} \quad (n = n_0 + 1, n_0 + 2, n_0 + 3, \dots) \quad (6.21)$$

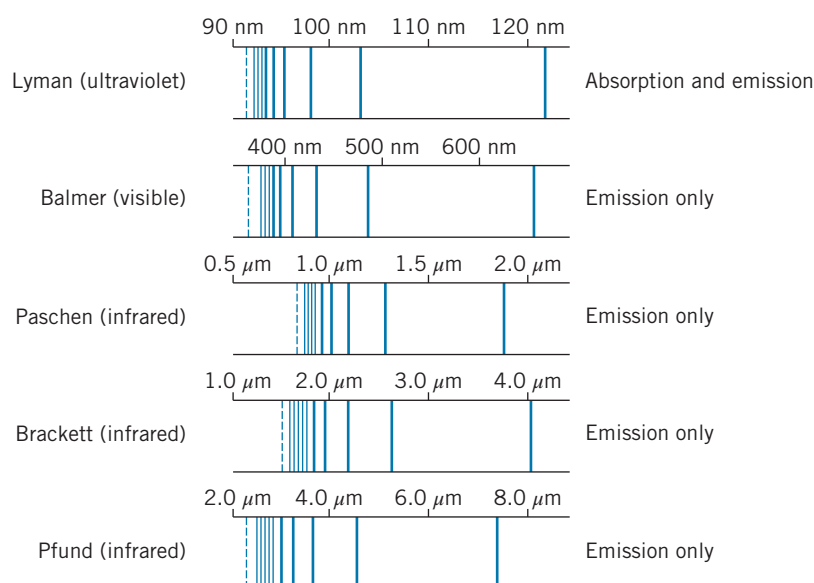


FIGURE 6.16 Emission and absorption spectral series of hydrogen. Note the regularities in the spacing of the spectral lines. The lines get closer together as the limit of each series (dashed line) is approached. Only the Lyman series appears in the absorption spectrum; all series are present in the emission spectrum.

where λ_{limit} is the wavelength of the appropriate series limit. For the Balmer series, $n_0 = 2$. The other series are today known as Lyman ($n_0 = 1$), Paschen ($n_0 = 3$), Brackett ($n_0 = 4$), and Pfund ($n_0 = 5$). These series of hydrogen spectral lines are shown in Figure 6.16.

Another interesting property of the hydrogen wavelengths is summarized in the *Ritz combination principle*. If we convert the hydrogen emission wavelengths to frequencies, we find the curious property that certain pairs of frequencies added together give other frequencies that appear in the spectrum.

Any successful model of the hydrogen atom must be able to explain the occurrence of these interesting arithmetic regularities in the emission spectra.

Example 6.5

The series limit of the Paschen series ($n_0 = 3$) is 820.1 nm. What are the three longest wavelengths of the Paschen series?

Solution

From Eq. 6.21,

$$\lambda = (820.1 \text{ nm}) \frac{n^2}{n^2 - 3^2} \quad (n = 4, 5, 6, \dots)$$

The three longest wavelengths are as follows:

$$n = 4 : \quad \lambda = (820.1 \text{ nm}) \frac{4^2}{4^2 - 3^2} = 1875 \text{ nm}$$

$$n = 5 : \quad \lambda = (820.1 \text{ nm}) \frac{5^2}{5^2 - 3^2} = 1281 \text{ nm}$$

$$n = 6 : \quad \lambda = (820.1 \text{ nm}) \frac{6^2}{6^2 - 3^2} = 1094 \text{ nm}$$

These transitions are in the infrared region of the electromagnetic spectrum.

Example 6.6

Show that the longest wavelength of the Balmer series and the longest *two* wavelengths of the Lyman series satisfy the Ritz combination principle. For the Lyman series, $\lambda_{\text{limit}} = 91.13 \text{ nm}$.

Solution

Using Eq. 6.20 with $n = 3$, we find the longest wavelength of the Balmer series to be 656.1 nm . Converting this to a frequency, we obtain

$$f = \frac{c}{\lambda} = \frac{2.998 \times 10^8 \text{ m/s}}{(656.1 \text{ nm})(10^{-9} \text{ m/nm})} = 4.57 \times 10^{14} \text{ Hz}$$

Using Eq. 6.21 for $n = 2$ and 3 with $n_0 = 1$, we find the longest two wavelengths of the Lyman series and their corresponding frequencies to be

$$n = 2 : \quad \lambda = (91.13 \text{ nm}) \frac{2^2}{2^2 - 1^2} = 121.5 \text{ nm}$$

$$f = \frac{c}{\lambda} = \frac{2.998 \times 10^8 \text{ m/s}}{(121.5 \text{ nm})(10^{-9} \text{ m/nm})} = 24.67 \times 10^{14} \text{ Hz}$$

$$n = 3 : \quad \lambda = (91.13 \text{ nm}) \frac{3^2}{3^2 - 1^2} = 102.5 \text{ nm}$$

$$f = \frac{c}{\lambda} = \frac{2.998 \times 10^8 \text{ m/s}}{(102.5 \text{ nm})(10^{-9} \text{ m/nm})} = 29.24 \times 10^{14} \text{ Hz}$$

Adding the smallest frequency of the Lyman series to the smallest frequency of the Balmer series gives the next smallest Lyman frequency:

$$24.67 \times 10^{14} \text{ Hz} + 4.57 \times 10^{14} \text{ Hz} = 29.24 \times 10^{14} \text{ Hz}$$

demonstrating the Ritz combination principle.

6.5 THE BOHR MODEL

Following Rutherford's proposal that the mass and positive charge are concentrated in a very small region at the center of the atom, the Danish physicist Niels Bohr in 1913 (while working in Rutherford's laboratory) suggested that the atom resembled a miniature planetary system, with the electrons circulating about the nucleus like planets circulating about the Sun. The atom thus doesn't collapse under the influence of the electrostatic Coulomb force of the nucleus on the electrons for the same reason that the solar system doesn't collapse under the influence of the gravitational force of the Sun on the planets. In both cases, the attractive force provides the centripetal acceleration necessary to maintain the orbital motion.

As we discuss later, the Bohr model does not give a correct view of the actual structure and properties of atoms, but it represents an important first step in achieving an understanding of atoms. The correct view requires methods of quantum mechanics, which we discuss in Chapter 7.

We consider for simplicity the hydrogen atom, with a single electron circulating about a nucleus that has a single positive charge, as in Figure 6.17. The radius of the circular orbit is r , and the electron (of mass m) moves with constant tangential speed v . The attractive Coulomb force provides the centripetal acceleration v^2/r , so

$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1||q_2|}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} = \frac{mv^2}{r} \quad (6.22)$$

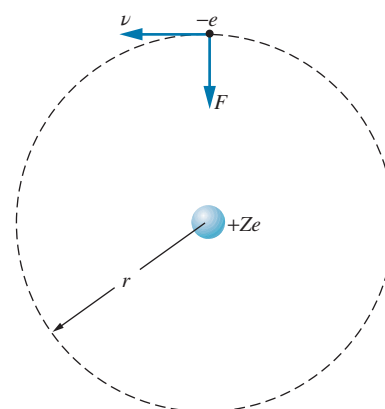


FIGURE 6.17 The Bohr model of the atom ($Z = 1$ for hydrogen).

Manipulating this equation, we can find the kinetic energy of the electron (we are assuming the more massive nucleus to remain at rest—more about this later):

$$K = \frac{1}{2}mv^2 = \frac{1}{8\pi\epsilon_0} \frac{e^2}{r} \quad (6.23)$$

The potential energy of the electron-nucleus system is the Coulomb potential energy:

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r} = -\frac{1}{4\pi\epsilon_0} \frac{e^2}{r} \quad (6.24)$$

The total energy $E = K + U$ is obtained by adding Eqs. 6.23 and 6.24:

$$E = K + U = \frac{1}{8\pi\epsilon_0} \frac{e^2}{r} + \left(-\frac{1}{4\pi\epsilon_0} \frac{e^2}{r} \right) = -\frac{1}{8\pi\epsilon_0} \frac{e^2}{r} \quad (6.25)$$

We have ignored one serious difficulty with this model thus far. Classical physics requires that an accelerated electric charge, such as our orbiting electron, must continuously radiate electromagnetic energy. As it radiates this energy, its total energy would decrease, the electron would spiral in toward the nucleus, and the atom would collapse. To overcome this difficulty, Bohr made a bold and daring hypothesis—he proposed that there are certain special states of motion, called *stationary states*, in which the electron may exist without radiating electromagnetic energy. In these states, according to Bohr, the angular momentum L of the electron takes values that are integer multiples of $\hbar$. In stationary states, the angular momentum of the electron may have magnitude $\hbar, 2\hbar, 3\hbar, \dots$, but never such values as $2.5\hbar$ or $3.1\hbar$. This is called the *quantization of angular momentum*.

In a circular orbit, the position vector $\vec{r}$ that locates the electron relative to the nucleus is always perpendicular to its linear momentum $\vec{p}$. The angular momentum, which is defined as $\vec{L} = \vec{r} \times \vec{p}$, has magnitude $L = rp = mvr$ when $\vec{r}$ is perpendicular to $\vec{p}$. Thus Bohr's postulate is

$$mvr = n\hbar \quad (6.26)$$

where n is an integer ($n = 1, 2, 3, \dots$). We can use this expression with Eq. 6.23 for the kinetic energy

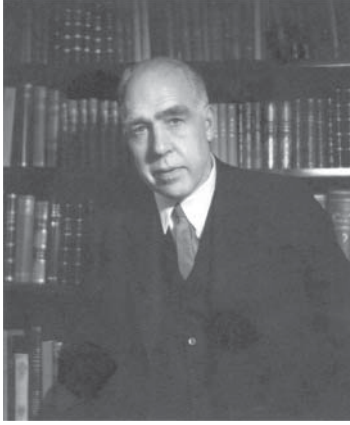
$$\frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{n\hbar}{mr}\right)^2 = \frac{1}{8\pi\epsilon_0} \frac{e^2}{r} \quad (6.27)$$

to find a series of allowed values of the radius r :

$$r_n = \frac{4\pi\epsilon_0\hbar^2}{me^2} n^2 = a_0 n^2 \quad (n = 1, 2, 3, \dots) \quad (6.28)$$

where the *Bohr radius* a_0 is defined as

$$a_0 = \frac{4\pi\epsilon_0\hbar^2}{me^2} = 0.0529 \text{ nm} \quad (6.29)$$



The Granger Collection, New York.

Niels Bohr (1885–1962, Denmark). He developed a successful theory of the radiation spectrum of atomic hydrogen and also contributed the concepts of stationary states and complementarity to quantum mechanics. Later he developed a successful theory of nuclear fission. The institute of theoretical physics he founded in Copenhagen attracts scholars from around the world.

This important result is very different from what we expect from classical physics. A satellite may be placed into Earth orbit at any desired radius by boosting it to the appropriate altitude and then supplying the proper tangential speed. This is not true for an electron's orbit—only certain radii are allowed by the Bohr model. The radius of the electron's orbit may be a_0 , $4a_0$, $9a_0$, $16a_0$, and so forth, but *never* $3a_0$ or $5.3a_0$.

Substituting Eq. 6.28 for r into Eq. 6.25 gives the energy:

$$E_n = -\frac{me^4}{32\pi^2\epsilon_0^2\hbar^2} \frac{1}{n^2} = \frac{-13.60 \text{ eV}}{n^2} \quad (n = 1, 2, 3, \dots) \quad (6.30)$$

The *energy levels* calculated from Eq. 6.30 are shown in Figure 6.18. The electron's energy is *quantized*—only certain energy values are possible. In its lowest level, with $n = 1$, the electron has energy $E_1 = -13.60 \text{ eV}$ and orbits with a radius of $r_1 = 0.0529 \text{ nm}$. This state is the *ground state*. The higher states ($n = 2$ with $E_2 = -3.40 \text{ eV}$, $n = 3$ with $E_3 = -1.51 \text{ eV}$, etc.) are the *excited states*.

The *excitation energy* of an excited state n is the energy above the ground state, $E_n - E_1$. Thus the first excited state ($n = 2$) has excitation energy

$$\Delta E = E_2 - E_1 = -3.40 \text{ eV} - (-13.60 \text{ eV}) = 10.20 \text{ eV}$$

the second excited state has excitation energy

$$\Delta E = E_3 - E_1 = -1.51 \text{ eV} - (-13.60 \text{ eV}) = 12.09 \text{ eV}$$

and so forth. The excitation energy can also be regarded as the amount of energy that the atom must absorb for the electron to make an upward jump. For example, if the atom absorbs an energy of 10.20 eV when the electron is in the ground state ($n = 1$), the electron will jump upward to the first excited state ($n = 2$).

The magnitude of an electron's energy $|E_n|$ is sometimes called its *binding energy*; for example, the binding energy of an electron in the $n = 2$ state is 3.40 eV . If the atom absorbs an amount of energy equal to the binding energy of the electron, the electron will be removed from the atom and become a free electron. The atom, minus its electron, is then called an *ion*. The amount of energy needed to remove an electron from an atom is also called the *ionization energy*. Usually the ionization energy of an atom indicates the energy to remove an electron from the ground state. If the atom absorbs more energy than the minimum necessary to remove the electron, the excess energy appears as the kinetic energy of the now free electron.

The binding energy can also be regarded as the energy that is released when the atom is assembled from an electron and a nucleus that are initially separated by a large distance. If we bring an electron from a large distance away (where $E = 0$) and place it in orbit in the state n where its energy has the negative value E_n , energy amounting to $|E_n|$ is released, usually in the form of one or more photons.

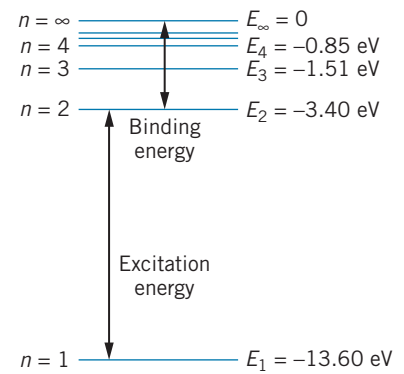


FIGURE 6.18 The energy levels of atomic hydrogen, showing the excitation energy of the electron from $n = 1$ to $n = 2$ and the binding energy of the $n = 2$ electron.

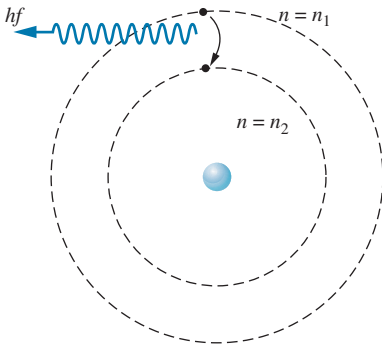


FIGURE 6.19 An electron jumps from the state n_1 to the state n_2 as a photon is emitted.

The Hydrogen Wavelengths in the Bohr Model

We previously discussed the emission and absorption spectra of atomic hydrogen, and our discussion of the Bohr model is not complete without an understanding of the origin of these spectra. Bohr postulated that even though the electron doesn't radiate when it remains in any particular stationary state, it can emit radiation when it moves to a lower energy level. In the lower level, the electron has less energy than in the original level, and the energy difference appears as a quantum of radiation whose energy hf is equal to the energy difference between the levels. That is, if the electron jumps from $n = n_1$ to $n = n_2$, as in Figure 6.19, a photon appears with energy

$$hf = E_{n_1} - E_{n_2} \quad (6.31)$$

or, using Eq. 6.30 for the energies,

$$f = \frac{me^4}{64\pi^3\epsilon_0^2\hbar^3} \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right) \quad (6.32)$$

The wavelength of the emitted radiation is

$$\lambda = \frac{c}{f} = \frac{64\pi^3\epsilon_0^2\hbar^3 c}{me^4} \left(\frac{n_1^2 n_2^2}{n_1^2 - n_2^2} \right) = \frac{1}{R_\infty} \left(\frac{n_1^2 n_2^2}{n_1^2 - n_2^2} \right) \quad (6.33)$$

where R_∞ is called the *Rydberg constant* and is given by

$$R_\infty = \frac{me^4}{64\pi^3\epsilon_0^2\hbar^3 c} \quad (6.34)$$

The presently accepted numerical value is

$$R_\infty = 1.097373 \times 10^7 \text{ m}^{-1}$$

Example 6.7

Find the wavelengths of the transitions from $n_1 = 3$ to $n_2 = 2$ and from $n_1 = 4$ to $n_2 = 2$ in atomic hydrogen.

Solution

For $n_1 = 3$ and $n_2 = 2$, Eq. 6.33 gives

$$\begin{aligned} \lambda &= \frac{1}{R_\infty} \left(\frac{n_1^2 n_2^2}{n_1^2 - n_2^2} \right) \\ &= \frac{1}{1.097 \times 10^7 \text{ m}^{-1}} \left(\frac{3^2 2^2}{3^2 - 2^2} \right) = 656.1 \text{ nm} \end{aligned}$$

$$\begin{aligned} \lambda &= \frac{1}{R_\infty} \left(\frac{n_1^2 n_2^2}{n_1^2 - n_2^2} \right) \\ &= \frac{1}{1.097 \times 10^7 \text{ m}^{-1}} \left(\frac{4^2 2^2}{4^2 - 2^2} \right) \\ &= 486.0 \text{ nm} \end{aligned}$$

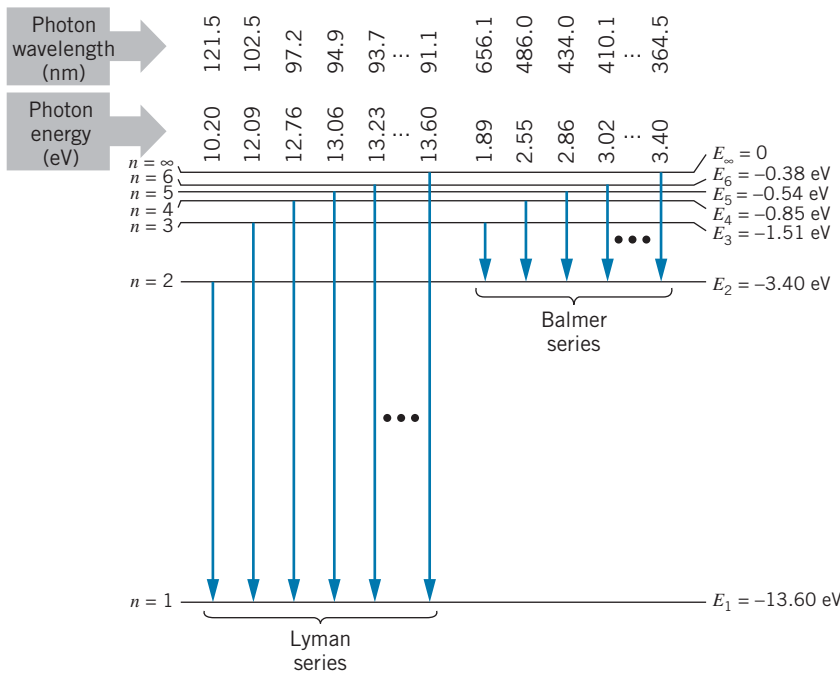


FIGURE 6.20 The transitions of the Lyman and Balmer series in hydrogen. The series limit is shown at the right of each group.

These wavelengths are remarkably close to the values of the two longest wavelengths of the Balmer series (Figure 6.16). In fact, Eq. 6.33 gives

$$\lambda = (364.5 \text{ nm}) \left(\frac{n_1^2}{n_1^2 - 4} \right)$$

for the wavelength of a transition from any state n_1 to $n_2 = 2$. This is identical with Eq. 6.21 for the Balmer series. Thus we see that the radiations identified as the Balmer series correspond to transitions from higher levels to the $n = 2$ level. Similar identifications can be made for other series of radiations, as shown in Figure 6.20. This association between the transitions expected according to the Bohr model and the observed wavelengths (as in Figure 6.16) represents a huge triumph for the model.

The Bohr formulas also explain the Ritz combination principle, according to which certain frequencies in the emission spectrum can be summed to give other frequencies. Let us consider a transition from a state n_3 to a state n_2 , that is followed by a transition from n_2 to n_1 . Equation 6.32 can be used for this case to give

$$f_{n_3 \rightarrow n_2} = cR_\infty \left(\frac{1}{n_2^2} - \frac{1}{n_3^2} \right)$$

$$f_{n_2 \rightarrow n_1} = cR_\infty \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

Thus

$$f_{n_3 \rightarrow n_2} + f_{n_2 \rightarrow n_1} = cR_\infty \left(\frac{1}{n_3^2} - \frac{1}{n_2^2} \right) + cR_\infty \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right) = cR_\infty \left(\frac{1}{n_3^2} - \frac{1}{n_1^2} \right)$$

which is equal to the frequency of the single photon emitted in a direct transition from n_3 to n_1 , so

$$f_{n_3 \rightarrow n_2} + f_{n_2 \rightarrow n_1} = f_{n_3 \rightarrow n_1} \quad (6.35)$$

The Bohr model is thus entirely consistent with the Ritz combination principle. The frequency of an emitted photon is related to its energy by $E = hf$, so the summing of frequencies is equivalent to the summing of energies. We may thus restate the Ritz combination principle in terms of energy: The energy of a photon emitted in a transition that skips or crosses over one or more states is equal to the step-by-step sum of the energies of the transitions connecting all of the individual states. (See Problem 26.)

The Bohr model also helps us understand why the atom doesn't absorb and emit radiation at all the same wavelengths. Isolated atoms are normally found only in the ground state; the excited states live for a very short time (less than 10^{-9} s) before decaying to the ground state. *The absorption spectrum therefore contains only transitions from the ground state.* From Figure 6.20, we see that only the radiations of the Lyman series can be found in the absorption spectrum of hydrogen. A hydrogen atom in its ground state can absorb radiation of 10.20 eV and reach the first excited state, or of 12.09 eV and reach the second excited state, and so forth. A hydrogen atom cannot absorb a photon of energy 1.89 eV (the first line of the Balmer series), because the atom is originally not in the $n = 2$ level. The Balmer series is therefore *not* found in the absorption spectrum.

Atoms with $Z > 1$

The Bohr theory for hydrogen can be used for any atom with a single electron, even if the nuclear charge Z is greater than 1. For example, we can calculate the energy levels of singly ionized helium (helium with one electron removed), doubly ionized lithium, and so on. The nuclear electric charge enters the Bohr theory in only one place—in the expression for the electrostatic force between nucleus and electron, Eq. 6.22. For a nucleus of charge Ze , the Coulomb force acting on the electron is

$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1||q_2|}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r^2} \quad (6.36)$$

That is, where we had e^2 previously, we now have Ze^2 . Making the same substitution in the final results, we can find the allowed radii:

$$r_n = \frac{4\pi\epsilon_0\hbar^2}{Ze^2m} n^2 = \frac{a_0 n^2}{Z} \quad (6.37)$$

and the energies become

$$E_n = -\frac{m(Ze^2)^2}{32\pi^2\epsilon_0^2\hbar^2} \frac{1}{n^2} = -(13.60 \text{ eV}) \frac{Z^2}{n^2} \quad (6.38)$$

The orbits in the higher- Z atoms are closer to the nucleus and have larger (negative) energies; that is, the electron is more tightly bound to the nucleus.

Example 6.8

Which of the five series of emitted radiations shown in Figure 6.16 would produce visible light for Be^{+++} ?

Solution

For Be, with $Z = 4$, the energies of the states would be $Z^2 = 16$ times the energy levels in H. The emitted wavelengths would correspondingly be smaller by $1/16$.

The largest wavelength of the hydrogen Brackett series, about 4000 nm, would shrink to about 250 nm in Be, in the ultraviolet region. Only the longest wavelength of the Pfund series ($n = 6$ to $n = 5$) would fall into the visible region ($7460 \text{ nm}/16 = 466 \text{ nm}$).

Example 6.9

Calculate the two longest wavelengths of the Balmer series of triply ionized beryllium ($Z = 4$).

Solution

The radiations of the Balmer series end with the $n = 2$ level, and so the two longest wavelengths are the radiations corresponding to $n = 3 \rightarrow n = 2$ and $n = 4 \rightarrow n = 2$. The energies of the radiations and their corresponding wavelengths are

$$E_3 - E_2 = -(13.60 \text{ eV})(4^2) \left(\frac{1}{3^2} - \frac{1}{2^2} \right) = 30.2 \text{ eV}$$

$$\lambda = \frac{hc}{E} = \frac{1240 \text{ eV} \cdot \text{nm}}{30.2 \text{ eV}} = 41.0 \text{ nm}$$

$$E_4 - E_2 = -(13.60 \text{ eV})(4^2) \left(\frac{1}{4^2} - \frac{1}{2^2} \right) = 40.8 \text{ eV}$$

$$\lambda = \frac{hc}{E} = \frac{1240 \text{ eV} \cdot \text{nm}}{40.8 \text{ eV}} = 30.4 \text{ nm}$$

These radiations are in the ultraviolet region.

Note that we cannot use Eq. 6.33 to find the wavelengths, because that equation applies only to hydrogen ($Z = 1$).

6.6 THE FRANCK–HERTZ EXPERIMENT

Let us imagine the following experiment, which is performed with the apparatus shown schematically in Figure 6.21. A filament heats the cathode, which then emits electrons. These electrons are accelerated toward the grid by the potential difference V , which we control. Electrons pass through the grid and reach the plate if V exceeds V_0 , a small retarding voltage between the grid and the plate. The current of electrons reaching the plate is measured using the ammeter A .

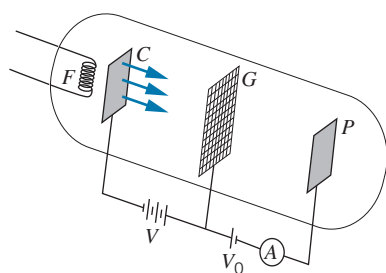


FIGURE 6.21 Franck–Hertz apparatus. Electrons leave the cathode C , are accelerated by the voltage V toward the grid G , and reach the plate P where they are recorded on the ammeter A .

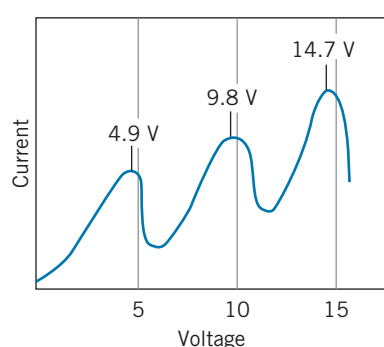
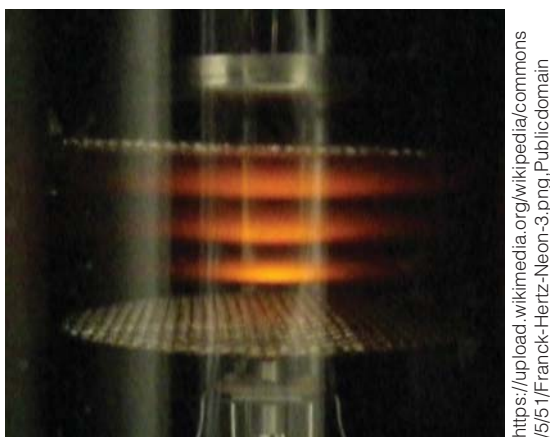


FIGURE 6.22 Result of Franck–Hertz experiment using mercury vapor, showing decreases in current at multiples of 4.9 V.

Now suppose the tube is filled with atomic hydrogen gas at a low pressure. As the voltage is increased from zero, more and more electrons reach the plate, and the current rises accordingly. The electrons inside the tube may make collisions with atoms of hydrogen, *but lose no energy in these collisions*—the collisions are perfectly elastic. The only way the electron can give up energy in a collision is if the electron has enough energy to cause the hydrogen atom to make a transition to an excited state. Thus, when the energy of the electrons reaches and barely exceeds 10.2 eV (or when the voltage reaches 10.2 V), the electrons can make *inelastic* collisions, leaving 10.2 eV of energy with the atom (now in the $n = 2$ level), and the original electron moves off with very little energy. If it should pass through the grid, the electron might not have sufficient energy to overcome the small retarding potential and reach the plate. Thus when $V = 10.2$ V, a drop in the current is observed. As V is increased further, we begin to see the effects of multiple collisions. That is, when $V = 20.4$ V, an electron can make an inelastic collision, leaving the atom in the $n = 2$ state. The electron loses 10.2 eV of energy in this process, and so it moves off after the collision with a remaining 10.2 eV of energy, which is sufficient to excite a second hydrogen atom in an inelastic collision. Thus, if a drop in the current is observed at V , similar drops are observed at $2V, 3V, \dots$

This experiment should thus give rather direct evidence for the existence of atomic excited states. Unfortunately, it is not easy to do this experiment with hydrogen, because hydrogen occurs naturally in the molecular form H_2 , rather than in atomic form. The molecules can absorb energy in a variety of ways, which would confuse the interpretation of the experiment. A similar experiment was done in 1914 by James Franck and Gustav Hertz, using a tube filled with mercury vapor. Their results are shown in Figure 6.22, which gives clear evidence for an excited state at 4.9 eV; whenever the voltage is a multiple of 4.9 V, a drop in the current appears. Coincidentally, the *emission* spectrum of mercury shows an intense ultraviolet line of wavelength 254 nm, which corresponds to an energy of 4.9 eV; this results from a transition between the same 4.9-eV excited state and the ground state. The Franck–Hertz experiment showed that an electron must have a certain minimum energy to make an inelastic collision with an atom; we now interpret that minimum energy as the energy of an excited state of the atom. Franck and Hertz were awarded the 1925 Nobel Prize in physics for this work.

In a modern version of the Franck–Hertz experiment that can be done in student laboratories, the tube is filled with neon gas. When the energy of the electrons reaches 18.7 eV, they can collide with neon atoms and raise the neon to an excited state at that energy. From there the atom transitions down to a state at 16.6 eV, in the process emitting a photon of energy 2.1 eV, which appears as orange light emitted by the gas. If the accelerating voltage is sufficiently high, the electrons can acquire enough energy as they move from the cathode toward the grid to excite a second and perhaps even a third neon atom. Figure 6.23 shows the results of an experiment in which the electrons, which move upward from the cathode near the bottom of the photo to the grid at the top, cause three glowing regions of the neon gas.



<https://upload.wikimedia.org/wikipedia/commons/5/51/Franck-Hertz-Neon-3.png>, Public domain

FIGURE 6.23 Electrons traveling upward from the cathode are accelerated to sufficient energy to produce three glowing regions of neon gas as they excite the neon atoms to discrete excited states.

*6.7 THE CORRESPONDENCE PRINCIPLE

We have seen how Bohr's model permits calculations of transition wavelengths in atomic hydrogen that are in excellent agreement with the wavelengths observed in the emission and absorption spectra. However, in order to obtain this agreement, Bohr had to introduce postulates that were radical departures from classical physics. In particular, according to classical physics an accelerated charged particle radiates electromagnetic energy, but an electron in Bohr's atomic model, accelerated as it moves in a circular orbit, does not radiate (unless it jumps to another orbit). Here we have a very different case than we did in our study of special relativity. You will recall, for example, that relativity gives us one expression for the kinetic energy, $K = E - E_0$, and classical physics gives us another, $K = \frac{1}{2}mv^2$; however, we showed that $E - E_0$ reduces to $\frac{1}{2}mv^2$ when $v \ll c$. Thus these two expressions are really not very different—one is merely a special case of the other. The dilemma associated with the accelerated electron is not simply a matter of atomic physics (as an example of quantum physics) being a special case of classical physics. Either the accelerated charge radiates, or it doesn't! Bohr's solution to this serious dilemma was to propose the *correspondence principle*, which states that

Quantum theory must agree with classical theory in the limit in which classical theory is known to agree with experiment,

*This is an optional section that may be skipped without loss of continuity.

or equivalently,

Quantum theory must agree with classical theory in the limit of large quantum numbers.

Let us see how we can apply this principle to the Bohr atom. According to classical physics, an electric charge moving in a circle radiates at a frequency equal to its frequency of rotation. For an atomic orbit, the period of revolution is the distance traveled in one orbit, $2\pi r$, divided by the orbital speed $v = \sqrt{2K/m}$, where K is the kinetic energy:

$$T = \frac{2\pi r}{\sqrt{2K/m}} = \frac{\sqrt{16\pi^3 \epsilon_0 m r^3}}{e} \quad (6.39)$$

where we use Eq. 6.23 for the kinetic energy. The frequency f is the inverse of the period:

$$f = \frac{1}{T} = \frac{e}{\sqrt{16\pi^3 \epsilon_0 m r^3}} \quad (6.40)$$

Using Eq. 6.28 for the radii of the allowed orbits, we find

$$f_n = \frac{me^4}{32\pi^3 \epsilon_0^2 \hbar^3} \frac{1}{n^3} \quad (6.41)$$

A “classical” electron moving in an orbit of radius r_n would radiate at this frequency f_n .

If we made the radius of the Bohr atom so large that it went from a quantum-sized object (10^{-10} m) to a laboratory-sized object (10^{-3} m), the atom should behave classically. The radius increases with increasing n like n^2 , so this classical behavior should occur for n in the range $10^3 - 10^4$. Let us then calculate the frequency of the radiation emitted by such an atom when the electron drops from the orbit n to the orbit $n - 1$. According to Eq. 6.32, the frequency is

$$f = \frac{me^4}{64\pi^3 \epsilon_0^2 \hbar^3} \left(\frac{1}{(n-1)^2} - \frac{1}{n^2} \right) = \frac{me^4}{64\pi^3 \epsilon_0^2 \hbar^3} \frac{2n-1}{n^2(n-1)^2} \quad (6.42)$$

If n is very large, then we can approximate $n - 1$ by n and $2n - 1$ by $2n$, which gives

$$f \cong \frac{me^4}{64\pi^3 \epsilon_0^2 \hbar^3} \frac{2n}{n^4} = \frac{me^4}{32\pi^3 \epsilon_0^2 \hbar^3} \frac{1}{n^3} \quad (6.43)$$

This is identical with Eq. 6.41 for the “classical” frequency. The “classical” electron spirals slowly in toward the nucleus, radiating at the frequency given by Eq. 6.41, while the “quantum” electron jumps from the orbit n to the orbit $n - 1$ and then to the orbit $n - 2$, and so forth, radiating at the frequency given by the identical Eq. 6.43. (When the circular orbits are very large, this jumping from one circular orbit to the next smaller one looks very much like a spiral, as in Figure 6.24.)

In the region of large n , where classical and quantum physics overlap, the classical and quantum expressions for the radiation frequencies are

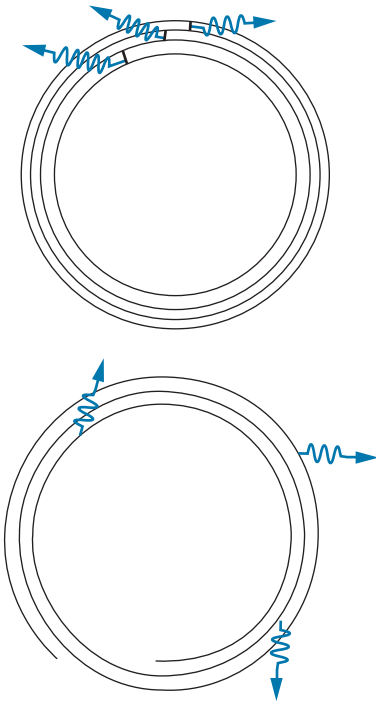


FIGURE 6.24 (Top) A large quantum atom. Photons are emitted in discrete transitions as the electron jumps to lower states. (Bottom) A classical atom. Photons are emitted continuously by the accelerated electron.

identical. This is an example of an application of Bohr's correspondence principle. The applications of the correspondence principle go far beyond the Bohr atom, and this principle is important in understanding how we get from the domain in which the laws of classical physics are valid to the domain in which the laws of quantum physics are valid.

6.8 DEFICIENCIES OF THE BOHR MODEL

The Bohr model gives us a picture of how electrons move about the nucleus, and many of our attempts to explain the behavior of atoms refer to this picture, even though it is not strictly correct. Our presentation ignored two effects that must be included to improve the accuracy of the model. Other deficiencies in the model cannot be so easily fixed, because they are inconsistent with the correct quantum-mechanical picture, which is presented in Chapter 7.

1. Motion of the Proton. Our model was based on an electron orbiting around a fixed proton, but actually the electron and proton both orbit about their center of mass (just as the Earth and the Sun orbit around their center of mass). The kinetic energy should thus include a term describing the motion of the proton. We can account for this effect if the mass that appears in the equation for the energy levels (Eq. 6.30) is not the electron mass but instead is the *reduced mass* of the proton–electron system, calculated from the electron mass m_e and proton mass m_p according to

$$m = \frac{m_e m_p}{m_e + m_p} \quad (6.44)$$

The reduced mass is just slightly smaller than the electron mass and has the effect of decreasing the energy and frequency or increasing the wavelength by about 0.05%. Equivalently, in Eq. 6.33 for the wavelengths we can replace the Rydberg constant R_∞ (so called because it would be correct if the proton mass were infinite) with the value $R = R_\infty(1 + m_e/m_p)$.

2. Wavelengths in Air. Another small but easily fixable error occurs when we convert the frequencies (Eq. 6.32) calculated directly from the Bohr energy levels to wavelengths (Eq. 6.33). The wavelength measurements are normally done in air, so we should calculate the wavelength as $\lambda = v_{\text{air}}/f$, where v_{air} is the speed of light in air. This has the effect of decreasing the calculated wavelengths by about 0.03% (to some extent offsetting the error we made by ignoring the motion of the proton).

3. Angular Momentum. A serious failure of the Bohr model is that it gives incorrect predictions for the angular momentum of the electron. In Bohr's theory, the orbital angular momentum is quantized in integer multiples of $\hbar$, which is correct. However, for the ground state of hydrogen ($n = 1$), the Bohr theory gives $L = \hbar$, while experiment clearly shows $L = 0$.

4. Uncertainty. Another deficiency of the model is that it violates the uncertainty relationship. (In Bohr's defense, remember that the model was developed a decade before the introduction of wave mechanics, with its accompanying ideas of uncertainty.) Suppose the electron orbits in the xy plane. In this case we know exactly its z coordinate (in the xy plane, $z = 0$ and so $\Delta z = 0$) and the z component of its momentum (also precisely zero, so $\Delta p_z = 0$). Such an atom would therefore violate the uncertainty relationship

$\Delta z \Delta p_z \geq \hbar$. In fact, as we discuss in Chapter 7, quantum mechanics introduces a degree of “fuzziness” to the behavior of electrons in atoms that is not consistent with any orbit in a single plane.

In spite of its successes, the Bohr model is at best an incomplete model. It is useful only for atoms that contain one electron (hydrogen, singly ionized helium, doubly ionized lithium, and so forth), but not for atoms with two or more electrons, because we have considered only the force on the electron due to the nucleus, and not the force of each electron on the others. Furthermore, if we look very carefully at the emission spectrum, we find that many lines are in fact not single lines, but very closely spaced combinations of two or more lines; the Bohr model is unable to account for these *doublets* of spectral lines. The model is also limited in its usefulness as a basis from which to calculate other properties of the atom; although we can accurately calculate the energies of the spectral lines, we cannot calculate their intensities. For example, how often will an electron in the $n = 3$ state jump directly to the $n = 1$ state, emitting the corresponding photon, and how often will it jump first to the $n = 2$ state and then to the $n = 1$ state, emitting two photons? A complete theory should provide a way to calculate this property.

We do not wish, however, to discard the model completely. The Bohr model provides a useful starting point in our study of atoms, and Bohr introduced several ideas (stationary states, quantization of angular momentum, correspondence principle) that carry over into the correct quantum-mechanical calculation. There are many atomic properties, especially those associated with magnetism, that can be simply modeled on the basis of Bohr orbits. Most remarkably, when we treat the hydrogen atom correctly in Chapter 7 using quantum mechanics, we find that the energy levels calculated by solving the Schrödinger equation are in fact identical with those of the Bohr model.

Chapter Summary

	Section		Section
Scattering impact parameter $b = \frac{zZ}{2K} \frac{e^2}{4\pi\epsilon_0} \cot \frac{1}{2}\theta$	6.3	Excitation energy $E_n - E_1$ of level n	6.5
Fraction scattered at angles $> \theta$ $f_{>\theta} = n\pi b^2$	6.3	Binding (or ionization) energy of level n $ E_n $	6.5
Rutherford scattering formula $N(\theta) = \frac{nt}{4r^2} \left(\frac{zZ}{2K} \right)^2 \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \frac{1}{\sin^4 \frac{1}{2}\theta}$	6.3	Hydrogen wavelengths in Bohr model $\lambda = \frac{64\pi^3\epsilon_0^2\hbar^3c}{me^4} \left(\frac{n_1^2n_2^2}{n_1^2 - n_2^2} \right) = \frac{1}{R_\infty} \left(\frac{n_1^2n_2^2}{n_1^2 - n_2^2} \right)$	6.5
Distance of closest approach $d = \frac{1}{4\pi\epsilon_0} \frac{zZe^2}{K}$	6.3	Single-electron atoms with $Z > 1$ $r_n = \frac{a_0 n^2}{Z}, E_n = -(13.60 \text{ eV}) \frac{Z^2}{n^2}$	6.5
Balmer formula $\lambda = (364.5 \text{ nm}) \frac{n^2}{n^2 - 4}$ ($n = 3, 4, 5, \dots$)	6.4	Reduced mass of proton–electron system $m = \frac{m_e m_p}{m_e + m_p}$	6.8
Radii of Bohr orbits in hydrogen $r_n = \frac{4\pi\epsilon_0\hbar^2}{me^2} n^2 = a_0 n^2$ ($n = 1, 2, 3, \dots$)	6.5		
Energies of Bohr orbits in hydrogen $E_n = -\frac{me^4}{32\pi^2\epsilon_0^2\hbar^2} \frac{1}{n^2} = \frac{-13.60 \text{ eV}}{2} (n = 1, 2, 3, \dots)$	6.5		

Questions

- Does the Thomson model fail at large scattering angles or at small scattering angles? Why?
- What principles of physics would be violated if we scattered a beam of alpha particles with a single impact parameter from a single target atom at rest?
- Could we use the Rutherford scattering formula to analyze the scattering of (a) protons incident on iron? (b) Alpha particles incident on lithium ($Z = 3$)? (c) Silver nuclei incident on gold? (d) Hydrogen atoms incident on gold? (e) Electrons incident on gold?
- What determines the angular range $d\theta$ in the alpha-particle scattering experiment (Figure 6.8)?
- Why didn't Bohr use the concept of de Broglie waves in his theory?
- In which Bohr orbit does the electron have the largest velocity? Are we justified in treating the electron nonrelativistically in that case?
- How does an electron in hydrogen get from $r = 4a_0$ to $r = a_0$ without being anywhere in between?
- How is the quantization of the energy in the hydrogen atom similar to the quantization of the systems discussed in Chapter 5? How is it different? Do the quantizations originate from similar causes?
- In a Bohr atom, an electron jumps from state n_1 , with angular momentum $n_1\hbar$, to state n_2 , with angular momentum $n_2\hbar$. How can an isolated system change its angular momentum? (In classical physics, a change in angular momentum requires an external torque.) Can the photon carry away the difference in angular momentum? Estimate the maximum angular momentum, relative to the center of the atom, that the photon can have. Does this suggest another failure of the Bohr model?
- The product $E_n r_n$ for the hydrogen atom is (1) independent of Planck's constant and (2) independent of the quantum number n . Does this observation have any significance? Is this a classical or a quantum effect?
- (a) How does a Bohr atom violate the position-momentum uncertainty relationship? (b) How does a Bohr atom violate the energy-time uncertainty relationship? (What is ΔE ? What does this imply about Δt ? What do you conclude about transitions between levels?)
- List the assumptions made in deriving the Bohr theory. Which of these are a result of neglecting small quantities? Which of these violate basic principles of relativity or quantum physics?
- List the assumptions made in deriving the Rutherford scattering formula. Which of these are a result of neglecting small quantities? Which of these violate basic principles of relativity or quantum physics?
- In both the Rutherford theory and the Bohr theory, we used the classical expression for the kinetic energy. Estimate the velocity of an electron in the Bohr atom and of an alpha particle in a typical scattering experiment, and decide whether the use of the classical formula is justified.
- In both the Rutherford theory and the Bohr theory, we neglected any wave properties of the particles. Estimate the de Broglie wavelength of an electron in a Bohr atom and compare it with the size of the atom. Estimate the de Broglie wavelength of an alpha particle and compare it with the size of the nucleus. Is the wave behavior expected to be important in either case?
- What is the distinction between binding energy and ionization energy? Between binding energy and excitation energy? If you were given the value of the binding energy of a level in hydrogen, could you find its excitation energy without knowing which level it is?
- Why are the decreases in current in the Franck-Hertz experiment not sharp?
- As indicated by the Franck-Hertz experiment, the first excited state of mercury is at an energy of 4.9 eV. Do you expect mercury to show absorption lines in the visible spectrum?
- Is the correspondence principle a necessary part of quantum physics or is it merely an accidental agreement of two formulas? Where do we draw the line between the world of quantum physics and the world of classical, nonquantum physics?

Problems

6.1 Basic Properties of Atoms

- Electrons in atoms are known to have kinetic energies in the range of a few electron-volts. Show that the uncertainty principle allows electrons of this energy to be confined in a region of space of the size of an atom (0.1 nm).

6.2 Scattering Experiments and the Thomson Model

- Consider an electron in Figure 6.1 subject to the electric field of the sphere of positive charge Ze in which it is

embedded. (a) Using Gauss's law, show that the electric field on the electron due to the positive charge is

$$E = \frac{1}{4\pi\epsilon_0} \frac{Ze}{R^3} r$$

(b) For this electric field, show that the force on the electron is given by Eq. 6.2.

- (a) Compute the oscillation frequency of the electron and the expected absorption or emission wavelength in

a Thomson-model hydrogen atom. Use $R = 0.053$ nm. Compare with the observed wavelength of the strongest emission and absorption line in hydrogen, 122 nm. (b) Repeat for sodium ($Z = 11$). Use $R = 0.18$ nm. Compare with the observed wavelength, 590 nm.

4. Consider the Thomson model for an atom with two electrons. Let the electrons be located along a diameter on opposite sides of the center of the sphere, each a distance x from the center. (a) Show that the configuration is stable if $x = R/2$. (b) Try to construct similar stable configurations for atoms with three, four, five, and six electrons.

6.3 The Rutherford Nuclear Atom

5. Alpha particles of kinetic energy 6.250 MeV are scattered at 90° by a gold foil. (a) What is the impact parameter? (b) What is the minimum distance between the alpha particles and the gold nucleus? (c) Find the kinetic and potential energies at that minimum distance.
6. How much kinetic energy must an alpha particle have before its distance of closest approach to a gold nucleus is equal to the nuclear radius (7.0×10^{-15} m)?
7. What is the distance of closest approach when alpha particles of kinetic energy 7.4 MeV are scattered by a thin copper foil?
8. Protons of energy 5.4 MeV are incident on a silver foil of thickness 3.6×10^{-6} m. What fraction of the incident protons is scattered at angles: (a) Greater than 90° ? (b) Greater than 10° ? (c) Between 5° and 10° ? (d) Less than 5° ?
9. Alpha particles of kinetic energy K are scattered either from a gold foil or a silver foil of identical thickness. What is the ratio of the number of particles scattered at angles greater than 90° by the gold foil to the same number for the silver foil?
10. The maximum kinetic energy given to the target nucleus will occur in a head-on collision with $b = 0$. (Why?) Estimate the maximum kinetic energy given to the target nucleus when 8.0-MeV alpha particles are incident on a gold foil. Are we justified in neglecting this energy?
11. The maximum kinetic energy that an alpha particle can transmit to an *electron* occurs during a head-on collision. Compute the kinetic energy lost by an alpha particle of kinetic energy 8.0 MeV in a head-on collision with an electron at rest. Are we justified in neglecting this energy in the Rutherford theory?
12. Alpha particles of energy 8.4 MeV are incident on a silver foil of thickness $6.5 \mu\text{m}$. For a certain value of the impact parameter, the alpha particles lose exactly half their incident kinetic energy when they reach their minimum separation from the nucleus. Find the minimum separation, the impact parameter, and the scattering angle.
13. Alpha particles of kinetic energy 6.0 MeV are incident at a rate of 3.0×10^7 per second on a gold foil of thickness 3.0×10^{-6} m. A circular detector of diameter 1.0 cm

is placed 12 cm from the foil at an angle of 30° with the direction of the incident alpha particles. At what rate does the detector measure scattered alpha particles?

14. An experiment is being done to observe scattering of 6.5-MeV alpha particles from a gold foil. A detector moves to different angles while keeping the same distance from the scattering foil. When the detector is at a 10° scattering angle, the counting rate is observed to be 11.3 per second. What would be the rate at 150° ?

6.4 Line Spectra

15. The shortest wavelength of the hydrogen Lyman series is 91.13 nm. Find the three longest wavelengths in this series.
16. One of the lines in the Brackett series (series limit = 1458 nm) has a wavelength of 1944 nm. Find the next higher and next lower wavelengths in this series.
17. The longest wavelength in the Pfund series is 7459 nm. Find the series limit.
18. A certain spectral line emitted by hydrogen has a wavelength of 1005 nm. What are the initial and final states for this line?

6.5 The Bohr Model

19. Find the electron's velocity, kinetic energy, and potential energy in the $n = 3$ state of hydrogen.
20. Use the Bohr theory to find the series wavelength limits of the Lyman and Paschen series of hydrogen.
21. (a) Show that the speed of an electron in the n th Bohr orbit of hydrogen is ac/n , where a is the fine structure constant. (b) What would be the speed in a hydrogenlike atom with a nuclear charge of Ze ?
22. Starting from the $n = 5$ state of hydrogen, to which states can the electron make transitions, and what are the energies of the emitted radiations?
23. Continue Figure 6.20, showing the transitions of the Paschen series and computing their energies and wavelengths.
24. A collection of hydrogen atoms in the ground state is illuminated with ultraviolet light of wavelength 59.0 nm. Find the kinetic energy of the emitted electrons.
25. Find the ionization energy of (a) the $n = 3$ level of hydrogen; (b) the $n = 2$ level of He^+ (singly ionized helium); (c) the $n = 4$ level of Li^{++} (doubly ionized lithium).
26. Use the Bohr formula to find the energy differences $E(n_1 \rightarrow n_2) = E_{n_1} - E_{n_2}$ and show that (a) $E(4 \rightarrow 2) = E(4 \rightarrow 3) + E(3 \rightarrow 2)$; (b) $E(4 \rightarrow 1) = E(4 \rightarrow 2) + E(2 \rightarrow 1)$. (c) Interpret these results based on the Ritz combination principle.
27. Find the shortest and the longest wavelengths of the Lyman series of singly ionized helium.
28. Draw an energy-level diagram showing the lowest four levels of singly ionized helium. Show all possible transitions from the levels and label each transition with its wavelength.

29. A long time ago, in a galaxy far, far away, electric charge had not yet been invented, and atoms were held together by gravitational forces. Compute the Bohr radius and the $n = 2$ to $n = 1$ transition energy in a gravitationally bound hydrogen atom.
30. An alternative development of the Bohr theory begins by assuming that the stationary states are those for which the circumference of the orbit is an integral number of de Broglie wavelengths. (a) Show that this condition leads to standing de Broglie waves around the orbit. (b) Show that this condition gives the angular momentum condition, Eq. 6.26, used in the Bohr theory.

6.6 The Franck–Hertz Experiment

31. A hypothetical atom has only two excited states, at 4.0 and 7.0 eV, and has a ground-state ionization energy of 9.0 eV. If we used a vapor of such atoms for the Franck–Hertz experiment, for what voltages would we expect to see decreases in the current? List all voltages up to 20 V.
32. The first excited state of sodium decays to the ground state by emitting a photon of wavelength 590 nm. If sodium vapor is used for the Franck–Hertz experiment, at what voltage will the first current drop be recorded?
33. Figure 6.25 shows the result of the Franck–Hertz experiment using argon gas in the tube. Analyze the data to find the best value for the energy of the excited state, and compare with known values of the energy states in argon. Based on the argon energy states, would you expect argon to emit visible lines (like neon in Figure 6.23) when the atoms are excited?

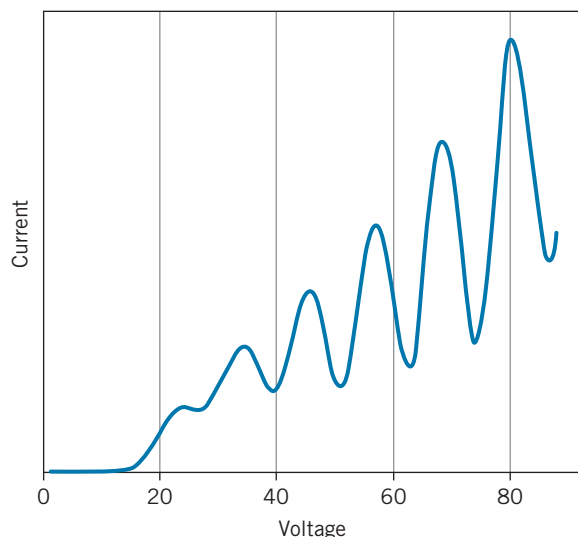


FIGURE 6.25 Problem 33.

6.7 The Correspondence Principle

34. Suppose all of the excited levels of hydrogen had lifetimes of 10^{-8} s. As we go to higher and higher excited states, they get closer and closer together, and soon they are so close in energy that the energy uncertainty of each state becomes as large as the energy spacing between states, and we can no longer resolve individual states. Find the value of n for which this occurs. What is the radius of such an atom?
35. Compare the frequency of revolution of an electron with the frequency of the photons emitted in transitions from n to $n - 1$ for (a) $n = 10$; (b) $n = 100$; (c) $n = 1000$; and (d) $n = 10,000$.

6.8 Deficiencies of the Bohr Model

36. What is the difference in wavelength between the first line of the Balmer series in ordinary hydrogen ($M = 1.007825$ u) and in “heavy” hydrogen ($M = 2.014102$ u)?

General Problems

37. A hydrogen atom is in the $n = 6$ state. (a) Counting all possible paths, how many different photon energies can be emitted if the atom ends up in the ground state? (b) Suppose only $\Delta n = 1$ transitions were allowed. How many different photon energies would be emitted? (c) How many different photon energies would occur in a Thomson-model hydrogen atom?
38. An electron is in the $n = 8$ level of ionized helium. (a) Find the three longest wavelengths that are emitted when the electron makes a transition from the $n = 8$ level to a lower level. (b) Find the shortest wavelength that can be emitted. (c) Find the three longest wavelengths at which the electron in the $n = 8$ level will absorb a photon and move to a higher state, if we could somehow keep it in that level long enough to absorb. (d) Find the shortest wavelength that can be absorbed.
39. The lifetimes of the levels in a hydrogen atom are of the order of 10^{-8} s. Find the energy uncertainty of the first excited state and compare it with the energy of the state.
40. The following wavelengths are found among the many radiations emitted by singly ionized helium: 24.30 nm, 25.63 nm, 102.5 nm, and 320.4 nm. If we group the transitions in helium as we did in hydrogen by identifying the final state n_0 and initial state n , to which series does each transition belong?
41. Adjacent wavelengths 72.90 nm and 54.00 nm are found in one series of transitions among the radiations emitted by doubly ionized lithium. Find the value of n_0 for this series and find the next wavelength in the series.
42. When an atom emits a photon in a transition from a state of energy E_1 to a state of energy E_2 , the photon energy is not precisely equal to $E_1 - E_2$. Conservation of momentum requires that the atom must recoil, and so some energy must go into recoil kinetic energy K_R .

Show that $K_R \cong (E_1 - E_2)^2 / 2Mc^2$, where M is the mass of the atom. Evaluate this recoil energy for the $n = 2$ to $n = 1$ transition of hydrogen.

43. In a *muonic atom*, the electron is replaced by a negatively charged particle called the *muon*. The muon mass is 207 times the electron mass. (a) Ignoring the correction for finite nuclear mass, what is the shortest wavelength of the Lyman series in a muonic hydrogen atom? In what region of the electromagnetic spectrum does this belong? (b) How large is the correction for the finite nuclear mass in this case? (See the discussion at the beginning of Section 6.8.)
44. Consider an atom in which the single electron is replaced by a negatively charged muon ($m_\mu = 207m_e$). What is the radius of the first Bohr orbit of a muonic lead atom ($Z = 82$)? Compare with the nuclear radius of about 7 fm.
45. Protons are incident on a copper foil 12 μm thick. (a) What should be the proton kinetic energy in order that the distance of closest approach equal the nuclear radius (5.0 fm)? (b) If the proton energy were 7.5 MeV, what is the impact parameter for scattering at 120° ? (c) What is the minimum distance between proton and nucleus for this case? (d) What fraction of the protons is scattered beyond 120° ?
46. Atoms with very high quantum numbers, so that the atom is laboratory sized, are known as *Rydberg atoms*. (a) What would be the quantum number of a hydrogen atom with a radius of 1 μm ? (b) What would be the ionization energy of this atom? (c) What would be the wavelength of the transition to the next lowest level?
47. The Humphreys series in hydrogen was discovered in 1953 by American physicist Curtis Humphreys. It is the next series above the Pfund series, consisting of transitions that terminate at the $n = 6$ level. Find the series limit and the three longest wavelengths of the Humphreys series. In what region of the electromagnetic spectrum do these occur?
48. Alpha particles are scattered from a thin aluminum foil. At what alpha-particle kinetic energy would you expect the Rutherford formula to fail because the alpha particle enters the aluminum nucleus? Assume $r = 1.9$ fm for the alpha particle and 3.6 fm for the aluminum nucleus.